



RTL Design Sherpa

RTL Common Library Module Reference — Rev 0.90

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1 RTLCommon Library Overview

The RTLCommon library provides a comprehensive collection of fundamental digital design building blocks, offering high-quality, parameterizable, and synthesis-friendly RTL modules for use across a wide range of digital systems.

1.1 Library Philosophy

1.1.1 Design Principles

The RTLCommon library is built on the following core principles:

Modularity: Each module focuses on a specific function and can be easily composed with others

Parameterizability: Configurable width and functionality parameters for maximum reusability

Synthesis Friendly: Optimized for modern synthesis tools with predictable area/timing results

Technology Independence: Clean RTL that works across ASIC and FPGA technologies

Comprehensive Coverage: From basic gates to complex arithmetic and control structures

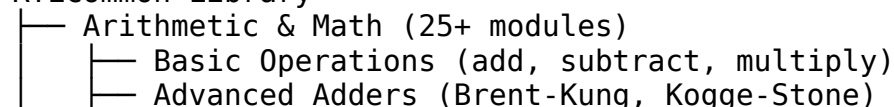
1.1.2 Quality Standards

- **Lint Clean:** All modules pass rigorous RTL linting checks
- **Simulation Verified:** Comprehensive testbenches validate functionality
- **Synthesis Proven:** Validated across multiple technology nodes and vendors
- **Documentation Complete:** Detailed documentation with usage examples

1.2 Architecture Overview

1.2.1 Module Hierarchy

RTLCommon Library



└─	Specialized Math Functions
─	Data Integrity (6 modules)
└─	Error Detection (CRC, Checksum, Parity)
└─	Error Correction (Hamming ECC)
─	Control & Arbitration (4 modules)
└─	Round-Robin Arbiters
└─	Priority Arbiters
─	Clock & Reset (4 modules)
└─	Clock Management (ICG, Dividers)
└─	Reset Synchronization
─	Counters & Sequences (6 modules)
└─	Binary & Gray Code Counters
└─	Specialized Counting
─	Data Conversion (8 modules)
└─	Code Converters (Binary/Gray/BCD)
└─	Display Decoders
─	Bit Operations (5 modules)
└─	Bit Searching & Counting
└─	Vector Manipulation
─	Shift & LFSR (4 modules)
└─	Barrel Shifters
└─	Pseudorandom Generation
─	Memory & Storage (4 modules)
└─	FIFO Implementations
└─	Content Addressable Memory
─	Utilities (2 modules)
└─	PWM Generation
└─	Sorting Algorithms

1.3 Functional Categories

1.3.1 1. Arithmetic and Mathematical Operations

The arithmetic category provides the foundation for digital signal processing and general computation.

1.3.1.1 Basic Arithmetic Building Blocks

- **Full Adders:** Single-bit and multi-bit addition with carry propagation
- **Half Adders:** Simplified 2-input addition without carry-in
- **Subtractors:** Complement-based subtraction with borrow handling
- **Add/Subtract Units:** Combined addition/subtraction functionality

1.3.1.2 High-Performance Parallel Adders

- **Kogge-Stone Adders:** Minimal depth parallel prefix adders for maximum speed
- **Brent-Kung Adders:** Area-efficient parallel prefix adders with reasonable speed
- **Carry-Lookahead:** Fast carry generation for moderate bit widths
- **Carry-Save:** Multiple operand addition with reduced carry propagation

1.3.1.3 Multiplication Architectures

- **Wallace Tree:** Fast parallel multiplier using carry-save addition
- **Dadda Tree:** Optimized multiplier with reduced partial product stages
- **Basic Cell:** Fundamental multiplication building blocks

Design Considerations: - Speed vs Area trade-offs clearly documented - Power consumption characteristics provided - Technology-specific optimization guidance included

1.3.2 2. Data Integrity and Error Management

Critical for reliable data storage and transmission in modern digital systems.

1.3.2.1 Cyclic Redundancy Check (CRC)

- **Generic CRC Engine:** Configurable polynomial, width, and initialization
- **Cascade Implementation:** High-throughput parallel CRC processing
- **XOR-Shift Variants:** Optimized for specific polynomial types

1.3.2.2 Error Correction Coding (ECC)

- **Hamming SEDED:** Single Error Correction, Double Error Detection
- **Configurable Width:** Support for various data widths
- **Encoder/Decoder Pairs:** Complete ECC implementation

1.3.2.3 Simple Integrity Checks

- **Checksum:** Software-compatible checksum algorithms
- **Parity:** Even/odd parity generation and checking

Key Features: - Standards-compliant implementations - Parameterizable for various data widths - Optimized for both latency and throughput scenarios

1.3.3 3. Control Logic and Arbitration

Essential components for managing shared resources and system coordination.

1.3.3.1 Round-Robin Arbitration

- **Simple Round-Robin:** Basic fair arbitration with minimal overhead
- **Weighted Round-Robin:** Priority-based fair arbitration
- **Advanced Features:** Configurable agent counts and weighting

1.3.3.2 Priority-Based Control

- **Priority Encoders:** Fixed priority arbitration schemes
- **Configurable Priority:** Runtime priority adjustment capability

Applications: - Bus arbitration in multi-master systems - Resource allocation in processors - Quality of Service (QoS) implementations

1.3.4 4. Clock and Reset Management

Power and timing management for robust digital system design.

1.3.4.1 Clock Control

- **Integrated Clock Gating (ICG):** Power optimization through controlled clock disabling
- **Clock Dividers:** Frequency synthesis and clock domain generation
- **Pulse Generators:** Controlled clock pulse generation

1.3.4.2 Reset and Synchronization

- **Reset Synchronizers:** Safe reset assertion/deassertion
- **Glitch-Free Logic:** Reliable signal conditioning
- **Debouncing:** Input signal stabilization

Power Management Benefits: - Up to 40% dynamic power reduction with clock gating - Proper reset sequencing for reliable startup - Signal integrity maintenance

1.3.5 5. Counting and Sequence Generation

Fundamental building blocks for timing, addressing, and control.

1.3.5.1 Binary Counters

- **Basic Counters:** Up/down counting with enable/reset
- **Gray Code Counters:** Glitch-free counting for async interfaces
- **Johnson Counters:** Shift-based sequence generation

1.3.5.2 Specialized Counting

- **Frequency-Invariant:** Clock-independent counting
- **Load/Clear:** Programmable counter initialization
- **Ring Counters:** One-hot sequence generation

Design Features: - Parameterizable width and functionality - Optional synchronous/asynchronous controls - Overflow/underflow detection

1.3.6 6. Data Conversion and Encoding

Code conversion and display interfacing utilities.

1.3.6.1 Number System Conversion

- **Binary ↔ Gray:** Safe async counter interfacing
- **Binary → BCD:** Decimal display conversion
- **Johnson Gray:** Specialized counting applications

1.3.6.2 Encoding/Decoding

- **Priority Encoders:** Multiple input to binary conversion
- **Binary Decoders:** Address decoding and control
- **7-Segment Display:** Hexadecimal character display

Applications: - Display interfaces - Address decoding - Safe async signal crossing

1.3.7 7. Bit Manipulation and Searching

Efficient bit-level operations for high-performance computing.

1.3.7.1 Bit Detection

- **Leading Zero Count:** Normalization and floating-point support
- **Find First/Last Set:** Efficient bit searching

- **Priority Detection:** Leading one and trailing one detection

1.3.7.2 Vector Operations

- **Bit Reversal:** Endianness conversion and FFT applications
- **Vector Manipulation:** Bit-level data reorganization

Performance Characteristics: - Logarithmic complexity algorithms where possible - Optimized for synthesis tool recognition - Support for various data widths

1.3.8 8. Shift Operations and LFSRs

Shift-based operations and pseudorandom sequence generation.

1.3.8.1 Shifters

- **Barrel Shifters:** Arbitrary shift/rotate in single cycle
- **Universal Shifters:** Combined left/right, logical/arithmetic shifting

1.3.8.2 Linear Feedback Shift Registers (LFSRs)

- **Fibonacci LFSRs:** External feedback topology
- **Galois LFSRs:** Internal feedback topology
- **Configurable Polynomials:** Various sequence lengths and properties

Applications: - Pseudorandom number generation - Built-in self-test (BIST) patterns - Encryption and scrambling

1.3.9 9. Memory and Storage

Buffering and storage solutions for data flow management.

1.3.9.1 FIFO Implementations

- **Synchronous FIFO:** Single clock domain buffering
- **Asynchronous FIFO:** Clock domain crossing with Gray code pointers
- **Specialized FIFOs:** Optimized for specific clock relationships

1.3.9.2 Associative Memory

- **Content Addressable Memory (CAM):** Tag matching and lookup

Features: - Configurable depth and width - Full/empty flag generation - Error handling and overflow protection

1.3.10 10. Utility and Specialized Functions

Specialized functions for specific application domains.

1.3.10.1 *Signal Generation*

- **PWM:** Pulse width modulation for analog control
- **Timing Control:** Precise pulse and timing generation

1.3.10.2 *Data Processing*

- **Hardware Sorting:** Parallel sorting implementations
- **Algorithmic Processing:** Specialized computational functions

1.4 Performance and Optimization

1.4.1 Synthesis Guidelines

1.4.1.1 *Area Optimization*

- Use ripple-carry arithmetic for low-area requirements
- Select appropriate FIFO depths based on data flow analysis
- Consider shared resources for infrequently used functions

1.4.1.2 *Speed Optimization*

- Use parallel prefix adders (Kogge-Stone) for critical paths
- Implement pipelining in high-throughput applications
- Utilize barrel shifters for single-cycle shift operations

1.4.1.3 *Power Optimization*

- Apply clock gating (ICG) to inactive functional blocks
- Use Gray code counters for reduced switching activity
- Implement frequency-invariant designs where applicable

1.4.2 Technology-Specific Considerations

1.4.2.1 *ASIC Implementation*

- Leverage dedicated adder and multiplier cells

- Use technology-specific clock gating cells
- Optimize for specific voltage and temperature ranges

1.4.2.2 FPGA Implementation

- Utilize DSP blocks for arithmetic operations
- Use BRAM for deep FIFOs
- Leverage dedicated clock management resources

1.5 Verification and Validation

1.5.1 Testbench Coverage

- **Functional Coverage:** All operational modes and edge cases
- **Directed Tests:** Specific functionality verification
- **Random Testing:** Stress testing with random inputs
- **Corner Cases:** Boundary conditions and error scenarios

1.5.2 Compliance Testing

- **Timing Analysis:** Setup/hold requirements validation
- **Power Analysis:** Dynamic and static power characterization
- **Area Analysis:** Resource utilization across technologies

1.6 Usage Guidelines

1.6.1 Module Selection

1. **Identify Requirements:** Performance, area, and power constraints
2. **Select Architecture:** Choose appropriate implementation (ripple vs. parallel, etc.)
3. **Configure Parameters:** Set widths and options for specific application
4. **Verify Integration:** Ensure proper interfacing and timing

1.6.2 Integration Best Practices

// Example: Building a complete arithmetic unit

```
module arithmetic_unit #(
    parameter int DATA_WIDTH = 32
) (
    input logic clk,
    input logic rst_n,
    input logic [DATA_WIDTH-1:0] operand_a,
    input logic [DATA_WIDTH-1:0] operand_b,
    input logic [1:0] operation,
    output logic [DATA_WIDTH-1:0] result,
    output logic overflow
);

    // Use appropriate math modules based on requirements
    logic [DATA_WIDTH-1:0] add_result, sub_result, mult_result;
    logic add_carry, sub_borrow;

    // High-speed addition using parallel prefix
    .N(DATA_WIDTH)
) u_adder (
    .a(operand_a), .b(operand_b), .cin(1'b0),
    .sum(add_result), .cout(add_carry)
);

    // Subtraction using complement method
    math_subtractor_full_nbit #(
        .N(DATA_WIDTH)
    ) u_subtractor (
        .a(operand_a), .b(operand_b), .bin(1'b0),
        .diff(sub_result), .bout(sub_borrow)
    );

    // Result selection based on operation
    always_comb begin
        case (operation)
            2'b00: begin result = add_result; overflow = add_carry;
end
            2'b01: begin result = sub_result; overflow = sub_borrow;
end
            default: begin result = '0; overflow = 1'b0; end
        endcase
    end

endmodule
```

1.7 Future Enhancements

1.7.1 Planned Additions

- **DSP Functions:** FFT, filtering, and signal processing primitives
- **Advanced Arithmetic:** Division, square root, and trigonometric functions
- **Security Primitives:** Hardware security modules and cryptographic functions
- **AI/ML Accelerators:** Matrix operations and neural network primitives

1.7.2 Technology Evolution

- **Emerging Technologies:** Support for new process nodes and device types
- **Tool Integration:** Enhanced synthesis and verification tool compatibility
- **Standards Compliance:** Updates for evolving industry standards

The RTLCommon library represents a comprehensive foundation for digital design, providing the building blocks necessary for implementing complex digital systems with confidence in their correctness, performance, and manufacturability.

2 Priority Encoder Arbiter

A high-speed combinational priority encoder optimized for arbitration applications, implementing fixed-priority arbitration where lower-indexed clients always have higher priority.

2.1 Overview

The `arbiter_priority_encoder` module provides single-cycle priority encoding with optimized implementations for common client counts (4, 8, 16, 32) using fully unrolled casez logic. For other client counts, it uses a synthesizable loop-based approach. The module supports dual request inputs (masked and unmasked) for use in sophisticated arbitration schemes.

2.2 Module Declaration

```
module arbiter_priority_encoder #(
    parameter int CLIENTS = 4,           // Number of requesting
clients
    parameter int N = $clog2(CLIENTS)    // Winner ID width (derived)
) (
    input logic [CLIENTS-1:0] requests_masked,    // Masked
request vector
    input logic [CLIENTS-1:0] requests_unmasked,  // Unmasked

```

```

request vector
  input logic          any_masked_requests, // Select masked
vs unmasked
  output logic [N-1:0] winner,             // Binary-
encoded winner ID
  output logic         winner_valid        // Winner output
is valid
);

```

2.3 Parameters

2.3.1 User-Settable Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Number of requesting clients (range: 2-1024)

2.3.2 Derived Localparam (Computed Internally)

Localparam	Computation	Description
N	\$clog2(CLIENTS)	Winner ID width in bits

Note: N is computed automatically and cannot be overridden by users.

2.4 Ports

2.4.1 Inputs

Port	Width	Description
requests_masked	CLIENTS	Masked request vector (higher priority source)
requests_unmasked	CLIENTS	Unmasked request vector (fallback source)
any_masked_requests	1	Select masked (1) or unmasked (0) requests

2.4.2 Outputs

Port	Width	Description
winner	N	Binary-encoded

Port	Width	Description
winner_valid	1	winner client ID Asserted when at least one request is active

2.5 Functionality

2.5.1 Priority Order

The module implements **fixed priority** arbitration: - **Client 0**: Highest priority (always wins if requesting) - **Client 1**: Second highest priority - ... - **Client CLIENTS-1**: Lowest priority

2.5.2 Request Selection

The module evaluates requests in priority order:

1. If **any_masked_requests** = 1: Use requests_masked input
2. If **any_masked_requests** = 0: Use requests_unmasked input
3. Find **lowest-indexed asserted bit** in selected request vector
4. If **no requests**: winner = 0 (don't care), winner_valid = 0

2.5.3 Combinational Operation

- **Zero cycles latency**: Purely combinational logic
- **No clock or reset**: Outputs update immediately with input changes
- **Single-cycle response**: Winner determined in one combinational path

2.6 Implementation Details

2.6.1 Optimized Implementations

The module uses different implementations based on client count:

Power-of-2 Counts (4, 8, 16, 32):

```
// Fully unrolled casez logic for maximum performance
always_comb begin
    casez (w_priority_requests)
        4'b???1: begin winner = 2'd0; winner_valid = 1'b1; end //
Client 0
        4'b???10: begin winner = 2'd1; winner_valid = 1'b1; end //
```



```

Client 1
    4'b?100: begin winner = 2'd2; winner_valid = 1'b1; end //
Client 2
    4'b1000: begin winner = 2'd3; winner_valid = 1'b1; end //
Client 3
    default: begin winner = 2'd0; winner_valid = 1'b0; end
    endcase
end

```

Other Client Counts:

```

// Generic loop-based implementation
always_comb begin
    winner = '0;
    winner_valid = 1'b0;

    // Find lowest-indexed requesting client
    for (int i = 0; i < CLIENTS; i++) begin
        if (w_priority_requests[i] && !winner_valid) begin
            winner = i[N-1:0];
            winner_valid = 1'b1;
        end
    end
end

```

2.6.2 Priority Selection Logic

```

// Internal priority request selection
logic [CLIENTS-1:0] w_priority_requests;
assign w_priority_requests = any_masked_requests ?
    requests_masked :
    requests_unmasked;

```

2.7 Performance Characteristics

2.7.1 Timing

Metric	Optimized (4,8,16,32)	Generic (<64)	Generic (>64)
Latency	0 cycles (combinational)	0 cycles	0 cycles
Propagation Delay	1-2 LUT levels	2-3 LUT levels	3-4 LUT levels
Max Frequency	>500 MHz (FPGA)	>400 MHz	>300 MHz

2.7.2 Resource Utilization

Client Count	Implementation	LUTs (Est.)	Area Complexity
4	Optimized	~8	O(CLIENTS)
8	Optimized	~16	O(CLIENTS)
16	Optimized	~32	O(CLIENTS)
32	Optimized	~64	O(CLIENTS)
Other	Generic	~CLIENTS×log(CLIENTS)	O(CLIENTS×log(CLIENTS))

2.8 Usage Examples

2.8.1 Standalone Priority Encoder

```
// Simple fixed-priority arbiter (single request input)
arbiter_priority_encoder #(
    .CLIENTS(8)
) u_pri_enc (
    .requests_masked    (8'b0),           // Unused
    .requests_unmasked (client_reqs),     // Primary request vector
    .any_masked_requests(1'b0),          // Always use unmasked
    .winner             (winner_id),
    .winner_valid       (grant_valid)
);

// Convert winner ID to one-hot grant signal
logic [7:0] grant_one_hot;
always_comb begin
    grant_one_hot = '0;
    if (grant_valid) begin
        grant_one_hot[winner_id] = 1'b1;
    end
end
```

2.8.2 Dual-Input Priority Encoder

```
// Masked + unmasked request inputs
arbiter_priority_encoder #(
    .CLIENTS(16)
) u_pri_enc_dual (
    .requests_masked    (masked_reqs),    // Higher priority requests
    .requests_unmasked (all_reqs),        // All requests
    .any_masked_requests(|masked_reqs),   // Select masked if any
present
    .winner             (winner_id),

```

```

        .winner_valid      (grant_valid)
    );

```

2.8.3 Integration in Round-Robin Arbiter

```

// Used within arbiter_round_robin.sv
// Masked requests = clients above last winner
// Unmasked requests = all clients
arbiter_priority_encoder #(
    .CLIENTS(CLIENTS)
) u_priority_encoder (
    .requests_masked      (w_requests_masked),
    .requests_unmasked    (w_requests_unmasked),
    .any_masked_requests  (w_any_masked_requests),
    .winner                (w_winner),
    .winner_valid         (w_winner_valid)
);

```

2.8.4 Multi-Level Priority Groups

```

// Implement priority groups: High (C0-C3), Med (C4-C7), Low (C8-C11)
logic [3:0] high_reqs, med_reqs, low_reqs;
logic [1:0] high_winner, med_winner, low_winner;
logic high_valid, med_valid, low_valid;

// Separate encoders for each priority group
arbiter_priority_encoder #(.CLIENTS(4)) u_high_pri (
    .requests_masked(4'b0),
    .requests_unmasked(high_reqs),
    .any_masked_requests(1'b0),
    .winner(high_winner),
    .winner_valid(high_valid)
);

arbiter_priority_encoder #(.CLIENTS(4)) u_med_pri (
    .requests_masked(4'b0),
    .requests_unmasked(med_reqs),
    .any_masked_requests(1'b0),
    .winner(med_winner),
    .winner_valid(med_valid)
);

arbiter_priority_encoder #(.CLIENTS(4)) u_low_pri (
    .requests_masked(4'b0),
    .requests_unmasked(low_reqs),
    .any_masked_requests(1'b0),
    .winner(low_winner),
    .winner_valid(low_valid)
);

```

```

// Select winner from highest priority group with requests
logic [3:0] final_winner;
always_comb begin
    if (high_valid)    final_winner = {2'b00, high_winner};
    else if (med_valid) final_winner = {2'b01, med_winner};
    else if (low_valid) final_winner = {2'b10, low_winner};
    else               final_winner = 4'b0;
end

```

2.9 Design Considerations

2.9.1 When to Use

✓ **Appropriate Use Cases:** - Fixed-priority arbitration systems - Interrupt controllers - Bus arbitration with priority levels - Resource allocation with service levels - Component within round-robin or weighted arbiters

2.9.2 Performance Optimization

For High-Speed Designs: - Use power-of-2 client counts (4, 8, 16, 32) for optimized implementation - Pipeline outputs if critical path is too long - Consider hierarchical encoding for >32 clients

For Area Optimization: - Non-power-of-2 counts use smaller generic implementation - Share encoder between multiple arbitration paths - Use smallest practical CLIENTS parameter

2.9.3 Synthesis Considerations

Timing Optimization:

```

// Pipeline option for high-speed designs
logic [N-1:0] r_winner;
logic r_winner_valid;

always_ff @(posedge clk) begin
    r_winner <= winner;
    r_winner_valid <= winner_valid;
end

```

Resource Optimization: - Optimized implementations use casez (efficient mux tree) - Generic implementation uses priority loop (more logic levels) - For CLIENTS > 32, consider hierarchical structure

2.10 Priority Arbitration Theory

2.10.1 Fixed Priority vs Round-Robin

Aspect	Fixed Priority	Round-Robin
Fairness	No (low-indexed clients favored)	Yes (all clients equal)
Latency	Deterministic for high-priority	Variable
Starvation	Possible for low-priority	Prevented
Simplicity	Very simple (this module)	More complex
Best Use	Interrupt handling, critical tasks	Data paths, fair scheduling

2.10.2 Priority Inversion

Issue: High-priority client blocked by low-priority client holding resource

Mitigation Strategies: 1. **Priority Inheritance:** Temporarily boost blocked client's priority 2. **Resource Preemption:** Force low-priority client to release resource 3. **Priority Ceiling:** Limit maximum priority of resource holders

2.11 Example Priority Scenarios

2.11.1 Scenario 1: Single Requester (Client 2)

```
requests_unmasked = 8'b0000_0100  
any_masked_requests = 0
```

```
→ winner = 3'd2  
→ winner_valid = 1
```

2.11.2 Scenario 2: Multiple Requesters (Clients 0, 2, 5)

```
requests_unmasked = 8'b0010_0101  
any_masked_requests = 0
```

```
→ winner = 3'd0 (lowest index wins)  
→ winner_valid = 1
```

2.11.3 Scenario 3: Masked Priority (Clients 3, 6 masked)

```
requests_masked    = 8'b0100_1000  
requests_unmasked = 8'b0110_1101  
any_masked_requests = 1
```

→ winner = 3'd3 (from masked, index 3 < 6)
→ winner_valid = 1

2.11.4 Scenario 4: No Requests

requests_unmasked = 8'b0000_0000
any_masked_requests = 0

→ winner = 3'd0 (don't care)
→ winner_valid = 0

2.12 Verification Notes

From comprehensive test suite (val/common/test_arbiter_priority_encoder.py):

- **Coverage:** 95%
- **Key Test Scenarios:**
 - Priority order verification (all bit patterns)
 - Masked vs unmasked selection
 - No requests (winner_valid = 0)
 - All clients requesting simultaneously
 - Single client at a time
 - Power-of-2 sizes (4, 8, 16, 32)
 - Non-power-of-2 sizes (5, 12)

Test Command: `pytest val/common/test_arbiter_priority_encoder.py -v`

2.13 Common Pitfalls

x Anti-Pattern 1: Assuming fairness

*// WRONG: Fixed priority is NOT fair
// Client 0 will always win if requesting*

x Anti-Pattern 2: Ignoring winner_valid

*// WRONG: Using winner when no requests
if (winner == 2'd0) ... // Ambiguous: valid client 0 or no requests?*

*// RIGHT: Check winner_valid first
if (winner_valid && winner == 2'd0) ...*

✗ **Anti-Pattern 3:** Mixing priority schemes

```
// WRONG: Trying to implement round-robin with priority encoder alone
// Use arbiter_round_robin.sv instead
```

2.14 Related Modules

- **arbiter_round_robin.sv** - Uses this module internally for winner selection
- **arbiter_round_robin_simple.sv** - Alternative simplified arbiter
- **arbiter_round_robin_weighted.sv** - Weighted arbitration with QoS
- **encoder.sv** - General-purpose binary encoder
- **priority_encoder.sv** - Alternative naming (may be same module)

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3 Round Robin Arbiter

3.1 Purpose

The `arbiter_round_robin` module implements a fair round-robin arbitration scheme for multiple clients. It ensures that all requesting clients receive equal access to a shared resource by cycling through them in order, preventing starvation of any client.

3.2 Parameters

- **CLIENTS**: Number of clients (default: 4)
- **WAIT_GNT_ACK**: When set to 1, waits for grant acknowledgment before moving to next client (default: 0)
- **N**: Internal parameter for address width calculation ($\lceil \log_2(\text{CLIENTS}) \rceil$)

3.3 Ports

3.3.1 Inputs

- **clk**: System clock

- `rst_n`: Active-low asynchronous reset
- `block_arb`: When asserted, blocks all arbitration (forces no grants)
- `request[CLIENTS-1:0]`: Request signals from each client
- `grant_ack[CLIENTS-1:0]`: Grant acknowledgment signals (used when `WAIT_GNT_ACK = 1`)

3.3.2 Outputs

- `grant_valid`: Indicates when a valid grant is being issued
- `grant[CLIENTS-1:0]`: One-hot encoded grant signals
- `grant_id[N-1:0]`: Binary encoded ID of the granted client
- `last_grant[CLIENTS-1:0]`: Previous cycle's grant (for debugging/history)

3.4 Key Internal Signals

- `r_last_grant_id`: Tracks the last winner's client ID (smaller than full mask)
- `r_last_valid`: Indicates if last winner should be used for mask generation
- `r_pending_ack`: ACK mode state (indicates ACK pending)
- `r_pending_client`: Which client has pending ACK (only in ACK mode)
- `w_requests_masked`: Requests with priority mask applied
- `w_requests_unmasked`: Raw gated requests without masking

3.5 Implementation Details

3.5.1 Priority Mechanism

The arbiter uses a pre-computed mask lookup table approach:

1. **Mask Lookup Tables**: Pre-computed at elaboration time (no logic cost)
 - `w_mask_decode[i]`: Mask for clients 0 through i (give priority after i)
 - `w_win_mask_decode[i]`: Mask to give priority to clients above i+1
2. **Fast Request Preprocessing**: Single LUT level for request gating

- Block arbitration immediately gates all requests
- Masked and unmasked request vectors computed in parallel

3. Last Winner Tracking: Uses client ID instead of full one-hot mask

- `r_last_grant_id`: More efficient than storing full grant vector
- `r_last_valid`: Indicates if mask should be applied

3.5.2 Priority Encoder

Uses `arbiter_priority_encoder` submodule for winner selection: - Takes both masked and unmasked request vectors - Returns binary-encoded winner ID - Outputs validity signal

3.5.3 Winner Selection Logic

```
// Priority encoder selects highest priority requester
arbiter_priority_encoder #(.CLIENTS(CLIENTS), .N(N))
u_priority_encoder (
    .requests_masked      (w_requests_masked),
    .requests_unmasked    (w_requests_unmasked),
    .any_masked_requests  (w_any_masked_requests),
    .winner               (w_winner),
    .winner_valid         (w_winner_valid)
);

// Grant decision with ACK permission check
assign w_should_grant = w_winner_valid && w_any_requests &&
w_can_grant;
```

3.5.4 Mask Update Logic

- **No-ACK Mode:** Mask updates immediately when grant issued (1-cycle round-robin)
- **ACK Mode:** Mask updates only when ACK received (prevents premature rotation)

3.6 Special Features

3.6.1 Block Arbitration

When `block_arb` is asserted, all requests are masked to zero, effectively disabling arbitration.

3.6.2 Grant Acknowledgment Support

When `WAIT_GNT_ACK` = 1, the arbiter waits for the granted client to acknowledge receipt before updating internal state and moving to the next client.

3.6.3 Fair Round-Robin Operation

The algorithm ensures fairness by: 1. Serving lower-indexed clients first within each round 2. After serving a client, giving priority to all lower-indexed clients 3. When no lower-indexed clients are requesting, wrapping around to serve from the top

3.7 Usage Notes

- The arbiter prioritizes lower-indexed clients when multiple requests arrive simultaneously
- The round-robin nature ensures long-term fairness across all clients
- Grant acknowledgment feature is useful in systems where the granted client needs time to process the grant

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4 arbiter_round_robin_simple

A generic rotating-priority arbiter with masking/rotation that provides fair arbitration among multiple requesting agents without using if/case ladders in the priority path.

4.1 Overview

The `arbiter_round_robin_simple` module implements a round-robin arbitration scheme that ensures fair access to a shared resource among N requesting agents. It maintains fairness by remembering the last granted agent and starting the next arbitration from the next agent in sequence.

4.2 Module Declaration

```
module arbiter_round_robin_simple #(
    parameter int unsigned N = 4
) (
    input logic clk,
    input logic rst_n,           // active-low reset
    input logic [N-1:0] request, // request bits [N-1:0]
    output logic grant_valid,    // any grant
    output logic [N-1:0] grant,  // one-hot grant
    output logic [$clog2(N)-1:0] grant_id // encoded grant
)
```

```
(undef if grant_valid==0)
);
```

4.3 Parameters

Parameter	Type	Default	Description
N	int unsigned	4	Number of requesting agents (N >= 1)

4.4 Ports

4.4.1 Inputs

Port	Width	Description
clk	1	Clock signal
rst_n	1	Active-low asynchronous reset
request	N	Request bits from N agents [N-1:0]

4.4.2 Outputs

Port	Width	Description
grant_valid	1	Indicates if any grant is active
grant	N	One-hot grant output [N-1:0]
grant_id	\$clog2(N)	Binary encoded grant ID (undefined if grant_valid==0)

4.5 Functionality

4.5.1 Round-Robin Algorithm

The arbiter implements a rotating priority scheme using the following approach:

1. **State Memory:** Maintains `r_last_grant` register storing the last granted agent ID
2. **Rotation:** Rotates the request vector to prioritize agents after the last granted one

3. **Priority Selection:** Uses lowest-set-bit isolation ($x \& (\sim x + 1)$) for efficient priority selection
4. **Restoration:** Rotates the result back to original bit positions

4.5.2 Key Features

- **Fair Arbitration:** Ensures all requesting agents get equal opportunity over time
- **Efficient Logic:** Uses bit manipulation instead of complex if/case structures
- **Single-Cycle Response:** Combinational grant generation with registered state update
- **Scalable:** Parameterizable for any number of agents ($N \geq 1$)

4.6 Implementation Details

4.6.1 Rotation Logic

```
// Shift amount = last_grant + 1 (mod N)
logic [$clog2(N)-1:0] w_shift_amount;
assign w_shift_amount = (r_last_grant == ($clog2(N))'(N-1)) ? '0 :
(r_last_grant + 1);

// Rotate-left request by w_shift_amount
if (w_shift_amount == '0) begin
    w_rot_req = request;
end else begin
    w_rot_req = (request << w_shift_amount) | (request >>
(($clog2(N))'(N) - w_shift_amount));
end
```

4.6.2 Priority Selection

The module uses lowest-set-bit isolation for efficient one-hot selection:

```
// Isolate lowest set bit (one-hot)
w_rot_sel = w_rot_req & ((~w_rot_req) + {{(N-1){1'b0}}, 1'b1});
```

4.6.3 Grant Encoding

A compact one-hot to binary encoder generates the grant ID:

```
always_comb begin
    w_grant_id = r_last_grant; // default to last grant
    for (int i = 0; i < N; i++) begin
        if (w_nxt_grant[i]) w_grant_id = i[$clog2(N)-1:0];
    end
end
```

end
end

4.7 Timing Characteristics

Characteristic	Value
Setup Time	1 clock cycle for state update
Response Time	Combinational (0 cycles)
Reset Recovery	1 clock cycle

4.8 Reset Behavior

- **Power-on Reset:** `r_last_grant` initialized to N-1, causing first arbitration to start from agent 0
- **Active Reset:** All state registers cleared, grant outputs become invalid
- **Reset Release:** Next arbitration cycle begins with agent 0 having highest priority

4.9 Usage Examples

4.9.1 Basic 4-Agent Arbiter

```
arbiter_round_robin_simple #(
    .N(4)
) u_arbiter (
    .clk          (clk),
    .rst_n        (rst_n),
    .request       (req_bits),           // [3:0]
    .grant_valid   (grant_valid),
    .grant         (grant_oh),           // [3:0] one-hot
    .grant_id      (grant_bin)           // [1:0] binary
);
```

4.9.2 8-Agent System

```
arbiter_round_robin_simple #(
    .N(8)
) u_8way_arbiter (
    .clk          (clk),
    .rst_n        (rst_n),
    .request       (agent_requests),    // [7:0]
    .grant_valid   (resource_granted),
    .grant         (agent_grants),      // [7:0]
```

```
); .grant_id (winning_agent) // [2:0]
```

4.10 Arbitration Sequence Example

For N=4 with requests [3:0]:

Cycle	Last Grant	Requests	Rotated	Selected	Grant	Grant ID
0	3 (init)	1111	1111	0001	0001	0
1	0	1110	1101	0001	0010	1
2	1	1101	0111	0001	0100	2
3	2	1011	1101	0001	1000	3
4	3	1111	1111	0001	0001	0

4.11 Design Considerations

4.11.1 Advantages

- **Simple Logic:** Uses efficient bit manipulation instead of complex priority encoders
- **Scalable:** Works for any number of agents with minimal logic overhead
- **Fair:** Guarantees round-robin fairness over time
- **Fast:** Single combinational path from request to grant

4.11.2 Limitations

- **Grant Undefined:** grant_id is undefined when grant_valid is low
- **Request Requirement:** At least one request must be active for meaningful arbitration
- **Power Consumption:** Rotational logic may consume more power than simpler priority schemes

4.12 Synthesis Considerations

- **Area:** Scales approximately as $O(N \log N)$ due to rotation logic
- **Timing:** Critical path through rotation and bit isolation logic
- **Power:** Barrel shifter logic may have higher switching activity

- **Optimization:** Consider using dedicated arbiter primitives for large N

4.13 Verification Notes

- **Reset Testing:** Verify proper initialization and first-grant behavior
- **Fairness Testing:** Confirm round-robin sequence over multiple cycles
- **Edge Cases:** Test behavior with single request, all requests, and no requests
- **Timing:** Verify setup/hold requirements for state register updates

4.14 Related Modules

- `arbiter_round_robin`: Enhanced round-robin arbiter with additional features
- `arbiter_priority_encoder`: Fixed priority arbiter implementation
- `encoder_priority_enable`: Priority encoding utility module

The `arbiter_round_robin_simple` module provides an efficient and fair arbitration solution suitable for most round-robin arbitration needs in digital systems.

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5 Weighted Round Robin Arbiter

A credit-based weighted round-robin arbiter with Quality of Service (QoS) support, providing proportional bandwidth allocation to multiple clients while maintaining starvation-free operation.

5.1 Overview

The `arbiter_round_robin_weighted` module combines fair round-robin arbitration with configurable per-client weights to provide proportional bandwidth allocation. Each client receives grants proportional to its assigned weight while preventing starvation. The module supports optional acknowledgment protocol and dynamic weight changes with atomic update semantics.

5.2 Module Declaration

```
module arbiter_round_robin_weighted #(
    parameter int MAX_LEVELS = 16,           // Maximum weight value per
client
    parameter int CLIENTS = 4,               // Number of requesting
clients
    parameter int WAIT_GNT_ACK = 0           // Enable ACK protocol (0=No-
ACK, 1=ACK)
) (
    // Derived localparam (computed internally - not user-settable):
    // localparam int MAX_LEVELS_WIDTH = $clog2(MAX_LEVELS); //
Credit counter width
    // localparam int N = $clog2(CLIENTS); //
Client ID width
    // localparam int C = CLIENTS; //
Convenience alias
    // localparam int MTW = MAX_LEVELS_WIDTH; //
Weight width alias
    // localparam int CXMTW = CLIENTS * MAX_LEVELS_WIDTH; //
Packed weight array width

    input logic clk,                        // Clock input
    input logic rst_n,                      // Active-low
asynchronous reset
    input logic block_arb,                  // Block all
arbitration
    input logic [CXMTW-1:0] max_thresh,     // Packed weight
values for all clients
    input logic [C-1:0] request,            // Request signals
from clients
    input logic [C-1:0] grant_ack,          // Grant
acknowledgment (ACK mode only)
    output logic grant_valid,               // Grant output valid
    output logic [C-1:0] grant,             // Grant vector (one-
hot)
    output logic [N-1:0] grant_id           // Grant client ID
(binary encoded)
);
```

5.3 Parameters

5.3.1 User-Settable Parameters

Parameter	Type	Default	Description
MAX_LEVELS	int	16	Maximum weight value per client (range: 1-256)

Parameter	Type	Default	Description
CLIENTS	int	4	Number of requesting clients (range: 2-32)
WAIT_GNT_ACK	int	0	Enable ACK protocol (0=No-ACK, 1=ACK required)

5.3.2 Derived Localparam (Computed Internally)

Localparam	Computation	Description
MAX_LEVELS_WIDTH	$\$clog2(\text{MAX_LEVELS_H})$	Credit counter width in bits
N	$\$clog2(\text{CLIENTS})$	Client ID width in bits
C	CLIENTS	Convenience alias for port widths
MTW	MAX_LEVELS_WIDTH	Convenience alias for weight width
CXMTW	$\text{CLIENTS} * \text{MAX_LEVELS_WIDTH}$	Total packed weight array width

Note: MAX_LEVELS_WIDTH, N, C, MTW, and CXMTW are computed internally and cannot be overridden by users.

5.4 Ports

5.4.1 Inputs

Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
block_arb	1	Block all arbitration (external gate)
max_thresh	CXMTW	Packed weight values: {weight[N-1], ..., weight[1], weight[0]}
request	C	Request signals from clients
grant_ack	C	Grant acknowledgment signals (ACK mode only)

5.4.2 Outputs

Port	Width	Description
grant_valid	1	Valid grant indicator
grant	C	Grant vector (one-hot encoded)
grant_id	N	Grant client ID (binary encoded)

5.5 Functionality

5.5.1 Credit-Based Weighting

The arbiter uses a credit-based system for weighted bandwidth allocation:

1. **Credit Initialization:** Each client's credit counter is initialized to its weight value
2. **Grant Completion:** When a grant completes, the client's credit counter decrements
3. **Credit Eligibility:** Only clients with non-zero credits are eligible for round-robin arbitration
4. **Global Replenishment:** When no requesting clients have credits, all credits are reloaded

5.5.2 Arbitration Stages

1. **Credit Eligibility:** Filter requests to only clients with credits > 0
2. **Round-Robin Selection:** Base arbiter (arbiter_round_robin) selects among eligible clients
3. **Grant Output:** Winning client receives one-hot grant signal
4. **Credit Update:** Winner's credit decrements on completion (immediate or ACK-based)

5.5.3 Weight Ratio Example

For weights [4, 2, 1, 1]: - **Client 0:** 4 credits → gets 4 consecutive grants - **Client 1:** 2 credits → gets 2 consecutive grants - **Client 2:** 1 credit → gets 1 grant - **Client 3:** 1 credit → gets 1 grant - **Pattern:** C0, C0, C0, C0, C1, C1, C2, C3, [replenish], repeat... - **Bandwidth:** C0=50%, C1=25%, C2=12.5%, C3=12.5%

5.6 Dynamic Weight Changes

5.6.1 Weight Management FSM

The module includes a finite state machine for atomic weight updates:

State	Description
WEIGHT_IDLE	Normal operation
WEIGHT_BLOCK	Detect weight change, block new grants
WEIGHT_DRAIN	Wait for pending ACK completion (2 cycles)
WEIGHT_UPDATE	Atomic weight update, reset credits
WEIGHT_STABILIZE	System stabilization (3 cycles)

5.6.2 Weight Update Sequence

1. **Detection:** FSM detects max_thresh change from shadow register
2. **Block:** New grants blocked, pending grants drain
3. **Drain:** Wait for ACK completion (ACK mode) or fixed delay
4. **Update:** Atomic update of weights, credit counters reset
5. **Stabilize:** System stabilization period
6. **Resume:** Return to normal operation

Total Latency: 5-15 cycles depending on pending ACKs

5.6.3 Safety Features

- **Shadow Register:** Active weights stored in r_safe_max_thresh for race-free updates
- **Timeout Protection:** 15-cycle timeout prevents FSM lockup
- **Credit Reset:** Credits reset to new weights during WEIGHT_STABILIZE state
- **ACK Awareness:** FSM waits for pending grant_ack before updating weights

5.7 Implementation Details

5.7.1 Credit Management

```
// Credit decrement on grant completion
if (w_grant_completed[i] && r_credit_counter[i] > 0) begin
    w_credit_counter[i] = r_credit_counter[i] - 1;
end
```

Grant Completion Modes: - **No-ACK Mode:** Completion when grant[i] && grant_valid (same cycle) - **ACK Mode:** Completion when grant[i] && grant_valid && grant_ack[i] (delayed)

5.7.2 Global Replenishment Logic

```
// Replenish when no requesting clients have credits
assign w_global_replenish = (w_normal_operation && !w_pending_grants
&&
                                (!request) && !
w_requesting_clients_with_credits);

// On replenish: reload all credit counters to their weights
if (w_global_replenish) begin
    w_credit_counter[i] = client_weight[i];
end
```

5.7.3 Request Masking for Fairness

```
// Only pass eligible clients to round-robin arbiter
assign w_requesting_eligible[i] = request[i] &&
                                ((w_has_crd[i]) ||
                                (w_global_replenish &&
client_weight[i] > 0));

// Apply masking for fairness among multiple eligible clients
assign w_mask_req[i] = (multiple_eligible) ?
                        (w_requesting_eligible[i] && !grant[i]) :
                        (w_requesting_eligible[i] &&
r_credit_counter[i] > 1);
```

5.7.4 Sub-Module: arbiter_round_robin

Uses base arbiter_round_robin module internally for fair selection among eligible clients:

```
arbiter_round_robin #(
    .CLIENTS          (CLIENTS),
    .WAIT_GNT_ACK     (WAIT_GNT_ACK)
) u_base_arbiter (
    .clk              (clk),
    .rst_n            (rst_n),
```

```

        .block_arb    (w_sub_block_arb),
        .request      (w_mask_req),
        .grant_ack     (grant_ack),
        .grant_valid  (grant_valid),
        .grant         (grant),
        .grant_id      (grant_id),
        .last_grant    (r_last_grant)
    );

```

5.8 Special Features

5.8.1 Zero-Weight Client Handling

Clients with `max_thresh = 0` are effectively disabled: - Never accumulate credits - Never participate in arbitration - Credit counters remain at zero

5.8.2 ACK Protocol Support

When `WAIT_GNT_ACK = 1`: - Grant remains asserted until `grant_ack` received - Credit decrement waits for ACK - Weight change FSM waits for pending ACKs - Enables pipelined/posted transaction masters

When `WAIT_GNT_ACK = 0`: - Grant completes immediately (1 cycle) - Credit decrements same cycle as grant - Optimal for simple masters (no pipeline)

5.8.3 Starvation Prevention

The arbiter guarantees starvation-free operation: - All non-zero weight clients eventually served - Maximum starvation time: Sum of all other clients' weights - Global replenishment ensures fairness

5.9 Usage Examples

5.9.1 Basic QoS Arbiter (No-ACK Mode)

```

// 4-client arbiter with QoS weights: 8:4:2:1
localparam int NUM_CLIENTS = 4;
localparam int MAX_WEIGHT = 16;
localparam int WEIGHT_W = $clog2(MAX_WEIGHT);

logic [NUM_CLIENTS-1:0] client_req, client_grant;
logic grant_vld;
logic [$clog2(NUM_CLIENTS)-1:0] grant_idx;
logic [NUM_CLIENTS*WEIGHT_W-1:0] qos_weights;

// Assign QoS weights (high priority = higher weight)
assign qos_weights = {4'd1, 4'd2, 4'd4, 4'd8}; // {C3, C2, C1, C0}

```

```

arbiter_round_robin_weighted #(
    .CLIENTS(NUM_CLIENTS),
    .MAX_LEVELS(MAX_WEIGHT),
    .WAIT_GNT_ACK(0)           // No-ACK mode
) u_qos_arbiter (
    .clk            (clk),
    .rst_n          (rst_n),
    .block_arb      (1'b0),
    .max_thresh     (qos_weights),
    .request        (client_req),
    .grant_ack      ('0),      // Unused in No-ACK mode
    .grant_valid    (grant_vld),
    .grant          (client_grant),
    .grant_id       (grant_idx)
);

```

5.9.2 Dynamic Weight Adjustment

```

// Adjust weights based on priority events
logic [NUM_CLIENTS*WEIGHT_W-1:0] r_qos_weights;

always_ff @(posedge clk) begin
    if (high_priority_event) begin
        r_qos_weights[7:0] <= 4'd15; // Boost Client 0 weight
    end else if (normal_priority) begin
        r_qos_weights[7:0] <= 4'd4;  // Reset Client 0 weight
    end
end

```

```

arbiter_round_robin_weighted #(
    .CLIENTS(NUM_CLIENTS),
    .MAX_LEVELS(MAX_WEIGHT),
    .WAIT_GNT_ACK(0)
) u_dynamic_arbiter (
    .clk            (clk),
    .rst_n          (rst_n),
    .block_arb      (1'b0),
    .max_thresh     (r_qos_weights), // Dynamic weights
    .request        (client_req),
    .grant_ack      ('0),
    .grant_valid    (grant_vld),
    .grant          (client_grant),
    .grant_id       (grant_idx)
);

```

5.9.3 ACK Mode for Pipelined Masters

```

// ACK mode arbiter for pipelined masters
arbiter_round_robin_weighted #(

```

```

        .CLIENTS(2),
        .MAX_LEVELS(8),
        .WAIT_GNT_ACK(1)                // ACK required
    ) u_ack_arbiter (
        .clk          (clk),
        .rst_n        (rst_n),
        .block_arb     (bus_busy),
        .max_thresh    ({4'd3, 4'd5}), // Weights [5, 3]
        .request       (m_req),
        .grant_ack      (m_done),       // Master completion signal
        .grant_valid    (m_grant_vld),
        .grant          (m_grant),
        .grant_id       (m_id)
    );

```

5.10 Timing Characteristics

5.10.1 Latency and Throughput

- **Latency:** 2 cycles (credit calculation + round-robin arbitration)
- **Throughput:** 1 grant per cycle (maximum)
- **Grant Hold:** No-ACK=1 cycle, ACK=Until grant_ack asserted
- **Weight Update:** 5-15 cycles (FSM: BLOCK→DRAIN→UPDATE→STABILIZE)
- **Reset:** Asynchronous (all credits→1, weights→default)

5.10.2 Critical Paths

1. **Credit Comparison:** Credit counter → eligibility logic
2. **Request Filtering:** Eligible clients → masked requests
3. **Round-Robin:** Base arbiter selection logic

5.11 Design Considerations

5.11.1 When to Use Each Mode

No-ACK Mode (WAIT_GNT_ACK=0): - ✓ Simple masters without pipelining - ✓ Combinational logic masters - ✓ Maximum throughput (1 grant/cycle) - ⚠ Grant completes same cycle (no hold)

ACK Mode (WAIT_GNT_ACK=1): - ✓ Pipelined masters (e.g., AXI) - ✓ Posted transaction buses - ✓ Multi-cycle operations - ⚠ Reduced throughput (grant held until ACK)

5.11.2 Weight Selection Guidelines

- **Finer QoS Control:** Increase MAX_LEVELS (but more credit counter bits)
- **Typical Range:** MAX_LEVELS=8 to 32 for most applications
- **Weight Ratio:** Should reflect desired bandwidth ratios
- **Zero Weights:** Effectively disable clients (never granted)

5.11.3 Dynamic Weight Change Best Practices

-  **DO NOT** change weights every cycle (causes FSM thrashing)
- ✓ **Recommended:** Change weights on policy updates (millisecond timescale)
- ✓ **Safe:** FSM ensures atomic updates without race conditions
- ✓ **Timeout:** 15-cycle timeout prevents lockup

5.12 Performance Characteristics

5.12.1 Resource Utilization

CLIENTS	MAX_LEVELS	FFs (Est.)	LUTs (Est.)
4	16	~40	~50
8	16	~70	~90
4	32	~45	~55
8	32	~80	~110

5.12.2 Maximum Frequency

- **Typical:** 300-500 MHz (modern FPGAs)
- **Limiting Factor:** Credit comparison and request filtering logic

5.13 Verification Notes

From the comprehensive test suite (val/common/test_arbiter_round_robin_weighted.py):

- **Coverage:** 94%
- **Key Test Scenarios:**
 - Weight ratio verification ([4,2,1,1] patterns)

- Global credit replenishment
- Dynamic weight changes (atomic updates)
- ACK mode timing and weight updates
- Zero weight handling (client disable)
- Single client optimization
- FSM timeout protection
- Starvation prevention

Test Command: `pytest val/common/test_arbiter_round_robin_weighted.py -v`

5.14 Related Modules

- **arbiter_round_robin.sv** - Base round-robin arbiter (used internally)
- **arbiter_round_robin_simple.sv** - Lightweight RR arbiter (no weights)
- **arbiter_priority_encoder.sv** - Fixed priority arbiter

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6 Single-Client Arbiter

Module: `arbiter_single_client.sv` **Location:** `rtl/common/` **Status:** Production Ready

6.1 Overview

`arbiter_single_client` is the degenerate, one-requester member of the round-robin arbiter family. It exists to give a single-channel datapath the **exact same registered, ack-held grant lifecycle** as `arbiter_round_robin` running in `WAIT_GNT_ACK` mode, without the parameter degeneracy that breaks the multi-client arbiter when `CLIENTS == 1`.

`arbiter_round_robin` underflows at `CLIENTS == 1` because `$clog2(1) = 0` produces a `grant_id[-1:0]` slice. Historically single-channel builds worked around this by substituting a purely *combinational* passthrough (`grant = request`). That passthrough does **not** reproduce the arbiter's registered grant/ack handshake, so it injects a bubble beat at every burst boundary in the

AXI read/write engines that depend on the “request, grant, hold-for-ack” timing. This module is the faithful single-client reduction of those rules.

6.1.1 Key Features

- **Timing-faithful:** Mirrors `arbiter_round_robin WAIT_GNT_ACK always_ff` rules exactly for one client
- **Registered grant:** Grant is a flip-flop output, held until acknowledged (no combinational passthrough bubble)
- **Ack-held lifecycle:** Grant persists until `grant_ack`, then clears and can re-grant the next cycle
- **Bubble-free bursts:** Preserves the grant timing the AXI read/write engines rely on for back-to-back burst beats
- **Optional ack mode:** `WAIT_GNT_ACK = 0` falls back to a simple registered request-to-grant with no ack hold
- **Trivial decode:** `grant` is a one-hot of width 1 (equal to `grant_valid`); `grant_id` is always 0

6.2 Module Purpose

The module provides a drop-in single-client arbiter so that a design parameterized down to one channel behaves identically to the same design with two or more channels. Rather than special-casing the `NUM_CHANNELS == 1` build with a passthrough, the datapath instantiates `arbiter_single_client` and gets the identical grant/ack sequencing the general arbiter would have produced for one requester.

Use Cases:

- Single-channel builds of a datapath that is otherwise parameterized for N channels
- AXI read/write engines that require the registered, ack-held grant handshake for correct burst sequencing
- Any place a `arbiter_round_robin` instance would be used, but with `CLIENTS == 1`
- Replacing a combinational `grant = request` passthrough that was injecting per-burst bubble beats

Key Benefit: A one-channel build behaves bit-for-bit like the multi-channel arbiter path, eliminating the burst-boundary bubble that a combinational passthrough introduced.

6.3 Parameters

Parameter	Type	Default	Description
WAIT_GNT_ACK	int	1	1 = registered grant is held until grant_ack is received (round-robin ACK lifecycle). 0 = simple registered request-to-grant with no ack hold.

6.4 Port Groups

6.4.1 Clock and Reset

Port	Direction	Width	Description
clk	Input	1	Clock
rst_n	Input	1	Active-low asynchronous reset

6.4.2 Control

Port	Direction	Width	Description
block_arb	Input	1	Block arbitration; when high the request is masked and no grant is issued
request	Input	1	The single client's request
grant_ack	Input	1	The single client's grant acknowledgment (used only when WAIT_GNT_ACK = 1)

6.4.3 Grant Outputs

Port	Direction	Width	Description
grant_val_id	Output	1	Grant valid (registered)

Port	Direction	Width	Description
grant	Output	1	One-hot grant of width 1; equal to grant_valid
grant_id	Output	1	Granted client index; always 0 for a single client

6.5 Functional Description

6.5.1 Request Qualification

The incoming request is first masked by block_arb:

```
assign w_req = request && !block_arb;
```

A blocked arbiter therefore behaves as if the client were not requesting.

6.5.2 Grant-Eligibility Logic (ACK mode)

When WAIT_GNT_ACK = 1, three combinational signals gate whether a new grant may be issued:

```
assign w_ack_received = r_pending_ack && grant_ack;           // ack
// seen for the outstanding grant
assign w_can_grant    = !r_pending_ack || w_ack_received;     // free
// to grant, or ack just closed the last grant
assign w_should_grant = w_req && w_can_grant;
```

r_pending_ack records that a grant has been issued and is awaiting its acknowledgment. A new grant can only be launched when there is no pending ack, or when the ack for the current grant arrives this cycle.

6.5.3 Grant Lifecycle (ACK mode)

The registered state machine is the single-client reduction of the arbiter_round_robin WAIT_GNT_ACK rules. Because there is only one client, w_other_requests is always 0, so the arbiter's Rule 4 (switch to another requester) never applies and Rule 3 simply clears:

State	Condition	Action	grant_valid next	r_pending_ack next
Rule 1	grant_val id == 0	Not granted; grant if request ing	w_should_grant	w_should_grant
Rule 2	granted, ! w_ack_rec	Hold the	1	1

State	Condition	Action	grant_valid next	r_pending_ack next
Rule 3	eived	grant		
	granted, w_ack_rec eived	Ack receive d; clear (re- grant next cycle)	0	0

The grant is thus **registered and held** across cycles until the client acknowledges, then it drops for one cycle before it can be re-issued.

6.5.4 No-ACK Fallback Mode

When WAIT_GNT_ACK = 0, the ack machinery is bypassed. grant_valid simply follows a registered version of the qualified request and r_pending_ack stays 0:

```
grant_valid    <= w_should_grant;    // w_can_grant is constant 1 in
this mode
r_pending_ack <= 1'b0;
```

This gives a one-cycle registered request-to-grant with no hold.

6.5.5 Output Mapping

```
assign grant    = grant_valid;    // one-hot of width 1
assign grant_id = 1'b0;           // single client
```

Because there is exactly one client, grant and grant_valid are identical, and grant_id is a constant 0.

6.6 Usage Example

```
// Single-channel build: use arbiter_single_client where a >1-channel
// build would instantiate arbiter_round_robin in WAIT_GNT_ACK mode.
arbiter_single_client #(
    .WAIT_GNT_ACK(1)
) u_arb (
    .clk          (aclk),
    .rst_n        (aresetn),
    .block_arb    (arb_blocked),
    .request       (engine_request),
    .grant_ack     (engine_grant_ack),
    .grant_valid   (engine_grant_valid),
    .grant         (engine_grant),    // width-1 one-hot
    .grant_id      ()                // always 0, usually unconnected
);
```

6.7 Design Notes

- **Why not a combinational passthrough?** `grant = request` looks correct for one client but loses the registered, ack-held timing. The AXI read/write engines interpret the missing hold as an early grant drop and insert a bubble beat at every burst boundary. This module preserves the exact cycle-by-cycle behavior.
- **grant_id degeneracy.** The general arbiter's `grant_id` is $\$clog2(CLIENTS)$ bits wide, which collapses to an illegal `[-1:0]` slice at `CLIENTS == 1`. Here `grant_id` is simply tied to 0, sidestepping the elaboration problem.
- **Block behavior.** `block_arb` masks the request combinatorially, so raising it prevents new grants; an already-pending ack-held grant continues to follow the state machine.
- **Reset.** Active-low asynchronous reset clears both `grant_valid` and `r_pending_ack`.

6.8 Related Modules

6.8.1 Used By

- Single-channel builds of AXI read/write datapath engines that otherwise instantiate `arbiter_round_robin`

6.8.2 Uses

- None (self-contained; no submodule instantiations)

6.8.3 See Also

- [arbiter_round_robin](#) - The general N-client arbiter whose `WAIT_GNT_ACK` timing this module reproduces for one client
- [arbiter_round_robin_simple](#) - Minimal-area round-robin variant
- [arbiter_round_robin_weighted](#) - Weighted / QoS round-robin variant
- [arbiter_priority_encoder](#) - Fixed-priority arbiter

6.9 References

6.9.1 Source Code

- `rtl/common/arbiter_single_client.sv`
- `rtl/common/arbiter_round_robin.sv` (timing reference)

6.9.2 Documentation

- `rtl/common/PRD.md`

Last Updated: 2026-07-15

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7 Binary to Gray Code Converter

7.1 Overview

The `bin2gray` module implements a purely combinational binary-to-Gray code converter. Gray code (also known as reflected binary code or unit distance code) is a binary numeral system where two successive values differ in only one bit. This property makes Gray code essential for reducing glitches and metastability in digital systems, particularly in asynchronous designs and clock domain crossings.

7.2 Module Declaration

```
module bin2gray #(
    parameter int WIDTH = 4
) (
    input wire [WIDTH-1:0] binary,
    output wire [WIDTH-1:0] gray
);
```

7.3 Parameters

7.3.1 WIDTH

- **Type:** `int`
- **Default:** 4
- **Description:** Bit width of both input and output

- **Range:** Any positive integer ≥ 1
- **Common Values:** 4, 8, 16, 32 for address counters and data buses
- **Impact:** Determines the number of XOR gates required

7.4 Ports

7.4.1 Inputs

Port	Width	Type	Description
binary	WIDTH	wire	Input binary value

7.4.2 Outputs

Port	Width	Type	Description
gray	WIDTH	wire	Output Gray code value

7.5 Gray Code Theory

7.5.1 Mathematical Definition

Gray code conversion follows a simple mathematical relationship: - **MSB:** $\text{gray}[\text{WIDTH}-1] = \text{binary}[\text{WIDTH}-1]$ (MSB unchanged) - **Other bits:** $\text{gray}[i] = \text{binary}[i] \oplus \text{binary}[i+1]$ for $i = 0$ to $\text{WIDTH}-2$

7.5.2 Why Gray Code?

Gray code solves several problems in digital systems:

1. **Single Bit Transitions:** Adjacent values differ by exactly one bit
2. **Glitch Reduction:** Eliminates intermediate states during transitions
3. **Metastability Prevention:** Safer for asynchronous clock domain crossings
4. **Mechanical Encoders:** Natural for optical/magnetic position encoders

7.5.3 Gray Code Sequence Examples

7.5.3.1 2-bit Gray Code

Decimal	Binary	Gray	Transition
0	00	00	-

Decimal	Binary	Gray	Transition
1	01	01	bit 0
2	10	11	bit 1
3	11	10	bit 0

7.5.3.2 3-bit Gray Code

Decimal	Binary	Gray	Changed Bit
0	000	000	-
1	001	001	0
2	010	011	1
3	011	010	0
4	100	110	2
5	101	111	0
6	110	101	1
7	111	100	0

7.5.3.3 4-bit Gray Code (Complete Table)

Dec	Binary	Gray	Dec	Binary	Gray
0	0000	0000	8	1000	1100
1	0001	0001	9	1001	1101
2	0010	0011	10	1010	1111
3	0011	0010	11	1011	1110
4	0100	0110	12	1100	1010
5	0101	0111	13	1101	1011
6	0110	0101	14	1110	1001
7	0111	0100	15	1111	1000

7.6 Implementation Analysis

7.6.1 Core Logic

```

genvar i;
generate
    for (i = 0; i < WIDTH - 1; i++) begin : gen_gray
        assign gray[i] = binary[i] ^ binary[i+1];
    end
end

```

endgenerate

assign gray[WIDTH-1] = binary[WIDTH-1];

7.6.2 Logic Structure

The implementation creates a chain of XOR gates: - **Lower bits**: Each gray bit is XOR of current and next binary bits - **MSB**: Directly assigned from binary MSB - **Propagation**: Single level of logic (all XORs in parallel)

7.6.3 Gate-Level Implementation

For WIDTH = 4:

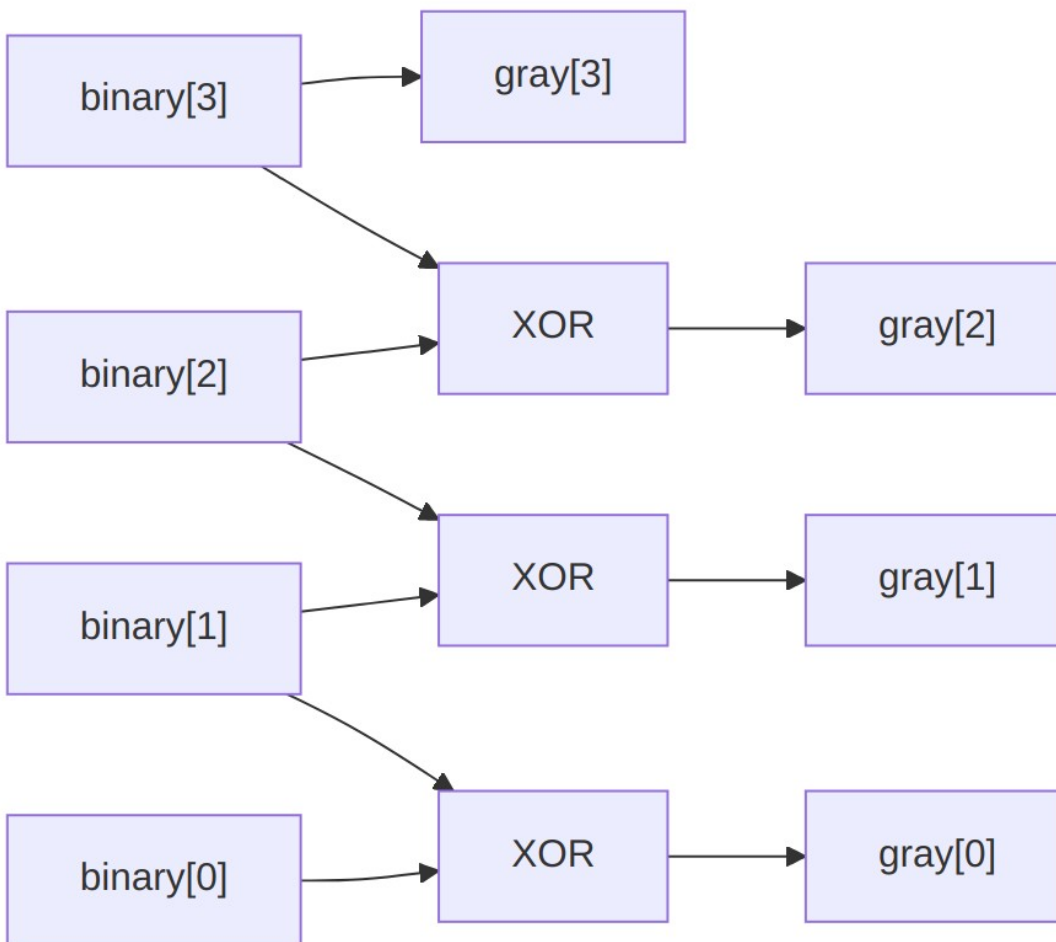


Diagram 1

7.6.4 Timing Characteristics

- **Propagation Delay:** Single XOR gate delay (typically 0.1-0.3ns in modern processes)
- **Critical Path:** Through one XOR gate only
- **Fan-out:** Each binary bit drives at most 2 XOR gates
- **Scalability:** Constant delay regardless of WIDTH

7.7 Design Examples and Applications

7.7.1 1. Asynchronous FIFO Pointer

```
module async_fifo_ptr #(
    parameter int ADDR_WIDTH = 4
) (
    input logic clk,
    input logic rst_n,
    input logic enable,
    output logic [ADDR_WIDTH:0] gray_ptr,
    output logic [ADDR_WIDTH-1:0] addr
);

    logic [ADDR_WIDTH:0] binary_ptr, binary_ptr_next;

    // Binary counter
    assign binary_ptr_next = enable ? (binary_ptr + 1) : binary_ptr;

    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n)
            binary_ptr <= 'b0;
        else
            binary_ptr <= binary_ptr_next;
    end

    // Binary to Gray conversion
    bin2gray #(
        .WIDTH(ADDR_WIDTH + 1)
    ) ptr_converter (
        .binary(binary_ptr),
        .gray(gray_ptr)
    );

    // Extract address (lower bits of binary counter)
    assign addr = binary_ptr[ADDR_WIDTH-1:0];
```

endmodule

7.7.2 2. Clock Domain Crossing Counter

```
module cross_domain_counter #(
    parameter int WIDTH = 8
) (
    // Source domain
    input logic          src_clk,
    input logic          src_rst_n,
    input logic          src_enable,

    // Destination domain
    input logic          dst_clk,
    input logic          dst_rst_n,
    output logic [WIDTH-1:0] dst_count_binary,
    output logic [WIDTH-1:0] dst_count_gray
);

    // Source domain counter
    logic [WIDTH-1:0] src_binary, src_gray;

    always_ff @(posedge src_clk or negedge src_rst_n) begin
        if (!src_rst_n)
            src_binary <= 'b0;
        else if (src_enable)
            src_binary <= src_binary + 1;
    end

    // Convert to Gray in source domain
    bin2gray #(.WIDTH(WIDTH)) src_converter (
        .binary(src_binary),
        .gray(src_gray)
    );

    // Synchronize Gray code to destination domain
    logic [WIDTH-1:0] dst_gray_sync;

    synchronizer #(.WIDTH(WIDTH)) gray_sync (
        .clk(dst_clk),
        .rst_n(dst_rst_n),
        .data_in(src_gray),
        .data_out(dst_gray_sync)
    );

    // Convert back to binary in destination domain
    gray2bin #(.WIDTH(WIDTH)) dst_converter (
```

```

        .gray(dst_gray_sync),
        .binary(dst_count_binary)
    );

    assign dst_count_gray = dst_gray_sync;

```

endmodule

7.7.3 3. Rotary Encoder Interface

```

module rotary_encoder_interface #(
    parameter int POSITION_WIDTH = 12
) (
    input logic                clk,
    input logic                rst_n,
    input logic [POSITION_WIDTH-1:0] encoder_binary,
    output logic [POSITION_WIDTH-1:0] encoder_gray,
    output logic [POSITION_WIDTH-1:0] position_filtered,
    output logic                position_changed
);

    // Convert encoder binary position to Gray
    bin2gray #(
        .WIDTH(POSITION_WIDTH)
    ) encoder_conv (
        .binary(encoder_binary),
        .gray(encoder_gray)
    );

    // Synchronize and filter
    logic [POSITION_WIDTH-1:0] gray_sync, gray_prev;

    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            gray_sync <= 'b0;
            gray_prev <= 'b0;
        end else begin
            gray_sync <= encoder_gray;
            gray_prev <= gray_sync;
        end
    end

    // Detect changes (only one bit should change in Gray code)
    logic valid_transition;
    assign valid_transition = ($countones(gray_sync ^ gray_prev) <=
1);

    // Convert back to binary for position output

```

```

        gray2bin #(.WIDTH(POSITION_WIDTH)) pos_converter (
            .gray(gray_sync),
            .binary(position_filtered)
        );

        assign position_changed = valid_transition && (gray_sync !=
gray_prev);

endmodule

```

7.7.4 4. Address Scrambling for Memory Testing

```

module memory_address_scrambler #(
    parameter int ADDR_WIDTH = 16
) (
    input logic [ADDR_WIDTH-1:0] linear_addr,
    output logic [ADDR_WIDTH-1:0] scrambled_addr
);

    logic [ADDR_WIDTH-1:0] gray_addr;

    // Convert linear address to Gray code for scrambling
    bin2gray #(.WIDTH(ADDR_WIDTH)) scrambler (
        .binary(linear_addr),
        .gray(gray_addr)
    );

    // Additional scrambling (optional)
    assign scrambled_addr = {gray_addr[0], gray_addr[ADDR_WIDTH-1:1]};

endmodule

```

7.7.5 5. Multi-Bit Synchronizer with Gray Code

Prefer the library module. The example below is written out in full to show the mechanism, but production code should instantiate `glitch_free_n_dff_arn` (rtl/common) or `cdc_synchronizer` (rtl/amba/cdc) rather than hand-rolling the flop chain.

Two conditions this example depends on, both easy to get wrong: - `src_gray` must be **registered in the source domain** before it crosses. The single-bit-change guarantee belongs to a registered counter sequence, not to the encoding in the abstract; a Gray value straight out of combinational logic can present a transient that was never a real state. - The source value must **increment by one**. A Gray value that jumps changes multiple bits and no encoding makes that safe.

```

module gray_code_synchronizer #(
    parameter int WIDTH = 8,
    parameter int SYNC_STAGES = 2
) (
    input logic          src_clk,
    input logic          src_rst_n,
    input logic [WIDTH-1:0] src_data,

    input logic          dst_clk,
    input logic          dst_rst_n,
    output logic [WIDTH-1:0] dst_data
);

    // Convert to Gray in source domain
    logic [WIDTH-1:0] src_gray;

    bin2gray #(.WIDTH(WIDTH)) src_conv (
        .binary(src_data),
        .gray(src_gray)
    );

    // Multi-stage synchronizer for Gray code
    logic [WIDTH-1:0] sync_regs [SYNC_STAGES];

    always_ff @(posedge dst_clk or negedge dst_rst_n) begin
        if (!dst_rst_n) begin
            for (int i = 0; i < SYNC_STAGES; i++) begin
                sync_regs[i] <= 'b0;
            end
        end else begin
            sync_regs[0] <= src_gray;
            for (int i = 1; i < SYNC_STAGES; i++) begin
                sync_regs[i] <= sync_regs[i-1];
            end
        end
    end

    // Convert back to binary in destination domain
    gray2bin #(.WIDTH(WIDTH)) dst_conv (
        .gray(sync_regs[SYNC_STAGES-1]),
        .binary(dst_data)
    );

endmodule

```

7.8 Companion Gray-to-Binary Converter

7.8.1 Implementation

```
module gray2bin #(
    parameter int WIDTH = 4
) (
    input wire [WIDTH-1:0] gray,
    output wire [WIDTH-1:0] binary
);

    genvar i;

    // MSB is unchanged
    assign binary[WIDTH-1] = gray[WIDTH-1];

    // Other bits: XOR accumulation from MSB down
    generate
        for (i = WIDTH-2; i >= 0; i--) begin : gen_binary
            assign binary[i] = binary[i+1] ^ gray[i];
        end
    endgenerate

endmodule
```

7.8.2 Gray-to-Binary Truth Table (4-bit)

Gray	Binary	Conversion Process
0000	0000	bin[3]=0, bin[2]=0 $\oplus$ 0=0, bin[1]=0 $\oplus$ 0=0, bin[0]=0 $\oplus$ 0=0
0001	0001	bin[3]=0, bin[2]=0 $\oplus$ 0=0, bin[1]=0 $\oplus$ 0=0, bin[0]=0 $\oplus$ 1=1
0011	0010	bin[3]=0, bin[2]=0 $\oplus$ 0=0, bin[1]=0 $\oplus$ 1=1, bin[0]=1 $\oplus$ 1=0
0010	0011	bin[3]=0, bin[2]=0 $\oplus$ 0=0, bin[1]=0 $\oplus$ 1=1,

Gray	Binary	Conversion Process
		$\text{bin}[0]=1\oplus 0=1$

7.9 Advanced Implementations

7.9.1 1. Parameterized Converter with Validation

```

module bin2gray_validated #(
    parameter int WIDTH = 4,
    parameter bit ENABLE_CHECKS = 1
) (
    input logic [WIDTH-1:0] binary,
    output logic [WIDTH-1:0] gray,
    output logic          valid
);

    // Basic conversion
    bin2gray #(.WIDTH(WIDTH)) converter (
        .binary(binary),
        .gray(gray)
    );

    // Optional validation
    generate
        if (ENABLE_CHECKS) begin : validation
            logic [WIDTH-1:0] binary_check;

            // Round-trip conversion check
            gray2bin #(.WIDTH(WIDTH)) check_converter (
                .gray(gray),
                .binary(binary_check)
            );

            assign valid = (binary == binary_check);
        end else begin : no_validation
            assign valid = 1'b1;
        end
    endgenerate

endmodule

```

7.9.2 2. Pipelined Converter for High Speed

```

module bin2gray_pipelined #(
    parameter int WIDTH = 32,
    parameter int PIPELINE_STAGES = 2
) (

```

```

    input  logic          clk,
    input  logic          rst_n,
    input  logic [WIDTH-1:0] binary_in,
    input  logic          valid_in,
    output logic [WIDTH-1:0] gray_out,
    output logic          valid_out
);

// Pipeline registers
logic [WIDTH-1:0] binary_pipe [PIPELINE_STAGES];
logic valid_pipe [PIPELINE_STAGES];
logic [WIDTH-1:0] gray_pipe [PIPELINE_STAGES];

// Stage 0: Input registration
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        binary_pipe[0] <= 'b0;
        valid_pipe[0] <= 1'b0;
    end else begin
        binary_pipe[0] <= binary_in;
        valid_pipe[0] <= valid_in;
    end
end

// Conversion in stage 0
bin2gray #(.WIDTH(WIDTH)) stage0_conv (
    .binary(binary_pipe[0]),
    .gray(gray_pipe[0])
);

// Additional pipeline stages
generate
    for (genvar s = 1; s < PIPELINE_STAGES; s++) begin :
pipeline_stages
        always_ff @(posedge clk or negedge rst_n) begin
            if (!rst_n) begin
                gray_pipe[s] <= 'b0;
                valid_pipe[s] <= 1'b0;
            end else begin
                gray_pipe[s] <= gray_pipe[s-1];
                valid_pipe[s] <= valid_pipe[s-1];
            end
        end
    end
endgenerate

// Output assignment
assign gray_out = gray_pipe[PIPELINE_STAGES-1];

```

```

    assign valid_out = valid_pipe[PIPELINE_STAGES-1];

endmodule

7.9.3 3. Bidirectional Converter
module bidirectional_gray_converter #(
    parameter int WIDTH = 8
) (
    input logic direction, // 0: bin→gray, 1: gray→bin
    input logic [WIDTH-1:0] data_in,
    output logic [WIDTH-1:0] data_out
);

    logic [WIDTH-1:0] bin_to_gray_out, gray_to_bin_out;

    // Both converters always active
    bin2gray #(.WIDTH(WIDTH)) b2g (
        .binary(data_in),
        .gray(bin_to_gray_out)
    );

    gray2bin #(.WIDTH(WIDTH)) g2b (
        .gray(data_in),
        .binary(gray_to_bin_out)
    );

    // Mux output based on direction
    assign data_out = direction ? gray_to_bin_out : bin_to_gray_out;

endmodule

```

7.10 Verification and Testing

7.10.1 Comprehensive Test Bench

```

module tb_bin2gray;

    parameter int WIDTH = 4;
    parameter int MAX_VAL = (1 << WIDTH) - 1;

    logic [WIDTH-1:0] binary, gray;
    logic [WIDTH-1:0] expected_gray;
    logic [WIDTH-1:0] binary_check;

    // DUT
    bin2gray #(.WIDTH(WIDTH)) dut (
        .binary(binary),

```

```

        .gray(gray)
    );

    // Reference converter for checking
    gray2bin #(.WIDTH(WIDTH)) check_conv (
        .gray(gray),
        .binary(binary_check)
    );

    // Test sequence
    initial begin
        $display("Testing %d-bit Binary to Gray converter", WIDTH);

        // Test all possible values
        for (int i = 0; i <= MAX_VAL; i++) begin
            binary = i;
            expected_gray = compute_gray_reference(i);

            #1; // Allow propagation

            // Check conversion correctness
            if (gray !== expected_gray) begin
                $error("Mismatch: binary=%b, expected_gray=%b,
actual_gray=%b",
                    binary, expected_gray, gray);
            end

            // Check round-trip conversion
            if (binary_check !== binary) begin
                $error("Round-trip failed: original=%b, recovered=%b",
                    binary, binary_check);
            end

            // Check single-bit change property
            if (i > 0) begin
                logic [WIDTH-1:0] prev_gray =
compute_gray_reference(i-1);
                int bit_changes = $countones(gray ^ prev_gray);
                if (bit_changes != 1) begin
                    $error("Multiple bit change: %d→%d, gray %b→%b,
changes=%d",
                        i-1, i, prev_gray, gray, bit_changes);
                end
            end

            $display("PASS: %d → binary:%b → gray:%b", i, binary,
gray);
        end
    end

```

```

        $display("All tests passed!");
        $finish;
    end

    // Reference Gray code computation
    function [WIDTH-1:0] compute_gray_reference(input [WIDTH-1:0]
bin);
        compute_gray_reference[WIDTH-1] = bin[WIDTH-1];
        for (int i = WIDTH-2; i >= 0; i--) begin
            compute_gray_reference[i] = bin[i] ^ bin[i+1];
        end
    endfunction

endmodule

```

7.10.2 Property-Based Verification

```

module bin2gray_properties #(
    parameter int WIDTH = 4
);

    logic [WIDTH-1:0] binary, gray;

    // Bind to DUT
    bind bin2gray bin2gray_properties #(WIDTH) props_inst (.*) ;

    // Property: MSB preservation
    property msb_unchanged;
        gray[WIDTH-1] == binary[WIDTH-1];
    endproperty

    // Property: Single bit change between consecutive values
    property single_bit_change;
        logic [WIDTH-1:0] bin_plus_one = binary + 1;
        logic [WIDTH-1:0] gray_plus_one;

        // Compute Gray code for binary+1
        assign gray_plus_one[WIDTH-1] = bin_plus_one[WIDTH-1];
        for (genvar i = 0; i < WIDTH-1; i++) begin
            assign gray_plus_one[i] = bin_plus_one[i] ^
bin_plus_one[i+1];
        end

        // Check single bit difference
        (binary < (2**WIDTH - 1)) |-> ($countones(gray ^
gray_plus_one) == 1);
    endproperty

```

```

// Property: Round-trip conversion
property round_trip_conversion;
    logic [WIDTH-1:0] recovered_binary;

    // Convert back to binary
    assign recovered_binary[WIDTH-1] = gray[WIDTH-1];
    for (genvar i = WIDTH-2; i >= 0; i--) begin
        assign recovered_binary[i] = recovered_binary[i+1] ^
gray[i];
    end

    recovered_binary == binary;
endproperty

// Assertions
assert property (msb_unchanged);
assert property (single_bit_change);
assert property (round_trip_conversion);

endmodule

```

7.10.3 Coverage Model

```

covergroup bin2gray_cg;

    cp_binary_values: coverpoint binary {
        bins zero = {0};
        bins powers_of_two[] = {1, 2, 4, 8, 16}; // For appropriate
WIDTH
        bins max_value = {2**WIDTH - 1};
        bins mid_range[] = {[1:2**(WIDTH-1)-1]};
        bins upper_range[] = {[2**(WIDTH-1):2**WIDTH-2]};
    }

    cp_gray_values: coverpoint gray {
        bins all_values[] = {[0:2**WIDTH-1]};
    }

    cp_bit_patterns: coverpoint binary {
        bins alternating_01 = {8'b01010101}; // For WIDTH=8
        bins alternating_10 = {8'b10101010};
        bins all_ones = {'1};
        bins all_zeros = {'0};
    }

    // Cross coverage between input patterns and outputs
    cross_binary_gray: cross cp_binary_values, cp_gray_values;

```

endcovergroup

7.11 Synthesis Optimization

7.11.1 Resource Utilization

For different widths, typical FPGA resource usage:

WIDTH	LUTs	Delay (ns)	Max Freq (MHz)
4	3	0.2	800+
8	7	0.2	800+
16	15	0.3	600+
32	31	0.4	500+

7.11.2 Timing Optimization

```
// For critical timing paths, add pipeline register
module bin2gray_registered #(
    parameter int WIDTH = 16
) (
    input logic          clk,
    input logic          rst_n,
    input logic [WIDTH-1:0] binary,
    output logic [WIDTH-1:0] gray
);

    logic [WIDTH-1:0] gray_comb;

    // Combinational conversion
    bin2gray #(.WIDTH(WIDTH)) conv (
        .binary(binary),
        .gray(gray_comb)
    );

    // Output register
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n)
            gray <= 'b0;
        else
            gray <= gray_comb;
    end
endmodule
```

7.11.3 Power Optimization

```
// Clock gating for power savings
module bin2gray_gated #(
    parameter int WIDTH = 8
) (
    input logic          clk,
    input logic          rst_n,
    input logic          enable,
    input logic [WIDTH-1:0] binary,
    output logic [WIDTH-1:0] gray
);

    logic gated_clk;

    // Clock gate
    clock_gate cg_inst (
        .clk(clk),
        .enable(enable),
        .gated_clk(gated_clk)
    );

    // Registered converter
    bin2gray_registered #(.WIDTH(WIDTH)) conv (
        .clk(gated_clk),
        .rst_n(rst_n),
        .binary(binary),
        .gray(gray)
    );

endmodule
```


7.12 Common Applications Summary

7.12.11. Asynchronous FIFOs: Prevent metastability in pointer comparisons

7.12.22. Clock Domain Crossing: Safe multi-bit signal transfer

7.12.33. Position Encoders: Mechanical/optical encoder interfaces

7.12.44. Memory Address Generation: Reduce EMI and power spikes

7.12.55. Test Pattern Generation: Controlled single-bit transitions

7.12.66. Error Detection: Single-bit error detection schemes

7.12.77. ADC/DAC Interfaces: Reduce glitches in conversion systems

7.13 Design Guidelines

7.13.11. Always Use for Async Boundaries: Gray code is essential for safe asynchronous transfers

7.13.22. Consider Timing: Purely combinational - no setup/hold concerns

7.13.33. Verify Single-Bit Property: Always verify adjacent values differ by one bit

7.13.44. Plan for Back-Conversion: Usually need Gray-to-Binary converter too

7.13.55. Check Width Requirements: Ensure sufficient bits for application range

The Binary-to-Gray converter is a fundamental building block in digital design, providing robust solutions for asynchronous interfaces and glitch-free operations. Its simplicity and reliability make it indispensable for safe digital system design.

7.14 Navigation

- [← Back to RTLCommon Index](#)
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8 Binary to BCD Converter - Comprehensive Documentation

8.1 Overview

The `bin_to_bcd` module implements the Double-Dabble algorithm (also known as the Add-3-Then-Shift algorithm) to convert binary numbers to Binary Coded Decimal (BCD) format. This is a fundamental operation in digital systems for displaying numeric values on seven-segment displays, LCD panels, and other decimal-based output devices.

8.2 Module Declaration

```
module bin_to_bcd #(
    parameter int WIDTH  = 8,
    parameter int DIGITS = 3
) (
    input logic          clk,          // Clock signal
    input logic          rst_n,        // Active-low reset signal
    input logic          start,
    input logic [WIDTH-1:0] binary,
    output logic [DIGITS*4-1:0] bcd,
    output logic         done
);
```

8.3 Parameters

8.3.1 WIDTH

- **Type:** int
- **Default:** 8
- **Description:** Bit width of the input binary number
- **Range:** 1 to 32 (practical range)
- **Impact:** Determines conversion complexity and timing
- **Sizing Rule:** For n-bit binary input, maximum decimal value is $2^n - 1$

8.3.2 DIGITS

- **Type:** int
- **Default:** 3

- **Description:** Number of BCD digits in the output
- **Range:** Must be sufficient for maximum input value
- **Calculation:** $\text{DIGITS} \geq \text{ceil}(\log_{10}(2^{\text{WIDTH}}))$
- **Output Width:** $\text{DIGITS} \times 4$ bits (4 bits per BCD digit)

8.3.3 Parameter Relationship Table

WIDTH	Max Binary Value	Max Decimal	Min DIGITS Required	BCD Width
4	15	15	2	8 bits
8	255	255	3	12 bits
10	1023	1023	4	16 bits
12	4095	4095	4	16 bits
16	65535	65535	5	20 bits
20	1048575	1048575	7	28 bits
24	16777215	16777215	8	32 bits

8.4 Ports

8.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	System clock
rst_n	1	logic	Active-low asynchronous reset
start	1	logic	Initiate conversion (pulse)
binary	WIDTH	logic	Binary number to convert

8.4.2 Outputs

Port	Width	Type	Description
bcd	DIGITS×4	logic	BCD result (packed)

Port	Width	Type	Description
done	1	logic	format) Conversion complete flag

8.5 Double-Dabble Algorithm Theory

8.5.1 Algorithm Principle

The Double-Dabble algorithm converts binary to BCD through iterative shifting and adjustment:

1. **Initialize:** BCD accumulator = 0, binary shift register = input value
2. **Check:** For each BCD digit, if value ≥ 5 , add 3
3. **Shift:** Shift entire register (BCD + binary) left by 1 bit
4. **Repeat:** Continue until all binary bits are processed
5. **Result:** BCD accumulator contains the decimal equivalent

8.5.2 Why Add 3?

The “add 3” operation prevents BCD overflow during shifting: - **BCD Range:** Each digit must stay within 0-9 (4 bits) - **Problem:** Shifting a BCD digit ≥ 5 creates values > 9 - **Solution:** Adding 3 before shifting ensures proper BCD after shift

8.5.3 Mathematical Proof

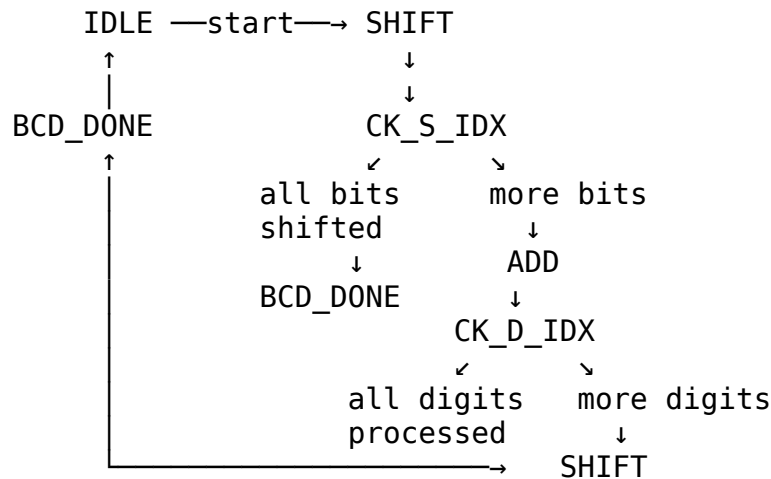
For BCD digit D where $D \geq 5$: - Before shift: $D + 3$ - After shift: $2(D + 3) = 2D + 6$ - For D = 5: $2(5) + 6 = 16 = 0x10$ (BCD: 1,6 - carry generated) - For D = 9: $2(9) + 6 = 24 = 0x18$ (BCD: 2,4 - carry generated)

8.6 FSM Implementation

8.6.1 State Definitions

```
typedef enum logic [5:0] {
    IDLE      = 6'b000001, // Wait for start signal
    SHIFT     = 6'b000010, // Shift binary bit into BCD
    CK_S_IDX  = 6'b000100, // Check if all bits shifted
    ADD       = 6'b001000, // Add 3 to digits > 4
    CK_D_IDX  = 6'b010000, // Check if all digits processed
    BCD_DONE  = 6'b100000 // Conversion complete
} fsm_state_t;
```

8.6.2 State Transition Diagram



8.6.3 Detailed State Descriptions

8.6.3.1 IDLE State

- **Function:** Wait for start signal
- **Duration:** Variable (until start asserted)
- **Actions:**
 - Monitor start input
 - Initialize counters and registers when start detected
- **Next State:** SHIFT when `start == 1'b1`

8.6.3.2 SHIFT State

- **Function:** Shift one bit from binary to BCD
- **Duration:** 1 clock cycle
- **Actions:**
 - Left-shift BCD accumulator by 1 bit
 - Move MSB of binary into LSB of BCD
 - Left-shift binary register by 1 bit
 - Increment loop counter

- **Next State:** Always CK_S_IDX

8.6.3.3 CK_S_IDX State

- **Function:** Check if all binary bits processed
- **Duration:** 1 clock cycle
- **Actions:**
 - Compare loop counter with WIDTH
 - Reset digit index for ADD phase
- **Next State:**
 - BCD_DONE if loop_count == WIDTH-1
 - ADD if more bits to process

8.6.3.4 ADD State

- **Function:** Add 3 to BCD digits ≥ 5
- **Duration:** 1 clock cycle
- **Actions:**
 - Extract current BCD digit
 - Add 3 if digit > 4
 - Update BCD accumulator
- **Next State:** Always CK_D_IDX

8.6.3.5 CK_D_IDX State

- **Function:** Check if all digits processed
- **Duration:** 1 clock cycle
- **Actions:**
 - Compare digit index with DIGITS
 - Increment digit index

- **Next State:**
 - SHIFT if `digit_index == DIGITS-1`
 - ADD if more digits to process

8.6.3.6 BCD_DONE State

- **Function:** Signal completion
- **Duration:** 1 clock cycle
- **Actions:**
 - Assert done flag
 - Prepare for next conversion
- **Next State:** Always IDLE

8.7 Implementation Details

8.7.1 Register Definitions

```
// FSM state register
fsm_state_t r_fsm_main;

// BCD accumulator - stores the converted result
logic [DIGITS*4-1:0] r_bcd;

// Binary shift register - input value being processed
logic [WIDTH-1:0] r_binary;

// Loop counter - tracks processed bits
localparam int LOOP_COUNT_WIDTH = $clog2(WIDTH + 1);
logic [LOOP_COUNT_WIDTH-1:0] r_loop_count;

// Digit index - tracks current BCD digit being processed
localparam int DIGIT_INDEX_WIDTH = $clog2(DIGITS);
logic [DIGIT_INDEX_WIDTH-1:0] r_digit_index;

// Data valid flag
logic r_dv;
```

8.7.2 Core Processing Logic

```
// Extract current BCD digit being processed
logic [3:0] w_bcd_digit;
assign w_bcd_digit = r_bcd[r_digit_index*4+:4];

// Main FSM
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_fsm_main    <= IDLE;
        r_bcd         <= '0;
        r_binary       <= '0;
        r_digit_index <= '0;
        r_loop_count   <= '0;
        r_dv           <= '0;
    end else begin
        r_dv <= 1'b0; // Default: done is low

        casez (r_fsm_main)
            IDLE: begin
                if (start == 1'b1) begin
                    r_binary    <= binary;
                    r_fsm_main  <= SHIFT;
                    r_bcd       <= '0;
                    r_loop_count <= '0;
                    r_digit_index <= '0;
                end else begin
                    r_fsm_main <= IDLE;
                end
            end

            SHIFT: begin
                // Shift BCD left and insert MSB of binary
                r_bcd    <= r_bcd << 1;
                r_bcd[0] <= r_binary[WIDTH-1];
                r_binary  <= r_binary << 1;
                r_fsm_main <= CK_S_IDX;
            end

            CK_S_IDX: begin
                if (r_loop_count == LOOP_COUNT_WIDTH'(WIDTH - 1))
begin
                    r_loop_count <= '0;
                    r_fsm_main  <= BCD_DONE;
                end else begin
                    r_loop_count <= r_loop_count + 1'b1;
                    r_digit_index <= '0;
                    r_fsm_main  <= ADD;
                end
            end
        end casez
    end
end
```



```

        end
    end

    ADD: begin
        if (w_bcd_digit > 4'd4) begin
            r_bcd[(r_digit_index*4)+:4] <= w_bcd_digit + 4'd3;
        end
        r_fsm_main <= CK_D_IDX;
    end

    CK_D_IDX: begin
        if (r_digit_index == DIGIT_INDEX_WIDTH'(DIGITS - 1))
            r_digit_index <= '0;
            r_fsm_main <= SHIFT;
        end else begin
            r_digit_index <= r_digit_index + 1'b1;
            r_fsm_main <= ADD;
        end
    end

    BCD_DONE: begin
        r_dv <= 1'b1;
        r_fsm_main <= IDLE;
    end

    default: begin
        r_fsm_main <= IDLE;
    end
endcase
end
end
end

```

8.8 Timing Analysis

8.8.1 Conversion Latency

The total conversion time depends on the input width and number of digits:

Formula: Latency = WIDTH + (WIDTH-1) × DIGITS clock cycles

8.8.2 Latency Breakdown

- **SHIFT operations:** WIDTH cycles (one per input bit)
- **ADD operations:** (WIDTH-1) × DIGITS cycles (check each digit after each shift except the last)

- **State transitions:** CK_S_IDX and CK_D_IDX states (included in above counts)
- **Completion:** 1 cycle for BCD_DONE state

8.8.3 Latency Examples

WIDTH	DIGITS	Total Cycles	@ 100MHz	@ 50MHz
4	2	$4 + 3 \times 2 = 10$	100ns	200ns
8	3	$8 + 7 \times 3 = 29$	290ns	580ns
12	4	$12 + 11 \times 4 = 56$	560ns	1.12μs
16	5	$16 + 15 \times 5 = 91$	910ns	1.82μs
20	7	$20 + 19 \times 7 = 153$	1.53μs	3.06μs

8.8.4 Pipeline Considerations

The conversion is inherently sequential, making pipelining challenging: - **Data Dependency:** Each iteration depends on previous results - **State-Based:** FSM prevents overlapping conversions - **Optimization:** Consider parallel converters for high throughput

8.9 BCD Format and Encoding

8.9.1 BCD Digit Encoding

Each BCD digit uses 4 bits to represent decimal values 0-9:

Decimal	BCD (4-bit)	Hex
0	0000	0x0
1	0001	0x1
2	0010	0x2
3	0011	0x3
4	0100	0x4
5	0101	0x5
6	0110	0x6
7	0111	0x7
8	1000	0x8

Decimal	BCD (4-bit)	Hex
9	1001	0x9
Invalid	1010-1111	0xA-0xF

8.9.2 Packed BCD Format

The output bcd uses packed BCD format where multiple digits are concatenated:

```
// For DIGITS = 3, bcd[11:0] contains:
// bcd[11:8] = hundreds digit
// bcd[7:4] = tens digit
// bcd[3:0] = ones digit

// Example: decimal 123 → bcd = 12'h123
```

8.9.3 BCD Digit Extraction

```
// Extract individual digits from packed BCD
logic [3:0] ones, tens, hundreds;

assign ones      = bcd[3:0];
assign tens      = bcd[7:4];
assign hundreds = bcd[11:8];

// Convert to ASCII for display
logic [7:0] ascii_ones, ascii_tens, ascii_hundreds;

assign ascii_ones   = 8'h30 + ones;    // Add ASCII '0'
assign ascii_tens   = 8'h30 + tens;
assign ascii_hundreds = 8'h30 + hundreds;
```

8.10 Worked Examples

8.10.1 Example 1: Converting 8-bit Binary 156 (WIDTH=8, DIGITS=3)

Initial State: - Binary: 8'b10011100 (156 decimal) - BCD: 12'b000000000000 - Loop count: 0

Iteration 1: 1. SHIFT: BCD = 000000000001, Binary = 00111000, Loop = 0 2. CK_S_IDX: More bits to process 3. ADD: Check digits [000, 000, 001] - none ≥ 5 4. CK_D_IDX: Check next digit 5. ADD: Check digits [000, 000, 001] - none ≥ 5 6. CK_D_IDX: Check next digit 7. ADD: Check digits [000, 000, 001] - none ≥ 5 8. CK_D_IDX: All digits checked, go to SHIFT

Iteration 2: 1. SHIFT: BCD = 000000000010, Binary = 01110000, Loop = 1 2. ADD phase: No digits ≥ 5

Iteration 3: 1. SHIFT: BCD = 000000000101, Binary = 11100000, Loop = 2

2. ADD phase: Ones digit = 5 $\geq$ 5, add 3 $\rightarrow$ BCD = 000000001000

Iteration 4: 1. SHIFT: BCD = 000000010001, Binary = 11000000, Loop = 3 2. ADD phase: No digits $\geq$ 5

Iteration 5: 1. SHIFT: BCD = 000000100011, Binary = 10000000, Loop = 4 2. ADD phase: No digits $\geq$ 5

Iteration 6: 1. SHIFT: BCD = 000001000111, Binary = 00000000, Loop = 5 2. ADD phase: Ones = 7 $\geq$ 5, add 3 $\rightarrow$ BCD = 000001001010 (ones = 10, invalid!) Actually: 7 + 3 = 10 = 1010₂, but this represents BCD "10" which will be handled in next shift

Iteration 7: 1. SHIFT: BCD = 000010010100, Loop = 6 2. ADD phase: Tens = 9, Ones = 4 - none $\geq$ 5

Iteration 8: 1. SHIFT: BCD = 000100101000, Loop = 7 2. CK_S_IDX: Loop count = 7 = WIDTH-1, conversion done

Final Result: BCD = 000100101000 = 4'h1, 4'h5, 4'h6 = 156 decimal $\checkmark$

8.10.2 Example 2: Converting 4-bit Binary 9 (WIDTH=4, DIGITS=2)

Step-by-step conversion of binary 1001₂ (9₁₀):

Cycle	State	Action	BCD	Binary	Notes
0	IDLE	Start	00000000	1001	Initialize
1	SHIFT	Shift bit	00000001	0010	MSB $\rightarrow$ BCD LSB
2	ADD	Check digits	00000001	0010	Ones=1 <5, no add
3	SHIFT	Shift bit	00000010	0100	Next bit
4	ADD	Check digits	00000010	0100	Ones=2 <5, no add
5	SHIFT	Shift bit	00000100	1000	Next bit
6	ADD	Check digits	00000100	1000	Ones=4 <5, no add
7	SHIFT	Shift bit	00001001	0000	Final bit
8	DONE	Complete	00001001	0000	Result: BCD=09

Result: BCD = 8'b00001001 = 09₁₆ = 9₁₀ ✓

8.11 Design Examples and Applications

8.11.11. Seven-Segment Display Driver

```
module display_controller #(
    parameter int WIDTH = 8
) (
    input logic clk,
    input logic rst_n,
    input logic [WIDTH-1:0] binary_value,
    input logic update,
    output logic [6:0] seg_hundreds,
    output logic [6:0] seg_tens,
    output logic [6:0] seg_ones,
    output logic display_ready
);

    // BCD converter
    logic [11:0] bcd_result;
    logic conversion_done;

    bin_to_bcd #(
        .WIDTH(WIDTH),
        .DIGITS(3)
    ) converter (
        .clk(clk),
        .rst_n(rst_n),
        .start(update),
        .binary(binary_value),
        .bcd(bcd_result),
        .done(conversion_done)
    );

    // Extract digits
    logic [3:0] digit_hundreds, digit_tens, digit_ones;
    assign digit_hundreds = bcd_result[11:8];
    assign digit_tens      = bcd_result[7:4];
    assign digit_ones      = bcd_result[3:0];

    // Seven-segment decoders
    seven_seg_decoder dec_hundreds
    (.digit(digit_hundreds), .segments(seg_hundreds));
    seven_seg_decoder dec_tens
    (.digit(digit_tens), .segments(seg_tens));
    seven_seg_decoder dec_ones
    (.digit(digit_ones), .segments(seg_ones));
```

```

    assign display_ready = conversion_done;

endmodule

// Seven-segment decoder
module seven_seg_decoder (
    input logic [3:0] digit,
    output logic [6:0] segments // {g,f,e,d,c,b,a}
);
    always_comb begin
        case (digit)
            4'd0: segments = 7'b0111111; // 0
            4'd1: segments = 7'b0000110; // 1
            4'd2: segments = 7'b1011011; // 2
            4'd3: segments = 7'b1001111; // 3
            4'd4: segments = 7'b1100110; // 4
            4'd5: segments = 7'b1101101; // 5
            4'd6: segments = 7'b1111101; // 6
            4'd7: segments = 7'b0000111; // 7
            4'd8: segments = 7'b1111111; // 8
            4'd9: segments = 7'b1101111; // 9
            default: segments = 7'b1000000; // Error (top segment
only)
        endcase
    end
endmodule

```

8.11.22. LCD/UART Decimal Display

```

module decimal_uart_tx #(
    parameter int WIDTH = 16,
    parameter int DIGITS = 5
) (
    input logic clk,
    input logic rst_n,
    input logic [WIDTH-1:0] value,
    input logic send_value,
    output logic uart_tx,
    output logic tx_busy
);

// BCD conversion
logic [DIGITS*4-1:0] bcd_value;
logic bcd_done;

bin_to_bcd #(
    .WIDTH(WIDTH),

```

```

        .DIGITS(DIGITS)
    ) bcd_conv (
        .clk(clk),
        .rst_n(rst_n),
        .start(send_value),
        .binary(value),
        .bcd(bcd_value),
        .done(bcd_done)
    );

    // ASCII conversion and transmission
    typedef enum {IDLE, CONVERT, TRANSMIT} state_t;
    state_t state;

    logic [2:0] digit_idx;
    logic [3:0] current_digit;
    logic [7:0] ascii_char;
    logic uart_start, uart_done;

    // Extract current digit (MSB first for display)
    assign current_digit = bcd_value[(DIGITS-1-digit_idx)*4+:4];
    assign ascii_char = 8'h30 + current_digit; // Convert to ASCII

    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            state <= IDLE;
            digit_idx <= 0;
            uart_start <= 0;
        end else begin
            uart_start <= 0; // Default

            case (state)
                IDLE: begin
                    if (bcd_done) begin
                        state <= TRANSMIT;
                        digit_idx <= 0;
                    end
                end

                TRANSMIT: begin
                    if (!uart_busy && !uart_start) begin
                        uart_start <= 1;
                        if (digit_idx == DIGITS-1) begin
                            state <= IDLE;
                            digit_idx <= 0;
                        end else begin
                            digit_idx <= digit_idx + 1;
                        end
                    end
                end
            endcase
        end
    end

```

```

        end
    endcase
end
end

// UART transmitter instance
uart_tx_module uart_tx_inst (
    .clk(clk),
    .rst_n(rst_n),
    .data(ascii_char),
    .start(uart_start),
    .tx(uart_tx),
    .busy(uart_busy),
    .done(uart_done)
);

assign tx_busy = (state != IDLE);

endmodule

```

8.11.33. Multi-Channel BCD Converter

```

module multi_channel_bcd #(
    parameter int CHANNELS = 4,
    parameter int WIDTH = 12,
    parameter int DIGITS = 4
) (
    input logic clk,
    input logic rst_n,
    input logic [CHANNELS-1:0] convert_request,
    input logic [WIDTH-1:0] binary_data [CHANNELS],
    output logic [DIGITS*4-1:0] bcd_results [CHANNELS],
    output logic conversion_done,
    output logic converter_busy
);

// Round-robin arbitration for converter access
logic [$clog2(CHANNELS)-1:0] current_channel;
logic [CHANNELS-1:0] pending_requests;
logic conversion_active;

// Single BCD converter instance
logic [WIDTH-1:0] conv_binary;
logic conv_start, conv_done;
logic [DIGITS*4-1:0] conv_bcd;

bin_to_bcd #(

```



```

        .WIDTH(WIDTH),
        .DIGITS(DIGITS)
    ) converter (
        .clk(clk),
        .rst_n(rst_n),
        .start(conv_start),
        .binary(conv_binary),
        .bcd(conv_bcd),
        .done(conv_done)
    );

    // Request management
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            pending_requests <= 0;
            current_channel <= 0;
            conversion_active <= 0;
            conv_start <= 0;
            conversion_done <= 0;
        end else begin
            conv_start <= 0;
            conversion_done <= 0;

            // Latch new requests
            pending_requests <= (pending_requests | convert_request) &
                                ~(conversion_done);

            if (!conversion_active && |pending_requests) begin
                // Find next channel to service
                for (int i = 0; i < CHANNELS; i++) begin
                    if (pending_requests[(current_channel + i) %
CHANNELS]) begin
                        current_channel <= (current_channel + i) %
CHANNELS;
                        conv_binary <= binary_data[(current_channel +
i) % CHANNELS];
                        conv_start <= 1;
                        conversion_active <= 1;
                        break;
                    end
                end
            end

            if (conv_done) begin
                bcd_results[current_channel] <= conv_bcd;
                conversion_done[current_channel] <= 1;
                pending_requests[current_channel] <= 0;
                conversion_active <= 0;
            end
        end
    end

```

```

        current_channel <= (current_channel + 1) % CHANNELS;
    end
end
end

assign converter_busy = conversion_active || |pending_requests;

endmodule

```

8.12 Optimization Techniques

8.12.11. Early Termination Optimization

```

// Optimize for small numbers by detecting leading zeros
logic [WIDTH-1:0] leading_zero_mask;
logic [$clog2(WIDTH):0] significant_bits;

```

```

// Count significant bits
count_leading_zeros #(.WIDTH(WIDTH)) clz_inst (
    .data(binary),
    .clz(leading_zeros)
);

```

```

assign significant_bits = WIDTH - leading_zeros;

```

```

// Modify loop termination condition
always_ff @(posedge clk) begin
    // ... other logic ...

    CK_S_IDX: begin
        if (r_loop_count == significant_bits - 1) begin
            r_fsm_main <= BCD_DONE; // Early termination
        end else begin
            // Continue normal processing
        end
    end
end
end

```

8.12.22. Parallel Digit Processing

```

// Process all digits in parallel during ADD state
logic [3:0] digit_array [DIGITS];
logic [3:0] adjusted_digits [DIGITS];

// Extract all digits
genvar i;
generate
    for (i = 0; i < DIGITS; i++) begin : extract_digits

```

```

        assign digit_array[i] = r_bcd[i*4+:4];
        assign adjusted_digits[i] = (digit_array[i] > 4) ?
                                   (digit_array[i] + 3) :
digit_array[i];
    end
endgenerate

// Update all digits simultaneously
always_ff @(posedge clk) begin
    // ... other states ...

    ADD: begin
        for (int j = 0; j < DIGITS; j++) begin
            r_bcd[j*4+:4] <= adjusted_digits[j];
        end
        r_fsm_main <= CK_S_IDX; // Skip CK_D_IDX state
    end
end

```

8.12.33. Pipelined Implementation

```

// Multi-stage pipeline for high throughput
module bin_to_bcd_pipelined #(
    parameter int WIDTH = 8,
    parameter int DIGITS = 3,
    parameter int STAGES = 4 // Pipeline depth
) (
    input logic clk,
    input logic rst_n,
    input logic valid_in,
    input logic [WIDTH-1:0] binary_in,
    output logic valid_out,
    output logic [DIGITS*4-1:0] bcd_out
);

// Pipeline stages
logic [STAGES-1:0] valid_pipe;
logic [WIDTH-1:0] binary_pipe [STAGES];
logic [DIGITS*4-1:0] bcd_pipe [STAGES];

// Stage processing
generate
    for (genvar s = 0; s < STAGES; s++) begin : pipeline_stages
        logic [7:0] bits_to_process;
        assign bits_to_process = WIDTH * s / STAGES;

        // Process subset of bits in each stage
        // Implementation depends on specific algorithm
    end
endgenerate

```

```
partitioning
    end
endgenerate
```

```
endmodule
```

8.13 Verification and Testing

8.13.1 Comprehensive Test Strategy

```
module tb_bin_to_bcd;
```

```
    // Test parameters
    parameter int WIDTH = 8;
    parameter int DIGITS = 3;
    parameter int MAX_VALUE = (1 << WIDTH) - 1;

    // DUT signals
    logic clk, rst_n, start, done;
    logic [WIDTH-1:0] binary;
    logic [DIGITS*4-1:0] bcd;

    // Test control
    logic [WIDTH-1:0] test_vector;
    logic [31:0] expected_decimal;
    logic [DIGITS*4-1:0] expected_bcd;

    // DUT instantiation
    bin_to_bcd #(
        .WIDTH(WIDTH),
        .DIGITS(DIGITS)
    ) dut (.*);

    // Clock generation
    initial clk = 0;
    always #5 clk = ~clk;

    // Test sequence
    initial begin
        rst_n = 0;
        start = 0;
        binary = 0;

        #100 rst_n = 1;

        // Test all possible values
        for (int i = 0; i <= MAX_VALUE; i++) begin
            test_conversion(i);
        end
    end
endmodule
```

```

    end

    // Boundary tests
    test_conversion(0);
    test_conversion(1);
    test_conversion(MAX_VALUE);

    // Random tests
    repeat (1000) begin
        test_conversion($random % (MAX_VALUE + 1));
    end

    $display("All tests completed successfully!");
    $finish;
end

// Test task
task test_conversion(input [WIDTH-1:0] value);
    begin
        binary = value;
        expected_decimal = value;
        expected_bcd = decimal_to_bcd(expected_decimal);

        @(posedge clk);
        start = 1;
        @(posedge clk);
        start = 0;

        wait(done);
        @(posedge clk);

        if (bcd !== expected_bcd) begin
            $error("Mismatch: binary=%d, expected_bcd=%h,
actual_bcd=%h",
                value, expected_bcd, bcd);
        end else begin
            $display("PASS: binary=%d → bcd=%h", value, bcd);
        end
    end
endtask

// Reference function
function [DIGITS*4-1:0] decimal_to_bcd(input [31:0] decimal);
    logic [31:0] temp;
    logic [DIGITS*4-1:0] result;
    begin
        temp = decimal;
        for (int i = 0; i < DIGITS; i++) begin

```

```

        result[i*4+:4] = temp % 10;
        temp = temp / 10;
    end
    decimal_to_bcd = result;
end
endfunction

```

```
endmodule
```

8.13.2 Coverage Model

```

covergroup bcd_conversion_cg;

    // Input value coverage
    cp_binary_value: coverpoint binary {
        bins zero = {0};
        bins small[] = {[1:9]};
        bins medium[] = {[10:99]};
        bins large[] = {[100:255]}; // For 8-bit
        bins max_val = {2**WIDTH - 1};
    }

    // State coverage
    cp_fsm_states: coverpoint dut.r_fsm_main {
        bins all_states[] = {IDLE, SHIFT, CK_S_IDX, ADD, CK_D_IDX,
BCD_DONE};
        bins state_transitions[] = (IDLE => SHIFT),
                                   (SHIFT => CK_S_IDX),
                                   (CK_S_IDX => ADD),
                                   (CK_S_IDX => BCD_DONE),
                                   (ADD => CK_D_IDX),
                                   (CK_D_IDX => ADD),
                                   (CK_D_IDX => SHIFT),
                                   (BCD_DONE => IDLE);
    }

    // BCD digit coverage
    cp_bcd_digits: coverpoint bcd {
        bins valid_bcd[] = {[12'h000:12'h999]}; // Valid BCD range
    }

    // Loop iteration coverage
    cp_loop_count: coverpoint dut.r_loop_count {
        bins all_counts[] = {[0:WIDTH-1]};
    }

    // Cross coverage
    cross_value_states: cross cp_binary_value, cp_fsm_states;

```

```
endcovergroup
```

8.13.3 Formal Verification Properties

```
// Properties for formal verification
module bin_to_bcd_properties #(
    parameter int WIDTH = 8,
    parameter int DIGITS = 3
);

    // Bind to DUT
    bind bin_to_bcd bin_to_bcd_properties #(WIDTH, DIGITS) props_inst
    (.*) ;

    // Basic properties
    property conversion_eventually_completes;
        @(posedge clk) disable iff (!rst_n)
        start |-> ##[1:200] done; // Adjust bound based on max latency
    endproperty

    property bcd_output_valid;
        @(posedge clk) disable iff (!rst_n)
        done |-> valid_bcd(bcd);
    endproperty

    property conversion_correctness;
        logic [31:0] expected_decimal;
        @(posedge clk) disable iff (!rst_n)
        (start, expected_decimal = binary) |->
        ##[1:200] (done && (bcd_to_decimal(bcd) == expected_decimal));
    endproperty

    // Helper functions
    function bit valid_bcd(logic [DIGITS*4-1:0] bcd_val);
        for (int i = 0; i < DIGITS; i++) begin
            if (bcd_val[i*4+:4] > 9) return 0;
        end
        return 1;
    endfunction

    function [31:0] bcd_to_decimal(logic [DIGITS*4-1:0] bcd_val);
        logic [31:0] result = 0;
        for (int i = 0; i < DIGITS; i++) begin
            result += bcd_val[i*4+:4] * (10**i);
        end
        return result;
    endfunction
```

```
// Assertions
assert property (conversion_eventually_completes);
assert property (bcd_output_valid);
assert property (conversion_correctness);
```

```
endmodule
```

8.14 Performance and Resource Analysis

8.14.1 Synthesis Results (Typical FPGA)

Configurat ion	LUTs	FFs	BRAM	Max Freq	Notes
8-bit, 3- digit	85	45	0	250 MHz	Basic config
12-bit, 4- digit	145	65	0	220 MHz	Medium size
16-bit, 5- digit	220	85	0	200 MHz	Larger design
20-bit, 7- digit	350	125	0	180 MHz	Complex design

8.14.2 Power Consumption Analysis

- **Dynamic Power:** Proportional to conversion frequency
- **Static Power:** Minimal (standard CMOS)
- **Optimization:** Clock gating during IDLE state
- **Trade-off:** Performance vs. power efficiency

8.15 Common Design Patterns

8.15.1 Pattern 1: Auto-Converting Display Interface

```
module auto_display_interface #(
    parameter int WIDTH = 10
) (
    input logic [WIDTH-1:0] sensor_data,
    input logic             data_valid,
    output logic [6:0]      display_segments[4],
    output logic            display_valid
```



```

);

// Auto-trigger conversion on new data
logic prev_data_valid;
logic start_conversion;

always_ff @(posedge clk) begin
    prev_data_valid <= data_valid;
end

assign start_conversion = data_valid && !prev_data_valid;

// BCD converter and display driver
// ... (implementation similar to previous examples)

endmodule

```

8.15.2 Pattern 2: Buffered Converter with FIFO

```

module buffered_bcd_converter #(
    parameter int WIDTH = 12,
    parameter int DIGITS = 4,
    parameter int BUFFER_DEPTH = 8
) (
    input logic clk,
    input logic rst_n,
    input logic [WIDTH-1:0] binary_in,
    input logic write_enable,
    output logic [DIGITS*4-1:0] bcd_out,
    output logic bcd_valid,
    output logic buffer_full,
    output logic buffer_empty
);

// Input FIFO
logic [WIDTH-1:0] fifo_data_out;
logic fifo_read_enable, fifo_empty_int;

sync_fifo #(
    .WIDTH(WIDTH),
    .DEPTH(BUFFER_DEPTH)
) input_buffer (
    .clk(clk),
    .rst_n(rst_n),
    .write_data(binary_in),
    .write_enable(write_enable),
    .read_data(fifo_data_out),
    .read_enable(fifo_read_enable),

```

```

        .full(buffer_full),
        .empty(fifo_empty_int)
    );

    // BCD converter control
    logic conversion_idle;

    assign fifo_read_enable = !fifo_empty_int && conversion_idle;
    assign buffer_empty = fifo_empty_int;

    // BCD converter instance
    bin_to_bcd #(
        .WIDTH(WIDTH),
        .DIGITS(DIGITS)
    ) converter (
        .clk(clk),
        .rst_n(rst_n),
        .start(fifo_read_enable),
        .binary(fifo_data_out),
        .bcd(bcd_out),
        .done(bcd_valid)
    );

    // Track converter state
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            conversion_idle <= 1;
        end else begin
            if (fifo_read_enable) begin
                conversion_idle <= 0;
            end else if (bcd_valid) begin
                conversion_idle <= 1;
            end
        end
    end
end

endmodule

```

8.16 Troubleshooting Guide

8.16.1 Common Issues and Solutions

8.16.1.1 *Issue 1: Incorrect BCD Output*

Symptoms: Output doesn't match expected decimal value **Causes:** - Insufficient DIGITS parameter
 - Width mismatch between binary input and internal registers - BCD digit corruption during processing

Debug Steps:

```
// Add internal signal monitoring
always_ff @(posedge clk) begin
    if (dut.r_fsm_main == ADD && dut.w_bcd_digit > 4) begin
        $display("ADD operation: digit=%d, new_value=%d",
            dut.w_bcd_digit, dut.w_bcd_digit + 3);
    end
end
```

8.16.1.2 Issue 2: Conversion Never Completes

Symptoms: done signal never asserts **Causes:** - FSM stuck in loop - Incorrect loop termination condition - Clock or reset issues

Debug Steps:

```
// Monitor FSM state transitions
always_ff @(posedge clk) begin
    $display("State: %s, Loop: %d, Digit: %d",
        dut.r_fsm_main.name(), dut.r_loop_count,
        dut.r_digit_index);
end
```

8.16.1.3 Issue 3: Timing Violations

Symptoms: Synthesis reports timing failures **Causes:** - Long combinational paths in ADD logic - High fan-out on BCD register - Insufficient pipeline stages

Solutions:

```
// Add pipeline registers
logic [DIGITS*4-1:0] bcd_pipe;
always_ff @(posedge clk) begin
    bcd_pipe <= r_bcd;
end

// Use parallel digit processing
// (Shown in optimization section)
```

This comprehensive documentation provides everything needed to understand, implement, verify, and optimize the Binary to BCD converter module. The Double-Dabble algorithm, while sequential in nature, provides reliable and predictable conversion suitable for a wide range of digital display applications.

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9 CAM Tag Module

9.1 Purpose

The `cam_tag` module implements a Content Addressable Memory (CAM) for tag storage and lookup. It provides fast associative lookup capabilities where data can be found based on content rather than address, commonly used in cache implementations, translation lookaside buffers (TLBs), and routing tables.

9.2 Parameters

- `ENABLE`: Enable/disable CAM functionality (default: 1)
- `N`: Width of each tag in bits (default: 8)
- `DEPTH`: Number of tag entries in the CAM (default: 16)

9.3 Ports

9.3.1 Inputs

- `clk`: System clock
- `rst_n`: Active-low asynchronous reset
- `tag_in_status[N-1:0]`: Tag to search for (lookup operation)
- `mark_valid`: Signal to add a new valid tag
- `tag_in_valid[N-1:0]`: Tag value to mark as valid
- `mark_invalid`: Signal to invalidate an existing tag
- `tag_in_invalid[N-1:0]`: Tag value to mark as invalid

9.3.2 Outputs

- `tags_empty`: Indicates when CAM contains no valid entries
- `tags_full`: Indicates when CAM is completely full
- `tag_status`: Result of lookup operation (1 if tag found, 0 if not found)

9.4 Key Internal Structures

9.4.1 Storage Arrays

```
logic [N-1:0]    r_tag_array [0:DEPTH-1]; // Tag storage array
logic [DEPTH-1:0] r_valid;               // Valid bit for each
entry
```

9.4.2 Search Logic

Three parallel search operations are implemented:

9.4.2.1 1. Next Available Location Search

```
always_comb begin
    w_next_valid_loc = -1;
    for (int i=DEPTH-1; i >= 0; i--)
        if (r_valid[i] == 1'b0)
            w_next_valid_loc = i;
end
```

- Searches from highest to lowest index
- Returns the highest-indexed free location
- Returns -1 if no free locations available

9.4.2.2 2. Valid Tag Match Search

```
always_comb begin
    w_match_loc = -1;
    for (int i = 0; i < DEPTH; i++)
        if (r_valid[i] == 1'b1 && tag_in_status == r_tag_array[i])
            w_match_loc = i;
end
```

- Searches for exact tag match among valid entries
- Returns index of matching entry or -1 if no match

9.4.2.3 3. Invalid Tag Match Search

```
always_comb begin
    w_match_invalid_loc = -1;
    for (int i = 0; i < DEPTH; i++)
        if (r_valid[i] == 1'b1 && tag_in_invalid == r_tag_array[i])
            w_match_invalid_loc = i;
end
```

- Used for invalidation operations

- Finds valid entries that match the invalidation tag

9.5 Operations

9.5.1 Tag Insertion

When mark_valid is asserted and CAM is not full: 1. Store tag_in_valid at w_next_valid_loc
2. Set corresponding valid bit 3. Only occurs when ENABLE != 0

9.5.2 Tag Invalidation

When mark_invalid is asserted and matching tag found: 1. Clear the tag value at matching location
2. Clear corresponding valid bit

9.5.3 Tag Lookup

Continuously compares tag_in_status against all valid entries: - Returns 1 if tag found among valid entries - Returns 0 if tag not found or entry is invalid

9.6 State Management

9.6.1 Update Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_valid <= 'b0;
        for (int i = 0; i < DEPTH; i++) begin
            r_tag_array[i] <= 'b0;
        end
    end else begin
        if (mark_valid && !tags_full && (ENABLE != 0)) begin
            r_tag_array[w_next_valid_loc] <= tag_in_valid;
            r_valid[w_next_valid_loc] <= 1'b1;
        end else if (mark_invalid && w_match_invalid_loc >= 0) begin
            r_tag_array[w_match_invalid_loc] <= 'b0;
            r_valid[w_match_invalid_loc] <= 1'b0;
        end
    end
end
```

9.6.2 Status Signals

- tags_empty = ~|r_valid: NOR of all valid bits
- tags_full = &r_valid: AND of all valid bits
- tag_status: Result of tag lookup operation

9.7 Special Features

9.7.1 Enable Control

When `ENABLE = 0`: - Tag insertion is disabled - Tag lookup and invalidation still function - Useful for disabling CAM without losing existing data

9.7.2 Allocation Strategy

- **First Available:** Uses highest-indexed free location
- **Replacement:** No automatic replacement policy (must manually invalidate)
- **Overflow Protection:** Prevents insertion when full

9.7.3 Debug Support

The module includes simulation-only logic for waveform viewing:

```
// synopsys translate_off
logic [(N*DEPTH)-1:0] flat_r_tag_array;
generate
    for (j = 0; j < DEPTH; j++) begin : gen_flatten_memory
        assign flat_r_tag_array[j*N+:N] = r_tag_array[j];
    end
endgenerate
// synopsys translate_on
```

9.8 Usage Considerations

- **Timing:** All lookups are combinational (single cycle)
- **Capacity:** Monitor `tags_full` before insertions
- **Conflicts:** No protection against duplicate tag insertion
- **Performance:** Search time is constant regardless of occupancy
- **Power:** All entries are searched simultaneously (high power for large depths)

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10 Clock Divider Module - Comprehensive Documentation

10.1 Overview

The `clock_divider` module generates multiple divided clock signals from a single input clock using configurable pick-off points from a shared master counter. This approach provides synchronized, phase-aligned divided clocks that are essential for multi-rate digital signal processing, communication systems, and power management applications.

⚠ WARNING: NOT recommended for functional clocks! Use PLL/MMCM/clock manager primitives instead. This module is intended for testbenches, debug outputs, and non-critical timing applications.

10.2 Module Declaration

```
module clock_divider #(
    parameter int N                = 4, // Number of output clocks
    parameter int PO_WIDTH        = 8, // Width of the Pick off
    registers
    parameter int COUNTER_WIDTH = 64 // Width of the counter
) (
    input logic clk, // Input clock
    signal
    input logic rst_n, // Active low
    reset signal
    input logic [(N*PO_WIDTH-1):0] pickoff_points, // the pick off
    point config registers
    output logic [N-1:0] divided_clk // Divided clock
    signals
);
```

10.3 Parameters

10.3.1 N

- **Type:** int
- **Default:** 4
- **Description:** Number of independently configurable output clocks
- **Range:** 1 to 16 (practical limit)
- **Impact:** Determines number of output clock domains and resource usage (N registers + N muxes)

10.3.2 PO_WIDTH

- **Type:** int
- **Default:** 8
- **Description:** Bit width of each pick-off point configuration register
- **Range:** 4 to 8 bits
- **Constraints:** Must be $> \lceil \log_2(\text{COUNTER_WIDTH}) \rceil$ to avoid truncation
 - Equivalently: $\text{PO_WIDTH} \geq \lceil \log_2(\text{COUNTER_WIDTH} + 1) \rceil$
 - This ensures PO_WIDTH can hold the value COUNTER_WIDTH without truncation
 - Examples: CW=16 needs $\text{PO} \geq 5$, CW=32 needs $\text{PO} \geq 6$, CW=64 needs $\text{PO} \geq 7$
- **Trade-off:** Larger width allows finer counter bit selection but uses more configuration storage
- **Typical:** PO_WIDTH=8 for COUNTER_WIDTH up to 128

10.3.3 COUNTER_WIDTH

- **Type:** int
- **Default:** 64
- **Description:** Bit width of the master binary counter
- **Range:** 2 to 64 bits (practical range)
- **Impact:** Determines maximum division ratio ($2^{\text{COUNTER_WIDTH}}$)
- **Example:** WIDTH=16 $\rightarrow$ max division ratio = 65536
- **Trade-off:** Larger counters enable slower frequencies but consume more resources

10.3.4 Parameter Relationships

- **Addressing Constraint:** $\text{PO_WIDTH} \geq \lceil \log_2(\text{COUNTER_WIDTH}) \rceil$
- **Division Range:** Each output can divide by 2^1 to $2^{\text{COUNTER_WIDTH}}$

- **Configuration Width:** Total configuration bits = $N \times PO_WIDTH$

10.4 Ports

10.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	Master input clock (source to be divided)
rst_n	1	logic	Active-low asynchronous reset
pickoff_points	$N \times PO_WIDTH$	logic	Packed pickoff point configuration array

pickoff_points Format:

Bit Packing: {po[N-1], ..., po[1], po[0]}

Each po[i] (PO_WIDTH bits): Counter bit index to sample for output i

10.4.2 Outputs

Port	Width	Type	Description
divided_clk	N	logic	Divided clock outputs (registered)

10.5 Architecture Details

10.5.1 Internal Signals

```

logic [COUNTER_WIDTH-1:0] r_divider_counters; // Free-running binary counter
localparam int ADDR_WIDTH = $clog2(COUNTER_WIDTH); // Address width

```

10.5.2 Counter Operation

- Free-running binary counter increments every input clock cycle
- Wraps at $2^{COUNTER_WIDTH}$ (no overflow flag)
- Each bit toggles at half the frequency of the previous bit
 - Bit 0: $clk/2$ (toggles every cycle)
 - Bit 1: $clk/4$ (toggles every 2 cycles)
 - Bit N: $clk/2^{(N+1)}$

10.5.3 Pickoff Point Selection

- Each output i samples `counter[pickoff_points[i]]`
- Division ratio = $2^{(\text{pickoff_point}+1)}$
- Example: `pickoff=3` → samples `counter[3]` → divides by 16

10.5.4 Out-of-Range Handling

- If `pickoff_points[i] ≥ COUNTER_WIDTH`, clamps to `COUNTER_WIDTH-1` (MSB)
- Prevents illegal bit indexing
- MSB provides slowest possible division

10.5.5 Output Registration

- All `divided_clk` outputs are registered for glitch-free operation
- Adds 1 cycle latency but prevents combinational glitches
- Reset clears all outputs to 0 (low phase)

10.6 Operation Principles

10.6.1 Division Ratio Table (100MHz Input Clock)

<code>pickoff_point</code>	Counter Bit	Division Ratio	Output Freq	Period
0	[0]	2	50MHz	20ns
1	[1]	4	25MHz	40ns
2	[2]	8	12.5MHz	80ns
3	[3]	16	6.25MHz	160ns
4	[4]	32	3.125MHz	320ns
5	[5]	64	1.5625MHz	640ns
10	[10]	2048	48.8kHz	20.5μs
15	[15]	65536	1.5kHz	655μs
19	[19]	1048576	95Hz	10.5ms
23	[23]	16777216	6Hz	168ms

10.6.2 Duty Cycle

- All divided clocks have 50% duty cycle
- Result of sampling counter bit (toggle behavior)
- High for $2^{(\text{pickoff_point})}$ input cycles
- Low for $2^{(\text{pickoff_point})}$ input cycles

10.7 Timing Characteristics

10.7.1 Latency

- **Output Latency:** 1 cycle (registered output)
- **Clock-to-Q:** Standard flip-flop delay
- **Propagation:** Counter increment → mux → output register

10.7.2 Phase Relationships

- All divided clocks share same counter → phase-locked to each other
- Divided clocks have phase offset relative to input clock
- Phase offset depends on counter reset value (0)

10.7.3 Jitter

- Input clock jitter propagates to outputs
- Additional quantization jitter: ± 1 input clock cycle
- For low-jitter clocks: Use dedicated PLL/clock synthesis

10.8 Design Examples

10.8.1 Example 1: Four Debug Clocks

```
// Generate clk/4, clk/16, clk/256, clk/65536
logic [31:0] pickoff_cfg;
assign pickoff_cfg = {8'd15, 8'd7, 8'd3, 8'd1}; // {po3, po2, po1, po0}
```

```
clock_divider #(
    .N(4),                // 4 outputs
    .PO_WIDTH(8),         // 8-bit pickoff point
```

```

        .COUNTER_WIDTH(16) // 16-bit counter (max div = 65536)
    ) u_clk_div (
        .clk            (clk_100mhz),
        .rst_n          (rst_n),
        .pickoff_points(pickoff_cfg),
        .divided_clk     ({clk_1p5khz, clk_390khz, clk_6p25mhz, clk_25mhz})
    );

```

10.8.2 Example 2: Baud Rate Generator (Approximate)

```

// For 9600 baud from 100MHz:
// Ideal: 100MHz / 9600 ≈ 10417 (not power-of-2)
// Closest: 213 = 8192 → 100MHz/8192 = 12.2kHz (≈9600 baud)

```

```

logic [7:0] baud_pickoff = 8'd12; // 213 = 8192 division

clock_divider #(
    .N(1),
    .PO_WIDTH(8),
    .COUNTER_WIDTH(16)
) u_baud_clk (
    .clk            (sys_clk),
    .rst_n          (rst_n),
    .pickoff_points(baud_pickoff),
    .divided_clk    (uart_baud_x16) // 16× oversampling clock
);

```

Note: For precise baud rates (not power-of-2), use counter_load_clear instead.

10.8.3 Example 3: Runtime-Programmable Divider

```

// APB-configurable clock divider
logic [7:0] cfg_pickoff_0, cfg_pickoff_1;

always_ff @(posedge pclk) begin
    if (pwrite && paddr == ADDR_PICKOFF_0) cfg_pickoff_0 <=
pdata[7:0];
    if (pwrite && paddr == ADDR_PICKOFF_1) cfg_pickoff_1 <=
pdata[7:0];
end

clock_divider #(
    .N(2),
    .PO_WIDTH(8),
    .COUNTER_WIDTH(24)
) u_cfg_clk_div (
    .clk            (core_clk),
    .rst_n          (core_rst_n),
    .pickoff_points({cfg_pickoff_1, cfg_pickoff_0}),

```

```

        .divided_clk    ({clk_out_1, clk_out_0})
    );

```

10.8.4 Example 4: Test Clock Generation

```

// Generate 8 test clocks for debug
clock_divider #(
    .N(8),                      // 8 debug clocks
    .PO_WIDTH(8),
    .COUNTER_WIDTH(32) // Support ultra-slow clocks
) u_test_clks (
    .clk            (ref_clk),
    .rst_n          (test_rst_n),
    .pickoff_points({8'd20, 8'd18, 8'd16, 8'd14, 8'd12, 8'd10, 8'd8,
8'd6}),
    .divided_clk    (test_clk_bus)
);

```

10.9 Advanced Usage

10.9.1 Changing Pickoff Points at Runtime

```

// To avoid glitches when changing pickoff configuration:
// 1. Assert rst_n (reset)
// 2. Change pickoff_points
// 3. Wait for configuration to settle
// 4. Deassert rst_n
// 5. Outputs start with new division ratios

```

```

always_ff @(posedge cfg_clk) begin
    if (cfg_write) begin
        rst_n <= 1'b0;                      // Assert reset
        new_pickoff <= cfg_data;

        @(posedge cfg_clk);
        pickoff_points <= new_pickoff;      // Update configuration

        @(posedge cfg_clk);
        rst_n <= 1'b1;                      // Release reset
    end
end

```

10.10 Synthesis Considerations

10.10.1 Resource Utilization

- **Flip-Flops:** COUNTER_WIDTH (counter) + N (output registers)

- **LUTs:** N muxes (pickoff selection) + clamping logic
- **Example (N=4, COUNTER_WIDTH=16):** ~20 FFs, ~8 LUTs

10.10.2 Timing Optimization

- **Critical Path:** Counter increment → mux → output register
- **For high-frequency operation:**
 - Reduce COUNTER_WIDTH (if possible)
 - Pipeline counter output (add register stage)
 - Use dedicated clock resources (PLL/MMCM)

10.10.3 Power Optimization

- **Clock Gating:** Gate counter when no divided clocks needed
- **Selective Outputs:** Only generate divided clocks actually used
- **Example:**

```

logic counter_enable;
assign counter_enable = |divided_clk_enable;  // Only run when needed

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) r_divider_counters <= 0;
    else if (counter_enable) r_divider_counters <= r_divider_counters
+ 1;
end

```

10.11 Important Warnings

10.11.1 NOT for Functional Clocks

Do NOT use for: - Primary system clocks - Clocking synchronous logic - Production designs requiring precise timing

Use PLL/MMCM/Clock Manager instead: - Dedicated clock synthesis primitives - Clean clock tree - Proper static timing analysis (STA) - Phase-locked loops for low jitter - Precise frequency generation

10.11.2 Derived Clock Hazards

Using divided_clk as a clock creates a **derived clock**: - Complicates timing analysis - STA tools may not properly constrain - Potential setup/hold violations - Clock domain crossing issues

10.11.3 ⚠️ Only Power-of-2 Divisions

- This module ONLY supports power-of-2 division ratios (2, 4, 8, 16, ...)
- For arbitrary ratios (e.g., divide by 7, 100, 1000): Use counter_load_clear

10.11.4 ⚠️ Glitches During Configuration Changes

- Changing pickoff_points at runtime can cause glitches
- Always reset module when changing configuration
- Do NOT change configuration while outputs are actively used

10.12 Verification Strategy

10.12.1 Test Scenarios

1. **Single Output:** N=1 with various pickoff points
2. **Multiple Outputs:** N=4 with different divisions
3. **Out-of-Range Clamping:** pickoff $\geq$ COUNTER_WIDTH
4. **Runtime Changes:** Modify pickoff_points dynamically
5. **Reset Behavior:** All outputs low after reset
6. **Division Verification:** Measure output frequencies
7. **Edge Cases:**
 - pickoff=0 (divide by 2)
 - pickoff=COUNTER_WIDTH-1 (slowest division)
 - pickoff > COUNTER_WIDTH (clamping)

10.12.2 Coverage Points

```
covergroup clock_divider_cg @(posedge clk);
  cp_pickoff_range: coverpoint pickoff_points[7:0] {
    bins low = {[0:3]};
    bins mid = {[4:11]};
    bins high = {[12:15]};
    bins oob = {[16:255]}; // Out-of-bounds (will clamp)
  }
```



```

    cp_outputs: coverpoint divided_clk {
        bins all_outputs[] = {[1:$]};
    }
endgroup

```

10.13 Performance Characteristics

10.13.1 Maximum Frequency

- **Limiting Factor:** Counter width and mux depth
- **Typical Performance:**
 - COUNTER_WIDTH=16: ~500 MHz (modern FPGAs)
 - COUNTER_WIDTH=32: ~400 MHz
 - COUNTER_WIDTH=64: ~300 MHz

10.13.2 Resource Scaling

COUNTER_W IDTH	N Outputs	FFs	LUTs	fmax (Est.)
16	4	20	8	500 MHz
32	4	36	10	400 MHz
64	4	68	12	300 MHz
16	8	24	16	500 MHz

10.14 Related Modules

- counter_bin.sv - Simple binary counter (used internally)
- counter_freq_invariant.sv - Time-based counter with 1MHz tick
- counter_load_clear.sv - Arbitrary division ratios (not power-of-2)
- clock_pulse.sv - Configurable pulse generator
- clock_gate_ctrl.sv - Clock gating for power management

10.15 Navigation

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11 Clock Gate Controller - Comprehensive Documentation

11.1 Overview

The `clock_gate_ctrl` module implements an intelligent clock gating controller that automatically stops the clock when a circuit block becomes idle, significantly reducing dynamic power consumption. It features configurable idle timeout, immediate wake-up capability, and safe clock gating using integrated clock gate (ICG) cells.

11.2 Module Declaration

```
module clock_gate_ctrl #(
    parameter int IDLE_CNTR_WIDTH = 4,
    parameter int N = IDLE_CNTR_WIDTH
) (
    // Inputs
    input logic      clk_in,
    input logic      aresetn,
    input logic      cfg_cg_enable,      // Global clock gate
enable
    input logic [N-1:0] cfg_cg_idle_count, // Idle countdown value
    input logic      wakeup,             // Signal to wake up the
block

    // Outputs
    output logic      clk_out,
    output logic      gating             // clock gating indicator
);
```

11.3 Parameters

11.3.1 IDLE_CNTR_WIDTH

- **Type:** int
- **Default:** 4
- **Description:** Bit width of the idle timeout counter
- **Range:** 2 to 16 bits (practical range)
- **Impact:** Determines maximum idle timeout ($2^{\text{IDLE_CNTR_WIDTH}} - 1$ cycles)

- **Power Trade-off:** Larger counters allow longer timeouts but consume more power

11.3.2 N

- **Type:** int
- **Default:** IDLE_CNTR_WIDTH
- **Description:** Alias for counter width (convenience parameter)
- **Usage:** Internal consistency check and code readability

11.4 Ports

11.4.1 Inputs

Port	Width	Type	Description
clk_in	1	logic	Input clock to be gated
aresetn	1	logic	Active-low asynchronous reset
cfg_cg_enable	1	logic	Global clock gating enable control
cfg_cg_idle_count	N	logic	Idle timeout value (configuration)
wakeup	1	logic	Immediate wake-up signal

11.4.2 Outputs

Port	Width	Type	Description
clk_out	1	logic	Gated output clock
gating	1	logic	Clock gating status indicator

11.5 Clock Gating Theory

11.5.1 Why Clock Gating?

Clock gating is one of the most effective power reduction techniques: - **Dynamic Power:** $P = C \times V^2 \times f \times \alpha$ (α = switching activity) - **Clock Networks:** Typically consume 20-40% of total chip power -

Gating Benefits: Can reduce power by 50-90% in idle blocks - **Area Overhead:** Minimal (< 5% for typical designs)

11.5.2 Clock Gating Safety

Proper clock gating requires: 1. **Glitch-Free:** No spurious clock edges during gating transitions 2. **Setup/Hold:** Adequate timing margins around gating decisions 3. **Reset Handling:** Proper behavior during reset conditions 4. **Tool Support:** Use of qualified ICG (Integrated Clock Gate) cells

11.6 Architecture and Implementation

11.6.1 Idle Counter Logic

```
logic [N-1:0] r_idle_counter;

always_ff @(posedge clk_in or negedge aresetn) begin
    if (!aresetn) begin
        r_idle_counter <= cfg_cg_idle_count;
    end else begin
        if (wakeup || !cfg_cg_enable) begin
            // On wakeup or global disable, reset counter
            r_idle_counter <= cfg_cg_idle_count;
        end else if (r_idle_counter != 'h0) begin
            // Normal counting operation - decrement when not zero
            r_idle_counter <= r_idle_counter - 1'b1;
        end
        // When counter reaches zero, it stays at zero
    end
end
```

11.6.2 Gating Control Logic

```
wire w_gate_enable = cfg_cg_enable && !wakeup && (r_idle_counter == 'h0);

// Instantiate the ICG cell
icg u_icg (
    .clk(clk_in),
    .en(~w_gate_enable), // Enable is active high in ICG
    .gclk(clk_out)
);

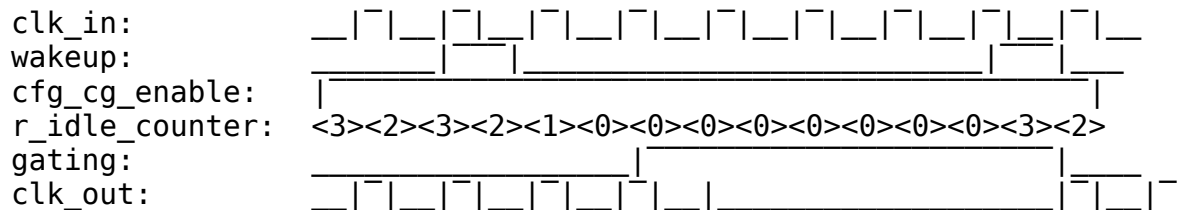
assign gating = w_gate_enable;
```

11.6.3 State Machine Behavior

The controller operates as a countdown timer with three states:

1. **ACTIVE State:** Counter loaded with `cfg_cg_idle_count`
2. **COUNTDOWN State:** Counter decrements each cycle when no activity
3. **GATED State:** Counter at zero, clock is gated off

11.6.4 Timing Diagram



11.7 ICG (Integrated Clock Gate) Cell

11.7.1 Standard ICG Interface

```

module icg (
    input logic clk,      // Input clock
    input logic en,       // Enable (active high)
    output logic gclk     // Gated clock output
);

    // Implementation varies by technology library
    // Typically includes:
    // - Latch to avoid glitches
    // - AND gate for gating
    // - Proper setup/hold timing

endmodule

```

11.7.2 ICG Implementation (Conceptual)

```

// Simplified ICG implementation (technology-specific)
module icg_generic (
    input logic clk,
    input logic en,
    output logic gclk
);

    logic en_latched;

    // Latch enable on falling edge to avoid glitches
    always_latch begin
        if (~clk)
            en_latched = en;
    end

endmodule

```

```

// Gate clock with latched enable
assign gclk = clk & en_latched;

```

```
endmodule
```

11.8 Design Examples and Applications

11.8.11. CPU Core Clock Gating

```

module cpu_clock_gating (
    input logic      clk_cpu,
    input logic      rst_n,

    // CPU control signals
    input logic      cpu_active,
    input logic      interrupt_pending,
    input logic      debug_mode,

    // Power management
    input logic [3:0] idle_threshold,
    input logic      power_save_enable,

    // Output clocks
    output logic      clk_cpu_gated,
    output logic      clk_debug,
    output logic      cpu_sleeping
);

    logic cpu_wakeup;
    logic debug_wakeup;

    // CPU wake-up conditions
    assign cpu_wakeup = cpu_active || interrupt_pending;

    // Debug interface always active in debug mode
    assign debug_wakeup = debug_mode || cpu_wakeup;

    // Main CPU clock gating
    clock_gate_ctrl #(
        .IDLE_CNTR_WIDTH(4)
    ) cpu_gate (
        .clk_in(clk_cpu),
        .aresetn(rst_n),
        .cfg_cg_enable(power_save_enable),
        .cfg_cg_idle_count(idle_threshold),
        .wakeup(cpu_wakeup),
        .clk_out(clk_cpu_gated),

```

```

        .gating(cpu_sleeping)
    );

    // Debug clock gating (shorter timeout)
    clock_gate_ctrl #(
        .IDLE_CNTR_WIDTH(3)
    ) debug_gate (
        .clk_in(clk_cpu),
        .aresetn(rst_n),
        .cfg_cg_enable(power_save_enable && !debug_mode),
        .cfg_cg_idle_count(3'd2), // Short timeout for debug
        .wakeup(debug_wakeup),
        .clk_out(clk_debug),
        .gating() // Unused
    );

```

endmodule

11.8.22. Peripheral Interface Clock Management

```

module peripheral_clock_manager (
    input logic      clk_system,
    input logic      rst_n,

    // Peripheral activity signals
    input logic      uart_active,
    input logic      spi_active,
    input logic      i2c_active,
    input logic      timer_active,

    // Configuration
    input logic [7:0] power_config, // [7:4] = enable bits, [3:0]
    = timeout

    // Gated clocks
    output logic      clk_uart,
    output logic      clk_spi,
    output logic      clk_i2c,
    output logic      clk_timer,

    // Status
    output logic [3:0] peripheral_sleeping
);

    logic [3:0] idle_timeout;
    logic [3:0] gate_enables;

    assign idle_timeout = power_config[3:0];

```

```

assign gate_enables = power_config[7:4];

// UART clock gating
clock_gate_ctrl #(4) uart_cg (
    .clk_in(clk_system),
    .aresetn(rst_n),
    .cfg_cg_enable(gate_enables[0]),
    .cfg_cg_idle_count(idle_timeout),
    .wakeup(uart_active),
    .clk_out(clk_uart),
    .gating(peripheral_sleeping[0])
);

// SPI clock gating
clock_gate_ctrl #(4) spi_cg (
    .clk_in(clk_system),
    .aresetn(rst_n),
    .cfg_cg_enable(gate_enables[1]),
    .cfg_cg_idle_count(idle_timeout),
    .wakeup(spi_active),
    .clk_out(clk_spi),
    .gating(peripheral_sleeping[1])
);

// I2C clock gating
clock_gate_ctrl #(4) i2c_cg (
    .clk_in(clk_system),
    .aresetn(rst_n),
    .cfg_cg_enable(gate_enables[2]),
    .cfg_cg_idle_count(idle_timeout),
    .wakeup(i2c_active),
    .clk_out(clk_i2c),
    .gating(peripheral_sleeping[2])
);

// Timer clock gating (different timeout)
clock_gate_ctrl #(4) timer_cg (
    .clk_in(clk_system),
    .aresetn(rst_n),
    .cfg_cg_enable(gate_enables[3]),
    .cfg_cg_idle_count(4'd1), // Quick timeout for timers
    .wakeup(timer_active),
    .clk_out(clk_timer),
    .gating(peripheral_sleeping[3])
);

```

endmodule

11.8.33. Memory Subsystem Clock Gating

```
module memory_clock_gating (
    input logic      clk_memory,
    input logic      rst_n,

    // Memory access signals
    input logic      mem_read,
    input logic      mem_write,
    input logic      mem_refresh_req,
    input logic      cache_miss,

    // Configuration
    input logic [2:0] mem_power_mode, // 0=always on, 7=aggressive
gating

    // Gated clocks
    output logic      clk_mem_ctrl,    // Memory controller
    output logic      clk_mem_data,    // Data path
    output logic      clk_mem_refresh, // Refresh logic

    // Status
    output logic [2:0] mem_gating_status
);

    logic [3:0] timeout_ctrl, timeout_data, timeout_refresh;
    logic mem_active, data_active, refresh_active;

    // Activity detection
    assign mem_active = mem_read || mem_write || cache_miss;
    assign data_active = mem_read || mem_write;
    assign refresh_active = mem_refresh_req;

    // Timeout configuration based on power mode
    always_comb begin
        case (mem_power_mode)
            3'd0: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd15, 4'd15, 4'd15}; // Always on
            3'd1: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd10, 4'd8, 4'd12}; // Conservative
            3'd2: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd8, 4'd6, 4'd10}; // Moderate
            3'd3: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd6, 4'd4, 4'd8}; // Balanced
            3'd4: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd4, 4'd2, 4'd6}; // Aggressive
            3'd5: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd2, 4'd1, 4'd4}; // Very aggressive
            3'd6: {timeout_ctrl, timeout_data, timeout_refresh} =
```

```

{4'd1, 4'd0, 4'd2};    // Maximum power save
    3'd7: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd0, 4'd0, 4'd1};    // Ultra power save
    endcase
end

// Memory controller clock gating
clock_gate_ctrl #(4) ctrl_cg (
    .clk_in(clk_memory),
    .aresetn(rst_n),
    .cfg_cg_enable(mem_power_mode != 0),
    .cfg_cg_idle_count(timeout_ctrl),
    .wakeup(mem_active),
    .clk_out(clk_mem_ctrl),
    .gating(mem_gating_status[0])
);

// Data path clock gating
clock_gate_ctrl #(4) data_cg (
    .clk_in(clk_memory),
    .aresetn(rst_n),
    .cfg_cg_enable(mem_power_mode != 0),
    .cfg_cg_idle_count(timeout_data),
    .wakeup(data_active),
    .clk_out(clk_mem_data),
    .gating(mem_gating_status[1])
);

// Refresh logic clock gating
clock_gate_ctrl #(4) refresh_cg (
    .clk_in(clk_memory),
    .aresetn(rst_n),
    .cfg_cg_enable(mem_power_mode > 2), // Only gate in aggressive
modes
    .cfg_cg_idle_count(timeout_refresh),
    .wakeup(refresh_active),
    .clk_out(clk_mem_refresh),
    .gating(mem_gating_status[2])
);

endmodule

```

11.8.44. Adaptive Clock Gating with Activity Monitoring

```

module adaptive_clock_gating #(
    parameter int ACTIVITY_WINDOW = 256 // Activity monitoring window
) (
    input logic      clk_in,
    input logic      rst_n,

```

```

// Activity signals
input logic [7:0] activity_vector,

// Configuration
input logic adaptive_enable,
input logic [3:0] base_timeout,

// Outputs
output logic clk_out,
output logic gating_active,
output logic [7:0] activity_level // Activity percentage
);

logic [15:0] activity_counter;
logic [7:0] activity_window_counter;
logic [3:0] adaptive_timeout;
logic block_active;

// Activity monitoring
always_ff @(posedge clk_in or negedge rst_n) begin
    if (!rst_n) begin
        activity_counter <= 0;
        activity_window_counter <= 0;
        activity_level <= 0;
    end else begin
        // Count activity in current window
        if (|activity_vector)
            activity_counter <= activity_counter + 1;

        activity_window_counter <= activity_window_counter + 1;

        // Update activity level at end of window
        if (activity_window_counter == ACTIVITY_WINDOW - 1) begin
            activity_level <= activity_counter[15:8]; // Scale to
0-255

            activity_counter <= 0;
            activity_window_counter <= 0;
        end
    end
end

// Adaptive timeout based on activity level
always_comb begin
    if (!adaptive_enable) begin
        adaptive_timeout = base_timeout;
    end else begin
        case (activity_level)

```

```

        8'h00: adaptive_timeout = base_timeout + 4'd8; // Very
low activity
        8'h01: adaptive_timeout = base_timeout + 4'd6; // Low
activity
        8'h02: adaptive_timeout = base_timeout + 4'd4; //
Medium-low activity
        8'h04: adaptive_timeout = base_timeout + 4'd2; //
Medium activity
        8'h08: adaptive_timeout = base_timeout; //
Normal activity
        default: adaptive_timeout = (base_timeout > 2) ?
base_timeout - 2 : 0; // High activity
    endcase
end
end

// Activity detection
assign block_active = |activity_vector;

// Clock gating controller
clock_gate_ctrl #(4) adaptive_cg (
    .clk_in(clk_in),
    .aresetn(rst_n),
    .cfg_cg_enable(1'b1),
    .cfg_cg_idle_count(adaptive_timeout),
    .wakeup(block_active),
    .clk_out(clk_out),
    .gating(gating_active)
);

endmodule

```

11.8.55. Hierarchical Clock Gating System

```

module hierarchical_clock_gating (
    input logic clk_root,
    input logic rst_n,

    // Global control
    input logic global_cg_enable,
    input logic [3:0] global_timeout,

    // Subsystem activity
    input logic cpu_active,
    input logic dsp_active,
    input logic peripheral_active,
    input logic memory_active,

    // Hierarchical clocks

```

```

output logic      clk_cpu_domain,
output logic      clk_dsp_domain,
output logic      clk_peripheral_domain,
output logic      clk_memory_domain,

// Power monitoring
output logic [3:0] domain_gating_status,
output logic [15:0] power_savings_estimate
);

logic system_active;
logic clk_system_gated;
logic system_gated;

// System-level activity detection
assign system_active = cpu_active || dsp_active ||
peripheral_active || memory_active;

// Top-level system clock gating
clock_gate_ctrl #(4) system_cg (
    .clk_in(clk_root),
    .aresetn(rst_n),
    .cfg_cg_enable(global_cg_enable),
    .cfg_cg_idle_count(global_timeout + 4'd4), // Longer timeout
for system level
    .wakeup(system_active),
    .clk_out(clk_system_gated),
    .gating(system_gated)
);

// CPU domain gating
clock_gate_ctrl #(4) cpu_cg (
    .clk_in(clk_system_gated),
    .aresetn(rst_n),
    .cfg_cg_enable(global_cg_enable && !system_gated),
    .cfg_cg_idle_count(global_timeout),
    .wakeup(cpu_active),
    .clk_out(clk_cpu_domain),
    .gating(domain_gating_status[0])
);

// DSP domain gating
clock_gate_ctrl #(4) dsp_cg (
    .clk_in(clk_system_gated),
    .aresetn(rst_n),
    .cfg_cg_enable(global_cg_enable && !system_gated),
    .cfg_cg_idle_count(global_timeout - 1), // Aggressive for DSP
    .wakeup(dsp_active),

```

```

        .clk_out(clk_dsp_domain),
        .gating(domain_gating_status[1])
    );

    // Peripheral domain gating
    clock_gate_ctrl #(4) peripheral_cg (
        .clk_in(clk_system_gated),
        .aresetn(rst_n),
        .cfg_cg_enable(global_cg_enable && !system_gated),
        .cfg_cg_idle_count(global_timeout + 1), // Conservative for
peripherals
        .wakeup(peripheral_active),
        .clk_out(clk_peripheral_domain),
        .gating(domain_gating_status[2])
    );

    // Memory domain gating
    clock_gate_ctrl #(4) memory_cg (
        .clk_in(clk_system_gated),
        .aresetn(rst_n),
        .cfg_cg_enable(global_cg_enable && !system_gated),
        .cfg_cg_idle_count(global_timeout + 2), // Conservative for
memory
        .wakeup(memory_active),
        .clk_out(clk_memory_domain),
        .gating(domain_gating_status[3])
    );

    // Power savings estimation (simplified)
    logic [3:0] gated_domains;
    assign gated_domains = domain_gating_status[3:0];

    always_ff @(posedge clk_root or negedge rst_n) begin
        if (!rst_n) begin
            power_savings_estimate <= 0;
        end else begin
            // Rough power savings estimate (percentage)
            case ({system_gated, gated_domains})
                5'b1_0000: power_savings_estimate <= 16'd9000; // 90%
savings
                5'b0_1111: power_savings_estimate <= 16'd7000; // 70%
savings
                5'b0_1110: power_savings_estimate <= 16'd6000; // 60%
savings
                5'b0_1100: power_savings_estimate <= 16'd5000; // 50%
savings
                5'b0_1000: power_savings_estimate <= 16'd3000; // 30%
savings
            end
        end
    end

```

```

savings        5'b0_0100: power_savings_estimate <= 16'd2000; // 20%
savings        5'b0_0010: power_savings_estimate <= 16'd1500; // 15%
savings        5'b0_0001: power_savings_estimate <= 16'd1000; // 10%
savings        default:   power_savings_estimate <= 16'd0000; // 0%
savings
    endcase
end
end
endmodule

```

11.9 Advanced Features

11.9.11. Glitch-Free Clock Switching

```

module glitch_free_clock_gate (
    input  logic clk_in,
    input  logic rst_n,
    input  logic gate_enable,
    input  logic wakeup,
    output logic clk_out
);

    logic gate_enable_sync;
    logic gate_request;

    // Synchronize gate enable to avoid glitches
    always_ff @(posedge clk_in or negedge rst_n) begin
        if (!rst_n) begin
            gate_enable_sync <= 0;
            gate_request <= 0;
        end else begin
            gate_enable_sync <= gate_enable;
            gate_request <= gate_enable_sync && !wakeup;
        end
    end

    // Use ICG with synchronized control
    icg u_icg (
        .clk(clk_in),
        .en(~gate_request),
        .gclk(clk_out)
    );

```

endmodule

11.9.22. Multi-Level Clock Gating

```
module multi_level_clock_gate #(
    parameter int LEVELS = 3
) (
    input    logic          clk_in,
    input    logic          rst_n,
    input    logic [LEVELS-1:0] level_enables,
    input    logic [LEVELS-1:0] level_wakeups,
    input    logic [4*LEVELS-1:0] level_timeouts,
    output   logic          clk_out,
    output   logic [LEVELS-1:0] level_gating_status
);

    logic [LEVELS:0] clk_levels; // Include input clock
    assign clk_levels[0] = clk_in;

    genvar i;
    generate
        for (i = 0; i < LEVELS; i++) begin : level_gates
            logic [3:0] timeout;
            assign timeout = level_timeouts[i*4+:4];

            clock_gate_ctrl #(4) level_cg (
                .clk_in(clk_levels[i]),
                .aresetn(rst_n),
                .cfg_cg_enable(level_enables[i]),
                .cfg_cg_idle_count(timeout),
                .wakeup(|level_wakeups[LEVELS-1:i]), // Wake if any
                higher level active
                .clk_out(clk_levels[i+1]),
                .gating(level_gating_status[i])
            );
        end
    endgenerate

    assign clk_out = clk_levels[LEVELS];

endmodule
```


11.10 Verification and Testing

11.10.1 Comprehensive Test Bench

```
module tb_clock_gate_ctrl;

    parameter IDLE_CNTR_WIDTH = 4;

    logic clk_in, aresetn;
    logic cfg_cg_enable;
    logic [IDLE_CNTR_WIDTH-1:0] cfg_cg_idle_count;
    logic wakeup;
    logic clk_out, gating;

    // Clock generation
    initial clk_in = 0;
    always #5ns clk_in = ~clk_in; // 100MHz

    // DUT instantiation
    clock_gate_ctrl #(
        .IDLE_CNTR_WIDTH(IDLE_CNTR_WIDTH)
    ) dut (.*);

    // Test sequence
    initial begin
        // Initialize
        aresetn = 0;
        cfg_cg_enable = 0;
        cfg_cg_idle_count = 4'd5;
        wakeup = 0;

        #100ns aresetn = 1;

        // Test basic gating
        test_basic_gating();

        // Test immediate wakeup
        test_immediate_wakeup();

        // Test global disable
        test_global_disable();

        // Test different timeouts
        test_timeout_values();

        $display("All tests completed!");
        $finish;
    end
end
```

```

// Test basic clock gating functionality
task test_basic_gating();
    begin
        $display("Testing basic clock gating...");

        cfg_cg_enable = 1;
        cfg_cg_idle_count = 4'd3;

        // Wait for gating to occur
        wait_for_gating();

        // Verify clock is gated
        check_clock_gated();

        // Wake up and verify clock restoration
        wakeup = 1;
        #20ns wakeup = 0;
        check_clock_active();
    end
endtask

// Wait for clock gating to occur
task wait_for_gating();
    int timeout_cycles;
    begin
        timeout_cycles = 0;
        while (!gating && timeout_cycles < 100) begin
            @(posedge clk_in);
            timeout_cycles++;
        end

        if (!gating) begin
            $error("Clock gating did not occur within timeout");
        end else begin
            $display("Clock gating occurred after %d cycles",
timeout_cycles);
        end
    end
endtask

// Check that clock is properly gated
task check_clock_gated();
    logic prev_clk_out;
    int stable_cycles;
    begin
        prev_clk_out = clk_out;
        stable_cycles = 0;

```

```

        repeat (10) begin
            @(posedge clk_in);
            if (clk_out == prev_clk_out) begin
                stable_cycles++;
            end else begin
                stable_cycles = 0;
                prev_clk_out = clk_out;
            end
        end

        if (stable_cycles >= 5) begin
            $display("PASS: Clock is properly gated");
        end else begin
            $error("FAIL: Clock is not properly gated");
        end
    end
endtask

// Check that clock is active
task check_clock_active();
    int edge_count;
    begin
        edge_count = 0;

        repeat (20) begin
            @(posedge clk_in);
            if ($rose(clk_out)) edge_count++;
        end

        if (edge_count >= 8) begin
            $display("PASS: Clock is active after wakeup");
        end else begin
            $error("FAIL: Clock not properly restored after
wakeup");
        end
    end
endtask

endmodule

```

11.10.2 Coverage Model

```

covergroup clock_gate_cg @(posedge clk_in);

    cp_idle_count: coverpoint cfg_cg_idle_count {
        bins short_timeout[] = {[0:3]};
        bins medium_timeout[] = {[4:10]};
        bins long_timeout[] = {[11:15]};
    }
endgroup

```

```

}

cp_counter_values: coverpoint dut.r_idle_counter {
    bins countdown[] = {[0:15]};
    bins zero = {0};
}

cp_gating_enable: coverpoint cfg_cg_enable {
    bins enabled = {1};
    bins disabled = {0};
}

cp_wakeup: coverpoint wakeup {
    bins active = {1};
    bins inactive = {0};
}

cp_gating_status: coverpoint gating {
    bins gated = {1};
    bins ungated = {0};
}

// Cross coverage
cross_timeout_gating: cross cp_idle_count, cp_gating_status;
cross_wakeup_gating: cross cp_wakeup, cp_gating_status;

```

endcovergroup

11.10.3 Formal Verification Properties

```

module clock_gate_properties;

    // Bind to DUT
    bind clock_gate_ctrl clock_gate_properties props_inst (.*);

    // Property: Immediate wakeup
    property immediate_wakeup;
        @(posedge clk_in) disable iff (!aresetn)
        wakeup |-> ##1 !gating;
    endproperty

    // Property: Gating after timeout
    property gating_after_timeout;
        @(posedge clk_in) disable iff (!aresetn)
        cfg_cg_enable && !wakeup && (dut.r_idle_counter == 0) |->
gating;
    endproperty

```

```

// Property: Counter behavior
property counter_countdown;
    @(posedge clk_in) disable iff (!aresetn)
        cfg_cg_enable && !wakeup && (dut.r_idle_counter > 0) | =>
            (dut.r_idle_counter == $past(dut.r_idle_counter) - 1);
endproperty

// Property: Global disable
property global_disable;
    @(posedge clk_in) disable iff (!aresetn)
        !cfg_cg_enable | -> !gating;
endproperty

// Assertions
assert property (immediate_wakeup);
assert property (gating_after_timeout);
assert property (counter_countdown);
assert property (global_disable);

endmodule

```

11.11 Power Analysis and Optimization

11.11.1 Power Savings Calculation

```

module power_analysis_wrapper (
    input logic      clk_in,
    input logic      rst_n,
    input logic      cfg_cg_enable,
    input logic [3:0] cfg_cg_idle_count,
    input logic      wakeup,
    output logic      clk_out,
    output logic      gating,
    output logic [15:0] power_savings_percent
);

// Clock gate controller
clock_gate_ctrl #(4) cg_ctrl (
    .clk_in(clk_in),
    .aresetn(rst_n),
    .cfg_cg_enable(cfg_cg_enable),
    .cfg_cg_idle_count(cfg_cg_idle_count),
    .wakeup(wakeup),
    .clk_out(clk_out),
    .gating(gating)
);

// Power monitoring

```

```

logic [31:0] total_cycles;
logic [31:0] gated_cycles;
logic [31:0] ungated_cycles;

always_ff @(posedge clk_in or negedge rst_n) begin
    if (!rst_n) begin
        total_cycles <= 0;
        gated_cycles <= 0;
        ungated_cycles <= 0;
    end else begin
        total_cycles <= total_cycles + 1;
        if (gating)
            gated_cycles <= gated_cycles + 1;
        else
            ungated_cycles <= ungated_cycles + 1;
    end
end

// Calculate power savings percentage
always_ff @(posedge clk_in) begin
    if (total_cycles > 1000) begin // Avoid division by small
numbers
        power_savings_percent <= (gated_cycles * 100) /
total_cycles;
    end else begin
        power_savings_percent <= 0;
    end
end

endmodule

```

11.11.2 Technology-Specific ICG Implementation

```

// Example for TSMC technology
module icg_tsmc (
    input logic clk,
    input logic en,
    output logic gclk
);

    // Technology-specific implementation
    CKGTPBUF_X1M_A9TH u_icg (
        .CK(clk),
        .E(en),
        .GCK(gclk)
    );

endmodule

```

```

// Example for Intel/Altera technology
module icg_intel (
    input logic clk,
    input logic en,
    output logic gclk
);

    // Intel FPGA implementation
    logic en_reg;

    always_ff @(negedge clk) begin
        en_reg <= en;
    end

    assign gclk = clk & en_reg;

endmodule

// Generic behavioral model
module icg_behavioral (
    input logic clk,
    input logic en,
    output logic gclk
);

    logic en_latch;

    // Latch enable on falling edge to prevent glitches
    always_latch begin
        if (!clk)
            en_latch = en;
    end

    // Gate the clock
    assign gclk = clk & en_latch;

endmodule

```

11.12 Synthesis Considerations

11.12.1 Resource Utilization

Configuration	LUTs	FFs	ICG Cells	Area Overhead
4-bit	8	4	1	< 2%

Configuration	LUTs	FFs	ICG Cells	Area Overhead
counter				
8-bit	15	8	1	< 3%
counter				
16-bit	30	16	1	< 5%
counter				

11.12.2 Timing Constraints

SDC constraints for clock gating

```
create_clock -name clk_in -period 10.0 [get_ports clk_in]
```

Clock gating timing

```
set_case_analysis 0 [get_pins u_icg/en]
```

```
set_case_analysis 1 [get_pins u_icg/en]
```

Setup/hold for gating control

```
set_input_delay -clock clk_in -max 2.0 [get_ports wakeup]
```

```
set_input_delay -clock clk_in -min 0.5 [get_ports wakeup]
```

Clock gating cell timing

```
set_clock_gating_check -setup 0.2 -hold 0.1 [get_pins u_icg/en]
```

11.12.3 Power Intent (UPF)

UPF power intent for clock gating

```
create_power_domain PD_MAIN
```

```
create_power_domain PD_GATED
```

Clock gating specification

```
set_domain_supply_net PD_MAIN -primary_power_net VDD -  
primary_ground_net VSS
```

```
set_domain_supply_net PD_GATED -primary_power_net VDD -  
primary_ground_net VSS
```

Clock gating control

```
create_power_switch PS_GATED \  
-domain PD_GATED \  
-input_supply_port {vin VDD} \  
-output_supply_port {vout VDD_GATED} \  
-control_port {ctrl gating} \  
-on_state {on vin {!ctrl}}
```


11.13 Design Guidelines and Best Practices

11.13.1 1. Timeout Selection

```
// Guidelines for timeout values
parameter TIMEOUT_GUIDELINES = {
    "CPU cores: 4-16 cycles",
    "Memory controllers: 8-32 cycles",
    "Peripherals: 16-64 cycles",
    "Debug interfaces: 2-8 cycles",
    "Always-on domains: Disable gating"
};
```

11.13.2 2. Activity Signal Design

```
// Good activity signal examples
logic cpu_active;
assign cpu_active = instruction_fetch ||
                    data_access ||
                    interrupt_processing ||
                    debug_access;

// Avoid overly broad activity signals
logic bad_activity;
assign bad_activity = |all_system_signals; // Too broad

// Avoid overly narrow activity signals
logic narrow_activity;
assign narrow_activity = specific_register_write; // Too narrow
```

11.13.3 3. Hierarchical Gating Strategy

```
System Level (Longest timeout)
├── CPU Domain (Medium timeout)
│   ├── ALU (Short timeout)
│   ├── Cache (Medium timeout)
│   └── Debug (Very short timeout)
├── DSP Domain (Short timeout)
├── Peripheral Domain (Long timeout)
└── Memory Domain (Medium timeout)
```

11.13.4 4. Verification Checklist

- ☐ Verify gating occurs after correct timeout
- ☐ Verify immediate wakeup functionality
- ☐ Test global enable/disable

- ☐ Check reset behavior
- ☐ Validate power savings measurements
- ☐ Test all timeout configurations
- ☐ Verify clock integrity (no glitches)
- ☐ Check setup/hold timing margins

11.14 Common Applications Summary

1. **CPU Power Management:** Core, cache, and pipeline clock gating
2. **Peripheral Controllers:** UART, SPI, I2C, timer clock gating
3. **Memory Subsystems:** Controller, data path, refresh logic gating
4. **Graphics/Video:** Pixel processing, display controller gating
5. **Communication Blocks:** Baseband, RF, protocol stack gating
6. **Test and Debug:** Scan chains, JTAG, debug interface gating
7. **AI/ML Accelerators:** Compute engine, memory interface gating

11.15 Troubleshooting Guide

11.15.1 Common Issues

1. **Clock Glitches:** Use qualified ICG cells, proper setup/hold margins
2. **Timing Violations:** Ensure adequate timing margins for gating signals
3. **Unexpected Gating:** Check activity signal connectivity and logic
4. **Power Not Saved:** Verify clock tree reaches gated logic
5. **Reset Issues:** Ensure proper reset sequencing and distribution

11.15.2 Debug Techniques

1. **Simulation:** Monitor gating signals and clock activity
2. **Power Analysis:** Use tools to measure actual power savings
3. **Timing Analysis:** Check setup/hold violations on gating paths

4. **Logic Analyzer:** Capture real-time gating behavior in hardware
5. **Power Profiling:** Measure current consumption with/without gating

The clock gate controller provides an essential power management capability, enabling significant dynamic power reduction with minimal area overhead and design complexity. Proper implementation requires careful attention to timing, safety, and verification to ensure reliable operation while maximizing power savings.

11.16 Navigation

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12 Clock Pulse Generator - Comprehensive Documentation

12.1 Overview

The `clock_pulse` module generates periodic pulse signals with configurable width (period). It creates a single-cycle pulse every `WIDTH` clock cycles, making it ideal for timing generation, heartbeat signals, sampling triggers, and periodic event generation in digital systems.

12.2 Module Declaration

```
module clock_pulse #(
    parameter int WIDTH = 10 // Width of the generated pulse in clock
    cycles
) (
    input logic clk, // Input clock signal
    input logic rst_n, // Input reset signal
    output logic pulse // Output pulse signal
);
```

12.3 Parameters

12.3.1 WIDTH

- **Type:** int
- **Default:** 10
- **Description:** Period of the pulse generation in clock cycles

- **Range:** 2 to $2^{32}-1$ (practical range)
- **Impact:** Determines pulse frequency ($f_{\text{clk}} / \text{WIDTH}$)
- **Pulse Width:** Always exactly 1 clock cycle
- **Duty Cycle:** $1/\text{WIDTH}$ (e.g., 10% for $\text{WIDTH}=10$)

12.4 Ports

12.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	System clock input
rst_n	1	logic	Active-low asynchronous reset

12.4.2 Outputs

Port	Width	Type	Description
pulse	1	logic	Periodic pulse output

12.5 Architecture and Implementation

12.5.1 Internal Counter

```

logic [WIDTH-1:0] r_counter;
logic [WIDTH-1:0] w_width_minus_one;

// Create a properly sized constant
assign w_width_minus_one = WIDTH[WIDTH-1:0] - 1'b1;

```

12.5.2 Core Logic

```

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_counter <= 'b0;
        pulse      <= 'b0;
    end else begin
        if (r_counter < w_width_minus_one)
            r_counter <= r_counter + 1'b1;
        else

```

```

        r_counter <= 'b0;

        pulse <= (r_counter == w_width_minus_one);
    end
end

```

12.5.3 Operation Principles

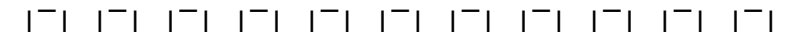
1. **Counter:** Free-running counter from 0 to WIDTH-1
2. **Pulse Generation:** Pulse asserted when counter reaches WIDTH-1
3. **Auto-Reset:** Counter wraps to 0 after reaching maximum
4. **Synchronous:** All operations synchronized to input clock
5. **Single Cycle:** Pulse duration is exactly one clock cycle

12.5.4 Timing Characteristics

- **Period:** WIDTH clock cycles
- **Frequency:** $f_{\text{clk}} / \text{WIDTH}$
- **Pulse Width:** 1 clock cycle
- **Duty Cycle:** $1/\text{WIDTH}$
- **Phase:** Pulse occurs on last count (WIDTH-1)

12.6 Timing Diagrams

12.6.1 Basic Operation (WIDTH=4)

Clock: 
 Counter: < 0 > 1 > 2 > 3 > 0 > 1 > 2 > 3 > 0 >
 Pulse: 

12.6.2 Reset Behavior

Clock: 
 Reset_n: 
 Counter: < 0 > 1 > 2 > 3 > 0 > 0 > 1 > 2 > 3 >
 Pulse: 

12.6.3 Different WIDTH Values

WIDTH	Period	Duty Cycle	Use Case
2	2 cycles	50%	Clock divider

WIDTH	Period	Duty Cycle	Use Case
4	4 cycles	25%	Sampling
10	10 cycles	10%	Timing reference
100	100 cycles	1%	Heartbeat
1000	1000 cycles	0.1%	Slow events

12.7 Design Examples and Applications

12.7.1 1. Heartbeat and Status Indicators

```

module system_heartbeat #(
    parameter int CLOCK_FREQ_HZ = 100_000_000, // 100MHz
    parameter int HEARTBEAT_FREQ_HZ = 1 // 1Hz heartbeat
) (
    input logic clk,
    input logic rst_n,
    input logic system_active,
    output logic heartbeat_led,
    output logic status_ok
);

    localparam int HEARTBEAT_WIDTH = CLOCK_FREQ_HZ /
HEARTBEAT_FREQ_HZ;

    logic heartbeat_pulse;

    // Generate 1Hz heartbeat pulse
    clock_pulse #(
        .WIDTH(HEARTBEAT_WIDTH)
    ) heartbeat_gen (
        .clk(clk),
        .rst_n(rst_n),
        .pulse(heartbeat_pulse)
    );

    // LED control with heartbeat
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            heartbeat_led <= 1'b0;
        end else if (heartbeat_pulse) begin
            heartbeat_led <= ~heartbeat_led; // Toggle LED
        end
    end

```

```

// System status based on heartbeat activity
logic [7:0] heartbeat_monitor;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        heartbeat_monitor <= 8'h00;
    end else begin
        heartbeat_monitor <= {heartbeat_monitor[6:0],
heartbeat_pulse};
    end
end

assign status_ok = system_active && |heartbeat_monitor;

endmodule

```

12.7.22. Sampling and Data Acquisition

```

module data_sampler #(
    parameter int SAMPLE_RATE_KHZ = 48,
    parameter int CLOCK_FREQ_MHZ = 100,
    parameter int DATA_WIDTH = 16
) (
    input logic clk,
    input logic rst_n,
    input logic enable,
    input logic [DATA_WIDTH-1:0] analog_data,

    output logic [DATA_WIDTH-1:0] sampled_data,
    output logic sample_valid,
    output logic buffer_full
);

    localparam int SAMPLE_WIDTH = (CLOCK_FREQ_MHZ * 1000) /
SAMPLE_RATE_KHZ;

    logic sample_trigger;

    // Generate sampling trigger
    clock_pulse #(
        .WIDTH(SAMPLE_WIDTH)
    ) sample_clock (
        .clk(clk),
        .rst_n(rst_n),
        .pulse(sample_trigger)
    );

    // Sample data on trigger

```

```

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        sampled_data <= '0;
        sample_valid <= 1'b0;
    end else if (enable && sample_trigger) begin
        sampled_data <= analog_data;
        sample_valid <= 1'b1;
    end else begin
        sample_valid <= 1'b0;
    end
end

// Buffer management (simplified)
logic [7:0] buffer_count;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        buffer_count <= 8'h00;
    end else if (sample_valid) begin
        buffer_count <= buffer_count + 1;
    end
end

assign buffer_full = &buffer_count; // Full at 255 samples

endmodule

```

12.7.33. Communication Protocol Timing

```

module uart_baud_generator #(
    parameter int CLOCK_FREQ_HZ = 100_000_000,
    parameter int BAUD_RATE = 115200
) (
    input logic clk,
    input logic rst_n,
    input logic enable,

    output logic baud_tick, // 1x baud rate
    output logic baud_tick_16x // 16x baud rate for oversampling
);

localparam int BAUD_WIDTH = CLOCK_FREQ_HZ / BAUD_RATE;
localparam int BAUD_16X_WIDTH = CLOCK_FREQ_HZ / (BAUD_RATE * 16);

logic baud_pulse, baud_16x_pulse;

// 1x baud rate generator
clock_pulse #(

```



```

        .WIDTH(BAUD_WIDTH)
    ) baud_1x_gen (
        .clk(clk),
        .rst_n(rst_n),
        .pulse(baud_pulse)
    );

    // 16x baud rate generator
    clock_pulse #(
        .WIDTH(BAUD_16X_WIDTH)
    ) baud_16x_gen (
        .clk(clk),
        .rst_n(rst_n),
        .pulse(baud_16x_pulse)
    );

    // Gate with enable
    assign baud_tick = enable ? baud_pulse : 1'b0;
    assign baud_tick_16x = enable ? baud_16x_pulse : 1'b0;

```

endmodule

12.7.44. Memory Refresh Controller

```

module dram_refresh_controller #(
    parameter int REFRESH_PERIOD_US = 64,           // 64µs refresh period
    parameter int CLOCK_FREQ_MHZ = 100,           // 100MHz system clock
    parameter int REFRESH_ROWS = 8192              // Number of rows to
refresh
) (
    input logic      clk,
    input logic      rst_n,
    input logic      refresh_enable,

    output logic     refresh_request,
    output logic [12:0] refresh_row_addr,
    output logic     refresh_active
);

    localparam int REFRESH_WIDTH = (REFRESH_PERIOD_US *
CLOCK_FREQ_MHZ) / REFRESH_ROWS;

    logic refresh_pulse;
    logic [12:0] row_counter;

    // Generate refresh timing
    clock_pulse #(
        .WIDTH(REFRESH_WIDTH)

```

```

) refresh_timer (
    .clk(clk),
    .rst_n(rst_n),
    .pulse(refresh_pulse)
);

// Row address counter
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        row_counter <= 13'h0000;
    end else if (refresh_pulse && refresh_enable) begin
        if (row_counter >= REFRESH_ROWS - 1)
            row_counter <= 13'h0000;
        else
            row_counter <= row_counter + 1;
    end
end

// Output assignments
assign refresh_request = refresh_pulse && refresh_enable;
assign refresh_row_addr = row_counter;
assign refresh_active = refresh_enable;

```

endmodule

12.7.55. Multi-Rate Pulse Generation

```

module multi_rate_pulse_generator (
    input logic      clk,
    input logic      rst_n,
    input logic [2:0] rate_select,    // Select pulse rate

    output logic      pulse_fast,     // Fast pulse output
    output logic      pulse_medium,   // Medium pulse output
    output logic      pulse_slow,     // Slow pulse output
    output logic      pulse_config    // Configurable pulse output
);

    logic [31:0] config_width;

    // Rate selection for configurable output
    always_comb begin
        case (rate_select)
            3'b000: config_width = 32'd10;        // Very fast
            3'b001: config_width = 32'd100;       // Fast
            3'b010: config_width = 32'd1000;      // Medium
            3'b011: config_width = 32'd10000;     // Slow
            3'b100: config_width = 32'd100000;    // Very slow
        endcase
    end

```

```

        default: config_width = 32'd1000; // Default medium
    endcase
end

// Fixed rate pulse generators
clock_pulse #(.WIDTH(50)) fast_gen (
    .clk(clk), .rst_n(rst_n), .pulse(pulse_fast)
);

clock_pulse #(.WIDTH(500)) medium_gen (
    .clk(clk), .rst_n(rst_n), .pulse(pulse_medium)
);

clock_pulse #(.WIDTH(5000)) slow_gen (
    .clk(clk), .rst_n(rst_n), .pulse(pulse_slow)
);

// Configurable rate generator (requires parameterizable module)
configurable_pulse_gen config_gen (
    .clk(clk),
    .rst_n(rst_n),
    .width_config(config_width),
    .pulse(pulse_config)
);

endmodule

// Configurable pulse generator module
module configurable_pulse_gen #(
    parameter int MAX_WIDTH = 1000000
) (
    input logic [31:0] width_config,
    input logic clk,
    input logic rst_n,
    output logic pulse
);

    logic [31:0] r_counter;
    logic [31:0] w_width_minus_one;

    assign w_width_minus_one = (width_config > 0) ? (width_config - 1)
: 0;

    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            r_counter <= 32'h00000000;
            pulse <= 1'b0;
        end else begin

```

```

        if (r_counter < w_width_minus_one)
            r_counter <= r_counter + 1;
        else
            r_counter <= 32'h00000000;

        pulse <= (r_counter == w_width_minus_one) && (width_config
> 0);
    end
end

endmodule

```

12.7.66. Test Pattern Generation

```

module test_pattern_generator (
    input logic clk,
    input logic rst_n,
    input logic test_enable,
    input logic [3:0] pattern_select,

    output logic [7:0] test_data,
    output logic test_valid,
    output logic pattern_complete
);

    logic update_pulse;
    logic [7:0] pattern_counter;

    // Generate test data update rate
    clock_pulse #(.WIDTH(16)) pattern_clock (
        .clk(clk),
        .rst_n(rst_n),
        .pulse(update_pulse)
    );

    // Pattern counter
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            pattern_counter <= 8'h00;
        end else if (test_enable && update_pulse) begin
            pattern_counter <= pattern_counter + 1;
        end
    end

    // Test pattern generation
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            test_data <= 8'h00;

```

```

        test_valid <= 1'b0;
    end else if (test_enable && update_pulse) begin
        case (pattern_select)
            4'h0: test_data <= pattern_counter;
// Incrementing
            4'h1: test_data <= ~pattern_counter;
// Decrementing
            4'h2: test_data <= {pattern_counter[0], 7'b00000000};
// Walking 1
            4'h3: test_data <= ~{pattern_counter[0], 7'b00000000};
// Walking 0
            4'h4: test_data <= 8'hAA;
// Alternating
            4'h5: test_data <= 8'h55;
// Alternating inv
            4'h6: test_data <= 8'hFF;
// All ones
            4'h7: test_data <= 8'h00;
// All zeros
            default: test_data <= pattern_counter;
// Default
        endcase
        test_valid <= 1'b1;
    end else begin
        test_valid <= 1'b0;
    end
end

assign pattern_complete = test_enable && (&pattern_counter) &&
update_pulse;

endmodule

```

12.8 Advanced Features and Variations

12.8.11. Enable-Controlled Pulse Generator

```

module pulse_generator_with_enable #(
    parameter int WIDTH = 10
) (
    input  logic clk,
    input  logic rst_n,
    input  logic enable,          // Pulse generation enable
    input  logic pause,          // Pause counting (hold current state)
    output logic pulse,
    output logic [WIDTH-1:0] count_value // Current counter value
);

```

```

logic [WIDTH-1:0] r_counter;
logic [WIDTH-1:0] w_width_minus_one;

assign w_width_minus_one = WIDTH[WIDTH-1:0] - 1'b1;
assign count_value = r_counter;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_counter <= 'b0;
        pulse <= 1'b0;
    end else if (enable && !pause) begin
        if (r_counter < w_width_minus_one)
            r_counter <= r_counter + 1'b1;
        else
            r_counter <= 'b0;

        pulse <= (r_counter == w_width_minus_one);
    end else begin
        pulse <= 1'b0; // No pulse when disabled or paused
    end
end

endmodule

```

12.8.22. Variable Width Pulse Generator

```

module variable_pulse_generator #(
    parameter int MAX_WIDTH = 65536
) (
    input logic                clk,
    input logic                rst_n,
    input logic [$clog2(MAX_WIDTH):0] pulse_width, // Configurable
width
    input logic                load_width, // Load new
width
    output logic                pulse,
    output logic                width_loaded
);

logic [$clog2(MAX_WIDTH):0] r_counter;
logic [$clog2(MAX_WIDTH):0] r_width;
logic [$clog2(MAX_WIDTH):0] w_width_minus_one;

assign w_width_minus_one = (r_width > 0) ? (r_width - 1) : 0;

// Width register
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin

```

```

        r_width <= MAX_WIDTH[$clog2(MAX_WIDTH):0]; // Default
width
        width_loaded <= 1'b0;
    end else if (load_width) begin
        r_width <= (pulse_width > 0) ? pulse_width : 1; //
Minimum width = 1
        width_loaded <= 1'b1;
    end else begin
        width_loaded <= 1'b0;
    end
end

// Counter and pulse generation
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_counter <= 'b0;
        pulse <= 1'b0;
    end else begin
        if (r_counter < w_width_minus_one)
            r_counter <= r_counter + 1;
        else
            r_counter <= 'b0;

        pulse <= (r_counter == w_width_minus_one) && (r_width >
0);
    end
end

endmodule

```

12.8.33. Multi-Phase Pulse Generator

```

module multi_phase_pulse_generator #(
    parameter int WIDTH = 12,
    parameter int PHASES = 4
) (
    input logic clk,
    input logic rst_n,
    input logic enable,
    output logic [PHASES-1:0] phase_pulses
);

    logic [WIDTH-1:0] r_counter;
    logic [WIDTH-1:0] w_width_minus_one;
    logic [WIDTH-1:0] phase_thresholds [PHASES];

    assign w_width_minus_one = WIDTH[WIDTH-1:0] - 1'b1;

```

```

// Calculate phase thresholds
genvar i;
generate
    for (i = 0; i < PHASES; i++) begin : calc_phases
        assign phase_thresholds[i] = (WIDTH * i) / PHASES;
    end
endgenerate

// Counter
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_counter <= 'b0;
    end else if (enable) begin
        if (r_counter < w_width_minus_one)
            r_counter <= r_counter + 1;
        else
            r_counter <= 'b0;
    end
end

// Phase pulse generation
generate
    for (i = 0; i < PHASES; i++) begin : gen_phases
        always_ff @(posedge clk or negedge rst_n) begin
            if (!rst_n) begin
                phase_pulses[i] <= 1'b0;
            end else begin
                phase_pulses[i] <= enable && (r_counter ==
phase_thresholds[i]);
            end
        end
    end
endgenerate

endmodule

```

12.8.44. Burst Pulse Generator

```

module burst_pulse_generator #(
    parameter int BURST_LENGTH = 8,
    parameter int BURST_PERIOD = 100,
    parameter int INTER_BURST_DELAY = 1000
) (
    input logic clk,
    input logic rst_n,
    input logic enable,
    input logic trigger,           // Start burst sequence

    output logic burst_pulse,     // Individual pulses in burst

```



```

    output logic burst_active,           // Burst sequence active
    output logic burst_complete         // Burst sequence completed
);

typedef enum logic [2:0] {
    IDLE,
    BURST_ACTIVE,
    INTER_PULSE_DELAY,
    INTER_BURST_DELAY_ST
} burst_state_t;

burst_state_t state;

logic [$clog2(BURST_LENGTH):0] pulse_count;
logic [$clog2(BURST_PERIOD):0] period_counter;
logic [$clog2(INTER_BURST_DELAY):0] delay_counter;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        state <= IDLE;
        pulse_count <= 0;
        period_counter <= 0;
        delay_counter <= 0;
        burst_pulse <= 1'b0;
        burst_complete <= 1'b0;
    end else begin
        burst_complete <= 1'b0; // Default

        case (state)
            IDLE: begin
                burst_pulse <= 1'b0;
                if (enable && trigger) begin
                    state <= BURST_ACTIVE;
                    pulse_count <= 0;
                    period_counter <= 0;
                    burst_pulse <= 1'b1; // First pulse
                end
            end

            BURST_ACTIVE: begin
                burst_pulse <= 1'b0;
                pulse_count <= pulse_count + 1;

                if (pulse_count >= BURST_LENGTH - 1) begin
                    state <= INTER_BURST_DELAY_ST;
                    delay_counter <= 0;
                    burst_complete <= 1'b1;
                end else begin

```

```

        state <= INTER_PULSE_DELAY;
        period_counter <= 0;
    end
end

INTER_PULSE_DELAY: begin
    period_counter <= period_counter + 1;
    if (period_counter >= BURST_PERIOD - 1) begin
        state <= BURST_ACTIVE;
        burst_pulse <= 1'b1;
    end
end

INTER_BURST_DELAY_ST: begin
    delay_counter <= delay_counter + 1;
    if (delay_counter >= INTER_BURST_DELAY - 1) begin
        state <= IDLE;
    end
end
endcase
end
end

assign burst_active = (state != IDLE);

endmodule

```

12.9 Verification and Testing

12.9.1 Comprehensive Test Bench

```

module tb_clock_pulse;

    parameter WIDTH = 10;

    logic clk, rst_n, pulse;

    // Clock generation
    initial clk = 0;
    always #5ns clk = ~clk; // 100MHz

    // DUT instantiation
    clock_pulse #(.WIDTH(WIDTH)) dut (.*);

    // Test sequence
    initial begin
        rst_n = 0;
        #100ns rst_n = 1;
    end
endmodule

```

```

    // Test basic pulse generation
    test_basic_pulse_generation();

    // Test reset behavior
    test_reset_behavior();

    // Test timing accuracy
    test_timing_accuracy();

    $display("All tests completed!");
    $finish;
end

// Test basic pulse generation
task test_basic_pulse_generation();
    int pulse_count = 0;
    int cycle_count = 0;
    begin
        $display("Testing basic pulse generation (WIDTH=%0d)...",
WIDTH);

        // Monitor for several pulse periods
        repeat (WIDTH * 3) begin
            @(posedge clk);
            cycle_count++;
            if (pulse) pulse_count++;
        end

        // Verify pulse count
        if (pulse_count == 3) begin
            $display("PASS: Correct number of pulses generated");
        end else begin
            $error("FAIL: Expected 3 pulses, got %0d",
pulse_count);
        end
    end
endtask

// Test timing accuracy
task test_timing_accuracy();
    time pulse_times[10];
    time pulse_periods[9];
    int i;
    real avg_period, expected_period;
    begin
        $display("Testing timing accuracy...");

```

```

    expected_period = WIDTH * 10.0; // 10ns clock period

    // Capture pulse timing
    for (i = 0; i < 10; i++) begin
        @(posedge pulse);
        pulse_times[i] = $time;
    end

    // Calculate periods
    for (i = 0; i < 9; i++) begin
        pulse_periods[i] = pulse_times[i+1] - pulse_times[i];
    end

    // Calculate average period
    avg_period = 0;
    for (i = 0; i < 9; i++) begin
        avg_period += pulse_periods[i];
    end
    avg_period = avg_period / 9.0;

    // Check accuracy (within 1% tolerance)
    if (abs(avg_period - expected_period) < expected_period *
0.01) begin
        $display("PASS: Average period = %.1f ns (expected
%.1f ns)",
                    avg_period, expected_period);
    end else begin
        $error("FAIL: Average period = %.1f ns (expected %.1f
ns)",
                    avg_period, expected_period);
    end
end
endtask

function real abs(real value);
    abs = (value >= 0) ? value : -value;
endfunction

endmodule

```

12.9.2 Coverage Model

```

covergroup clock_pulse_cg @(posedge clk);

    cp_counter_values: coverpoint dut.r_counter {
        bins zero = {0};
        bins low[] = {[1:WIDTH/4]};
        bins mid[] = {[WIDTH/4+1:3*WIDTH/4]};
        bins high[] = {[3*WIDTH/4+1:WIDTH-2]};
    }
endcovergroup

```

```

    bins max = {WIDTH-1};
}

cp_pulse: coverpoint pulse {
    bins asserted = {1};
    bins deasserted = {0};
}

cp_reset: coverpoint rst_n {
    bins reset = {0};
    bins normal = {1};
}

// Transition coverage
cp_pulse_edges: coverpoint pulse {
    bins rising = (0 ==> 1);
    bins falling = (1 ==> 0);
    bins stable_low = (0 ==> 0);
    bins stable_high = (1 ==> 1);
}

// Cross coverage
cross_counter_pulse: cross cp_counter_values, cp_pulse;

```

endcovergroup

12.9.3 Formal Properties

module clock_pulse_properties;

```

// Bind to DUT
bind clock_pulse clock_pulse_properties props_inst (.*)

// Property: Pulse occurs at correct count
property pulse_at_max_count;
    @(posedge clk) disable iff (!rst_n)
        (dut.r_counter == WIDTH-1) |-> pulse;
endproperty

// Property: Pulse only occurs at max count
property pulse_only_at_max;
    @(posedge clk) disable iff (!rst_n)
        pulse |-> (dut.r_counter == WIDTH-1);
endproperty

// Property: Counter wraps correctly
property counter_wrap;
    @(posedge clk) disable iff (!rst_n)

```

```

        (dut.r_counter == WIDTH-1) |>= (dut.r_counter == 0);
    endproperty

    // Property: Counter increments
    property counter_increment;
        @(posedge clk) disable iff (!rst_n)
            (dut.r_counter < WIDTH-1) |>=
                (dut.r_counter == $past(dut.r_counter) + 1);
    endproperty

    // Property: Reset behavior
    property reset_behavior;
        @(posedge clk)
            !rst_n |>= (dut.r_counter == 0) && !pulse;
    endproperty

    // Assertions
    assert property (pulse_at_max_count);
    assert property (pulse_only_at_max);
    assert property (counter_wrap);
    assert property (counter_increment);
    assert property (reset_behavior);

endmodule

```

12.10 Synthesis Considerations

12.10.1 Resource Utilization

WIDTH	Counter Bits	LUTs	FFs	Max Freq
8	3	8	4	500MHz
16	4	12	5	450MHz
32	5	18	6	400MHz
1024	10	28	11	350MHz

12.10.2 Timing Optimization

```

// For high-frequency applications, pipeline the comparison
module clock_pulse_pipelined #(
    parameter int WIDTH = 1000
) (
    input logic clk,
    input logic rst_n,
    output logic pulse
);

```

```

logic [WIDTH-1:0] r_counter;
logic [WIDTH-1:0] w_width_minus_one;
logic compare_result;

assign w_width_minus_one = WIDTH[WIDTH-1:0] - 1'b1;

// Pipeline the comparison
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        compare_result <= 1'b0;
    end else begin
        compare_result <= (r_counter == w_width_minus_one);
    end
end

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_counter <= 'b0;
        pulse <= 1'b0;
    end else begin
        if (compare_result)
            r_counter <= 'b0;
        else
            r_counter <= r_counter + 1;
        pulse <= compare_result;
    end
end

endmodule

```

12.11 Common Applications Summary

1. **Timing References:** System heartbeats, watchdog timers
2. **Sampling Systems:** ADC triggers, data acquisition timing
3. **Communication:** Baud rate generation, protocol timing
4. **Memory Systems:** Refresh timing, access scheduling
5. **Test Equipment:** Pattern generation, stimulus timing
6. **Power Management:** Activity monitoring, timeout generation
7. **Display Systems:** Refresh rates, sync generation

12.12 Design Guidelines

1. **Choose Appropriate WIDTH:** Balance between resolution and resource usage
2. **Consider Reset Timing:** Ensure clean startup behavior
3. **Validate Timing:** Verify pulse frequency matches requirements
4. **Plan for Variation:** Consider process, voltage, temperature effects
5. **Monitor Resource Usage:** Large counters can impact timing and area
6. **Use Enables:** Add enable signals for conditional operation
7. **Document Timing:** Clearly specify pulse rates and relationships

The clock pulse generator provides a simple, reliable solution for periodic timing generation, essential for many digital system functions requiring precise, regular timing events.

12.13 Navigation

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13 Binary-Gray Counter Module

13.1 Overview

The counter_bingray module is a dual-output counter that simultaneously provides both binary and Gray code representations of the same count value. This is essential for asynchronous FIFO implementations where Gray code is required to prevent metastability when crossing clock domains, while binary values are needed for arithmetic operations.

13.2 Module Declaration

```
module counter_bingray #(
    parameter int WIDTH = 4
) (
    input logic clk,
    input logic rst_n,
    input logic enable,
    output logic [WIDTH-1:0] counter_bin,
    output logic [WIDTH-1:0] counter_bin_next,
```



```

        output logic [WIDTH-1:0] counter_gray
    );

```

13.3 Parameters

13.3.1 WIDTH

- **Type:** int
- **Default:** 4
- **Description:** Bit width of both binary and Gray code outputs
- **Range:** Any positive integer ≥ 1
- **Impact:** Determines maximum count value ($2^{\text{WIDTH}} - 1$)

13.4 Ports

13.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	System clock input
rst_n	1	logic	Active-low asynchronous reset
enable	1	logic	Counter enable control

13.4.2 Outputs

Port	Width	Type	Description
counter_bin	WIDTH	logic	Binary counter output (registered)
counter_bin_next	WIDTH	logic	Next binary value (combinational)
counter_gray	WIDTH	logic	Gray code output (registered)

13.5 Architecture Details

13.5.1 Internal Logic

```
logic [WIDTH-1:0] w_counter_bin, w_counter_gray;
```

```
assign w_counter_bin = enable ? (counter_bin + 1) : counter_bin;
```

```
assign w_counter_gray = w_counter_bin ^ (w_counter_bin >> 1);
```

```
assign counter_bin_next = w_counter_bin;
```

13.5.2 Gray Code Conversion

The module implements the standard binary-to-Gray conversion: - **MSB**: `gray[WIDTH-1] = binary[WIDTH-1]` - **Other bits**: `gray[i] = binary[i] ^ binary[i+1]` for `i = 0` to `WIDTH-2`

13.6 Gray Code Theory

13.6.1 Why Gray Code?

Gray code (reflected binary code) has the property that adjacent values differ by exactly one bit. This is crucial for:

1. **Metastability Prevention:** When crossing clock domains, only one bit changes at a time
2. **Glitch Elimination:** Reduces intermediate states during transitions
3. **Asynchronous Safety:** Safe for use in asynchronous circuits

13.6.2 Gray Code Sequence Example (4-bit)

Decimal	Binary	Gray	Changes
0	0000	0000	-
1	0001	0001	bit 0
2	0010	0011	bit 1
3	0011	0010	bit 0
4	0100	0110	bit 2
5	0101	0111	bit 0
6	0110	0101	bit 1
7	0111	0100	bit 0
8	1000	1100	bit 3
9	1001	1101	bit 0

Decimal	Binary	Gray	Changes
10	1010	1111	bit 1
11	1011	1110	bit 0
12	1100	1010	bit 2
13	1101	1011	bit 0
14	1110	1001	bit 1
15	1111	1000	bit 0

13.7 Implementation Analysis

13.7.1 Combinational Logic

```
// Next binary calculation
w_counter_bin = enable ? (counter_bin + 1) : counter_bin;

// Gray code conversion
w_counter_gray = w_counter_bin ^ (w_counter_bin >> 1);
```

13.7.2 Sequential Logic

```
always_ff @(posedge clk, negedge rst_n) begin
    if (!rst_n) begin
        counter_bin <= 'b0;
        counter_gray <= 'b0;
    end else begin
        counter_bin <= w_counter_bin;
        counter_gray <= w_counter_gray;
    end
end
```

13.8 Timing Characteristics

13.8.1 Critical Paths

1. **Binary Increment:** Standard binary addition timing
2. **Gray Conversion:** XOR tree with $\log(\text{WIDTH})$ levels
3. **Combined Path:** Increment + conversion in same cycle

13.8.2 Propagation Delays

- **Clock-to-Q:** Standard flip-flop delay
- **Combinational:** Depends on addition and XOR logic depth

- **Setup Time:** Must account for longest combinational path

13.9 Application: Asynchronous FIFO

13.9.1 FIFO Pointer Implementation

```
// Write domain counter
counter_bingray #(.WIDTH(ADDR_WIDTH+1)) wr_counter (
    .clk(wr_clk),
    .rst_n(wr_rst_n),
    .enable(wr_enable),
    .counter_bin(wr_bin),
    .counter_bin_next(wr_bin_next),
    .counter_gray(wr_gray)
);

// Read domain counter
counter_bingray #(.WIDTH(ADDR_WIDTH+1)) rd_counter (
    .clk(rd_clk),
    .rst_n(rd_rst_n),
    .enable(rd_enable),
    .counter_bin(rd_bin),
    .counter_bin_next(rd_bin_next),
    .counter_gray(rd_gray)
);
```

13.9.2 Cross-Domain Synchronization

```
// Synchronize Gray code pointers across domains
logic [ADDR_WIDTH:0] wr_gray_sync, rd_gray_sync;

// Write domain: synchronize read Gray pointer
synchronizer #(.WIDTH(ADDR_WIDTH+1)) rd_sync (
    .clk(wr_clk),
    .rst_n(wr_rst_n),
    .data_in(rd_gray),
    .data_out(rd_gray_sync)
);

// Read domain: synchronize write Gray pointer
synchronizer #(.WIDTH(ADDR_WIDTH+1)) wr_sync (
    .clk(rd_clk),
    .rst_n(rd_rst_n),
    .data_in(wr_gray),
    .data_out(wr_gray_sync)
);
```

13.9.3 FIFO Status Generation

```
// FIFO empty: Gray pointers equal
assign fifo_empty = (rd_gray == wr_gray_sync);

// FIFO full: Binary addresses equal, MSBs different
wire [ADDR_WIDTH-1:0] wr_addr = wr_bin[ADDR_WIDTH-1:0];
wire [ADDR_WIDTH-1:0] rd_addr_sync = gray2bin(rd_gray_sync)
[ADDR_WIDTH-1:0];
wire wr_msb = wr_bin[ADDR_WIDTH];
wire rd_msb_sync = gray2bin(rd_gray_sync)[ADDR_WIDTH];

assign fifo_full = (wr_addr == rd_addr_sync) && (wr_msb !=
rd_msb_sync);
```

13.10 Advanced Features

13.10.1 Almost Full/Empty Flags

```
// Calculate occupancy using binary values
wire [ADDR_WIDTH:0] occupancy = wr_bin - gray2bin(rd_gray_sync);

// Generate status flags
assign almost_full = (occupancy >= ALMOST_FULL_THRESH);
assign almost_empty = (occupancy <= ALMOST_EMPTY_THRESH);
```

13.10.2 Gray-to-Binary Conversion Function

```
function automatic [WIDTH-1:0] gray2bin;
    input [WIDTH-1:0] gray;
    integer i;
    begin
        gray2bin[WIDTH-1] = gray[WIDTH-1];
        for (i = WIDTH-2; i >= 0; i--) begin
            gray2bin[i] = gray2bin[i+1] ^ gray[i];
        end
    end
endfunction
```

13.11 Design Examples

13.11.1 1. 8-bit Asynchronous FIFO Pointer

```
parameter FIFO_DEPTH = 256;
parameter ADDR_WIDTH = $clog2(FIFO_DEPTH);
parameter PTR_WIDTH = ADDR_WIDTH + 1; // Extra bit for full detection
```

```
counter_bingray #(
```

```

        .WIDTH(PTR_WIDTH)
    ) fifo_wr_ptr (
        .clk(wr_clk),
        .rst_n(async_rst_n),
        .enable(wr_enable && !fifo_full),
        .counter_bin(wr_ptr_bin),
        .counter_bin_next(wr_ptr_bin_next),
        .counter_gray(wr_ptr_gray)
    );

```

13.11.2 2. Clock Domain Crossing Counter

// Source domain

```

counter_bingray #(.WIDTH(8)) src_counter (
    .clk(src_clk),
    .rst_n(src_rst_n),
    .enable(src_enable),
    .counter_bin(src_bin),
    .counter_bin_next(),
    .counter_gray(src_gray)
);

```

// Destination domain receives synchronized Gray value

```

logic [7:0] dest_gray_sync;
synchronizer #(.WIDTH(8)) sync_inst (
    .clk(dest_clk),
    .rst_n(dest_rst_n),
    .data_in(src_gray),
    .data_out(dest_gray_sync)
);

```

13.12 Verification Strategy

13.12.1 Test Scenarios

1. **Sequential Counting:** Verify both outputs increment correctly
2. **Gray Code Properties:** Ensure single-bit changes between adjacent values
3. **Reset Behavior:** Confirm both outputs reset to zero
4. **Enable Control:** Test hold behavior when disabled
5. **Rollover:** Verify proper wrap-around from maximum value

13.12.2 Coverage Points

```

covergroup counter_bingray_cg @(posedge clk);
    cp_binary: coverpoint counter_bin {

```

```

        bins all_values[] = {[0:2**WIDTH-1]};
    }

    cp_gray: coverpoint counter_gray {
        bins all_values[] = {[0:2**WIDTH-1]};
    }

    cp_enable: coverpoint enable {
        bins enabled = {1};
        bins disabled = {0};
    }

    cp_transitions: coverpoint counter_bin {
        bins transitions[] = ([0:2**WIDTH-2] => [1:2**WIDTH-1]);
        bins rollover = (2**WIDTH-1 => 0);
    }
endgroup

```

13.12.3 Assertions

```

// Verify Gray code has single bit changes
property gray_single_bit_change;
    @(posedge clk) disable iff (!rst_n)
    enable && !$isunknown(counter_gray) |->
    $countones(counter_gray ^ $past(counter_gray)) <= 1;
endproperty

```

```

assert property (gray_single_bit_change);

```

```

// Verify binary and Gray relationship
property bin_gray_relationship;
    @(posedge clk) disable iff (!rst_n)
    counter_gray == (counter_bin ^ (counter_bin >> 1));
endproperty

```

```

assert property (bin_gray_relationship);

```

13.13 Synthesis Considerations

13.13.1 Resource Utilization

- **Flip-Flops:** 2×WIDTH (separate for binary and Gray)
- **LUTs:** Increment logic + XOR tree for Gray conversion
- **Critical Path:** Through binary increment and Gray conversion

13.13.2 Optimization Techniques

```
// Optional: Pipeline Gray conversion for high speed  
logic [WIDTH-1:0] counter_gray_pipe;  
always_ff @(posedge clk) begin  
    counter_gray_pipe <= w_counter_gray;  
end
```

13.13.3 Tool-Specific Attributes

```
(* ASYNC_REG = "TRUE" *) logic [WIDTH-1:0] counter_gray; // Xilinx  
// synthesis attribute ASYNC_REG of counter_gray is "TRUE" //  
Altera/Intel
```

13.14 Performance Characteristics

13.14.1 Maximum Frequency

- **Typical:** 200-400 MHz in modern FPGAs
- **Limitation:** Binary increment + Gray conversion logic depth
- **Optimization:** Pipeline for higher frequencies

13.14.2 Power Consumption

- **Dynamic:** Proportional to switching activity
- **Static:** Minimal for CMOS technology
- **Optimization:** Clock gating when enable is inactive

13.15 Common Applications

1. **Asynchronous FIFO Pointers:** Primary use case
2. **Clock Domain Crossing Counters:** Safe multi-domain counting
3. **Position Encoders:** Mechanical/optical encoder interfaces
4. **State Machine Counters:** When glitch-free operation required
5. **Memory Address Generation:** For dual-port memory systems

13.16 Troubleshooting Guide

13.16.1 Common Issues

1. **Metastability:** Ensure proper synchronization of Gray values
2. **Timing Violations:** Pipeline Gray conversion if needed
3. **Incorrect FIFO Status:** Verify Gray-to-binary conversion
4. **Reset Skew:** Use proper reset synchronization

13.16.2 Debug Techniques

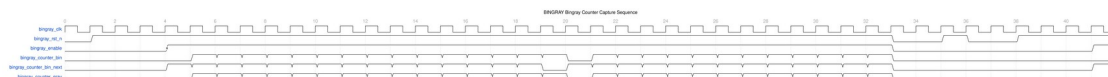
1. **Simulation:** Verify Gray code properties with assertions
2. **Logic Analyzer:** Capture actual transitions in hardware
3. **Timing Analysis:** Check setup/hold requirements
4. **Cross-Domain Checks:** Verify synchronizer timing

13.17 WaveDrom Visualization

High-quality waveforms showcasing Binary-Gray counter operation are available!

The following timing diagrams demonstrate the dual-output counter design across 4 key scenarios:

13.17.1 Scenario 1: Binary vs Gray Code Comparison

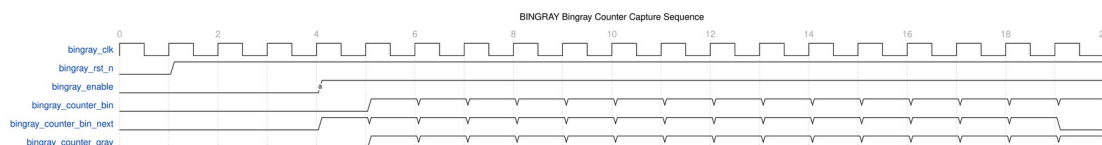


BinGray Comparison

WaveJSON: [bingray_counter_binary_vs_gray.json](#)

Side-by-side comparison of both outputs: - Binary: Normal sequential counting (0→1→2→3→...) - Gray: Single-bit transitions between values - Shows full cycle demonstrating encoding differences - Illustrates why Gray code is CDC-safe

13.17.2 Scenario 2: Single-Bit Transitions (CDC Safety) ★ KEY FEATURE

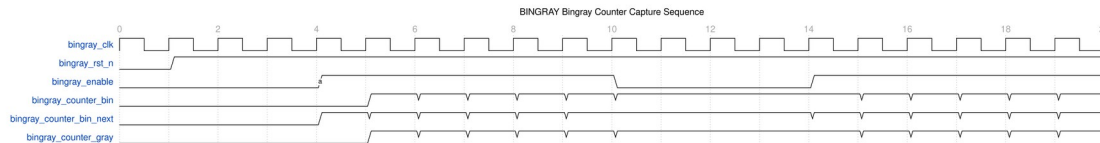


BinGray Single-Bit Transitions

WaveJSON: [bingray_counter_single_bit_transitions.json](#)

Gray code CDC safety property: - Each Gray transition changes EXACTLY one bit - Hamming distance = 1 between adjacent values - **Critical for fifo_async CDC mechanism** - Prevents metastability in clock domain crossing

13.17.3 Scenario 3: Lookahead Signal (counter_bin_next)

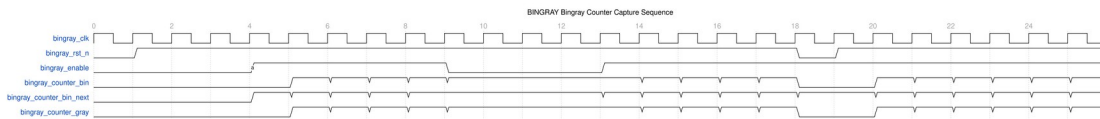


BinGray Lookahead

WaveJSON: [bingray_counter_lookahead.json](#)

Combinational lookahead feature: - Predicts next binary value one cycle ahead - Useful for FIFO full/empty prediction - Shows enable gating behavior - Enables early decision-making

13.17.4 Scenario 4: Enable and Reset Control



BinGray Enable and Reset

WaveJSON: [bingray_counter_enable_reset.json](#)

Control signal behavior: - Both outputs hold when enable=0 - Asynchronous reset to zero - Clean state transitions - Reset during counting demonstration

To regenerate these waveforms:

```
pytest val/common/test_counter_bingray_wavedrom.py -v
# Then convert JSON to PNG:
cd docs/markdown/assets/WAVES/counter_bingray
for f in *.json; do wavedrom-cli -i "$f" -p "${f%.json}.png"; done
```

What Makes Binary-Gray Counters Special:

The waveforms highlight the unique dual-output design: - **Dual Encoding:** Binary for arithmetic, Gray for CDC safety - **Single-Bit Transitions:** Gray code changes one bit at a time - **Lookahead:**

counter_bin_next provides one-cycle prediction - **CDC-Safe**: Safe for asynchronous clock domain crossing

Relationship to fifo_async:

Binary-Gray counters are the foundation of the standard fifo_async module: - **fifo_async** uses this counter for read/write pointers - Gray outputs cross clock domains safely - Binary outputs used for arithmetic (occupancy, address generation) - Logarithmic width ($\lceil \log_2(\text{DEPTH}) \rceil + 1$) for resource efficiency

Comparison Table:

Feature	BinGray Counter	Johnson Counter
Output Width	$\lceil \log_2(\text{DEPTH}) \rceil + 1$ bits	DEPTH bits
CDC Method	Standard Gray code	Johnson code
Depth Support	Power-of-2 only	Any even number
Resource Efficiency	Logarithmic (better for large depths)	Linear
Conversion	XOR tree (simple)	Position detection (complex)
Used In	fifo_async USE_JOHNSON=0	fifo_async USE_JOHNSON=1

Comparison with Other Modules:

- test_counter_johnson_wavedrom.py - Johnson counter (even depths, linear width)
- test_fifo_async_wavedrom.py - BinGray counter in action (async FIFO, power-of-2)

13.18 Test and Verification

Comprehensive Test Suite: - val/common/test_counter_bingray.py - Full functional verification - val/common/test_counter_bingray_wavedrom.py - WaveDrom timing diagrams



Run Tests:

```
# Full functional test (basic/medium/full levels)
pytest val/common/test_counter_bingray.py -v
```

```
# WaveDrom waveform generation
pytest val/common/test_counter_bingray_wavedrom.py -v
```

13.19 Navigation

- [← Back to RTLCommon Index](#)
- [← Back to Main Documentation Index](#)

14 Binary Counter with Load and Variable Increment

A FIFO-optimized binary counter supporting standard increment, variable increment, and direct load operations with configurable wraparound behavior for efficient FIFO pointer management.

14.1 Overview

The counter_bin_load module extends basic binary counting with three distinct operating modes: standard +1 increment, variable-amount increment, and direct load operations. It is specifically optimized for FIFO pointer management where drop/flush operations require jumping the read pointer by arbitrary amounts. The module implements FIFO-style wraparound at $2 \times \text{MAX}$ with MSB toggle for full/empty detection.

14.2 Module Declaration

```
module counter_bin_load #(
    parameter int WIDTH = 5,           // Counter bit width
    parameter int MAX = 10             // Maximum count value (wraps at MAX-1)
) (
    // Derived localparam (computed internally - not user-settable):
    // localparam int WRAP_BOUNDARY = 2 * MAX; // Wraparound boundary

    input logic clk,                   // Clock input
    input logic rst_n,                 // Active-low
    asynchronous reset
    input logic enable,                // Count +1 enable
    input logic add_enable,            // Variable increment
    enable
    input logic [WIDTH-1:0] add_value, // Value to add
    input logic load,                  // Load enable
    (highest priority)
    input logic [WIDTH-1:0] load_value, // Value to load
    output logic [WIDTH-1:0] counter_bin_curr, // Current count
    (registered)
    output logic [WIDTH-1:0] counter_bin_next // Next count
    (combinational)
);
```

14.3 Parameters

14.3.1 User-Settable Parameters

Parameter	Type	Default	Description
WIDTH	int	5	Counter bit width (range: 2-64)
MAX	int	10	Maximum count value before wraparound (range: 2 to $2^{(WIDTH-1)}$)

WIDTH Guidelines: - **Minimum:** 2 bits (required for MSB inversion behavior) - **Typical:** FIFO address width + 1 (e.g., 4 for 8-entry FIFO: 3 addr + 1 MSB) - **Maximum:** 64 bits (practical limit for synthesis)

MAX Constraints: - Must fit within WIDTH-1 bits (excludes MSB) - Example: WIDTH=4 $\rightarrow$ MAX $\leq$ 8 (3 bits for addresses 0-7, 1 MSB for full/empty) - FIFO depth is typically MAX (e.g., MAX=8 $\rightarrow$ 8-entry FIFO)

14.3.2 Derived Localparam (Computed Internally)

Localparam	Computation	Description
WRAP_BOUNDARY	$2 \times \text{MAX}$	Wraparound boundary for add operations

Note: WRAP_BOUNDARY is computed internally and cannot be overridden by users.

14.4 Ports

14.4.1 Inputs

Port	Width	Description
clk	1	System clock (rising edge active)
rst_n	1	Active-low asynchronous reset
enable	1	Standard +1 increment enable
add_enable	1	Variable increment enable

add_value	WIDTH	Value to add when add_enable=1
load	1	Load enable (highest priority operation)
load_value	WIDTH	Value to load when load=1

14.4.2 Outputs

Port	Width	Description
counter_bin_curr	WIDTH	Current counter value (registered output)
counter_bin_next	WIDTH	Next counter value (combinational preview)

14.5 Functionality

14.5.1 Operation Priority

The module implements a three-level priority hierarchy:

- Load Operation** (Highest Priority): load = 1
 - Directly sets counter_bin_next = load_value
 - Used for: Flush all, jump to specific position
- Variable Increment**: add_enable = 1 (when load = 0)
 - Adds add_value to current count: counter_bin_next = counter_bin_curr + add_value
 - Wraps at 2×MAX: Subtracts 2×MAX if sum ≥ 2×MAX
 - Used for: Drop multiple entries, bulk operations
- Standard Increment**: enable = 1 (when load = 0 and add_enable = 0)
 - Adds 1 to current count: counter_bin_next = counter_bin_curr + 1
 - Special FIFO wraparound: MSB toggles at MAX-1, lower bits clear
 - Used for: Normal FIFO read/write pointer increment

4. **Hold:** All enables = 0

- Maintains current value: `counter_bin_next = counter_bin_curr`

14.5.2 FIFO-Optimized Wraparound

14.5.2.1 *Standard Increment Mode (enable=1):*

For MAX=8, WIDTH=4:

Count sequence: 0, 1, 2, 3, 4, 5, 6, 7, 8 (toggle MSB, clear lower)

Binary: 0000 → 0001 → ... → 0111 → 1000 (MSB inverts, lower bits = 0)

0000 → 0001 → ... → 0111 → 1000 → 1001 → ... → 1111 → 0000

Key Feature: MSB toggles when lower bits reach MAX-1 - This creates a 2×MAX count range: 0 to (2×MAX - 1) - MSB different between write/read pointers → FIFO full or empty

14.5.2.2 *Variable Increment Mode (add_enable=1):*

Wraparound at 2×MAX:

Example: `counter_bin_curr = 14, add_value = 5, MAX = 8, WRAP_BOUNDARY = 16`

`Sum = 14 + 5 = 19 ≥ 16 → counter_bin_next = 19 - 16 = 3`

Key Feature: Full 0 to (2×MAX - 1) range with modulo arithmetic

14.5.3 Full/Empty Detection Pattern

// In FIFO controller:

logic full, empty;

assign full = (`wr_ptr[WIDTH-2:0] == rd_ptr[WIDTH-2:0]`) &&
(`wr_ptr[WIDTH-1] != rd_ptr[WIDTH-1]`); *// MSB different*

assign empty = (`wr_ptr[WIDTH-2:0] == rd_ptr[WIDTH-2:0]`) &&
(`wr_ptr[WIDTH-1] == rd_ptr[WIDTH-1]`); *// MSB same*

Why This Works: - Normal operation: `wr_ptr` advances, MSBs eventually differ → full - After wraparound: `rd_ptr` catches up, MSBs match again → empty - MSB acts as “lap counter” for circular buffer

14.6 Implementation Details

14.6.1 Priority Logic

always_comb begin

`counter_bin_next = counter_bin_curr;` *// Default: hold*

if (load) **begin**

// Highest priority: Direct load

```

        counter_bin_next = load_value;

    end else if (add_enable) begin
        // Variable increment with wraparound at 2*MAX
        w_sum_ext = {1'b0, counter_bin_curr} + {1'b0, add_value};

        if (w_sum_ext >= WRAP_BOUNDARY) begin
            counter_bin_next = w_sum_ext[WIDTH-1:0] -
WIDTH'(WRAP_BOUNDARY);
        end else begin
            counter_bin_next = w_sum_ext[WIDTH-1:0];
        end

    end else if (enable) begin
        // Standard +1 with MSB toggle at MAX-1
        if (counter_bin_curr[WIDTH-2:0] == (MAX - 1)) begin
            counter_bin_next = {~counter_bin_curr[WIDTH-1], {(WIDTH-1)
{1'b0}}};
        end else begin
            counter_bin_next = counter_bin_curr + 1;
        end
    end
    // else: hold (default assignment)
end

```

14.6.2 Registered Output

```

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n)
        counter_bin_curr <= 'b0; // Reset to 0
    else
        counter_bin_curr <= counter_bin_next;
end

```

Reset Behavior: Counter initializes to 0 on `rst_n = 0`

14.6.3 Combinational Next-Value Preview

The `counter_bin_next` output provides a one-cycle lookahead of the counter value, useful for: -
FIFO full/empty prediction - Pre-computing next address - Pipelined operations

14.7 Usage Examples

14.7.1 FIFO Read Pointer with Drop Support

```

// 8-entry FIFO with drop capability
localparam int FIFO_DEPTH = 8;
localparam int PTR_WIDTH = $clog2(FIFO_DEPTH) + 1; // +1 for MSB

```



```

logic [PTR_WIDTH-1:0] rd_ptr, rd_ptr_next;
logic rd_en, drop_valid;
logic [PTR_WIDTH-1:0] drop_count;

counter_bin_load #(
    .WIDTH(PTR_WIDTH),          // 4 bits: 3 for addr (0-7) + 1 MSB
    .MAX(FIFO_DEPTH)           // Wrap at 8
) u_rd_ptr (
    .clk                        (clk),
    .rst_n                     (rst_n),
    .enable                    (rd_en & !fifo_empty),           // Normal read
    .add_enable                (drop_valid),                   // Drop multiple
    .add_value                 (drop_count),                   // How many to
    drop                       (flush_all),                     // Flush operation
    .load                      (flush_all),                     // Jump to write
    .load_value               (wr_ptr),
    ptr                       (rd_ptr),
    .counter_bin_curr          (rd_ptr),
    .counter_bin_next          (rd_ptr_next)
);

// Use lower bits for memory address
assign rd_addr = rd_ptr[PTR_WIDTH-2:0];

```

14.7.2 FIFO Write Pointer

```

logic [PTR_WIDTH-1:0] wr_ptr, wr_ptr_next;
logic wr_en;

counter_bin_load #(
    .WIDTH(PTR_WIDTH),
    .MAX(FIFO_DEPTH)
) u_wr_ptr (
    .clk                        (clk),
    .rst_n                     (rst_n),
    .enable                    (wr_en & !fifo_full),
    .add_enable                (1'b0),                         // No variable increment
    for write                 (flush_all),
    .add_value                 ('0),                             // Reset to read pointer
    .load                      (flush_all),
    .load_value               (rd_ptr),
    .counter_bin_curr          (wr_ptr),
    .counter_bin_next          (wr_ptr_next)
);

assign wr_addr = wr_ptr[PTR_WIDTH-2:0];

```

14.7.3 Full/Empty Status Logic

```
// Full: Lower bits match, MSB different
assign fifo_full = (wr_ptr[PTR_WIDTH-2:0] == rd_ptr[PTR_WIDTH-2:0]) &&
                    (wr_ptr[PTR_WIDTH-1] != rd_ptr[PTR_WIDTH-1]);

// Empty: All bits match
assign fifo_empty = (wr_ptr == rd_ptr);

// Count calculation (handles wraparound)
logic [PTR_WIDTH-1:0] fifo_count;
assign fifo_count = (wr_ptr >= rd_ptr) ? (wr_ptr - rd_ptr) :
                                         (2*FIFO_DEPTH - rd_ptr +
wr_ptr);
```

14.7.4 Packet Drop Controller

```
// Drop N packets from FIFO
logic [7:0] packets_to_drop;
logic drop_start;

// State machine for multi-cycle drop
typedef enum logic [1:0] {
    IDLE,
    DROPPING
} drop_state_t;
drop_state_t drop_state;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        drop_state <= IDLE;
        drop_valid <= 1'b0;
        drop_count <= '0;
    end else begin
        case (drop_state)
            IDLE: begin
                if (drop_start && packets_to_drop > 0) begin
                    drop_valid <= 1'b1;
                    drop_count <= packets_to_drop;
                    drop_state <= DROPPING;
                end
            end

            DROPPING: begin
                drop_valid <= 1'b0; // Single-cycle pulse
                drop_state <= IDLE;
            end
        endcase
    end
```

```

end
end

// Connect to counter
counter_bin_load #(
    .WIDTH(PTR_WIDTH),
    .MAX(FIFO_DEPTH)
) u_rd_ptr (
    .clk          (clk),
    .rst_n        (rst_n),
    .enable        (rd_en & !fifo_empty),
    .add_enable    (drop_valid),
    .add_value     (drop_count),
    .load          (1'b0),
    .load_value    ('0),
    .counter_bin_curr(rd_ptr),
    .counter_bin_next(rd_ptr_next)
);

```

14.8 Timing Characteristics

14.8.1 Latency and Throughput

- **Increment Latency:** 1 cycle (registered output)
- **Next-Value Availability:** 0 cycles (combinational counter_bin_next)
- **Throughput:** 1 operation per cycle
- **Reset Recovery:** 1 cycle (asynchronous reset)

14.8.2 Operation Timing Diagrams

14.8.2.1 Standard Increment (enable=1):

Cycle:	0	1	2	3	4	5	6	7	8	9
clk:	— — — — — — — — — — — —									
enable:	— — — — — — — — — — — —									
curr:	0000	0000	0001	0010	0011	0100	0101	0110		
next:	0000	0001	0010	0011	0100	0101	0110	0111		

14.8.2.2 Variable Increment (add_enable=1, add_value=3):

Cycle:	0	1	2	3
clk:	— — — — — — — — — — — —			
add_en:	— — — — — — — — — — — —			
add_val:	-----[3]-----			
curr:	0010	0010	0101	

```
next:      0010  0101  0110
           ↑ +3
```

14.8.2.3 Load Operation (load=1, load_value=12):

```
Cycle:      0      1      2
clk:        ┌─┐┌─┐┌─┐┌─┐
load:        ┌──┐┌──┐┌──┐
load_val:    -----[C]---
curr:        0011  0011  1100
next:        0011  1100  1101
           ↑ direct load
```

14.9 Design Considerations

14.9.1 When to Use counter_bin_load

✓ **Appropriate Use Cases:** - FIFO read/write pointers with drop/flush - Circular buffer management with skip operations - Packet queue pointers with bulk discard - Ring buffer pointers with reset capability - Any counter needing jump-to-position capability

14.9.2 When to Use Alternatives

Use counter_bin.sv if: - Only need standard +1 increment - No load or variable increment required - Simpler interface preferred

Use counter_load_clear.sv if: - Need load/clear but not FIFO wraparound - Don't need MSB toggle behavior - Variable increment not required

14.9.3 Parameter Selection Guidelines

WIDTH Calculation:

```
// For FIFO with N entries:
localparam int FIFO_DEPTH = 16;
localparam int PTR_WIDTH = $clog2(FIFO_DEPTH) + 1; // +1 for MSB
```

```
// Example: 16-entry FIFO
// FIFO_DEPTH = 16
// PTR_WIDTH = 4 + 1 = 5 bits
// Lower 4 bits = address (0-15)
// MSB = full/empty indicator
```

MAX Selection: - Set MAX equal to FIFO depth - MAX must be power of 2 for efficient address decoding - MAX must fit in WIDTH-1 bits

14.9.4 Critical Paths

1. **Add Operation:** Extended addition + wraparound comparison

- Longest path: counter_bin_curr + add_value → overflow detection → modulo
2. **Standard Increment:** Comparison + MSB toggle
 - Path: Lower bits comparison → MSB invert + clear
 3. **Load Operation:** Direct assignment
 - Shortest path: Mux select

Optimization Tip: If add operation is critical path, consider pipelining:

```
// Pipeline add operation
logic [WIDTH-1:0] r_add_result;
always_ff @(posedge clk) begin
    r_add_result <= counter_bin_curr + add_value;
end
```

14.10 Performance Characteristics

14.10.1 Resource Utilization

Configuration	FFs	LUTs	Description
WIDTH=4, MAX=8	4	~12	Minimal (8-entry FIFO)
WIDTH=5, MAX=16	5	~16	Small (16-entry FIFO)
WIDTH=9, MAX=256	9	~30	Medium (256-entry FIFO)
WIDTH=13, MAX=4096	13	~45	Large (4K-entry FIFO)

Area Scaling: Approximately linear with WIDTH due to adder and comparator

14.10.2 Maximum Frequency

- **Typical:** 300-500 MHz (modern FPGAs)
- **Limiting Factor:** Add operation with wraparound detection
- **Optimization:** Pipeline add path for >500 MHz

14.11 Verification Notes

Test suite location: `val/common/test_counter_bin_load.py`

Key Test Scenarios: - Standard increment (`enable=1`) - Variable increment with various `add_value` - Load operation priority verification - Wraparound behavior at $2 \times \text{MAX}$ - MSB toggle at $\text{MAX}-1$ for standard increment - Full/empty condition verification - Priority hierarchy (`load > add > enable`) - Reset behavior

Test Command:

```
pytest val/common/test_counter_bin_load.py -v
```

14.12 Common Pitfalls

x Anti-Pattern 1: Incorrect WIDTH for FIFO depth

```
// WRONG: WIDTH = $clog2(FIFO_DEPTH)
localparam int WIDTH = $clog2(16); // = 4 bits
// Problem: No MSB for full/empty detection!

// RIGHT: WIDTH = $clog2(FIFO_DEPTH) + 1
localparam int WIDTH = $clog2(16) + 1; // = 5 bits
// Lower 4 bits = address, MSB = full/empty indicator
```

x Anti-Pattern 2: Ignoring operation priority

```
// WRONG: Trying to increment while load is active
load <= 1'b1;
enable <= 1'b1; // This will be IGNORED!
// Result: Counter loads load_value, enable has no effect

// RIGHT: Ensure only one operation active
load <= drop_all;
enable <= rd_en & !drop_all; // Mutually exclusive
```

x Anti-Pattern 3: `add_value` exceeds $2 \times \text{MAX}$

```
// WRONG: add_value larger than valid range
// MAX = 8, add_value = 20
// Result: Multiple wraparounds, unpredictable behavior

// RIGHT: Constrain add_value
assert property (@(posedge clk) add_enable |-> add_value < 2*MAX);
```

x Anti-Pattern 4: Using wrong bits for memory address

```
// WRONG: Using full counter including MSB
assign mem_addr = wr_ptr; // Includes MSB!
```

```
// Result: Addresses 0-15 appear as 0-15 and 16-31

// RIGHT: Use only lower bits
assign mem_addr = wr_ptr[WIDTH-2:0]; // Exclude MSB
```

14.13 Related Modules

- **counter_bin.sv** - Simple binary counter (no load or variable increment)
- **counter_load_clear.sv** - Counter with load/clear (no FIFO wraparound)
- **counter_freq_invariant.sv** - Time-based counter (millisecond/microsecond timing)
- **gaxi_drop_fifo_sync.sv** - Uses counter_bin_load for drop-capable FIFO
- **gaxi_fifo_sync.sv** - Standard synchronous FIFO with this counter

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15 Binary Counter for FIFOs

15.1 Overview

The counter_bin module is a specialized binary counter designed specifically for FIFO (First-In-First-Out) buffer applications. Unlike traditional counters, it implements a unique wrap-around behavior where the MSB is inverted and lower bits are cleared when reaching the maximum value, creating a pattern suitable for circular buffer management.

15.2 Module Declaration

```
module counter_bin #(
    parameter int WIDTH = 5,
    parameter int MAX    = 10
) (
    input logic          clk,
    input logic          rst_n,
    input logic          enable,
    output logic [WIDTH-1:0] counter_bin_curr,
    output logic [WIDTH-1:0] counter_bin_next
);
```

15.3 Parameters

15.3.1 WIDTH

- **Type:** int
- **Default:** 5
- **Description:** Total width of the counter in bits
- **Range:** Must be ≥ 2 for proper MSB inversion operation
- **Usage:** Determines the counter's bit width and maximum representable value

15.3.2 MAX

- **Type:** int
- **Default:** 10
- **Description:** Maximum count value before special wrap behavior
- **Range:** Should be $< 2^{(WIDTH-1)}$ for proper operation
- **Constraint:** Must fit within the lower WIDTH-1 bits

15.4 Ports

15.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	System clock input
rst_n	1	logic	Active-low asynchronous reset
enable	1	logic	Counter enable control

15.4.2 Outputs

Port	Width	Type	Description
counter_bin_curr	WIDTH	logic	Current counter value

Port	Width	Type	Description
counter_bin_next	WIDTH	logic	(registered) Next counter value (combinational)

15.5 Architecture Details

15.5.1 Internal Signals

```
logic [WIDTH-2:0] w_max_val;
assign w_max_val = (WIDTH-1)'(MAX - 1);
```

15.5.2 Special Wrap Logic

The counter implements a unique wrap-around behavior:

```
always_comb begin
    if (enable) begin
        if (counter_bin_curr[WIDTH-2:0] == w_max_val)
            counter_bin_next = {~counter_bin_curr[WIDTH-1], {(WIDTH - 1){1'b0}}};
        else
            counter_bin_next = counter_bin_curr + 1;
        end else begin
            counter_bin_next = counter_bin_curr;
        end
    end
```

15.6 Operation Principles

15.6.1 Normal Counting

When counter_bin_curr[WIDTH-2:0] < MAX-1: - Counter increments normally by 1 - MSB remains unchanged - Standard binary counting behavior

15.6.2 Wrap Condition

When counter_bin_curr[WIDTH-2:0] == MAX-1: - MSB is inverted:
~counter_bin_curr[WIDTH-1] - Lower bits are cleared: {(WIDTH-1){1'b0}} - Creates a “folded” counting pattern

15.6.3 Enable Control

- When enable = 1: Counter operates normally
- When enable = 0: Counter holds current value

15.7 Counting Sequence Example

15.7.1 Configuration: WIDTH=4, MAX=6

Initial: 0000 (MSB=0, lower=000)

0001

0010

0011

0100

0101 <- MAX-1 reached

Wrap: 1000 <- MSB inverted (0→1), lower bits cleared

1001

1010

1011

1100

1101 <- MAX-1 reached again

Wrap: 0000 <- MSB inverted (1→0), lower bits cleared

15.7.2 Pattern Analysis

- **First Half:** MSB=0, counts 0 to MAX-1
- **Second Half:** MSB=1, counts 0 to MAX-1
- **Full Cycle:** 2×MAX states total

15.8 FIFO Application Context

15.8.1 Address Generation

This counter pattern is ideal for FIFO addressing because: 1. **Circular Addressing:** Lower bits provide circular buffer addresses 2. **Overflow Detection:** MSB toggles indicate buffer wrap-around 3. **Full/Empty Logic:** Comparing MSBs between read/write pointers

15.8.2 FIFO Pointer Comparison

// FIFO empty when pointers exactly equal

assign fifo_empty = (rd_ptr == wr_ptr);

// FIFO full when addresses equal but MSBs differ

assign fifo_full = (rd_ptr[WIDTH-2:0] == wr_ptr[WIDTH-2:0]) &&
 (rd_ptr[WIDTH-1] != wr_ptr[WIDTH-1]);

15.9 Timing Characteristics

15.9.1 Combinational Paths

- **Enable to Next:** Single logic level
- **Current to Next:** Increment/wrap logic depth
- **Critical Path:** Wrap detection and MSB inversion

15.9.2 Sequential Behavior

- **Setup Time:** Standard flip-flop requirements
- **Clock-to-Q:** Standard flip-flop propagation
- **Reset Recovery:** One cycle after reset deassertion

15.10 Design Examples

15.10.1 1. Small FIFO Buffer (8 entries)

```
counter_bin #(
    .WIDTH(4),      // 4-bit counter
    .MAX(8)         // 8 entries (0-7)
) wr_ptr_counter (
    .clk(clk),
    .rst_n(rst_n),
    .enable(wr_enable),
    .counter_bin_curr(wr_ptr),
    .counter_bin_next(wr_ptr_next)
);
```

15.10.2 2. Large FIFO Buffer (1024 entries)

```
counter_bin #(
    .WIDTH(11),     // 11-bit counter (10 bits address + 1 MSB)
    .MAX(1024)      // 1024 entries (0-1023)
) rd_ptr_counter (
    .clk(clk),
    .rst_n(rst_n),
    .enable(rd_enable),
    .counter_bin_curr(rd_ptr),
    .counter_bin_next(rd_ptr_next)
);
```

15.10.3 3. Asymmetric FIFO

```
// Write pointer - 16 entries
counter_bin #(.WIDTH(5), .MAX(16)) wr_cnt (...);

// Read pointer - different size for width conversion
counter_bin #(.WIDTH(6), .MAX(32)) rd_cnt (...);
```

15.11 Advanced Usage

15.11.1 Address Extraction

```
// Extract memory address (lower bits only)
assign mem_addr = counter_bin_curr[WIDTH-2:0];

// Extract wrap flag (MSB only)
assign wrap_flag = counter_bin_curr[WIDTH-1];
```

15.11.2 Gray Code Conversion

For asynchronous FIFOs, convert to Gray code:

```
logic [WIDTH-1:0] gray_ptr;
bin2gray #(.WIDTH(WIDTH)) b2g (
    .binary(counter_bin_curr),
    .gray(gray_ptr)
);
```

15.12 Synthesis Considerations

15.12.1 Resource Utilization

- **Flip-Flops:** WIDTH registers for current counter
- **LUTs:** Increment, wrap detection, and MSB inversion logic
- **Optimization:** Synthesis tools typically optimize the wrap logic efficiently

15.12.2 Timing Optimization

```
// Optional: Pipeline for high-frequency operation
logic [WIDTH-1:0] counter_bin_pipe;
always_ff @(posedge clk) begin
    counter_bin_pipe <= counter_bin_next;
end
```

15.12.3 Parameter Validation

```
initial begin
    assert (WIDTH >= 2)
```

```

        else $error("WIDTH must be at least 2 for MSB operation");
    assert (MAX <= (2**(WIDTH-1)))
        else $error("MAX too large for given WIDTH");
    assert (MAX > 0)
        else $error("MAX must be positive");
end

```

15.13 Verification Strategy

15.13.1 Test Scenarios

1. **Basic Increment:** Verify normal counting behavior
2. **Wrap Boundary:** Test MSB inversion at MAX-1
3. **Enable Control:** Verify hold behavior when disabled
4. **Reset Operation:** Confirm proper reset to zero
5. **Full Sequence:** Complete cycle through all states

15.13.2 Coverage Points

```

covergroup counter_bin_cg @(posedge clk);
    cp_lower_bits: coverpoint counter_bin_curr[WIDTH-2:0] {
        bins all_values[] = {[0:MAX-1]};
    }
    cp_msb: coverpoint counter_bin_curr[WIDTH-1] {
        bins low = {0};
        bins high = {1};
    }
    cp_wrap: coverpoint (counter_bin_curr[WIDTH-2:0] == w_max_val);
    cp_enable: coverpoint enable;
endgroup

```

15.14 Common Issues and Solutions

15.14.1 Issue 1: Incorrect FIFO Size

Problem: MAX parameter doesn't match actual FIFO depth **Solution:** Ensure MAX equals the number of FIFO entries

15.14.2 Issue 2: Width Mismatch

Problem: WIDTH too small for MAX value **Solution:** Set $WIDTH \geq \lceil \log_2(2*MAX) \rceil$ for full range

15.14.3 Issue 3: Timing Violations

Problem: Wrap logic creates long combinational paths **Solution:** Pipeline the next value calculation or reduce MAX

15.15 Performance Characteristics

15.15.1 Maximum Frequency

- **Limiting Factor:** Wrap detection and MSB inversion logic
- **Typical Performance:** 300-500 MHz in modern FPGAs
- **Optimization:** Consider registering intermediate signals for high speed

15.15.2 Power Consumption

- **Dynamic Power:** Proportional to toggle rate
- **Optimization:** Use enable signal to reduce unnecessary toggles
- **Clock Gating:** Gate clock when FIFO operations are idle

15.16 Related Modules

- `counter_bingray`: Provides both binary and Gray code outputs
- `counter`: Basic counter without wrap behavior
- `counter_load_clear`: Loadable counter with different wrap behavior
- Standard FIFO controllers that use this addressing scheme

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16 Frequency Invariant Counter Module

16.1 Overview

The `counter_freq_invariant` module is an advanced timing counter that maintains consistent time-based counting regardless of the input clock frequency. It uses a configurable prescaler to

divide the input clock and provides a frequency-independent output counter, making it ideal for applications requiring consistent timing across different clock domains or frequencies.

16.2 Module Declaration

```
module counter_freq_invariant #(
    parameter int COUNTER_WIDTH = 5,      // Width of the output
    counter
    parameter int PRESCALER_MAX = 65536   // Maximum value of the pre-
    scaler
) (
    input logic clk,                      // Input clock
    input logic rst_n,                   // Active low
    reset
    input logic sync_reset_n, // Synchronous
    reset signal
    input logic [3:0] freq_sel,          // Frequency
    selection (configurable)
    output logic [COUNTER_WIDTH-1:0] counter, // 5-bit output
    counter
    output logic tick                    // Pulse every
    time counter increments
);
```

16.3 Parameters

16.3.1 COUNTER_WIDTH

- **Type:** int
- **Default:** 5
- **Description:** Bit width of the main output counter
- **Range:** 1 to 32 (practical range)
- **Impact:** Determines maximum count value ($2^{\text{COUNTER_WIDTH}} - 1$)

16.3.2 PRESCALER_MAX

- **Type:** int
- **Default:** 65536
- **Description:** Maximum division factor for the prescaler
- **Range:** Must be $\geq$ largest division factor (10000)

- **Usage:** Determines internal prescaler counter width

16.4 Ports

16.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	Input clock signal (any frequency)
rst_n	1	logic	Active-low asynchronous reset
sync_reset_n	1	logic	Synchronous reset control
freq_sel	4	logic	Frequency selection (0-15)

16.4.2 Outputs

Port	Width	Type	Description
counter	COUNTER_WID TH	logic	Main counter output
tick	1	logic	Pulse when counter increments

16.5 Frequency Selection Table

The module provides 16 predefined frequency divisions based on a 1GHz reference:

freq_sel	Division Factor	Target Frequency	Period	Use Case
4'b0000	1	1000MHz (1GHz)	1ns	High-speed processing
4'b0001	10	100MHz	10ns	Standard FPGA operations
4'b0010	20	50MHz	20ns	Memory

freq_sel	Division Factor	Target Frequency	Period	Use Case
				interfaces
4'b0011	25	40MHz	25ns	Display controllers
4'b0100	40	25MHz	40ns	Audio sampling
4'b0101	50	20MHz	50ns	Serial communications
4'b0110	80	12.5MHz	80ns	Sensor interfaces
4'b0111	100	10MHz	100ns	ADC/DAC timing
4'b1000	125	8MHz	125ns	Microcontroller speeds
4'b1001	200	5MHz	200ns	Slow peripherals
4'b1010	250	4MHz	250ns	Power-optimized
4'b1011	500	2MHz	500ns	Ultra-low power
4'b1100	1000	1MHz	1μs	Timing references
4'b1101	2000	500kHz	2μs	Human interface
4'b1110	5000	200kHz	5μs	LED refresh rates
4'b1111	10000	100kHz	10μs	Very slow timing

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16.7.2 Change Detection Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_prev_freq_sel <= 4'b0;
        r_clear_pulse <= 1'b0;
    end else begin
        r_prev_freq_sel <= freq_sel;

        // Generate clear pulse when freq_sel changes or sync reset
        r_clear_pulse <= (freq_sel != r_prev_freq_sel) || !
sync_reset_n;
    end
end
```

16.7.3 Prescaler Implementation

The prescaler uses the counter_load_clear module to implement configurable division:

```
counter_load_clear #(
    .MAX(PRESCALER_MAX)
) prescaler_counter(
    .clk(clk),
    .rst_n(rst_n),
    .clear(r_clear_pulse),           // Clear on frequency change
    .increment(1'b1),               // Always increment
    .load(1'b1),                    // Always load new value
    .loadval(w_division_factor[$clog2(PRESCALER_MAX)-1:0] - 1'b1),
    .done(w_prescaler_done),
    .count()                         // Unused
);
```

16.7.4 Main Counter and Tick Generation

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        counter <= 'b0;
        tick <= 'b0;
    end else begin
        // Generate tick when prescaler completes and counter at max
        if (w_prescaler_done && counter)
            tick <= 'b1;
        else
            tick <= 'b0;

        // Update main counter
        if (r_clear_pulse)
            counter <= 'b0;
        else if (w_prescaler_done)
            counter <= counter + 1'b1; // Natural rollover
    end
end
```

end
end

16.8 Timing Analysis

16.8.1 Prescaler Operation

- **Load Value:** $\text{division_factor} - 1$
- **Count Cycles:** division_factor clock cycles per prescaler tick
- **Frequency:** $f_{\text{input}} / \text{division_factor}$

16.8.2 Main Counter Operation

- **Increment Rate:** Prescaler tick rate
- **Roll-over:** At $2^{\text{COUNTER_WIDTH}}$
- **Tick Generation:** One cycle when counter is all 1's and prescaler completes

16.8.3 Overall Timing

- **Counter Period:** $\text{division_factor} \times 2^{\text{COUNTER_WIDTH}}$ input clock cycles
- **Tick Period:** division_factor input clock cycles
- **Frequency Independence:** Same tick rate regardless of input frequency

16.9 Design Examples

16.9.1 1. 1MHz Tick Generator (Any Input Clock)

```
counter_freq_invariant #(
    .COUNTER_WIDTH(8),           // 8-bit counter (0-255)
    .PRESCALER_MAX(65536)       // Support up to 65536 division
) tick_1mhz (
    .clk(clk_any_freq),         // Could be 100MHz, 200MHz, etc.
    .rst_n(rst_n),
    .sync_reset_n(1'b1),
    .freq_sel(4'b1100),         // Select 1MHz (1000 division)
    .counter(timer_count),
    .tick(tick_1mhz)
);
```

16.9.22. Configurable Baud Rate Generator

```
logic [3:0] baud_sel;
logic baud_tick;

// Baud rate selection mapping
// 4'b0111 = 10MHz → 115200 baud (10MHz/87 ≈ 115200)
// 4'b1001 = 5MHz → 57600 baud (5MHz/87 ≈ 57600)
// 4'b1100 = 1MHz → 9600 baud (1MHz/104 ≈ 9600)

counter_freq_invariant #(
    .COUNTER_WIDTH(4),
    .PRESCALER_MAX(10000)
) baud_gen (
    .clk(sys_clk),
    .rst_n(rst_n),
    .sync_reset_n(1'b1),
    .freq_sel(baud_sel),
    .counter(), // Unused
    .tick(baud_tick)
);
```

16.9.33. Multi-Rate Timer System

```
// Different timers for different purposes
counter_freq_invariant #(.COUNTER_WIDTH(6)) fast_timer (
    .freq_sel(4'b0001), // 100MHz rate
    .tick(fast_tick), // Every 10ns
    ...
);

counter_freq_invariant #(.COUNTER_WIDTH(8)) med_timer (
    .freq_sel(4'b0111), // 10MHz rate
    .tick(med_tick), // Every 100ns
    ...
);

counter_freq_invariant #(.COUNTER_WIDTH(10)) slow_timer (
    .freq_sel(4'b1100), // 1MHz rate
    .tick(slow_tick), // Every 1μs
    ...
);
```

16.10 Advanced Features

16.10.1 Dynamic Frequency Switching

// Example: Switch between fast and slow modes

```
logic power_save_mode;
logic [3:0] dynamic_freq_sel;

always_comb begin
    if (power_save_mode)
        dynamic_freq_sel = 4'b1111; // 100kHz for power save
    else
        dynamic_freq_sel = 4'b0001; // 100MHz for performance
end

counter_freq_invariant adaptive_timer (
    .freq_sel(dynamic_freq_sel),
    .sync_reset_n(!power_save_mode), // Reset when switching modes
    ...
);
```

16.10.2 Synchronous Reset Applications

// Reset timer when entering specific system state

```
logic enter_calibration_mode;

counter_freq_invariant cal_timer (
    .sync_reset_n(!enter_calibration_mode),
    .freq_sel(4'b1010), // 4MHz for calibration
    ...
);
```

16.11 Verification Strategy

16.11.1 Test Scenarios

1. **Frequency Selection:** Test all 16 freq_sel values
2. **Dynamic Switching:** Change freq_sel during operation
3. **Reset Behavior:** Test both async and sync reset
4. **Tick Generation:** Verify tick timing at various frequencies
5. **Counter Rollover:** Test main counter wrap-around
6. **Prescaler Boundary:** Test prescaler terminal count

16.11.2 Testbench Example

```
module tb_counter_freq_invariant;
    logic clk, rst_n, sync_reset_n;
    logic [3:0] freq_sel;
    logic [4:0] counter;
    logic tick;

    // Clock generation - 100MHz
    initial clk = 0;
    always #5ns clk = ~clk;

    // Test sequence
    initial begin
        rst_n = 0;
        sync_reset_n = 1;
        freq_sel = 4'b0000;

        #100ns rst_n = 1;

        // Test different frequencies
        #1000ns freq_sel = 4'b0001; // 100MHz → 10MHz
        #1000ns freq_sel = 4'b0111; // 10MHz
        #1000ns freq_sel = 4'b1100; // 1MHz

        // Test synchronous reset
        #1000ns sync_reset_n = 0;
        #50ns sync_reset_n = 1;

        #10000ns $finish;
    end

    // Monitor tick frequency
    real tick_period, last_tick_time;
    always @(posedge tick) begin
        tick_period = $realtime - last_tick_time;
        last_tick_time = $realtime;
        $display("Tick period: %.2f ns (freq_sel=%b)", tick_period,
freq_sel);
    end
endmodule
```

16.11.3 Coverage Points

```
covergroup freq_invariant_cg @(posedge clk);
    cp_freq_sel: coverpoint freq_sel {
        bins all_freqs[] = {[0:15]};
    }
}
```

```

cp_counter: coverpoint counter {
    bins low = {[0:7]};
    bins mid = {[8:23]};
    bins high = {[24:31]};
}

cp_freq_change: coverpoint (freq_sel != $past(freq_sel)) {
    bins change = {1};
    bins stable = {0};
}

cp_sync_reset: coverpoint sync_reset_n {
    bins active = {0};
    bins inactive = {1};
}
endgroup

```

16.12 Performance Characteristics

16.12.1 Resource Utilization

- **Prescaler Counter:** $\lceil \log_2(\text{PRESCALER_MAX}) \rceil$ bits
- **Main Counter:** COUNTER_WIDTH bits
- **Logic:** Division factor lookup table + change detection
- **Total FFs:** ~20-25 flip-flops for typical configuration

16.12.2 Timing Performance

- **Maximum Frequency:** Limited by prescaler counter increment
- **Typical:** 200-400 MHz in modern FPGAs
- **Critical Path:** Prescaler increment + done detection

16.12.3 Power Consumption

- **Dynamic:** Proportional to clock frequency and toggle rate
- **Optimization:** Higher division factors reduce overall power
- **Clock Gating:** Consider gating prescaler when not needed

16.13 Applications

16.13.1 Communication Systems

```
// UART baud rate generation
counter_freq_invariant uart_baud (
    .freq_sel(baud_rate_sel), // From configuration register
    .tick(baud_x16_tick),     // 16x oversampling
    ...
);
```

16.13.2 Display Controllers

```
// Pixel clock generation for different video modes
counter_freq_invariant pixel_clk (
    .freq_sel(video_mode_sel), // Different pixel rates
    .tick(pixel_enable),
    ...
);
```

16.13.3 ADC/DAC Timing

```
// Sample rate control independent of system clock
counter_freq_invariant sample_rate (
    .freq_sel(sample_rate_sel), // Audio sample rates
    .tick(conversion_trigger),
    ...
);
```

16.13.4 Power Management

```
// Frequency scaling for power optimization
counter_freq_invariant power_timer (
    .freq_sel(power_mode_sel), // Fast/medium/slow/sleep
    .tick(power_event),
    ...
);
```

16.14 Design Guidelines

16.14.1 Parameter Selection

1. **COUNTER_WIDTH**: Choose based on maximum count needed
2. **PRESCALER_MAX**: Must be $\geq$ largest division factor (10000)
3. **freq_sel Mapping**: Customize division table for your application

16.14.2 Reset Strategy

1. **Asynchronous Reset:** For system-wide reset
2. **Synchronous Reset:** For mode changes and state machine control
3. **Clear Pulse:** Automatically generated on frequency changes

16.14.3 Frequency Planning

1. **Input Clock:** Can be any stable frequency
2. **Division Factors:** Choose based on target frequencies
3. **Jitter:** Higher division factors may accumulate jitter

16.15 Troubleshooting

16.15.1 Common Issues

1. **Incorrect Tick Rate:** Verify division factor calculation
2. **Missing Ticks:** Check for proper reset sequencing
3. **Frequency Switching Glitches:** Ensure clean freq_sel transitions
4. **Timing Violations:** Pipeline if prescaler counter is too wide

16.15.2 Debug Techniques

1. **Simulation:** Use time-based measurements to verify tick rates
2. **Logic Analyzer:** Capture actual tick timing in hardware
3. **Frequency Counter:** Measure actual output frequency
4. **Reset Analysis:** Check reset propagation and synchronization

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17 Johnson Counter Module

17.1 Overview

The `counter_johnson` module implements a Johnson counter (also known as a twisted-ring counter or switch-tail counter). This is a special type of shift register where the inverted output of the last stage is fed back to the first stage, creating a sequence with $2 \times \text{WIDTH}$ unique states. Johnson counters are widely used for generating multi-phase clock signals, state machine control, and sequential timing applications.

17.2 Module Declaration

```
module counter_johnson #(
    parameter int WIDTH = 4
) (
    input logic          clk,
    input logic          rst_n,
    input logic          enable,
    output logic [WIDTH - 1:0] counter_gray
);
```

17.3 Parameters

17.3.1 WIDTH

- **Type:** int
- **Default:** 4
- **Description:** Number of stages in the Johnson counter
- **Range:** Any positive integer ≥ 1
- **States Generated:** $2 \times \text{WIDTH}$ unique states
- **Impact:** Determines sequence length and number of output phases

17.4 Ports

17.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	System clock input
rst_n	1	logic	Active-low

Port	Width	Type	Description
enable	1	logic	asynchronous reset Counter enable control

17.4.2 Outputs

Port	Width	Type	Description
counter_gray	WIDTH	logic	Johnson counter output register

17.5 Theory of Operation

17.5.1 Johnson Counter Principle

A Johnson counter is a shift register with inverted feedback: - **Normal Operation:** Bits shift left (or right) each clock cycle - **Feedback:** The complement of the MSB feeds back to the LSB - **Sequence Length:** $2 \times \text{WIDTH}$ states before repeating

17.5.2 Mathematical Representation

For a WIDTH-bit Johnson counter:

Next_State[i] = Current_State[i-1] for i = 1 to WIDTH-1
Next_State[0] = ~Current_State[WIDTH-1]

17.6 Implementation Details

17.6.1 Core Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n)
        counter_gray <= {WIDTH{1'b0}}; // Reset to all zeros
    else if (enable) begin
        counter_gray <= {counter_gray[WIDTH-2:0], ~counter_gray[WIDTH-1]};
    end
end
```

17.6.2 Operation Breakdown

1. **Shift Operation:** counter_gray[WIDTH-2:0] → lower bits of next state

2. **Feedback:** $\sim\text{counter_gray}[\text{WIDTH}-1] \rightarrow$ MSB complement becomes new LSB
3. **Enable Control:** Only advance when enable is asserted
4. **Reset:** All bits cleared on reset

17.7 State Sequences

17.7.1 4-bit Johnson Counter (WIDTH=4)

Step	counter_gray	Decimal	Description
0	0000	0	Reset state
1	0001	1	First '1' enters
2	0011	3	'1' propagates
3	0111	7	'1' propagates
4	1111	15	All '1's
5	1110	14	First '0' enters
6	1100	12	'0' propagates
7	1000	8	'0' propagates
8	0000	0	Back to start

17.7.2 3-bit Johnson Counter (WIDTH=3)

Step	counter_gray	Binary Pattern
0	000	All zeros
1	001	One bit set
2	011	Two bits set
3	111	All ones
4	110	Two bits set
5	100	One bit set
6	000	Cycle repeats

17.7.3 2-bit Johnson Counter (WIDTH=2)

Step	counter_gray	State Name
0	00	State A
1	01	State B

Step	counter_gray	State Name
2	11	State C
3	10	State D
4	00	Back to A

17.8 Unique Properties

17.8.1 Self-Starting

Johnson counters are self-starting from most invalid states: - **Valid States:** $2 \times \text{WIDTH}$ states in the normal sequence - **Invalid States:** $(2^{\text{WIDTH}} - 2 \times \text{WIDTH})$ states not in sequence - **Recovery:** Most invalid states naturally converge to valid sequence

17.8.2 Symmetrical Patterns

The sequence has inherent symmetry: - **First Half:** Filling with '1's ($0 \rightarrow 1 \rightarrow 11 \rightarrow 111 \dots$) - **Second Half:** Filling with '0's ($111 \rightarrow 110 \rightarrow 100 \rightarrow 000 \dots$) - **Phase Shift:** Each bit represents a different phase

17.8.3 Single-Bit Transition Properties

Johnson counter outputs share the one-bit-change property that makes Gray code safe to synchronize, but they are **not Gray code**:

- **Adjacent Differences:** consecutive states differ by exactly **one** bit, including the wrap from the last state back to state 0
- **Predictable Transitions:** well-defined state progression
- **Decode Simplicity:** easy to decode specific states

Johnson is not Gray code. Johnson achieves single-bit transitions through sequential shift-register operation; Gray code achieves them through a mathematical encoding ($\text{gray} = \text{bin} \oplus (\text{bin} \gg 1)$). Consequences: - Johnson yields $2N$ states from N bits; Gray yields 2^N . - Johnson is safe only for a counter stepping through its own sequence. It cannot encode an arbitrary value, and there is no closed-form binary conversion – decoding needs `johnson2bin`, which detects position. - Do not substitute one for the other in general-purpose CDC. For an arbitrary binary value crossing domains, use Gray (`bin2gray/gray2bin`).

17.9 Design Examples

17.9.1 1. 4-Phase Clock Generator

```
counter_johnson #(
    .WIDTH(4)
```

```

) phase_gen (
    .clk(master_clk),
    .rst_n(rst_n),
    .enable(1'b1),
    .counter_gray(phases)
);

// Decode individual phases
assign phase_0 = (phases == 4'b0001); // Step 1
assign phase_1 = (phases == 4'b0011); // Step 2
assign phase_2 = (phases == 4'b0111); // Step 3
assign phase_3 = (phases == 4'b1111); // Step 4

```

17.9.22. LED Chaser Effect

```

counter_johnson #(
    .WIDTH(8)
) led_chaser (
    .clk(slow_clk),          // ~1Hz clock
    .rst_n(rst_n),
    .enable(chase_enable),
    .counter_gray(led_pattern)
);

// LEDs show walking pattern
assign leds[7:0] = led_pattern;

```

17.9.33. State Machine Controller

```

counter_johnson #(
    .WIDTH(3)
) state_ctrl (
    .clk(clk),
    .rst_n(rst_n),
    .enable(state_advance),
    .counter_gray(state_code)
);

// Decode states for control logic
always_comb begin
    case (state_code)
        3'b001: current_state = INIT;
        3'b011: current_state = LOAD;
        3'b111: current_state = PROCESS;
        3'b110: current_state = STORE;
        3'b100: current_state = CLEANUP;
        3'b000: current_state = IDLE;
        default: current_state = ERROR;
    endcase
end

```

```
    endcase
end
```

17.9.44. Quadrature Signal Generation

```
// Generate quadrature encoder-like signals
counter_johnson #(
    .WIDTH(4)
) quad_gen (
    .clk(position_clk),
    .rst_n(rst_n),
    .enable(motor_moving),
    .counter_gray(quad_state)
);

// Extract A and B channels with 90° phase shift
assign quad_a = quad_state[0] ^ quad_state[1];
assign quad_b = quad_state[1] ^ quad_state[2];
```

17.10 Advantages

17.10.1 1. Simplicity

- **Minimal Logic:** Just a shift register with inverted feedback
- **No Decode Logic:** States can be used directly
- **Self-Clocking:** No complex timing requirements

17.10.2 2. Reliability

- **Self-Starting:** Recovers from most error states
- **No Invalid States:** All reachable states are functional
- **Glitch-Free:** Clean transitions between states

17.10.3 3. Performance

- **High Speed:** Simple logic allows high clock frequencies
- **Low Power:** Minimal switching activity
- **Predictable Timing:** Regular state progression

17.10.4 4. Flexibility

- **Scalable:** Any width supported

- **Multiple Outputs:** Each bit provides different phase
- **Easy Modification:** Simple parameter changes

17.11 Disadvantages

17.11.1 1. Limited States

- **Count Efficiency:** Only $2 \times \text{WIDTH}$ states vs 2^{WIDTH} for binary
- **Sequence Constraint:** Fixed progression pattern
- **No Random Access:** Must step through sequence

17.11.2 2. Decode Complexity

- **State Detection:** May need comparators for specific states
- **Non-Binary:** Not directly compatible with binary logic
- **Overlap Prevention:** Care needed to avoid state conflicts

17.12 Applications

17.12.1 Clock Generation

```
// Multi-phase clock for pipeline stages
counter_johnson #(.WIDTH(4)) pipeline_phases (
    .clk(master_clk),
    .enable(pipeline_enable),
    .counter_gray(phase_vector)
);
```

```
// Generate phase-shifted clocks
assign clk_ph0 = master_clk & phase_vector[0];
assign clk_ph1 = master_clk & phase_vector[1];
assign clk_ph2 = master_clk & phase_vector[2];
assign clk_ph3 = master_clk & phase_vector[3];
```

17.12.2 Sequential Control

```
// Memory refresh controller
counter_johnson #(.WIDTH(3)) refresh_ctrl (
    .clk(clk),
    .enable(refresh_request),
    .counter_gray(refresh_phase)
);
```

```
// Control refresh operations
assign precharge = (refresh_phase == 3'b001);
assign activate  = (refresh_phase == 3'b011);
assign refresh   = (refresh_phase == 3'b111);
assign restore   = (refresh_phase == 3'b110);
```

17.12.3 Pattern Generation

```
// Test pattern generator
counter_johnson #(.WIDTH(8)) pattern_gen (
    .clk(test_clk),
    .enable(pattern_enable),
    .counter_gray(test_pattern)
);
```

```
// Use for built-in self-test (BIST)
assign test_data = test_pattern;
```

17.13 Verification Strategy

17.13.1 Functional Tests

1. **Sequence Verification:** Check complete $2 \times \text{WIDTH}$ state cycle
2. **Reset Behavior:** Verify initialization to all zeros
3. **Enable Control:** Test hold behavior when disabled
4. **Self-Starting:** Verify recovery from invalid states

17.13.2 Coverage Points

```
covergroup johnson_cg @(posedge clk);
    cp_states: coverpoint counter_gray {
        // Valid Johnson counter states for WIDTH=4
        bins valid_states[] = {4'b0000, 4'b0001, 4'b0011, 4'b0111,
                               4'b1111, 4'b1110, 4'b1100, 4'b1000};
        bins invalid_states[] = default;
    }

    cp_transitions: coverpoint counter_gray {
        bins valid_trans[] = (4'b0000 => 4'b0001),
                               (4'b0001 => 4'b0011),
                               (4'b0011 => 4'b0111),
                               (4'b0111 => 4'b1111),
                               (4'b1111 => 4'b1110),
                               (4'b1110 => 4'b1100),
                               (4'b1100 => 4'b1000),
                               (4'b1000 => 4'b0000);
    }
endgroup
```

```

    }

    cp_enable: coverpoint enable {
        bins enabled = {1};
        bins disabled = {0};
    }
endgroup

```

17.13.3 Assertions

// Verify valid state progression

```

property johnson_sequence;
    @(posedge clk) disable iff (!rst_n)
    enable && (counter_gray == 4'b0000) | =>
    counter_gray == 4'b0001;
endproperty

```

```

assert property (johnson_sequence);

```

// Check for invalid states

```

property no_invalid_states;
    @(posedge clk) disable iff (!rst_n)
    counter_gray inside {4'b0000, 4'b0001, 4'b0011, 4'b0111,
                        4'b1111, 4'b1110, 4'b1100, 4'b1000};
endproperty

```

```

assert property (no_invalid_states);

```

17.14 Synthesis Considerations

17.14.1 Resource Utilization

- **Flip-Flops:** WIDTH registers
- **LUTs:** Minimal - just for shift and invert logic
- **Routing:** Simple connections between adjacent stages

17.14.2 Optimization

// For high-speed applications, consider pipeline stages

```

logic [WIDTH-1:0] johnson_pipe;
always_ff @(posedge clk) begin
    johnson_pipe <= {counter_gray[WIDTH-2:0], ~counter_gray[WIDTH-1]};
end

```

17.14.3 Tool Attributes

```
(* SHREG_EXTRACT = "NO" *) logic [WIDTH-1:0] counter_gray; // Prevent  
SRL inference
```

17.15 Performance Characteristics

17.15.1 Maximum Frequency

- **Typical:** 400-600 MHz in modern FPGAs
- **Limitation:** Shift register timing
- **Optimization:** Pipeline for extreme speeds

17.15.2 Power Consumption

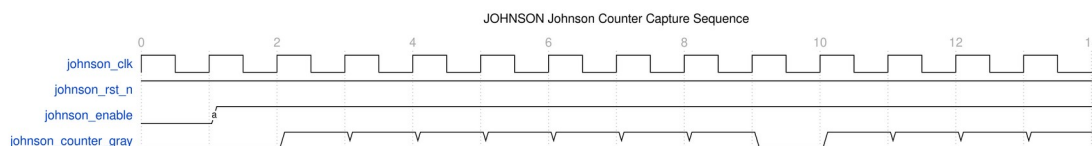
- **Low Dynamic Power:** Only one bit changes per cycle
- **Efficient:** Minimal logic switching
- **Clock Gating:** Use enable for power savings

17.16 WaveDrom Visualization

High-quality waveforms showcasing Johnson counter operation are available!

The following timing diagrams demonstrate the unique properties of Johnson counters across 4 key scenarios:

17.16.1 Scenario 1: Walking Ones and Walking Zeros Pattern

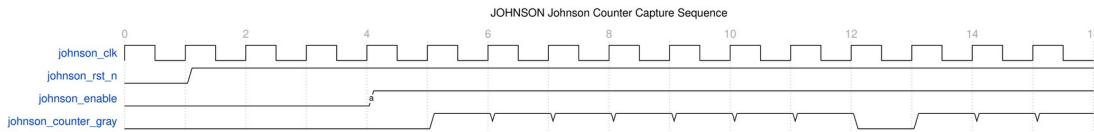


Johnson Walking Pattern

WaveJSON: [johnson_counter_walking_pattern.json](#)

Complete $2 \times \text{WIDTH}$ state cycle (8 states for $\text{WIDTH}=4$): - Walking ones: 0000 → 0001 → 0011 → 0111 → 1111 - Walking zeros: 1111 → 1110 → 1100 → 1000 → 0000 - Demonstrates predictable sequential progression

17.16.2 Scenario 2: Single-Bit Transitions (CDC Safety) ★ KEY FEATURE

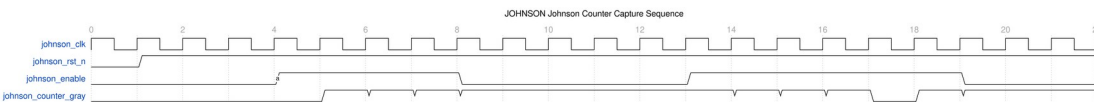


Johnson Single-Bit Transitions

WaveJSON: [johnson_counter_single_bit_transitions.json](#)

Each transition changes only ONE bit: - CDC-safe like Gray codes - **Critical for the fifo_async USE_JOHNSON=1 CDC mechanism** - Prevents metastability in clock domain crossing - Hamming distance = 1 between all adjacent states

17.16.3 Scenario 3: Enable Control

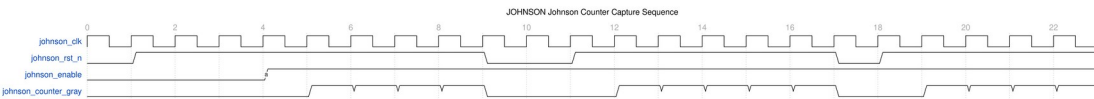


Johnson Enable Control

WaveJSON: [johnson_counter_enable_control.json](#)

Enable control and state holding: - Counter advances when enable=1 - Counter holds state when enable=0 - Clean enable/disable transitions - Demonstrates conditional counting

17.16.4 Scenario 4: Reset Behavior



Johnson Reset Behavior

WaveJSON: [johnson_counter_reset_behavior.json](#)

Reset and initialization: - Asynchronous reset to all zeros (0000) - Immediate reset effect - Clean restart from reset state - Reset during counting operation

To regenerate these waveforms:

```
pytest val/common/test_counter_johnson_wavedrom.py -v
# Then convert JSON to PNG:
cd docs/markdown/assets/WAVES/counter_johnson
for f in *.json; do wavedrom-cli -i "$f" -p "${f%.json}.png"; done
```

What Makes Johnson Counters Special:

The waveforms highlight the unique properties that make Johnson counters useful: - **Single-Bit Transitions:** Only one bit changes per state, making them CDC-safe - **2×WIDTH States:** More efficient than one-hot (N states with N/2 flip-flops) - **Walking Pattern:** Natural “fill then empty” sequence useful for visual effects - **Self-Starting:** Recovers from invalid states automatically

Relationship to fifo_async (USE_JOHNSON=1):

Johnson counters are the foundation of the fifo_async USE_JOHNSON=1 CDC mechanism: - **fifo_async with USE_JOHNSON=1** uses Johnson counters for pointer synchronization - Single-bit transitions enable safe clock domain crossing - Linear width scaling (DEPTH bits) allows flexible even depths

Comparison with Other Counters:

- test_counter_bingray_wavedrom.py - Binary-Gray counter (power-of-2 depths, logarithmic width)
- test_fifo_async_wavedrom.py - Gray code in action (async FIFO)

17.17 Related Modules

- counter_ring: Different shift register pattern
- counter_bingray: Binary and Gray code counter
- counter: Simple binary counter
- Standard shift register implementations

17.18 Test and Verification

Comprehensive Test Suite: - val/common/test_counter_johnson.py - Full functional verification - val/common/test_counter_johnson_wavedrom.py - WaveDrom timing diagrams



Run Tests:

```
# Full functional test (basic/medium/full levels)
pytest val/common/test_counter_johnson.py -v
```

```
# WaveDrom waveform generation
pytest val/common/test_counter_johnson_wavedrom.py -v
```

17.19 Navigation

- [← Back to RTLCommon Index](#)
- [← Back to Main Documentation Index](#)

18 Counter Load Clear Module

18.1 Overview

The counter_load_clear module is a versatile counter with loadable terminal count and clear functionality. It provides precise control over counting behavior with configurable match values, making it ideal for timing applications, protocol implementations, and configurable delay generation.

18.2 Module Declaration

```
module counter_load_clear #(
    parameter int MAX = 32'd32
) (
    input logic          clk,
    input logic          rst_n,
    input logic          clear,
    input logic          increment,
    input logic          load,
    input logic          loadval,
    input logic [$clog2(MAX)-1:0] count,
    output logic [$clog2(MAX)-1:0] done
);
```

18.3 Parameters

18.3.1 MAX

- **Type:** int
- **Default:** 32'd32
- **Description:** Maximum possible count value
- **Range:** Any positive 32-bit integer
- **Usage:** Determines counter width and maximum loadable value
- **Width Calculation:** Counter width = $\$clog2(MAX)$

18.4 Ports

18.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	System clock
rst_n	1	logic	Active-low asynchronous reset
clear	1	logic	Synchronous clear (highest priority)
increment	1	logic	Enable counter increment
load	1	logic	Load new match value
loadval	$\$clog2(\text{MAX})$	logic	New terminal count value

18.4.2 Outputs

Port	Width	Type	Description
count	$\$clog2(\text{MAX})$	logic	Current counter value
done	1	logic	Terminal count reached flag

18.5 Architecture Details

18.5.1 Internal Registers

```
logic [$clog2(MAX)-1:0] count;           // Current count value
logic [$clog2(MAX)-1:0] r_match_val;    // Terminal count register
```

18.5.2 Priority Logic

The module implements a priority-based control scheme:

1. **Asynchronous Reset** (Highest Priority)

- Resets both count and r_match_val to 0
 - Independent of clock edge
2. **Load Operation** (High Priority)
 - Updates r_match_val with loadval
 - Can occur simultaneously with other operations
 3. **Clear Operation** (Medium Priority)
 - Immediately sets count to 0
 - Overrides increment operation
 4. **Increment Operation** (Lowest Priority)
 - Advances counter or wraps to 0 at terminal count
 - Only when increment is asserted

18.6 Implementation Details

18.6.1 Main Control Logic

```

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        count <= 'b0;
        r_match_val <= 'b0;
    end else begin
        // Load has independent priority
        if (load) r_match_val <= loadval;

        // Clear has highest operational priority
        if (clear) begin
            count <= 'b0;
        end else if (increment) begin
            count <= (count == r_match_val) ? 'b0 : count + 'b1;
        end
    end
end

```

18.6.2 Done Signal Generation

```

assign done = (count == r_match_val);

```

18.7 Operation Modes

18.7.1 1. Basic Counting Mode

```
// Simple timer: count to 100
counter_load_clear #(.MAX(1000)) timer (
    .clk(clk),
    .rst_n(rst_n),
    .clear(1'b0),
    .increment(1'b1),
    .load(1'b1),
    .loadval(8'd99),      // Count 0-99 (100 cycles)
    .count(timer_count),
    .done(timer_done)
);
```

18.7.22. Configurable Delay Mode

```
// Variable delay generator
logic [7:0] delay_setting;
logic start_delay, delay_complete;

counter_load_clear #(.MAX(256)) delay_gen (
    .clk(clk),
    .rst_n(rst_n),
    .clear(start_delay),      // Start new delay
    .increment(1'b1),
    .load(start_delay),      // Load new value when starting
    .loadval(delay_setting),
    .count(),
    .done(delay_complete)
);
```

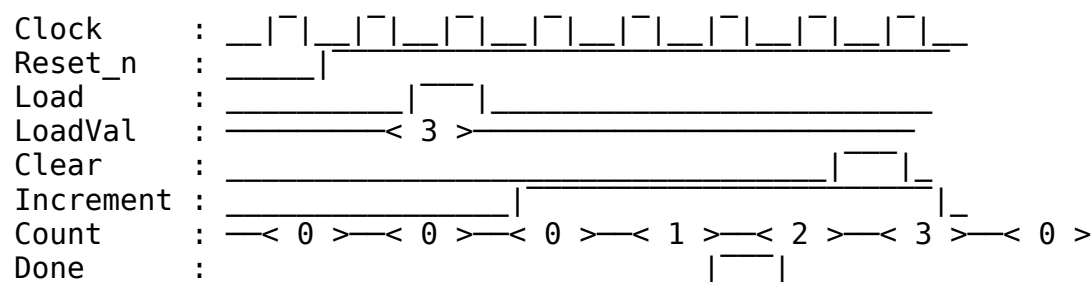
18.7.33. Retriggerable Timer Mode

```
// Watchdog timer that resets on activity
logic activity_detected, timeout;

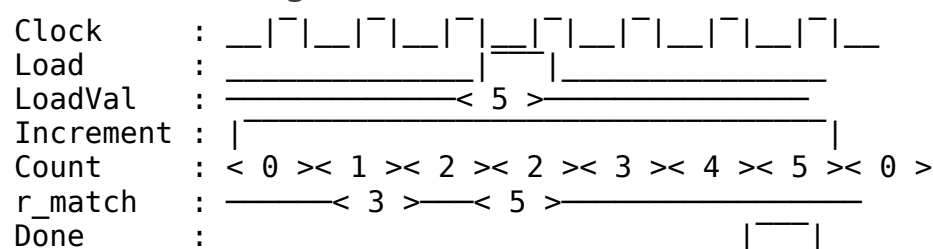
counter_load_clear #(.MAX(10000)) watchdog (
    .clk(clk),
    .rst_n(rst_n),
    .clear(activity_detected), // Reset on activity
    .increment(1'b1),
    .load(1'b1),
    .loadval(16'd9999),      // 10000 cycle timeout
    .count(),
    .done(timeout)
);
```

18.8 Timing Diagrams

18.8.1 Basic Operation



18.8.2 Load During Count



18.9 Advanced Usage Examples

18.9.1 1. Dual-Rate Timer

// Timer with fast/slow modes

logic fast_mode;

logic [15:0] timer_period;

always_comb begin

timer_period = fast_mode ? 16'd99 : 16'd999; *// 100 vs 1000*
cycles

end

```
counter_load_clear #(.MAX(65536)) dual_timer (  
    .clk(clk),  
    .rst_n(rst_n),  
    .clear(timer_done),           // Auto-restart  
    .increment(1'b1),  
    .load(timer_done),           // Reload on completion  
    .loadval(timer_period),  
    .count(timer_value),  
    .done(timer_done)  
);
```

18.9.22. PWM Generator

```
// Simple PWM with configurable duty cycle
logic [7:0] duty_cycle; // 0-255
logic pwm_out;

counter_load_clear #(.MAX(256)) pwm_counter (
    .clk(clk),
    .rst_n(rst_n),
    .clear(1'b0),
    .increment(1'b1),
    .load(1'b1),
    .loadval(8'd255),           // Always count 0-255
    .count(pwm_count),
    .done()
);

assign pwm_out = (pwm_count < duty_cycle);
```

18.9.33. Burst Generator

```
// Generate bursts of N pulses
logic [7:0] burst_length;
logic start_burst, burst_active, burst_complete;

counter_load_clear #(.MAX(256)) burst_counter (
    .clk(clk),
    .rst_n(rst_n),
    .clear(start_burst),
    .increment(burst_active),
    .load(start_burst),
    .loadval(burst_length - 1),
    .count(burst_count),
    .done(burst_complete)
);

// State machine for burst control
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        burst_active <= 1'b0;
    end else begin
        if (start_burst)
            burst_active <= 1'b1;
        else if (burst_complete)
            burst_active <= 1'b0;
    end
end
```

18.9.44. Communication Timeout

```
// UART receive timeout
logic rx_active, rx_timeout, char_received;
parameter RX_TIMEOUT_CYCLES = 1000;

counter_load_clear #(.MAX(2048)) rx_timer (
    .clk(clk),
    .rst_n(rst_n),
    .clear(char_received),      // Reset on new character
    .increment(rx_active),     // Only count when receiving
    .load(rx_active && !char_received),
    .loadval(RX_TIMEOUT_CYCLES-1),
    .count(),
    .done(rx_timeout)
);
```

18.10 Design Patterns

18.10.1 Pattern 1: Auto-Reloading Timer

```
wire auto_reload = timer_done;
counter_load_clear timer (
    .clear(auto_reload),
    .load(auto_reload),
    .loadval(PERIOD-1),
    .done(timer_done),
    ...
);
```

18.10.2 Pattern 2: Conditional Counting

```
wire count_enable = condition_met && !paused;
counter_load_clear conditional_timer (
    .increment(count_enable),
    ...
);
```

18.10.3 Pattern 3: Dynamic Period Adjustment

```
logic period_changed;
wire reload_needed = timer_done || period_changed;

counter_load_clear adaptive_timer (
    .clear(reload_needed),
    .load(reload_needed),
    .loadval(current_period),
    ...
);
```

18.11 Verification Strategy

18.11.1 Test Scenarios

1. **Basic Counting:** Verify normal increment operation
2. **Terminal Count:** Test done signal generation
3. **Load Operation:** Verify match value updates
4. **Clear Priority:** Test clear overrides increment
5. **Simultaneous Operations:** Load + clear + increment
6. **Boundary Conditions:** Count = 0, count = match_val
7. **Reset Behavior:** Async reset during various operations

18.11.2 Coverage Model

```
covergroup load_clear_cg @(posedge clk);
  cp_count: coverpoint count {
    bins zero = {0};
    bins low = {[1:MAX/4]};
    bins mid = {[MAX/4+1:3*MAX/4]};
    bins high = {[3*MAX/4+1:MAX-1]};
  }

  cp_match_val: coverpoint r_match_val {
    bins zero = {0};
    bins low = {[1:MAX/4]};
    bins mid = {[MAX/4+1:3*MAX/4]};
    bins high = {[3*MAX/4+1:MAX-1]};
  }

  cp_operations: coverpoint {clear, increment, load} {
    bins clear_only = {3'b100};
    bins inc_only = {3'b010};
    bins load_only = {3'b001};
    bins clear_inc = {3'b110};
    bins clear_load = {3'b101};
    bins inc_load = {3'b011};
    bins all_ops = {3'b111};
    bins no_ops = {3'b000};
  }

  cp_done: coverpoint done;
```

```

    // Cross coverage
    cross_done_ops: cross cp_operations, done;
endgroup

```

18.11.3 Assertions

```

// Count should never exceed match value

```

```

property count_bounds;
    @(posedge clk) disable iff (!rst_n)
    count <= r_match_val;
endproperty

```

```

// Done should be true when count equals match

```

```

property done_when_match;
    @(posedge clk) disable iff (!rst_n)
    (count == r_match_val) |-> done;
endproperty

```

```

// Clear should reset count to 0

```

```

property clear_resets_count;
    @(posedge clk) disable iff (!rst_n)
    clear |=> (count == 0);
endproperty

```

```

// Load should update match value

```

```

property load_updates_match;
    @(posedge clk) disable iff (!rst_n)
    load |=> (r_match_val == $past(loadval));
endproperty

```

```

assert property (count_bounds);
assert property (done_when_match);
assert property (clear_resets_count);
assert property (load_updates_match);

```

18.12 Synthesis Considerations

18.12.1 Resource Usage

- **Flip-Flops:** $\$clog2(\text{MAX})$ for count + $\$clog2(\text{MAX})$ for match = $2 \times \$clog2(\text{MAX})$
- **LUTs:** Increment logic, comparator, and control logic
- **Typical:** 15-25 LUTs for 16-bit counter

18.12.2 Timing Optimization

```
// For high-speed operation, pipeline the comparison  
logic done_pipe;  
always_ff @(posedge clk) begin  
    done_pipe <= (count == r_match_val);  
end
```

18.12.3 Parameter Validation

```
initial begin  
    assert (MAX > 0) else $error("MAX must be positive");  
    assert (MAX <= 2**30) else $warning("Large MAX may impact  
synthesis");  
end
```

18.13 Performance Characteristics

18.13.1 Maximum Frequency

- **Typical:** 300-500 MHz in modern FPGAs
- **Critical Path:** Increment + comparison logic
- **Optimization:** Consider pipelining for extreme speeds

18.13.2 Latency

- **Load Operation:** 1 cycle to update match value
- **Clear Operation:** 1 cycle to reset count
- **Done Signal:** Combinational (0 cycles)

18.14 Common Applications

1. **Timer Modules:** Configurable timing generation
2. **Protocol Engines:** Timeout and retry mechanisms
3. **PWM Controllers:** Variable duty cycle generation
4. **Burst Generators:** Programmable pulse trains
5. **Watchdog Timers:** System monitoring and reset
6. **Clock Dividers:** Variable frequency division
7. **Test Pattern Generators:** Configurable test sequences

18.15 Related Modules

- counter: Basic counter without load/clear
- counter_freq_invariant: Uses this module for prescaling
- Standard timer and PWM controller implementations

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19 Basic Counter Module

19.1 Overview

The counter module is a simple, configurable up-counter that generates a tick pulse when reaching its maximum value. It's designed for basic timing applications where a periodic signal is needed.

19.2 Module Declaration

```
module counter #(
    parameter int MAX = 32767
) (
    input logic clk,
    input logic rst_n,
    output logic tick
);
```

19.3 Parameters

19.3.1 MAX

- **Type:** int
- **Default:** 32767
- **Description:** Maximum count value before wrapping to zero
- **Range:** Any positive integer
- **Width Calculation:** Counter width is automatically calculated as $\lceil \log_2(\text{MAX}+1) \rceil$

19.4 Ports

19.4.1 Inputs

Port	Width	Type	Description
clk	1	logic	System clock input
rst_n	1	logic	Active-low asynchronous reset

19.4.2 Outputs

Port	Width	Type	Description
tick	1	logic	Pulse output when counter reaches MAX

19.5 Internal Architecture

19.5.1 Counter Register

logic [**\$clog2**(MAX+1)-1:0] r_count;

- **Width:** Automatically sized to accommodate MAX value
- **Reset Value:** '0 (zero)
- **Operation:** Increments every clock cycle until reaching MAX

19.5.2 State Machine Behavior

The counter operates as a simple state machine:

1. **Reset State:** r_count = 0, tick = 0
2. **Counting State:** r_count increments each cycle
3. **Maximum State:** r_count = MAX, tick = 1 for one cycle
4. **Wrap State:** r_count returns to 0, cycle repeats

19.6 Implementation Details

19.6.1 Main Counter Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_count <= '0;
    end else begin
        if (r_count == MAX[$clog2(MAX+1)-1:0]) begin
            r_count <= '0;
        end else begin
            r_count <= r_count + 1'b1;
        end
    end
end
```

19.6.2 Tick Generation

```
assign tick = (r_count == MAX[$clog2(MAX+1)-1:0]);
```

19.6.3 Key Features

- **Automatic Width Sizing:** Counter width adapts to MAX parameter
- **Wrap-around Operation:** Automatically returns to zero after MAX
- **Single Cycle Tick:** Tick pulse lasts exactly one clock cycle
- **Asynchronous Reset:** Immediate reset capability

19.7 Timing Characteristics

19.7.1 Period Calculation

- **Total Period:** MAX + 1 clock cycles
- **Tick Frequency:** $f_{\text{clk}} / (\text{MAX} + 1)$
- **Duty Cycle:** $1 / (\text{MAX} + 1)$

19.7.2 Examples

MAX Value	Counter Width	Period (cycles)	Tick Frequency
7	3 bits	8	$f_{\text{clk}}/8$
15	4 bits	16	$f_{\text{clk}}/16$
99	7 bits	100	$f_{\text{clk}}/100$
999	10 bits	1000	$f_{\text{clk}}/1000$

19.8 Usage Examples

19.8.11. Basic Timer (1ms tick at 100MHz)

```
counter #(
    .MAX(99999) // 100,000 cycles = 1ms at 100MHz
) timer_1ms (
    .clk(clk_100mhz),
    .rst_n(rst_n),
    .tick(tick_1ms)
);
```

19.8.22. Heartbeat Generator (1Hz at 50MHz)

```
counter #(
    .MAX(49999999) // 50M cycles = 1 second at 50MHz
) heartbeat (
    .clk(clk_50mhz),
    .rst_n(rst_n),
    .tick(heartbeat_1hz)
);
```

19.8.33. Simple Divider (Divide by 8)

```
counter #(
    .MAX(7) // 8 cycles total
) div8 (
    .clk(clk_in),
    .rst_n(rst_n),
    .tick(clk_div8)
);
```

19.9 Design Considerations

19.9.1 Resource Usage

- **LUTs:** Approximately $\lceil \log_2(\text{MAX}+1) \rceil$ LUTs for the counter
- **FFs:** $\lceil \log_2(\text{MAX}+1) \rceil$ flip-flops for the counter register
- **Logic Levels:** Single level for tick generation

19.9.2 Performance

- **Maximum Frequency:** Limited by counter increment logic
- **Reset Recovery:** One clock cycle after reset deassertion
- **Propagation Delay:** Tick output has minimal combinational delay

19.9.3 Power Optimization

- **Clock Gating:** Consider gating when tick output not needed
- **Reset Strategy:** Asynchronous reset reduces reset tree loading
- **Parameter Selection:** Smaller MAX values reduce power consumption

19.10 Synthesis Considerations

19.10.1 Parameter Constraints

```
initial begin
    assert (MAX > 0) else $error("MAX must be positive");
    assert (MAX < 2**31) else $warning("Large MAX may impact
synthesis");
end
```

19.10.2 Optimization Hints

- **Binary MAX Values:** Powers of 2 minus 1 optimize better (7, 15, 31, etc.)
- **Tool Directives:** Consider synthesis attributes for timing-critical paths
- **Reset Inference:** Ensure reset is properly inferred as asynchronous

19.11 Common Applications

1. **Periodic Timers:** Generating regular time intervals
2. **Clock Division:** Creating slower clock enables
3. **Timeout Generation:** Watchdog timers and timeouts
4. **Sampling Control:** Periodic data sampling triggers
5. **LED Blinking:** Visual indicators and status displays
6. **Protocol Timing:** Communication protocol timeouts

19.12 Verification Considerations

19.12.1 Test Scenarios

1. **Basic Counting:** Verify counting from 0 to MAX
2. **Wrap Behavior:** Confirm return to zero after MAX

3. **Tick Generation:** Verify single-cycle tick pulse
4. **Reset Functionality:** Test asynchronous reset operation
5. **Parameter Sweep:** Test various MAX values

19.12.2 Coverage Points

- Counter reaches MAX value
- Counter wraps from MAX to 0
- Tick pulse generation timing
- Reset during counting operation

19.13 Limitations

1. **No Enable Control:** Always counting when not in reset
2. **Up-Counter Only:** Cannot count down or bidirectionally
3. **Fixed Increment:** Always increments by 1
4. **No Load Capability:** Cannot load arbitrary values
5. **Single Output:** Only provides tick signal, not count value

19.14 Related Modules

- `counter_load_clear`: Adds load and clear functionality
- `counter_bin`: Specialized for FIFO applications
- `counter_freq_invariant`: Frequency-independent timing
- `clock_pulse`: Similar functionality with different interface

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20 Ring Counter Module

20.1 Overview

The counter_ring module implements a classic ring counter where a single '1' bit circulates through a shift register. This creates a one-hot encoded sequence that's ideal for generating sequential enable signals, creating walking LED patterns, or controlling multi-phase operations where only one stage should be active at a time.

20.2 Module Declaration

```
module counter_ring #(
    parameter int WIDTH = 4
) (
    input wire      clk,
    input wire      rst_n,
    input wire      enable,
    output reg [WIDTH-1:0] ring_out
);
```

20.3 Parameters

20.3.1 WIDTH

- **Type:** int
- **Default:** 4
- **Description:** Number of stages in the ring counter
- **Range:** Any positive integer ≥ 1
- **States Generated:** WIDTH unique states
- **Pattern:** Single '1' bit rotates through WIDTH positions

20.4 Ports

20.4.1 Inputs

Port	Width	Type	Description
clk	1	wire	System clock input
rst_n	1	wire	Active-low asynchronous

Port	Width	Type	Description
enable	1	wire	reset Counter enable control

20.4.2 Outputs

Port	Width	Type	Description
ring_out	WIDTH	reg	Ring counter output (one-hot encoded)

20.5 Theory of Operation

20.5.1 Ring Counter Principle

A ring counter is a circular shift register where: - **Initialization:** Single '1' bit in LSB position - **Operation:** The '1' bit rotates right each clock cycle - **Feedback:** MSB connects back to LSB (no inversion) - **States:** WIDTH unique one-hot states

20.5.2 Mathematical Representation

For each clock cycle when enabled:

```
ring_out[i] = ring_out[i-1] for i = 1 to WIDTH-1
ring_out[0] = ring_out[WIDTH-1]
```

20.6 Implementation Details

20.6.1 Core Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        ring_out <= 'b1; // Initialize with LSB set
    end else if (enable) begin
        // Right rotate operation
        ring_out <= {ring_out[0], ring_out[WIDTH-1:1]};
    end
end
```

20.6.2 Operation Breakdown

1. **Reset State:** ring_out = 'b1 (only LSB set)
2. **Rotation:** {ring_out[0], ring_out[WIDTH-1:1]}

- MSB becomes new LSB
 - All other bits shift right
3. **Enable Control:** Only advances when enable is high
 4. **Circular:** After WIDTH cycles, returns to initial state

20.7 State Sequences

20.7.1 4-bit Ring Counter (WIDTH=4)

Step	ring_out	Binary	One-Hot Position
0	0001	Reset state	Position 0 active
1	1000	After 1 clock	Position 3 active
2	0100	After 2 clocks	Position 2 active
3	0010	After 3 clocks	Position 1 active
4	0001	After 4 clocks	Back to start

20.7.2 3-bit Ring Counter (WIDTH=3)

Step	ring_out	Active Stage
0	001	Stage 0
1	100	Stage 2
2	010	Stage 1
3	001	Stage 0 (repeat)

20.7.3 8-bit Ring Counter Example

Step 0: 00000001 (Stage 0)
 Step 1: 10000000 (Stage 7)
 Step 2: 01000000 (Stage 6)
 Step 3: 00100000 (Stage 5)
 Step 4: 00010000 (Stage 4)
 Step 5: 00001000 (Stage 3)
 Step 6: 00000100 (Stage 2)
 Step 7: 00000010 (Stage 1)
 Step 8: 00000001 (Stage 0) - Cycle repeats

20.8 Design Examples

20.8.1 1. Sequential Enable Generator

```
counter_ring #(
    .WIDTH(4)
) seq_enable (
    .clk(clk),
    .rst_n(rst_n),
    .enable(sequence_active),
    .ring_out(stage_enables)
);

// Use each bit as an enable for different stages
assign stage0_enable = stage_enables[0];
assign stage1_enable = stage_enables[1];
assign stage2_enable = stage_enables[2];
assign stage3_enable = stage_enables[3];

// Example: Pipeline stage control
always_ff @(posedge clk) begin
    if (stage0_enable) stage0_data <= input_data;
    if (stage1_enable) stage1_data <= process_stage0(stage0_data);
    if (stage2_enable) stage2_data <= process_stage1(stage1_data);
    if (stage3_enable) output_data <= process_stage2(stage2_data);
end
```

20.8.22. LED Chaser Display

```
// Create walking LED pattern
counter_ring #(
    .WIDTH(8)
) led_chaser (
    .clk(slow_clk),           // ~2Hz for visible effect
    .rst_n(rst_n),
    .enable(chase_enabled),
    .ring_out(led_pattern)
);

// Connect directly to LEDs
assign leds[7:0] = led_pattern;

// Optional: Bidirectional chaser
logic direction;
logic [7:0] led_display;

always_comb begin
    if (direction)
```

```

        led_display = led_pattern;
    else
        led_display = {led_pattern[0], led_pattern[7:1]}; // Reverse
end

```

20.8.33. Memory Bank Sequencer

```

counter_ring #(
    .WIDTH(4)
) bank_sequencer (
    .clk(clk),
    .rst_n(rst_n),
    .enable(memory_active),
    .ring_out(bank_select)
);

// Memory bank control
logic [3:0] bank_cs_n;
assign bank_cs_n = ~bank_select; // Active low chip selects

// Example: Round-robin memory access
always_comb begin
    mem_addr = base_addr;
    mem_data_out = 32'h0;

    case (bank_select)
        4'b0001: begin
            mem_addr = bank0_addr;
            mem_data_out = bank0_data;
        end
        4'b0010: begin
            mem_addr = bank1_addr;
            mem_data_out = bank1_data;
        end
        4'b0100: begin
            mem_addr = bank2_addr;
            mem_data_out = bank2_data;
        end
        4'b1000: begin
            mem_addr = bank3_addr;
            mem_data_out = bank3_data;
        end
    endcase
end

```

20.8.44. State Machine Controller

```

// Simple state machine using ring counter
counter_ring #(

```

```

        .WIDTH(5)
    ) state_machine (
        .clk(clk),
        .rst_n(rst_n),
        .enable(state_advance),
        .ring_out(current_state)
    );

// Decode states
assign state_idle    = current_state[0];
assign state_request = current_state[1];
assign state_wait    = current_state[2];
assign state_active  = current_state[3];
assign state_done    = current_state[4];

// State machine logic
always_ff @(posedge clk) begin
    case (1'b1) // One-hot case statement
        state_idle:    state_advance <= start_request;
        state_request: state_advance <= request_granted;
        state_wait:    state_advance <= ready_signal;
        state_active:  state_advance <= operation_complete;
        state_done:    state_advance <= 1'b1; // Always advance from
done
        default:      state_advance <= 1'b0;
    endcase
end

```

20.8.55. Clock Domain Multiplexer

```

// Sequentially select different clock domains
counter_ring #(
    .WIDTH(3)
) clk_mux_ctrl (
    .clk(control_clk),
    .rst_n(rst_n),
    .enable(mux_advance),
    .ring_out(clk_select)
);

// Clock multiplexer (use proper clock muxing in real design)
logic selected_clk;
always_comb begin
    case (clk_select)
        3'b001: selected_clk = clk_domain_0;
        3'b010: selected_clk = clk_domain_1;
        3'b100: selected_clk = clk_domain_2;
        default: selected_clk = 1'b0;
    endcase
end

```

```
    endcase
end
```

20.9 Properties and Characteristics

20.9.1 One-Hot Encoding

Ring counters naturally provide one-hot encoding: - **Exactly One Bit Set**: Always exactly one bit is '1' - **Unique States**: Each state is distinctly different - **Easy Decode**: No additional decoding logic needed - **Glitch-Free**: Clean transitions between states

20.9.2 Self-Correcting Behavior

Ring counters have some self-correcting properties: - **All Zeros**: Will remain stuck (not self-correcting) - **Multiple Ones**: Will maintain multiple bits indefinitely - **Single One**: Will operate correctly

20.9.3 Comparison with Other Counters

Counter Type	States	Decode Logic	One-Hot	Power
Binary	2^N	Complex	No	Low
Gray	2^N	Moderate	No	Low
Johnson	$2 \times N$	Moderate	No	Low
Ring	N	None	Yes	Medium

20.10 Advanced Usage

20.10.1 1. Self-Correcting Ring Counter

```
// Add error correction for stuck states
logic error_detected;
logic [WIDTH-1:0] corrected_ring;

assign error_detected = (ring_out == 'b0) || ($countones(ring_out) != 1);

always_comb begin
    if (error_detected)
        corrected_ring = 'b1; // Force to valid state
    else
        corrected_ring = ring_out;
end
```

20.10.2 2. Bidirectional Ring Counter

```
logic direction; // 0 = right, 1 = left

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        ring_out <= 'b1;
    end else if (enable) begin
        if (direction)
            // Left rotate
            ring_out <= {ring_out[WIDTH-2:0], ring_out[WIDTH-1]};
        else
            // Right rotate
            ring_out <= {ring_out[0], ring_out[WIDTH-1:1]};
        end
    end
end
```

20.10.3 3. Programmable Ring Counter

```
logic [WIDTH-1:0] init_pattern;
logic load_pattern;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        ring_out <= 'b1;
    end else if (load_pattern) begin
        ring_out <= init_pattern;
    end else if (enable) begin
        ring_out <= {ring_out[0], ring_out[WIDTH-1:1]};
    end
end
```

20.11 Verification Strategy

20.11.1 Test Scenarios

1. **Reset Behavior:** Verify initialization to single '1' bit
2. **Rotation Sequence:** Check complete WIDTH-step cycle
3. **Enable Control:** Test hold behavior when disabled
4. **One-Hot Property:** Ensure exactly one bit set always
5. **Cycle Completion:** Verify return to initial state

20.11.2 Coverage Model

```
covergroup ring_cg @(posedge clk);
  cp_ring_state: coverpoint ring_out {
    bins valid_states[] = {1, 2, 4, 8, 16, 32, 64, 128}; // For
WIDTH=8
    bins invalid_states[] = default;
  }

  cp_enable: coverpoint enable {
    bins enabled = {1};
    bins disabled = {0};
  }

  cp_one_hot: coverpoint $countones(ring_out) {
    bins exactly_one = {1};
    bins zero = {0};
    bins multiple = {[2:WIDTH]};
  }

  // Verify complete cycle
  cp_transitions: coverpoint ring_out {
    bins cycle[] = (1 ==> 8), (8 ==> 4), (4 ==> 2), (2 ==> 1); // For
WIDTH=4
  }
endgroup
```

20.11.3 Assertions

```
// Exactly one bit should be set (except during reset)
property one_hot_property;
  @(posedge clk) disable iff (!rst_n)
  $countones(ring_out) == 1;
endproperty

// After reset, should start with LSB set
property reset_state;
  @(posedge clk)
  $rose(rst_n) |-> ring_out == 1;
endproperty

// When enabled, should rotate right
property rotation_property;
  @(posedge clk) disable iff (!rst_n)
  enable |=> ring_out == {$past(ring_out[0]), $past(ring_out[WIDTH-
1:1])});
endproperty

assert property (one_hot_property);
```

```
assert property (reset_state);  
assert property (rotation_property);
```

20.12 Synthesis Considerations

20.12.1 Resource Utilization

- **Flip-Flops:** WIDTH registers
- **LUTs:** Minimal - just shift logic
- **Routing:** Simple interconnections

20.12.2 Optimization Tips

```
// Use SRL (Shift Register LUT) inference when appropriate  
(* SRL_STYLE = "register" *) reg [WIDTH-1:0] ring_out;
```

```
// For very wide counters, consider using dedicated shift register primitives
```

20.12.3 Timing Considerations

- **Clock Skew:** Ensure clean clock distribution
- **Setup/Hold:** Standard flip-flop requirements
- **Routing Delay:** Minimal due to simple structure

20.13 Performance Characteristics

20.13.1 Maximum Frequency

- **Typical:** 400-600 MHz in modern FPGAs
- **Limitation:** Shift register timing
- **Advantage:** Very simple logic path

20.13.2 Power Consumption

- **Dynamic:** Proportional to switching frequency
- **Optimization:** Higher switching than binary counters
- **Trade-off:** Simplicity vs. power efficiency

20.14 Common Pitfalls

20.14.1 1. Invalid Initial States

```
// Problem: Multiple bits set  
ring_out <= 4'b1001; // Invalid - two bits set  
  
// Solution: Use proper initialization  
ring_out <= 4'b0001; // Valid - exactly one bit
```

20.14.2 2. Stuck at Zero

```
// Problem: All zeros state  
if (ring_out == 'b0) begin  
    ring_out <= 'b1; // Force recovery  
end
```

20.14.3 3. Width Mismatch

```
// Problem: Initialization doesn't match width  
parameter WIDTH = 8;  
ring_out <= 4'b0001; // Wrong - should be 8'b00000001  
  
// Solution: Use proper width  
ring_out <= 'b1; // Correct - sized automatically
```

20.15 Related Modules

- counter_johnson: Twisted-ring counter with $2 \times \text{WIDTH}$ states
- counter_bingray: Binary/Gray counter for different applications
- Standard shift register implementations
- One-hot state machines

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21 Count Leading Zeros Module

21.1 Overview

The `count_leading_zeros` module implements a leading zero counter that determines how many consecutive zero bits appear at the beginning (MSB side) of a data word. This is a fundamental operation in computer arithmetic, used extensively in floating-point normalization, priority encoding, and bit manipulation algorithms.

The scan starts at `data[WIDTH-1]` and proceeds downward, stopping at the first set bit:

```
data = 32'h8000_0000 -> clz = 0    (MSB is already set)
data = 32'h0000_0001 -> clz = 31  (31 zeros above the LSB)
data = 32'h0000_0000 -> clz = WIDTH
```

For the complementary count from the LSB upward, use [count_trailing_zeros](#).

History - do not reintroduce input reversal. Earlier revisions of this module scanned from bit 0 upward, which computes *trailing* zeros, not leading zeros, despite the module name. Callers compensated by bit-reversing their input before connecting it. Both the module and every caller have been corrected: the module now performs a true MSB-down scan, and callers pass their operand straight in. Any reversal generate block or `w_*_reversed` wire feeding this module is stale and must not be re-added. If you actually want the LSB-up count, instantiate `count_trailing_zeros` instead of reversing the input.

21.2 Module Declaration

```
module count_leading_zeros #(
    parameter int WIDTH = 32
) (
    input logic [WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] clz
);
```

21.3 Parameters

21.3.1 WIDTH

- **Type:** int
- **Default:** 32
- **Description:** Bit width of the input data

- **Range:** Any positive integer ≥ 1
- **Common Values:** 8, 16, 32, 64 for standard data widths
- **Impact:** Determines output width and algorithm complexity

WIDTH is the only parameter.

21.4 Ports

21.4.1 Inputs

Port	Width	Type	Description
data	WIDTH	logic	Input data word to analyze

21.4.2 Outputs

Port	Width	Type	Description
clz	$\$clog2(WIDTH) + 1$	logic	Count of leading zeros

21.4.3 Output Width Explanation

The output width $\$clog2(WIDTH) + 1$ ensures it can represent all possible counts: - **Range:** 0 to WIDTH (inclusive) - **Example:** For WIDTH=32, output is 6 bits (0-32 requires 6 bits) - **Special Case:** All zeros input produces CLZ = WIDTH

21.5 Algorithm Implementation

21.5.1 Function-Based Approach

```
function automatic [$clog2(WIDTH):0] clz_func;
    input [WIDTH-1:0] input_data;
    logic found;
    begin
        clz_func = 0;
        found = 1'b0;
        // Scan from the MSB downward.
        for (int i = WIDTH - 1; i >= 0; i--) begin
            if (!input_data[i] && !found) begin
                clz_func += 1;
            end else begin
                found = 1'b1; // Stop counting when first '1' found
            end
        end
    end
end
```

```

        end
    end
endfunction

```

21.5.2 Bit Scanning Process

The algorithm scans from MSB to LSB: 1. **Initialize:** `clz_func = 0`, `found = 0` 2. **Scan Loop:** For each bit position `i` from `WIDTH-1` down to `0`: - If `data[i] == 0` AND no '1' found yet: increment count - If `data[i] == 1`: set found flag, stop counting 3. **Result:** Final count is the number of zeros above the highest set bit

21.5.3 Why MSB-to-LSB Scanning?

“Leading” refers to the bits that lead the word when it is written out, which is the MSB side. The loop therefore starts at `data[WIDTH-1]` and walks down: - **Bit Order:** `data[0]` is LSB, `data[WIDTH-1]` is MSB - **Termination:** The found flag latches at the highest set bit, so all zeros below it are ignored - only the contiguous run above it is counted - **Count Logic:** `clz = (WIDTH-1) - position_of_highest_set_one`, and `clz = WIDTH` when no bit is set

Because the scan direction already matches the definition, the input must be fed in directly. Reversing data before connecting it turns this module back into a trailing zero counter - see the history note above.

21.6 Examples and Truth Tables

21.6.1 8-bit Examples (WIDTH=8)

The count is set by the **highest** set bit; bits below it never affect the result.

Input (data)	Binary	Highest set bit	Leading Zeros	CLZ Output
8'b00000000	00000000	None	8	8
0				
8'b00000001	00000001	Bit 0	7	7
1				
8'b00000010	00000010	Bit 1	6	6
0				
8'b00000100	00000100	Bit 2	5	5
0				
8'b00001000	00001000	Bit 3	4	4
0				
8'b00010000	00010000	Bit 4	3	3
0				

Input (data)	Binary	Highest set bit	Leading Zeros	CLZ Output
8'b0010000 0	00100000	Bit 5	2	2
8'b0100000 0	01000000	Bit 6	1	1
8'b1000000 0	10000000	Bit 7	0	0
8'b1111111 1	11111111	Bit 7	0	0
8'b1010101 0	10101010	Bit 7	0	0
8'b0001010 1	00010101	Bit 4	3	3
8'b0011000 0	00110000	Bit 5	2	2

21.6.2 32-bit Examples

Input	Hex	Leading Zeros	CLZ
32'h00000000	0x00000000	32	32
32'h00000001	0x00000001	31	31
32'h00000080	0x00000080	24	24
32'h00008000	0x00008000	16	16
32'h00800000	0x00800000	8	8
32'h80000000	0x80000000	0	0
32'hFFFFFFFF	0xFFFFFFFF	0	0

21.7 Applications

21.7.1 1. Floating-Point Normalization

```
// IEEE 754 floating-point normalization
logic [31:0] mantissa_raw;
logic [5:0] shift_amount;
logic [31:0] normalized_mantissa;
```

```
count_leading_zeros #(.WIDTH(32)) norm_clz (
```

```

        .data(mantissa_raw),
        .clz(shift_amount)
    );

    // Normalize mantissa by shifting left
    assign normalized_mantissa = mantissa_raw << shift_amount;

    // Adjust exponent accordingly
    logic [7:0] adjusted_exponent;
    assign adjusted_exponent = raw_exponent - shift_amount;

```

21.7.22. Priority Encoder

```

    // Find highest priority request
    logic [15:0] request_vector;
    logic [4:0] leading_zeros;
    logic [3:0] highest_priority;
    logic any_request;

    count_leading_zeros #(.WIDTH(16)) priority_enc (
        .data(request_vector),
        .clz(leading_zeros)
    );

    assign any_request = (leading_zeros != 16);
    assign highest_priority = any_request ? (15 - leading_zeros) : 4'b0;

    // Example: request_vector = 16'b0000_0100_1000_0000
    // Highest set bit is bit 10
    // CLZ = 5 (five leading zeros above bit 10)
    // highest_priority = 15 - 5 = 10

```

21.7.33. Bit Width Calculation

```

    // Determine minimum bits needed to represent a number
    logic [31:0] number;
    logic [5:0] min_bits_needed;

    count_leading_zeros #(.WIDTH(32)) bit_width_calc (
        .data(number),
        .clz(leading_zeros)
    );

    assign min_bits_needed = (number == 0) ? 1 : (32 - leading_zeros);

    // Examples:
    // number = 0      → min_bits = 1

```

```
// number = 1      → min_bits = 1
// number = 255    → min_bits = 8
// number = 256    → min_bits = 9
// number = 65535  → min_bits = 16
```

21.7.44. Log2 Calculation

```
// Calculate floor(log2(x)) for x > 0
logic [31:0] input_value;
logic [4:0] log2_result;
logic valid;

count_leading_zeros #(.WIDTH(32)) log2_calc (
    .data(input_value),
    .clz(leading_zeros)
);

assign valid = (input_value != 0);
assign log2_result = valid ? (31 - leading_zeros) : 5'b0;

// Examples:
// input = 1      → log2 = 0
// input = 2      → log2 = 1
// input = 4      → log2 = 2
// input = 7      → log2 = 2 (floor)
// input = 8      → log2 = 3
```

21.7.55. Data Compression - Run Length Encoding

```
// Count leading zeros for compression
logic [63:0] data_block;
logic [6:0] zero_run_length;

count_leading_zeros #(.WIDTH(64)) rle_counter (
    .data(data_block),
    .clz(zero_run_length)
);

// Use in compression algorithm
if (zero_run_length > THRESHOLD) begin
    // Encode as zero run
    compressed_data = {RLE_CODE, zero_run_length};
end else begin
    // Encode as literal data
    compressed_data = {LITERAL_CODE, data_block};
end
```

21.8 Advanced Implementations

21.8.1 1. Pipeline Version for High Speed

```
module count_leading_zeros_pipelined #(
    parameter int WIDTH = 32,
    parameter int STAGES = 2
) (
    input logic clk,
    input logic rst_n,
    input logic [WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] clz
);

// Stage 1: Partial counting
logic [WIDTH-1:0] data_s1;
logic [$clog2(WIDTH):0] partial_clz_s1;

always_ff @(posedge clk) begin
    data_s1 <= data;
    // Count leading zeros in upper half
    partial_clz_s1 <= clz_upper_half(data);
end

// Stage 2: Complete counting
always_ff @(posedge clk) begin
    if (partial_clz_s1 == WIDTH/2) begin
        // All upper bits are zero, count in lower half
        clz <= WIDTH/2 + clz_lower_half(data_s1);
    end else begin
        // Found '1' in upper half
        clz <= partial_clz_s1;
    end
end

endmodule
```

21.8.2. Hierarchical Implementation

```
// Divide and conquer approach for large widths
module count_leading_zeros_hierarchical #(
    parameter int WIDTH = 64
) (
    input logic [WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] clz
);

localparam int HALF_WIDTH = WIDTH/2;
```



```

logic [$clog2(HALF_WIDTH):0] clz_upper, clz_lower;
logic upper_all_zeros;

// Count in upper half
count_leading_zeros #(.WIDTH(HALF_WIDTH)) upper_clz (
    .data(data[WIDTH-1:HALF_WIDTH]),
    .clz(clz_upper)
);

// Count in lower half
count_leading_zeros #(.WIDTH(HALF_WIDTH)) lower_clz (
    .data(data[HALF_WIDTH-1:0]),
    .clz(clz_lower)
);

assign upper_all_zeros = (clz_upper == HALF_WIDTH);

// Combine results
assign clz = upper_all_zeros ? (HALF_WIDTH + clz_lower) : clz_upper;

endmodule

```

21.8.33. LUT-Based Implementation (Small Widths)

```

// Optimized for small widths using a wildcard case statement.
// casez is required so that '?' is treated as a don't-care.
module count_leading_zeros_lut #(
    parameter int WIDTH = 8
) (
    input logic [WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] clz
);

always_comb begin
    casez (data)
        8'b00000000: clz = 8;
        8'b00000001: clz = 7;
        8'b0000001?: clz = 6;
        8'b000001?: clz = 5;
        8'b00001??? : clz = 4;
        8'b0001???? : clz = 3;
        8'b001????? : clz = 2;
        8'b01?????? : clz = 1;
        8'b1??????? : clz = 0;
        default:      clz = 0;
    endcase

```

end

endmodule

21.9 Performance Analysis

21.9.1 Resource Utilization

WIDTH	LUTs (Typical)	Delay Levels	Max Frequency
8	15-20	3-4	500+ MHz
16	30-40	4-5	400+ MHz
32	60-80	5-6	300+ MHz
64	120-150	6-7	250+ MHz

21.9.2 Timing Characteristics

- **Combinational Delay:** $O(\log(\text{WIDTH}))$ for tree implementations
- **Critical Path:** Through the priority encoding logic
- **Scalability:** Linear increase in logic for function implementation

21.10 Verification Strategy

21.10.1 Test Scenarios

1. **Boundary Cases:** All zeros, all ones, single bit patterns
2. **Random Patterns:** Comprehensive random testing
3. **Systematic Sweep:** Test all possible leading zero counts
4. **Corner Cases:** Maximum width values, alternating patterns

21.10.2 Coverage Model

```
covergroup clz_cg;
  cp_leading_zeros: coverpoint clz {
    bins zero_clz = {0};
    bins low_clz = {[1:WIDTH/4]};
    bins mid_clz = {[WIDTH/4+1:3*WIDTH/4]};
    bins high_clz = {[3*WIDTH/4+1:WIDTH-1]};
    bins all_zeros = {WIDTH};
  }

  cp_data_patterns: coverpoint data {
```

```

    bins all_zeros = {'0'};
    bins all_ones = {'1'};
    bins single_bit[] = {1, 2, 4, 8, 16, 32}; // For appropriate
WIDTH
    bins msb_set = {1 << (WIDTH-1)};
    bins mixed[] = {[1:2** (WIDTH-1)-1]};
}
endgroup

```

21.10.3 Assertions

```

// CLZ should never exceed WIDTH
property clz_bounds;
    clz <= WIDTH;
endproperty

// All zeros should give CLZ = WIDTH
property all_zeros_case;
    (data == '0) |-> (clz == WIDTH);
endproperty

// MSB set should give CLZ = 0
property msb_set_case;
    data[WIDTH-1] |-> (clz == 0);
endproperty

// Relationship between CLZ and first '1' position
property clz_correctness;
    logic [WIDTH-1:0] shifted;
    assign shifted = data << clz;
    (data != '0) |-> shifted[WIDTH-1];
endproperty

assert property (clz_bounds);
assert property (all_zeros_case);
assert property (msb_set_case);
assert property (clz_correctness);

```

21.11 Synthesis Optimization

21.11.1 Tool-Specific Attributes

```

// Xilinx: Prevent SRL inference for better timing
(* SRL_STYLE = "register" *) logic [WIDTH-1:0] data_reg;

// Intel/Altera: Use specific LUT implementation
// synthesis attribute LUT_SIZE of clz_func is 6;

```

21.11.2 Area vs. Speed Trade-offs

// For area optimization: Use iterative approach
// For speed optimization: Use tree-based or LUT approach
// For power optimization: Add enable signals and clock gating

21.12 Common Use Cases Summary

1. **CPU/DSP Cores:** Instruction implementation (CLZ instruction)
2. **Floating-Point Units:** Mantissa normalization
3. **Network Processors:** Longest prefix matching
4. **Compression Engines:** Run-length encoding optimization
5. **Memory Controllers:** Priority arbitration
6. **Graphics Processors:** Texture coordinate processing
7. **Cryptographic Units:** Bit manipulation operations

21.13 Related Modules and Functions

- [count_trailing_zeros](#) - the LSB-up counterpart. Use CLZ when you care about the magnitude of a value (normalization, log2, bit width); use CTZ when you care about alignment or the lowest pending request.
- Population count (number of '1' bits)
- Find first set (FFS) / Find last set (FLS)
- Priority encoders
- Barrel shifters

21.14 Navigation

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22 Count Trailing Zeros Module

22.1 Overview

The `count_trailing_zeros` module implements a trailing zero counter that determines how many consecutive zero bits appear at the end (LSB side) of a data word. It is the mirror image of `count_leading_zeros` and is the natural primitive for alignment checks, lowest-set-bit extraction, and picking the least significant pending request out of a vector.

The scan starts at `data[0]` and proceeds upward, stopping at the first set bit:

```
data = 32'h0000_0001 -> ctz = 0    (bit 0 is already set)
data = 32'h8000_0000 -> ctz = 31   (31 zeros below the MSB)
data = 32'h0000_0000 -> ctz = WIDTH
```

For the complementary count from the MSB downward, use [count_leading_zeros](#).

22.1.1 Choosing Between CLZ and CTZ

Question you are asking	Module
How big is this value? (normalization, log2, minimum bit width)	<code>count_leading_zeros</code>
How is this value aligned? (address alignment, burst size, power-of-two stride)	<code>count_trailing_zeros</code>
Which is the highest priority request set?	<code>count_leading_zeros</code>
Which is the lowest index request set?	<code>count_trailing_zeros</code>

Never emulate one by bit-reversing the input to the other. Both modules exist; picking the right one costs no extra logic and keeps intent visible in the netlist.

22.2 Module Declaration

```
module count_trailing_zeros #(
    parameter int WIDTH = 32
) (
    input logic [WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] ctz
);
```

22.3 Parameters

22.3.1 WIDTH

- **Type:** int

- **Default:** 32
- **Description:** Bit width of the input data
- **Range:** Any positive integer ≥ 1
- **Common Values:** 8, 16, 32, 64 for standard data widths
- **Impact:** Determines output width and algorithm complexity

WIDTH is the only parameter.

22.4 Ports

22.4.1 Inputs

Port	Width	Type	Description
data	WIDTH	logic	Input data word to analyze

22.4.2 Outputs

Port	Width	Type	Description
ctz	$\$clog2(WIDTH)+1$	logic	Count of trailing zeros

22.4.3 Output Width Explanation

The output width $\$clog2(WIDTH)+1$ ensures it can represent all possible counts: - **Range:** 0 to WIDTH (inclusive) - **Example:** For WIDTH=32, output is 6 bits (0-32 requires 6 bits) - **Special Case:** All zeros input produces CTZ = WIDTH

22.5 Algorithm Implementation

22.5.1 Function-Based Approach

```
function automatic [$clog2(WIDTH):0] ctz_func;
    input [WIDTH-1:0] input_data;
    logic found;
    begin
        ctz_func = 0;
        found = 1'b0;
        // Scan from the LSB upward.
        for (int i = 0; i < WIDTH; i++) begin
            if (!input_data[i] && !found) begin
```

```

        ctz_func += 1;
    end else begin
        found = 1'b1; // Stop counting when first '1' found
    end
end
end
endfunction

```

22.5.2 Bit Scanning Process

The algorithm scans from LSB to MSB: 1. **Initialize:** `ctz_func = 0, found = 0` 2. **Scan Loop:** For each bit position `i` from 0 to `WIDTH-1`: - If `data[i] == 0` AND no '1' found yet: increment count - If `data[i] == 1`: set found flag, stop counting 3. **Result:** Final count is the number of zeros below the lowest set bit

22.5.3 Why LSB-to-MSB Scanning?

“Trailing” refers to the bits that trail the word when it is written out, which is the LSB side. The loop therefore starts at `data[0]` and walks up: - **Bit Order:** `data[0]` is LSB, `data[WIDTH-1]` is MSB - **Termination:** The found flag latches at the lowest set bit, so all bits above it are ignored - only the contiguous run below it is counted - **Count Logic:** `ctz = position_of_lowest_set_one`, and `ctz = WIDTH` when no bit is set

22.6 Examples and Truth Tables

22.6.1 8-bit Examples (WIDTH=8)

The count is set by the **lowest** set bit; bits above it never affect the result.

Input (data)	Binary	Lowest set bit	Trailing Zeros	CTZ Output
8'b00000000	00000000	None	8	8
8'b00000001	00000001	Bit 0	0	0
8'b00000010	00000010	Bit 1	1	1
8'b00000100	00000100	Bit 2	2	2
8'b00001000	00001000	Bit 3	3	3
8'b00010000	00010000	Bit 4	4	4
8'b00100000	00100000	Bit 5	5	5

Input (data)	Binary	Lowest set bit	Trailing Zeros	CTZ Output
	0			
8'b01000000	0100000	Bit 6	6	6
	0			
8'b10000000	1000000	Bit 7	7	7
	0			
8'b11111111	1111111	Bit 0	0	0
	1			
8'b10101010	1010101	Bit 1	1	1
	0			
8'b00110000	0011000	Bit 4	4	4
	0			

22.6.2 32-bit Examples

Input	Hex	Trailing Zeros	CTZ
32'h00000000	0x00000000	32	32
32'h00000001	0x00000001	0	0
32'h00000080	0x00000080	7	7
32'h00008000	0x00008000	15	15
32'h00800000	0x00800000	23	23
32'h80000000	0x80000000	31	31
32'hFFFFFFFF	0xFFFFFFFF	0	0

22.6.3 Relationship to CLZ

For a single-bit input data == (1 < k), the two modules are complementary:

```
ctz = k
clz = WIDTH - 1 - k
ctz + clz = WIDTH - 1
```

That identity holds **only** for one-hot inputs. For general data the two counts are independent, because CLZ is decided by the highest set bit and CTZ by the lowest. For example 8'b00110000 gives clz = 2 and ctz = 4, which sum to 6, not 7.

22.7 Applications

22.7.1 1. Address Alignment Detection

```
// Determine the largest power-of-two boundary an address is aligned
to
logic [31:0] address;
logic [5:0] alignment_log2;

count_trailing_zeros #(.WIDTH(32)) align_ctz (
    .data(address),
    .ctz(alignment_log2)
);

// alignment_log2 = 0  -> byte aligned only
// alignment_log2 = 2  -> 4-byte aligned
// alignment_log2 = 6  -> 64-byte aligned (cache line)
// alignment_log2 = 32 -> address is zero, aligned to everything
```

22.7.22. Maximum Burst Size for a DMA Transfer

```
// A burst may not cross its natural alignment boundary, so the
alignment of the
// start address caps the burst size.
logic [31:0] start_addr;
logic [5:0] addr_align;
logic [5:0] max_burst_log2;

count_trailing_zeros #(.WIDTH(32)) burst_ctz (
    .data(start_addr),
    .ctz(addr_align)
);

// Clamp against the protocol maximum (e.g. 4KB = 12)
assign max_burst_log2 = (addr_align > 6'd12) ? 6'd12 : addr_align;
```

22.7.33. Lowest-Index Request Arbitration

```
// Fixed priority arbiter favouring the lowest requester index
logic [15:0] request_vector;
logic [4:0] trailing_zeros;
logic [3:0] granted_index;
logic any_request;

count_trailing_zeros #(.WIDTH(16)) req_ctz (
    .data(request_vector),
    .ctz(trailing_zeros)
);
```

```

assign any_request  = (trailing_zeros != 16);
assign granted_index = any_request ? trailing_zeros[3:0] : 4'b0;

// Example: request_vector = 16'b0000_0100_1000_0000
// Lowest set bit is bit 7
// CTZ = 7, so requester 7 is granted

```

22.7.44. Isolating and Clearing the Lowest Set Bit

```

// Iterate over set bits one at a time (software-style "x & -x" in
// hardware)
logic [31:0] pending;
logic [5:0]  next_index;
logic [31:0] lowest_bit_mask;
logic [31:0] pending_next;

count_trailing_zeros #(.WIDTH(32)) pend_ctz (
    .data(pending),
    .ctz(next_index)
);

assign lowest_bit_mask = pending & (~pending + 32'd1); // isolates
// bit next_index
assign pending_next    = pending & ~lowest_bit_mask;  // clears it

```

22.7.55. Power-of-Two Detection

```

// A non-zero value is a power of two exactly when its only set bit is
// the lowest one
logic [31:0] value;
logic [5:0]  value_ctz;
logic        is_power_of_two;

count_trailing_zeros #(.WIDTH(32)) pow2_ctz (
    .data(value),
    .ctz(value_ctz)
);

assign is_power_of_two = (value != 0) && (value == (32'd1 <<
value_ctz));

// value = 64    -> ctz = 6,  1<<6 = 64    -> true
// value = 96    -> ctz = 5,  1<<5 = 32    -> false
// value = 0      -> false by the explicit guard

```

22.8 Advanced Implementations

22.8.1 1. Hierarchical Implementation

```
// Divide and conquer approach for large widths
module count_trailing_zeros_hierarchical #(
    parameter int WIDTH = 64
) (
    input logic [WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] ctz
);

localparam int HALF_WIDTH = WIDTH/2;

logic [$clog2(HALF_WIDTH):0] ctz_upper, ctz_lower;
logic lower_all_zeros;

// Count in lower half
count_trailing_zeros #(.WIDTH(HALF_WIDTH)) lower_ctz (
    .data(data[HALF_WIDTH-1:0]),
    .ctz(ctz_lower)
);

// Count in upper half
count_trailing_zeros #(.WIDTH(HALF_WIDTH)) upper_ctz (
    .data(data[WIDTH-1:HALF_WIDTH]),
    .ctz(ctz_upper)
);

assign lower_all_zeros = (ctz_lower == HALF_WIDTH);

// Combine results
assign ctz = lower_all_zeros ? (HALF_WIDTH + ctz_upper) : ctz_lower;

endmodule
```

22.8.22. LUT-Based Implementation (Small Widths)

```
// Optimized for small widths using a wildcard case statement.
// casez is required so that '?' is treated as a don't-care.
module count_trailing_zeros_lut #(
    parameter int WIDTH = 8
) (
    input logic [WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] ctz
);
```

```

always_comb begin
    casez (data)
        8'b???????1: ctz = 0;
        8'b??????10: ctz = 1;
        8'b?????100: ctz = 2;
        8'b????1000: ctz = 3;
        8'b???10000: ctz = 4;
        8'b??100000: ctz = 5;
        8'b?1000000: ctz = 6;
        8'b10000000: ctz = 7;
        default:      ctz = 8;    // all zeros
    endcase
end

endmodule

```

22.8.33. Mask-Based Implementation

```

// Isolate the lowest set bit, then one-hot encode its position.
// Often maps well to carry-chain logic on FPGAs.
logic [WIDTH-1:0] w_lowest_one;

assign w_lowest_one = data & (~data + 1'b1);
// w_lowest_one is one-hot (or all zero), so a plain one-hot encoder
yields ctz

```

22.9 Performance Analysis

22.9.1 Resource Utilization

WIDTH	LUTs (Typical)	Delay Levels	Max Frequency
8	15-20	3-4	500+ MHz
16	30-40	4-5	400+ MHz
32	60-80	5-6	300+ MHz
64	120-150	6-7	250+ MHz

22.9.2 Timing Characteristics

- **Combinational Delay:** $O(\log(\text{WIDTH}))$ for tree implementations
- **Critical Path:** Through the priority encoding logic
- **Scalability:** Linear increase in logic for the function implementation

22.10 Verification Strategy

22.10.1 Test Scenarios

1. **Boundary Cases:** All zeros, all ones, single bit patterns
2. **Random Patterns:** Comprehensive random testing
3. **Systematic Sweep:** Test all possible trailing zero counts
4. **Corner Cases:** Maximum width values, alternating patterns
5. **Cross-Check:** For one-hot inputs, confirm $\text{ctz} + \text{clz} == \text{WIDTH} - 1$ against `count_leading_zeros`

22.10.2 Coverage Model

```
covergroup ctz_cg;
  cp_trailing_zeros: coverpoint ctz {
    bins zero_ctz    = {0};
    bins low_ctz     = {[1:WIDTH/4]};
    bins mid_ctz     = {[WIDTH/4+1:3*WIDTH/4]};
    bins high_ctz    = {[3*WIDTH/4+1:WIDTH-1]};
    bins all_zeros   = {WIDTH};
  }

  cp_data_patterns: coverpoint data {
    bins all_zeros   = {'0'};
    bins all_ones    = {'1'};
    bins single_bit[] = {1, 2, 4, 8, 16, 32}; // For appropriate
WIDTH
    bins lsb_set     = {1};
    bins msb_set     = {1 << (WIDTH-1)};
    bins mixed[]     = {[1:2** (WIDTH-1) - 1]};
  }
endgroup
```

22.10.3 Assertions

```
// CTZ should never exceed WIDTH
property ctz_bounds;
  ctz <= WIDTH;
endproperty

// All zeros should give CTZ = WIDTH
property all_zeros_case;
  (data == '0') |-> (ctz == WIDTH);
endproperty
```

```

// LSB set should give CTZ = 0
property lsb_set_case;
    data[0] |-> (ctz == 0);
endproperty

// Relationship between CTZ and the lowest '1' position
property ctz_correctness;
    logic [WIDTH-1:0] shifted;
    assign shifted = data >> ctz;
    (data != '0) |-> shifted[0];
endproperty

// CTZ must select exactly the isolated lowest set bit
property ctz_isolates_lowest;
    (data != '0) |-> ((data & (~data + 1)) == (1 << ctz));
endproperty

assert property (ctz_bounds);
assert property (all_zeros_case);
assert property (lsb_set_case);
assert property (ctz_correctness);
assert property (ctz_isolates_lowest);

```

22.11 Synthesis Optimization

22.11.1 Area vs. Speed Trade-offs

```

// For area optimization: Use the iterative function approach
// For speed optimization: Use the hierarchical or LUT approach
// For power optimization: Add enable signals and clock gating

```

22.12 Common Use Cases Summary

1. **CPU/DSP Cores:** Instruction implementation (CTZ / FFS instruction)
2. **Memory Controllers:** Address alignment and burst-size derivation
3. **DMA Engines:** Legal transfer size given a start address
4. **Schedulers and Arbiters:** Lowest-index pending request selection
5. **Allocators:** Free-list and bitmap scanning
6. **Interrupt Controllers:** Lowest pending interrupt vector

22.13 Related Modules and Functions

- [count_leading_zeros](#) - the MSB-down counterpart. See the selection table at the top of this page before choosing between them.
- Population count (number of '1' bits)
- Find first set (FFS) / Find last set (FLS)
- Priority encoders
- Barrel shifters

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23 dataint_checksum Module Documentation

23.1 Purpose

The dataint_checksum module implements a simple accumulating checksum calculator. It continuously adds incoming data values to maintain a running checksum that can be used for basic data integrity verification.

23.2 Module Declaration

```
module dataint_checksum #(
    parameter int WIDTH = 8
) (
    input logic          clk,
    rst_n,
    input logic          reset,
    input logic          valid,
    input logic [WIDTH-1:0] data,
    output logic [WIDTH-1:0] checksum
);
```

23.3 Parameters

Parameter	Default	Description
WIDTH	8	Width of the data and checksum in bits

23.4 Ports

Port	Direction	Width	Description
clk	Input	1	Clock signal
rst_n	Input	1	Active-low asynchronous reset
reset	Input	1	Synchronous reset signal
valid	Input	1	Data valid signal - when high, data is accumulated
data	Input	WIDTH	Input data to be added to checksum
chksum	Output	WIDTH	Current checksum value

23.5 Functionality

23.5.1 Operation

The module maintains an internal counter (`r_count`) that accumulates incoming data: - On reset (either asynchronous `rst_n` or synchronous `reset`), the counter is cleared to zero - When `valid` is asserted, the input data is added to the current counter value - The checksum output is simply the current counter value

23.5.2 Reset Behavior

The module supports dual reset mechanisms: 1. **Asynchronous Reset:** `rst_n` (active-low) immediately clears the counter 2. **Synchronous Reset:** `reset` (active-high) clears the counter on the next clock edge

23.6 Implementation Details

23.6.1 Key Features

- **Simple Accumulation:** Uses basic addition for checksum calculation
- **Parameterizable Width:** Supports any data width through the `WIDTH` parameter
- **Dual Reset Support:** Both asynchronous and synchronous reset capabilities
- **Enable Control:** Data accumulation only occurs when `valid` is asserted

23.6.2 Special Considerations

- **Overflow Handling:** The checksum will wrap around when the accumulated value exceeds the maximum value representable in WIDTH bits
- **Continuous Output:** The checksum output is always valid and reflects the current accumulated value
- **Reset Priority:** Asynchronous reset (rst_n) takes precedence over synchronous reset (reset)

23.7 State Machine

This module does not implement a formal FSM but operates as a simple accumulator with the following states: - **RESET:** Counter is zero - **ACCUMULATING:** Counter adds valid data inputs

23.8 Usage Example

```
// 16-bit checksum calculator
dataint_checksum #(
    .WIDTH(16)
) checksum_calc (
    .clk(clk),
    .rst_n(rst_n),
    .reset(soft_reset),
    .valid(data_valid),
    .data(input_data),
    .chksum(calculated_checksum)
);
```

23.9 Applications

- Simple data integrity checking
- Packet checksum calculation
- Basic error detection in data streams
- Accumulation of data values for verification purposes

23.10 Navigation

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24 dataint_crc

A generic Cyclic Redundancy Check (CRC) computation engine that provides flexible, high-performance CRC calculation with configurable polynomials, data widths, and processing options.

24.1 Overview

The `dataint_crc` module implements a comprehensive CRC calculation engine suitable for data integrity applications, communication protocols, and storage systems. It supports configurable polynomial widths, data widths, input/output reflection, and cascade processing for high-throughput applications.

24.2 Module Declaration

```
module dataint_crc #(
    parameter string ALGO_NAME = "DEADF1F0", // Algorithm identifier
    parameter int DATA_WIDTH = 64,          // Input data width
    parameter int CRC_WIDTH = 64,            // CRC polynomial width
    parameter int REF_IN = 1,                // Reflect input data
    parameter int REF_OUT = 1                // Reflect output data
) (
    // Derived localparam (computed internally - not user-settable):
    // localparam int CHUNKS = DATA_WIDTH / 8; // Number of byte
chunks
    // localparam int CW = CRC_WIDTH;          // CRC width alias
    // localparam int DW = DATA_WIDTH;        // Data width alias
    // localparam int CH = CHUNKS;             // Chunks alias
    input logic [CRC_WIDTH-1:0] POLY,          // CRC polynomial
    input logic [CRC_WIDTH-1:0] POLY_INIT,     // Initial CRC
value
    input logic [CRC_WIDTH-1:0] XOROUT,        // Output XOR mask
    input logic clk,                          // Clock
    input logic rst_n,                        // Active-low
reset
    input logic load_crc_start,               // Load initial
CRC
    input logic load_from_cascade,            // Load from
cascade
    input logic [CHUNKS-1:0] cascade_sel,     // Cascade
selection (one-hot)
    input logic [DATA_WIDTH-1:0] data,        // Input data
    output logic [CRC_WIDTH-1:0] crc          // CRC output
);
```

24.3 Parameters

24.3.1 User-Settable Parameters

Parameter	Type	Default	Description
ALGO_NAME	string	“DEADF1F0”	Algorithm identifier for documentation
DATA_WIDTH	int	64	Input data width in bits (must be multiple of 8)
CRC_WIDTH	int	64	CRC polynomial width in bits (8, 16, 32, or 64)
REFIN	int	1	Reflect input data bytes (1=reflect, 0=direct)
REFOUT	int	1	Reflect output CRC (1=reflect, 0=direct)

24.3.2 Derived Localparam (Computed Internally)

Localparam	Computation	Description
CHUNKS (CH)	DATA_WIDTH/8	Number of 8-bit processing chunks per cycle
CW	CRC_WIDTH	Convenience alias for CRC width
DW	DATA_WIDTH	Convenience alias for data width

Note: CHUNKS, CW, DW, and CH are computed internally and cannot be overridden by users.

24.4 Ports

24.4.1 Inputs

Port	Width	Description
POLY	CRC_WIDTH	CRC polynomial coefficient
POLY_INIT	CRC_WIDTH	Initial CRC value (seed)
XOROUT	CRC_WIDTH	Final XOR mask for

Port	Width	Description
		output
clk	1	System clock
rst_n	1	Active-low asynchronous reset
load_crc_start	1	Load initial CRC value
load_from_cascade	1	Load CRC from cascade output
cascade_sel	CHUNKS	One-hot cascade stage selection
data	DATA_WIDTH	Input data for CRC calculation

24.4.2 Outputs

Port	Width	Description
crc	CRC_WIDTH	Computed CRC value (registered)

24.5 Functionality

24.5.1 CRC Algorithm Implementation

The module implements a parallel CRC calculation using the following stages:

1. **Input Data Conditioning:** Optional bit reflection per REFIN parameter
2. **Cascade Processing:** Parallel processing through multiple XOR-shift stages
3. **Output Conditioning:** Optional bit reflection per REFOUT parameter
4. **Final XOR:** Application of XOROUT mask to final result

24.5.2 Key Features

- **Parallel Processing:** Processes multiple bytes simultaneously for high throughput
- **Configurable Reflection:** Support for MSB-first and LSB-first bit ordering
- **Cascade Architecture:** Enables partial CRC computation and continuation

- **Standards Compliance:** Configurable for standard CRC algorithms (CRC-32, CRC-64, etc.)

24.6 Implementation Details

24.6.1 Input Data Reflection

```
// Input reflection based on REFIN parameter
if (REFIN != 0) begin : gen_reflect_inputs
    for (genvar i = 0; i < CH; i++) begin : gen_ch_reflect
        for (genvar j = 0; j < 8; j++) begin : gen_bit_reflect
            assign w_block_data[i][j] = data[i*8+7-j]; // Bit
reversal
        end
    end
end else begin : gen_direct_assign_inputs
    for (genvar i = 0; i < CH; i++) begin : gen_ch_direct
        assign w_block_data[i] = data[i*8+:8]; // Direct assignment
    end
end
```

24.6.2 Cascade Processing Architecture

```
// Chain of XOR-shift cascade stages
for (i = 0; i < CH; i++) begin : gen_xor_shift_cascade
    if (i == 0) begin : gen_xor_cascade_0
        dataint_crc_xor_shift_cascade #(.CRC_WIDTH(CRC_WIDTH))
cascade_0 (
            .block_input(r_crc_value), // Start with current CRC
            .poly(w_poly),
            .data_input(w_block_data[i]),
            .block_output(w_cascade[i])
        );
    end else begin : gen_xor_cascade_N
        dataint_crc_xor_shift_cascade #(.CRC_WIDTH(CRC_WIDTH))
cascade_N (
            .block_input(w_cascade[i-1]), // Chain from previous
stage
            .poly(w_poly),
            .data_input(w_block_data[i]),
            .block_output(w_cascade[i])
        );
    end
end
```

24.6.3 Cascade Selection Logic

```
// Select appropriate cascade output based on data length
always_comb begin
```

```

w_selected_cascade_output = POLY_INIT; // Default
for (int i = 0; i < CH; i++) begin
    if (cascade_sel[i]) begin
        w_selected_cascade_output = w_cascade[i];
    end
end
end

```

24.7 CRC Standards Compatibility

24.7.1 Common CRC Configurations

24.7.1.1 CRC-32 (IEEE 802.3)

```

dataint_crc #(
    .ALGO_NAME("CRC32_IEEE"),
    .DATA_WIDTH(32),
    .CRC_WIDTH(32),
    .REFIN(1),
    .REFOUT(1)
) u_crc32 (
    .POLY(32'h04C11DB7),
    .POLY_INIT(32'hFFFFFFFF),
    .XOROUT(32'hFFFFFFFF),
    // ... other connections
);

```

24.7.1.2 CRC-16 (CCITT)

```

dataint_crc #(
    .ALGO_NAME("CRC16_CCITT"),
    .DATA_WIDTH(16),
    .CRC_WIDTH(16),
    .REFIN(0),
    .REFOUT(0)
) u_crc16 (
    .POLY(16'h1021),
    .POLY_INIT(16'hFFFF),
    .XOROUT(16'h0000),
    // ... other connections
);

```

24.7.1.3 CRC-64 (ECMA-182)

```

dataint_crc #(
    .ALGO_NAME("CRC64_ECMA"),
    .DATA_WIDTH(64),
    .CRC_WIDTH(64),
    .REFIN(1),
    .REFOUT(1)
);

```

```

) u_crc64 (
    .POLY(64'h42F0E1EBA9EA3693),
    .POLY_INIT(64'h0000000000000000),
    .XOROUT(64'hFFFFFFFFFFFFFFFF),
    // ... other connections
);

```

24.8 Usage Examples

24.8.1 Basic CRC Calculation

```

logic [63:0] input_data;
logic [31:0] crc_result;
logic data_valid, crc_ready;

dataint_crc #(
    .DATA_WIDTH(64),
    .CRC_WIDTH(32),
    .REFIN(1),
    .REFOUT(1)
) u_crc_engine (
    .POLY(32'h04C11DB7),
    .POLY_INIT(32'hFFFFFFFF),
    .XOROUT(32'hFFFFFFFF),
    .clk(clk),
    .rst_n(rst_n),
    .load_crc_start(data_valid),
    .load_from_cascade(1'b0),
    .cascade_sel(8'hFF), // Use all chunks
    .data(input_data),
    .crc(crc_result)
);

```

24.8.2 Streaming CRC with Cascade

```

// Process variable-length packets using cascade selection
logic [63:0] packet_data;
logic [7:0] byte_count; // Number of valid bytes (1-8)
logic [7:0] cascade_select;

// Convert byte count to one-hot cascade selection
always_comb begin
    cascade_select = '0;
    if (byte_count > 0 && byte_count <= 8) begin
        cascade_select[byte_count-1] = 1'b1;
    end
end

```

```

dataint_crc #(
    .DATA_WIDTH(64),
    .CRC_WIDTH(32)
) u_stream_crc (
    .clk(clk),
    .rst_n(rst_n),
    .load_crc_start(packet_start),
    .load_from_cascade(packet_continue),
    .cascade_sel(cascade_select),
    .data(packet_data),
    .crc(streaming_crc),
    // ... other connections
);

```

24.8.3 Multi-Stage CRC Pipeline

```

// Pipeline multiple CRC engines for high throughput
logic [63:0] stage_data [0:3];
logic [31:0] stage_crc [0:3];

genvar stage;
generate
    for (stage = 0; stage < 4; stage++) begin : gen_crc_pipeline
        dataint_crc #(
            .DATA_WIDTH(64),
            .CRC_WIDTH(32),
            .ALGO_NAME($sprintf("STAGE_%0d", stage))
        ) u_pipeline_crc (
            .clk(clk),
            .rst_n(rst_n),
            .data(stage_data[stage]),
            .crc(stage_crc[stage]),
            // ... configure for pipeline operation
        );
    end
endgenerate

```

24.9 Performance Characteristics

24.9.1 Throughput Analysis

Data Width	CRC Width	Max Frequency	Throughput
32-bit	32-bit	~400 MHz	12.8 Gbps
64-bit	32-bit	~350 MHz	22.4 Gbps
128-bit	64-bit	~300 MHz	38.4 Gbps

24.9.2 Latency Characteristics

- **Pipeline Stages:** 2 cycles (input conditioning + CRC computation)
- **Reset Recovery:** 1 cycle
- **Cascade Selection:** 0 cycles (combinational)

24.10 Timing Considerations

24.10.1 Setup Requirements

```
// Data must be stable before clock edge
timing_spec: assert property (
    @(posedge clk) $stable(data) throughout (load_crc_start)
);

// Cascade selection must be stable
cascade_timing: assert property (
    @(posedge clk) $stable(cascade_sel) throughout (load_from_cascade)
);
```

24.10.2 Critical Paths

1. **Cascade Chain:** Data flows through all cascade stages
2. **Output Reflection:** Bit reversal for REFOUT
3. **Register Setup:** Loading CRC state register

24.11 Design Considerations

24.11.1 Area Optimization

- **Polynomial Optimization:** Use polynomials with fewer terms
- **Width Reduction:** Match CRC_WIDTH to actual requirements
- **Cascade Stages:** Reduce CHUNKS for lower throughput applications

24.11.2 Power Optimization

- **Clock Gating:** Gate unused cascade stages
- **Data Gating:** Prevent unnecessary switching when not processing

24.11.3 Verification Notes

- **Polynomial Validation:** Verify against known CRC test vectors
- **Cascade Testing:** Test all cascade selection combinations
- **Reflection Testing:** Verify bit ordering for REFIN/REFOUT modes
- **Standards Compliance:** Validate against published CRC standards

24.12 Related Modules

- **dataint_crc_xor_shift_cascade:** Building block for cascade stages
- **dataint_crc_xor_shift:** Simple XOR-shift CRC implementation
- **dataint_checksum:** Alternative integrity checking method
- **dataint_parity:** Simpler parity-based error detection

The `dataint_crc` module provides a comprehensive, high-performance solution for CRC-based data integrity checking with the flexibility to support a wide range of CRC standards and applications.

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25 dataint_crc_xor_shift_cascade Module Documentation

25.1 Purpose

The `dataint_crc_xor_shift_cascade` module processes 8 bits of data through a cascade of CRC calculation stages. It chains together 8 instances of `dataint_crc_xor_shift` to efficiently process a full byte of data in a single combinational operation.

25.2 Module Declaration

```
module dataint_crc_xor_shift_cascade #(
    parameter int CRC_WIDTH = 32
) (
    input  [CRC_WIDTH-1:0] block_input,
    input  [CRC_WIDTH-1:0] poly,
    input  [          7:0] data_input,
```

```

        output [CRC_WIDTH-1:0] block_output
    );

```

25.3 Parameters

Parameter	Default	Description
CRC_WIDTH	32	Width of the CRC register and polynomial

25.4 Ports

Port	Direction	Width	Description
block_input	Input	CRC_WIDTH	Initial CRC state for the byte
poly	Input	CRC_WIDTH	CRC polynomial
data_input	Input	8	8-bit data byte to process
block_output	Output	CRC_WIDTH	Final CRC state after processing 8 bits

25.5 Functionality

25.5.1 Cascade Operation

The module creates a pipeline of 8 CRC calculation stages: 1. **Stage 0**: Processes MSB of data byte with initial CRC state 2. **Stages 1-6**: Each processes next data bit with previous stage output 3. **Stage 7**: Processes LSB and produces final result

25.5.2 Data Bit Ordering

Data bits are processed from MSB to LSB: - data_input[7] is processed first - data_input[0] is processed last - This matches typical serial transmission order

25.6 Implementation Details

25.6.1 Architecture Overview

```

block_input → [Stage 0] → [Stage 1] → ... → [Stage 7] → block_output
               data[7]    data[6]                data[0]

```

25.6.2 Generate Block Structure

The module uses a parameterized generate block to create the cascade:

```

generate
  for (i = 0; i < 8; i++) begin : gen_xor_shift_cascade
    if (i == 0) begin : gen_xor_shift_0
      // First stage uses block_input
      dataint_crc_xor_shift #(.CRC_WIDTH(CRC_WIDTH)) stage (
        .stage_input(block_input),
        .poly(poly),
        .new_bit(data_input[7-i]),
        .stage_output(w_cascade[i])
      );
    end else begin : gen_xor_shift_N
      // Subsequent stages chain previous output
      dataint_crc_xor_shift #(.CRC_WIDTH(CRC_WIDTH)) stage (
        .stage_input(w_cascade[i-1]),
        .poly(poly),
        .new_bit(data_input[7-i]),
        .stage_output(w_cascade[i])
      );
    end
  end
endgenerate

```

25.6.3 Intermediate Signals

- **w_cascade[0:7]**: Array holding intermediate CRC states
- **Fixed Width**: Each element is CRC_WIDTH bits wide
- **Sequential Chain**: Each stage feeds the next

25.7 Key Features

25.7.1 Parallel Byte Processing

- Processes 8 bits in one combinational operation
- No clock required - purely combinational
- Suitable for high-speed applications

25.7.2 Bit Order Handling

- MSB-first processing (network byte order)
- Bit index calculation: data_input[7-i]
- Matches standard CRC bit processing order

25.7.3 Scalable Architecture

- Easy to modify for different byte sizes
- Generate blocks enable clean scaling
- Parameterized for different CRC widths

25.8 Timing Characteristics

25.8.1 Combinational Delay

Total delay is the sum of 8 CRC stage delays: - Each stage: ~2-3 gate delays - Total cascade: ~16-24 gate delays - May require pipelining for very high frequencies

25.8.2 Critical Path

The critical path runs through all 8 stages:

block_input → Stage0 → Stage1 → ... → Stage7 → block_output

25.9 Usage Examples

25.9.1 Basic Usage

```
dataint_crc_xor_shift_cascade #(
    .CRC_WIDTH(16)
) crc_byte_processor (
    .block_input(current_crc_state),
    .poly(16'h1021),           // CRC-16-CCITT
    .data_input(data_byte),
    .block_output(updated_crc_state)
);
```

25.9.2 Multi-Byte Processing

```
// Process 4 bytes (32 bits) of data
wire [CRC_WIDTH-1:0] stage_output[0:3];

genvar i;
generate
    for (i = 0; i < 4; i++) begin : gen_multi_byte
        dataint_crc_xor_shift_cascade #(.CRC_WIDTH(CRC_WIDTH)) cascade
        (
            .block_input(i == 0 ? initial_crc : stage_output[i-1]),
            .poly(polynomial),
            .data_input(data_bytes[i]),
            .block_output(stage_output[i])
        );
    end
endgenerate
```

```
end  
endgenerate
```

```
assign final_crc = stage_output[3];
```

25.10 Integration with CRC System

25.10.1 Role in CRC Module

This module is used by `dataint_crc` for: - Processing data in byte-sized chunks - Building wider data processing capability - Maintaining processing efficiency

25.10.2 Interface with Parent Module

- Multiple cascade instances handle wide data
- Cascade selection allows intermediate results
- Enables flexible data width support

25.11 Performance Optimization

25.11.1 Pipelining Considerations

For high-frequency operation: - Consider adding pipeline registers between cascades - Balance latency vs. throughput requirements - May need to pipeline within the 8-stage cascade

25.11.2 Area Optimization

- Each stage requires $\sim \text{CRC_WIDTH}$ XOR gates
- Total area scales with $\text{CRC_WIDTH} \times 8$
- Consider time-multiplexed alternatives for area-critical applications

25.12 Applications

- High-speed packet processing
- Real-time data integrity checking
- Communication protocol implementations
- Storage system error detection
- Building block for wider CRC processors

25.13 Related Modules

- **dataint_crc_xor_shift**: Single-bit processing building block
- **dataint_crc**: Parent CRC calculation module
- Used together to create flexible CRC calculation systems

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26 dataint_crc_xor_shift Module Documentation

26.1 Purpose

The `dataint_crc_xor_shift` module performs CRC calculation for a single bit of input data. It implements the fundamental CRC operation: shifting the current CRC state and XORing with the polynomial when the feedback bit is set.

26.2 Module Declaration

```
module dataint_crc_xor_shift #(
    parameter int CRC_WIDTH = 32
) (
    input [CRC_WIDTH-1:0] stage_input,
    input [CRC_WIDTH-1:0] poly,
    input new_bit,
    output [CRC_WIDTH-1:0] stage_output
);
```

26.3 Parameters

Parameter	Default	Description
CRC_WIDTH	32	Width of the CRC register and polynomial

26.4 Ports

Port	Direction	Width	Description
stage_input	Input	CRC_WIDTH	Current CRC

Port	Direction	Width	Description
poly	Input	CRC_WIDTH	state CRC polynomial
new_bit	Input	1	New data bit to process
stage_output	Output	CRC_WIDTH	Updated CRC state

26.5 Functionality

26.5.1 Core CRC Operation

The module implements the basic CRC shift-and-XOR operation: 1. **Feedback Calculation:** XOR the new input bit with the MSB of current CRC 2. **Shift Operation:** Shift CRC register left by one position 3. **Polynomial XOR:** If feedback bit is 1, XOR with polynomial

26.5.2 Mathematical Foundation

The CRC operation follows this algorithm:

```
feedback = new_bit ⊕ CRC[MSB]
CRC_new[0] = feedback
CRC_new[MSB:1] = CRC[MSB-1:0] ⊕ (polynomial[MSB:1] & {WIDTH-1{feedback}})
```

26.6 Implementation Details

26.6.1 Key Features

- **Single Bit Processing:** Processes exactly one bit per operation
- **Combinational Logic:** Pure combinational implementation (no clock)
- **Parameterizable Width:** Supports any CRC width
- **Optimized Design:** Breaks circular dependency with intermediate signal

26.6.2 Circular Dependency Solution

The original implementation could have a circular dependency. This is solved by:

```
logic feedback_bit;

// Calculate feedback first
```



```

assign feedback_bit = new_bit ^ stage_input[CRC_WIDTH-1];

// Then use it for output calculation
assign stage_output[0] = feedback_bit;
assign stage_output[CRC_WIDTH-1:1] = stage_input[CRC_WIDTH-2:0] ^
    ({CRC_WIDTH-1{feedback_bit}} & poly[CRC_WIDTH-1:1]);

```

26.6.3 Bit-wise Operations

- **LSB Assignment:** Always gets the feedback bit
- **MSBs Assignment:** Shifted previous bits XORed with polynomial when feedback is active
- **Conditional XOR:** Uses bit replication to create conditional XOR mask

26.7 Special Implementation Features

26.7.1 Feedback Bit Logic

The feedback bit is critical for CRC operation: - Determines whether polynomial is applied - Calculated as XOR of input bit and current MSB - Used to gate polynomial application

26.7.2 Polynomial Application

The polynomial is applied conditionally: - Only bits [MSB:1] of polynomial are used (bit 0 is implicit) - Applied via AND mask created by replicating feedback bit - XORed with shifted CRC bits

26.7.3 No State Elements

This module is purely combinational: - No clock or reset signals needed - Instantaneous response to input changes - Can be cascaded for multi-bit processing

26.8 Usage in CRC Chain

This module is typically used as a building block:

26.8.1 Single Bit Processing

```

dataint_crc_xor_shift #(.CRC_WIDTH(16)) crc_stage (
    .stage_input(current_crc),
    .poly(crc_polynomial),
    .new_bit(data_bit),
    .stage_output(next_crc)
);

```

26.8.2 Cascaded for Multiple Bits

Multiple instances can be chained to process multiple bits:

```

// Process 8 bits
wire [CRC_WIDTH-1:0] stage[0:7];
assign stage[0] = initial_crc;

genvar i;
generate
    for (i = 0; i < 8; i++) begin
        dataint_crc_xor_shift #(.CRC_WIDTH(CRC_WIDTH)) stage_inst (
            .stage_input(i == 0 ? initial_crc : stage[i-1]),
            .poly(polyynomial),
            .new_bit(data_byte[7-i]),
            .stage_output(stage[i])
        );
    end
endgenerate

assign final_crc = stage[7];

```

26.9 Mathematical Verification

For a polynomial $P(x)$ and data bit d : - If feedback = 0: Simple left shift - If feedback = 1: Left shift XOR polynomial

This implements the mathematical CRC division in hardware.

26.10 Applications

- Building block for CRC calculation engines
- Serial CRC processors
- Component in parallel CRC architectures
- Educational CRC implementations
- Custom protocol CRC calculators

26.11 Performance Characteristics

- **Latency:** Combinational (zero clock cycles)
- **Throughput:** One bit per clock when used in sequential system
- **Area:** Minimal - few XOR gates and multiplexers
- **Power:** Low - simple combinational logic

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27 dataint_ecc_hamming_decode_secded Module Documentation

27.1 Purpose

The `dataint_ecc_hamming_decode_secded` module implements a Hamming decoder with Single Error Correction, Double Error Detection (SECDDED) capability. It decodes Hamming-encoded data, detects and corrects single-bit errors, and flags double-bit errors.

27.2 Module Declaration

```
module dataint_ecc_hamming_decode_secded #(
    parameter int WIDTH = 4,
    parameter int DEBUG = 0
) (
    input logic                clk,
    input logic                rst_n,
    input logic                enable,
    input logic [WIDTH+ParityBits:0] hamming_data,
    output logic [WIDTH-1:0] data,
    output logic                error_detected,
    output logic                double_error_detected
);
```

27.3 Parameters

Parameter	Default	Description
WIDTH	4	Width of original data in bits
DEBUG	0	Enable debug output (1=enabled, 0=disabled)

27.4 Calculated Parameters

Parameter	Formula	Description
ParityBits	$\lceil \log_2(WIDTH + 1) \rceil$	Number of Hamming parity

Parameter	Formula	Description
	$\$clog2(WIDTH) + 1$	bits
TotalWidth	$WIDTH + ParityBits + 1$	Total encoded data width

27.5 Ports

Port	Direction	Width	Description
clk	Input	1	Clock signal
rst_n	Input	1	Active-low asynchronous reset
enable	Input	1	Enable decoding operation
hamming_data	Input	TotalWidth	Encoded data with ECC bits
data	Output	WIDTH	Decoded original data
error_detected	Output	1	Single or double error detected
double_error_detected	Output	1	Double error specifically detected

27.6 Functionality

27.6.1 Decoding Process

1. **Syndrome Generation:** Calculate parity check syndrome
2. **Overall Parity Check:** Verify SECDED bit
3. **Error Classification:** Determine error type and location
4. **Error Correction:** Fix single-bit errors
5. **Data Extraction:** Extract corrected original data

27.6.2 Error Detection Logic

Overall Parity	Syndrome	Error Type	Action
Match	Zero	No Error	Pass data through

Overall Parity	Syndrome	Error Type	Action
Mismatch	Non-zero	Single Error (Hamming)	Correct using syndrome
Mismatch	Zero	Single Error (SECDED)	No correction needed
Match	Non-zero	Double Error	Flag error, no correction

27.7 Implementation Details

27.7.1 Key Functions

27.7.1.1 *bit_position(k)* Function

Calculates the position of data bit *k* in the encoded stream:

```
function automatic integer bit_position(input integer k);
    integer j, pos;
    begin
        pos = k + 1; // Account for parity bit at position 0
        for (j = 0; j < ParityBits; j++) begin
            if (pos >= (2 ** j)) pos = pos + 1; // Skip parity
positions
        end
        bit_position = pos - 1; // Convert to 0-based index
    end
endfunction
```

27.7.1.2 *get_covered_bits(parity_bit)* Function

Returns bitmask of positions covered by a specific parity bit:

```
function automatic [TotalWidth-1:0] get_covered_bits(input integer
parity_bit);
    integer j;
    begin
        get_covered_bits = 0;
        // FIXED: Exclude SECDED bit from Hamming parity calculations
        for (j = 0; j < TotalWidth - 1; j++) begin
            if (|(((j + 1) >> parity_bit) & 1))
                get_covered_bits[j] = 1'b1;
        end
    end
endfunction
```

27.7.2 Syndrome Generation

```
always_comb begin : create_syndrome_covered_bits
    if (enable) begin
        for (i = 0; i < ParityBits; i++) begin
            parity_pos = (2 ** i) - 1; // Parity bit positions
            w_syndrome_in[i] = hamming_data[parity_pos]; // Stored
            parity
            w_syndrome[i] = 1'b0;

            w_covered_bits = get_covered_bits(i);
            for (bit_index = 0; bit_index < TotalWidth; bit_index++)
                begin
                    // Calculate parity over covered bits (excluding
                    parity bit itself)
                    if (w_covered_bits[bit_index] && (bit_index !=
                    parity_pos))
                        w_syndrome[i] = w_syndrome[i] ^
                        hamming_data[bit_index];
                    end

                    // Compare with stored parity to get syndrome
                    w_syndrome[i] = w_syndrome[i] ^ w_syndrome_in[i];
                end
            end
        end
    end
end
```

27.7.3 Overall Parity Check

```
always_comb begin : check_overall_parity
    w_overall_parity_in = hamming_data[TotalWidth-1]; // SECDED bit
    if (enable) begin
        w_overall_parity = ^hamming_data[TotalWidth-2:0]; //
        Calculated parity
    end else begin
        w_overall_parity = 'b0;
    end
end
```

27.8 State Machine Operation

27.8.1 Control States

The module operates under enable control with these logical states:

1. **IDLE:** enable = 0 - No processing
2. **DECODE:** enable = 1 - Active decoding and error correction

27.8.2 Error Correction Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_data_with_parity <= 'b0;
        error_detected <= 'b0;
        double_error_detected <= 'b0;
    end else if (enable) begin
        r_data_with_parity <= hamming_data; // Default assignment
        error_detected <= 'b0;
        double_error_detected <= 'b0;

        // SECEDED error detection logic
        if (w_overall_parity != w_overall_parity_in) begin
            error_detected <= 1'b1;

            if (w_syndrome != {ParityBits{1'b0}}) begin
                // Single-bit error in Hamming data - correct it
                r_data_with_parity[w_syndrome_0_based] <=
                    ~hamming_data[w_syndrome_0_based];
            end
            // Single-bit error in SECEDED bit - no correction needed

        end else if (w_syndrome != {ParityBits{1'b0}}) begin
            // Double-bit error detected
            error_detected <= 1'b1;
            double_error_detected <= 1'b1;
        end
    end
end
```

27.9 Key Features

27.9.1 Error Correction Capabilities

- **Single Error Correction:** Automatically fixes single-bit errors
- **Double Error Detection:** Flags but cannot correct double-bit errors
- **Error Location:** Syndrome points to exact error location
- **SECEDED Protection:** Additional parity bit improves error detection

27.9.2 Performance Characteristics

- **Latency:** One clock cycle for error correction
- **Throughput:** One word per clock cycle

- **Detection Rate:** 100% for single and double errors

27.9.3 Debug Support

When `DEBUG != 0`: - Displays parity calculations - Shows syndrome values - Reports error detection details - Useful for verification and debugging

27.10 Error Handling Details

27.10.1 Syndrome Interpretation

- **Syndrome = 0:** No error in Hamming bits
- **Syndrome $\neq 0$:** Points to error bit position (1-based)
- **Zero-based Syndrome:** `w_syndrome_0_based = w_syndrome - 1`

27.10.2 SECDED Bit Function

The SECDED bit enables distinguishing between: - Single-bit errors (correctable) - Double-bit errors (detectable but uncorrectable) - No errors

27.10.3 Error Correction Process

1. Calculate syndrome from received parity bits
2. Check overall parity (SECDED bit)
3. If single error detected, flip bit at syndrome position
4. If double error detected, flag but don't correct

27.11 Usage Examples

27.11.1 Basic Decoder

```
dataint_ecc_hamming_decode_secDED #(
    .WIDTH(8),
    .DEBUG(0)
) hamming_decoder (
    .clk(clk),
    .rst_n(rst_n),
    .enable(decode_enable),
    .hamming_data(encoded_input),
    .data(decoded_data),
    .error_detected(single_or_double_error),
    .double_error_detected(uncorrectable_error)
);
```


27.11.2 Error Handling System

```
always_ff @(posedge clk) begin
    if (error_detected) begin
        if (double_error_detected) begin
            // Log uncorrectable error
            error_log <= error_log + 1;
            // Request retransmission
            request_retransmit <= 1'b1;
        end else begin
            // Log corrected error
            corrected_errors <= corrected_errors + 1;
            // Continue with corrected data
        end
    end
end
```

27.12 Data Extraction

27.12.1 Generate Block for Data Output

```
genvar j;
generate
    for (j = 0; j < WIDTH; j++) begin : gen_block
        assign data[j] = r_data_with_parity[bit_position(j)];
    end
endgenerate
```

This extracts only the original data bits from the corrected encoded word.

27.13 Applications

- ECC memory systems
- Communication error correction
- Storage system integrity
- Real-time data processing
- Safety-critical systems
- Network protocol implementation

27.14 Related Modules

- **dataint_ecc_hamming_encode_secdded**: Corresponding encoder

- Used together for complete ECC system implementation

27.15 Performance Considerations

- **Critical Path:** Syndrome calculation and error correction
- **Pipeline Stages:** Consider pipelining for high-frequency operation
- **Area:** Scales with data width and parity requirements
- **Power:** Moderate due to combinational parity calculations

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28 dataint_ecc_hamming_encode_secded Module Documentation

28.1 Purpose

The `dataint_ecc_hamming_encode_secded` module implements a Hamming encoder with Single Error Correction, Double Error Detection (SECDDED) capability. It generates error correction codes that can detect up to 2-bit errors and correct single-bit errors in data.

28.2 Module Declaration

```
module dataint_ecc_hamming_encode_secded #(
    parameter int WIDTH = 4,
    parameter int DEBUG = 0
) (
    input logic [WIDTH-1:0] data,
    output logic [TotalWidth-1:0] encoded_data
);
```

28.3 Parameters

Parameter	Default	Description
WIDTH	4	Width of input data in bits
DEBUG	0	Enable debug output

Parameter	Default	Description
		(1=enabled, 0=disabled)

28.4 Calculated Parameters

Parameter	Formula	Description
ParityBits	$\$clog2(WIDTH + \$clog2(WIDTH) + 1)$	Number of Hamming parity bits
TotalWidth	$WIDTH + ParityBits + 1$	Total encoded data width (includes SECDED bit)

28.5 Ports

Port	Direction	Width	Description
data	Input	WIDTH	Original data to be encoded
encoded_data	Output	TotalWidth	Encoded data with parity and SECDED bits

28.6 Functionality

28.6.1 Hamming Code Structure

The encoded data contains: 1. **Data Bits**: Original data placed at specific positions 2. **Parity Bits**: Hamming parity bits at power-of-2 positions (1, 2, 4, 8, ...) 3. **SECDED Bit**: Additional parity bit for double error detection

28.6.2 Bit Position Layout

```
Position: 1  2  3  4  5  6  7  8  9 10 11 12 ...
Content:  P1 P2 D1 P4 D2 D3 D4 P8 D5 D6 D7 D8 ...
```

Where: P = Parity bit, D = Data bit

28.7 Implementation Details

28.7.1 Key Functions

28.7.1.1 *bit_position(k)* Function

Calculates where data bit k should be placed in the encoded stream:

```

function automatic integer bit_position(input integer k);
    integer j, pos;
    begin
        pos = k + 1; // Start at k+1 to account for parity at
        position 0
        for (j = 0; j < ParityBits; j++) begin
            if (pos >= (2 ** j)) pos = pos + 1; // Skip parity
            positions
        end
        bit_position = pos - 1; // Convert to 0-based index
    end
endfunction

```

28.7.1.2 get_covered_bits(parity_bit) Function

Returns a bitmask of positions covered by a specific parity bit:

```

function automatic [TotalWidth-1:0] get_covered_bits(input integer
parity_bit);
    integer j;
    begin
        get_covered_bits = 'b0;
        for (j = 0; j < TotalWidth; j++) begin
            // Check if position j+1 has parity_bit set in its binary
            representation
            if (((j + 1) >> parity_bit) & 1)
                get_covered_bits[j] = 1'b1;
        end
    end
endfunction

```

28.7.2 Encoding Algorithm

28.7.2.1 Step 1: Data Placement

```

// Initialize with zeros
w_data_with_parity = {TotalWidth{1'b0}};

// Insert data bits into correct positions
for (i = 0; i < WIDTH; i++) begin
    w_data_with_parity[bit_position(i)] = data[i];
end

```

28.7.2.2 Step 2: Parity Calculation

```

// Calculate each Hamming parity bit
for (i = 0; i < ParityBits; i++) begin
    parity_pos = (2 ** i) - 1; // Positions 0, 1, 3, 7, 15, ...
    w_data_with_parity[parity_pos] = 1'b0; // Initialize

    // XOR all covered positions

```

```

    w_covered_bits = get_covered_bits(i);
    for (bit_index = 0; bit_index < TotalWidth; bit_index++) begin
        if (w_covered_bits[bit_index]) begin
            w_data_with_parity[parity_pos] =
                w_data_with_parity[parity_pos] ^
w_data_with_parity[bit_index];
            end
        end
    end
end

```

28.7.2.3 Step 3: SECDED Bit Calculation

```

// Calculate overall parity for SECDED
w_data_with_parity[TotalWidth-1] = ^w_data_with_parity[TotalWidth-
2:0];

```

28.8 Key Features

28.8.1 Combinational Design

- **No Clock Required:** Purely combinational logic
- **Immediate Output:** Encoded data available immediately after input change
- **Stateless Operation:** No internal state or memory

28.8.2 Scalable Architecture

- **Parameterizable Width:** Supports any data width
- **Automatic Sizing:** Parity bits calculated automatically
- **Efficient Layout:** Optimal bit positioning for Hamming code

28.8.3 Debug Support

When `DEBUG != 0`: - Function calls display bit position calculations - Covered bits masks are displayed - Useful for verification and understanding

28.9 Example Configurations

28.9.1 4-bit Data (Common Example)

- Data bits: 4
- Parity bits: 3 (positions 0, 1, 3)
- SECDED bit: 1 (position 7)

- Total width: 8 bits

28.9.28-bit Data

- Data bits: 8
- Parity bits: 4 (positions 0, 1, 3, 7)
- SECDDED bit: 1 (position 12)
- Total width: 13 bits

28.9.3 16-bit Data

- Data bits: 16
- Parity bits: 5 (positions 0, 1, 3, 7, 15)
- SECDDED bit: 1 (position 21)
- Total width: 22 bits

28.10 Usage Example

28.10.1 Basic Encoding

```
// 8-bit data encoder
dataint_ecc_hamming_encode_secded #(
    .WIDTH(8),
    .DEBUG(0)
) hamming_encoder (
    .data(input_data),
    .encoded_data(ecc_encoded_data)
);
```

28.10.2 With Debug

```
// Debug-enabled encoder for verification
dataint_ecc_hamming_encode_secded #(
    .WIDTH(4),
    .DEBUG(1)
) debug_encoder (
    .data(test_data),
    .encoded_data(encoded_output)
);
```

28.11 Error Correction Capabilities

28.11.1 Single Error Correction

- Can correct any single-bit error in the encoded data
- Uses Hamming parity bits to identify error position
- Decoder can flip the erroneous bit

28.11.2 Double Error Detection

- SECEDED bit enables detection of double-bit errors
- Cannot correct double errors, but can flag them
- Prevents silent data corruption

28.11.3 Error Detection Matrix

Error Type	Detection	Correction
No Error	✓	N/A
Single Bit Error	✓	✓
Double Bit Error	✓	✗
Triple+ Bit Error	Partial	✗

28.12 Applications

- Memory systems (ECC RAM)
- Communication protocols
- Storage systems
- Critical data processing
- Aerospace and automotive systems
- Any application requiring data integrity

28.13 Related Modules

- `dataint_ecc_hamming_decode_secDED`: Corresponding decoder module
- Used together for complete SECEDED error correction system

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29 dataint_parity Module Documentation

29.1 Purpose

The dataint_parity module implements a generic parity generator and checker. It can generate parity bits for data chunks and verify existing parity bits, supporting both even and odd parity schemes across multiple data segments.

29.2 Module Declaration

```
module dataint_parity #(
    parameter int CHUNKS = 4, // Number of chunks to check parity
    parameter int WIDTH  = 32 // Total width of the data
) (
    input logic [WIDTH-1:0] data_in, // Data input
    input logic [CHUNKS-1:0] parity_in, // Parity input for
checking
    input logic parity_type, // 1=even, 0=odd
    output logic [CHUNKS-1:0] parity, // Generated parity bits
    output logic [CHUNKS-1:0] parity_err // Error indicators
);
```

29.3 Parameters

Parameter	Default	Description
CHUNKS	4	Number of data chunks to process
WIDTH	32	Total width of input data in bits

29.4 Calculated Parameters

Parameter	Formula	Description
ChunkSize	WIDTH / CHUNKS	Size of each chunk in bits
ExtraBits	WIDTH % CHUNKS	Remaining bits if WIDTH not evenly divisible

29.5 Ports

Port	Direction	Width	Description
data_in	Input	WIDTH	Input data to calculate/check parity
parity_in	Input	CHUNKS	Expected parity bits for verification
parity_type	Input	1	Parity type (1=even, 0=odd)
parity	Output	CHUNKS	Calculated parity bits
parity_err	Output	CHUNKS	Error flags (1=parity mismatch)

29.6 Functionality

29.6.1 Dual Operation Modes

The module simultaneously: 1. **Generates Parity**: Calculates parity for each data chunk 2. **Checks Parity**: Compares calculated parity with input parity

29.6.2 Parity Types

- **Even Parity** (parity_type = 1): Parity bit makes total 1s even
- **Odd Parity** (parity_type = 0): Parity bit makes total 1s odd

29.6.3 Data Chunking

Data is divided into chunks with special handling for non-even divisions: - Most chunks: ChunkSize bits each - Last chunk: May include extra bits if $WIDTH \% CHUNKS \neq 0$

29.7 Implementation Details

29.7.1 Generate Block Architecture

```
genvar i;  
generate  
  for (i = 0; i < CHUNKS; i++) begin : gen_parity
```

```

// Calculate bounds for each chunk
localparam int LowerBound = i * ChunkSize;
localparam int UpperBound = (i < CHUNKS - 1) ?
    ((i + 1) * ChunkSize) - 1 : WIDTH - 1;

// Calculate parity for this chunk
wire calculated_parity = parity_type ?
    ^data_in[UpperBound:LowerBound] :           // Even parity
    ~^data_in[UpperBound:LowerBound];           // Odd parity

assign parity[i] = calculated_parity;
assign parity_err[i] = (calculated_parity != parity_in[i]);
end
endgenerate

```

29.7.2 Chunk Boundary Calculation

Each chunk's boundaries are calculated statically: - **Regular Chunks** ($i < \text{CHUNKS}-1$): Fixed size chunks - **Last Chunk** ($i = \text{CHUNKS}-1$): Includes any remaining bits

29.7.3 Parity Calculation Logic

For each chunk: 1. **XOR Reduction**: $\text{^data_in}[\text{UpperBound}:\text{LowerBound}]$ 2. **Parity Type Application**: - Even parity: Use XOR result directly - Odd parity: Invert XOR result

29.7.4 Error Detection

Error detection compares calculated vs. expected parity:

```
assign parity_err[i] = (calculated_parity != parity_in[i]);
```

29.8 Key Features

29.8.1 Scalable Architecture

- **Parameterizable Chunks**: Any number of data segments
- **Flexible Data Width**: Supports any total data width
- **Automatic Sizing**: Handles non-even chunk divisions

29.8.2 Simultaneous Operation

- **Generation and Checking**: Both operations happen concurrently
- **Independent Chunks**: Each chunk operates independently
- **Parallel Processing**: All chunks processed simultaneously

29.8.3 Parity Flexibility

- **Runtime Configurable:** Parity type can change during operation
- **Per-Module Setting:** All chunks use same parity type
- **Standard Support:** Supports both common parity schemes

29.9 Example Configurations

29.9.1 32-bit Data, 4 Chunks (8 bits each)

```
Chunk 0: data_in[7:0]    → parity[0], parity_err[0]
Chunk 1: data_in[15:8]   → parity[1], parity_err[1]
Chunk 2: data_in[23:16]  → parity[2], parity_err[2]
Chunk 3: data_in[31:24]  → parity[3], parity_err[3]
```

29.9.2 30-bit Data, 4 Chunks (uneven division)

```
Chunk 0: data_in[6:0]    (7 bits) → parity[0], parity_err[0]
Chunk 1: data_in[13:7]   (7 bits) → parity[1], parity_err[1]
Chunk 2: data_in[20:14]  (7 bits) → parity[2], parity_err[2]
Chunk 3: data_in[29:21]  (9 bits) → parity[3], parity_err[3]
```

29.10 Usage Examples

29.10.1 Basic Parity Generation

```
dataint_parity #(
    .CHUNKS(8),
    .WIDTH(64)
) parity_gen (
    .data_in(input_data),
    .parity_in(8'b0),           // Not used for generation
    .parity_type(1'b1),        // Even parity
    .parity(generated_parity),
    .parity_err()              // Not used for generation
);
```

29.10.2 Parity Checking

```
dataint_parity #(
    .CHUNKS(4),
    .WIDTH(32)
) parity_check (
    .data_in(received_data),
    .parity_in(received_parity),
    .parity_type(1'b0),        // Odd parity
    .parity(recalc_parity),
    .parity_err(error_flags)
```

```
);
```

```
// Check for any errors
```

```
wire any_parity_error = |error_flags;
```

29.10.3 Error Correction System

```
logic [CHUNKS-1:0] error_location;
```

```
logic [WIDTH-1:0] corrected_data;
```

```
dataint_parity #(.CHUNKS(CHUNKS), .WIDTH(WIDTH)) parity_checker (  
    .data_in(received_data),  
    .parity_in(received_parity),  
    .parity_type(even_parity),  
    .parity(recalc_parity),  
    .parity_err(error_location)  
);
```

```
// Simple error correction (single-bit errors only)
```

```
always_comb begin
```

```
    corrected_data = received_data;
```

```
    for (int i = 0; i < CHUNKS; i++) begin
```

```
        if (error_location[i]) begin
```

```
            // Flip bits in the erroneous chunk
```

```
            // (Simplified - real correction needs more sophisticated
```

```
logic)
```

```
        end
```

```
    end
```

```
end
```

29.11 Performance Characteristics

29.11.1 Timing

- **Combinational Logic:** Zero clock delay
- **Critical Path:** XOR tree depth $\approx \log_2(\text{ChunkSize})$
- **Propagation Delay:** Minimal for typical chunk sizes

29.11.2 Area

- **Linear Scaling:** Area increases with CHUNKS and WIDTH
- **Efficient Implementation:** Simple XOR trees
- **Low Overhead:** Minimal logic per chunk

29.11.3 Power

- **Low Power:** Simple combinational logic
- **Activity Dependent:** Power scales with data switching

29.12 Applications

29.12.1 Error Detection

- Memory parity checking
- Communication protocol verification
- Data bus integrity monitoring
- Storage system error detection

29.12.2 Data Integrity

- Real-time parity generation
- Concurrent error checking
- Multi-segment data validation
- Protocol compliance verification

29.12.3 System Integration

- Building block for ECC systems
- Component in larger error correction schemes
- Interface protection
- Debug and validation tools

29.13 Design Considerations

29.13.1 Chunk Size Selection

- **Power of 2:** Often preferred for alignment
- **Protocol Requirements:** Match system requirements
- **Error Granularity:** Smaller chunks = finer error localization

29.13.2 Parity Type Selection

- **Even Parity:** More common in digital systems
- **Odd Parity:** Provides all-zeros detection
- **System Compatibility:** Match existing standards

29.13.3 Width Considerations

- **Alignment:** Consider data bus alignment
- **Extra Bits:** Handle remainder bits appropriately
- **Performance:** Balance chunk count vs. granularity

29.14 Related Applications

- **ECC Systems:** Building block for Hamming codes
- **Network Protocols:** Ethernet, UART parity
- **Memory Systems:** DRAM/SRAM parity protection
- **Communication:** RS-232, SPI, I2C error detection

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30 Debounce Module (debounce . sv)

30.1 Purpose

Eliminates mechanical switch bounce by sampling button inputs at regular intervals and requiring a stable state for multiple consecutive samples before considering the button state as valid.

30.2 Ports

30.2.1 Input Ports

- **clk** - System clock signal

- **rst_n** - Active-low reset signal
- **long_tick** - ~10ms sampling tick signal (controls sampling rate)
- **button_in[N-1:0]** - Raw button input signals to be debounced

30.2.2 Output Ports

- **button_out[N-1:0]** - Debounced button output signals

30.2.3 Parameters

- **N** - Number of buttons/input signals (default: 4)
- **DEBOUNCE_DELAY** - Number of consecutive stable samples required (default: 4)
- **PRESSED_STATE** - Logic level when button is pressed (1 for normally open, 0 for normally closed)

30.3 Implementation Details

30.3.1 Core Algorithm

The module uses shift registers to track the history of each button's state:

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        for (int i = 0; i < N; i++) begin
            r_shift_regs[i] <= {DEBOUNCE_DELAY{1'b0}};
        end
    end else if (long_tick) begin
        for (int i = 0; i < N; i++) begin
            r_shift_regs[i] <= {
                r_shift_regs[i][DEBOUNCE_DELAY-2:0],
                PRESSED_STATE ? button_in[i] : ~button_in[i]
            };
        end
    end
end
```

30.3.2 Key Features

30.3.2.1 Sampling Control

- Only samples inputs when **long_tick** is asserted

- Prevents over-sampling which could cause false triggering
- Typical long_tick period: ~10ms

30.3.2.2 *Button Type Support*

- **Normally Open (NO):** PRESSED_STATE = 1
 - Button reads '0' when not pressed, '1' when pressed
- **Normally Closed (NC):** PRESSED_STATE = 0
 - Button reads '1' when not pressed, '0' when pressed
- Input inversion handles NC buttons transparently

30.3.2.3 *Debounce Logic*

- Maintains DEBOUNCE_DELAY samples for each button
- Output goes high only when all samples in shift register are '1'
- This ensures the button has been stable in pressed state for required duration

```
always_comb begin
    for (int i = 0; i < N; i++) begin
        // Output 1 when shift register shows stable pressed state
        (all 1s)
        w_debounced_signals[i] = &r_shift_regs[i];
    end
end
```

30.3.2.4 *Output Registration*

- Final output is registered to provide clean, glitch-free transitions
- Resets to all zeros on system reset

30.4 Timing Characteristics

30.4.1 Debounce Delay

- **Total delay** = DEBOUNCE_DELAY × long_tick period
- **Default:** 4 × 10ms = 40ms debounce time
- **Minimum stable time:** Button must remain stable for full delay period

30.4.2 Response Time

- **Press detection:** DEBOUNCE_DELAY ticks after button stabilizes
- **Release detection:** DEBOUNCE_DELAY ticks after button releases
- **Asymmetric:** Same delay for both press and release

30.5 Use Cases

- Mechanical pushbuttons and switches
- Rotary encoder inputs (with appropriate delay settings)
- Any digital input subject to contact bounce
- Both normally open and normally closed switch types

30.6 Design Considerations

- **Delay tuning:** Adjust DEBOUNCE_DELAY based on switch characteristics
- **Tick frequency:** long_tick should be much slower than bounce duration
- **Multiple buttons:** All buttons share same debounce parameters
- **Reset behavior:** All outputs go low on reset, regardless of button states

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31 Decoder Module (decoder . sv)

31.1 Purpose

Converts an M-bit binary encoded input to a one-hot N-bit output where $N = 2^M$. This is a standard binary-to-one-hot decoder implementation.

31.2 Ports

31.2.1 Input Ports

- **encoded[M-1:0]** - Binary encoded input value

31.2.2 Output Ports

- **data[N-1:0]** - One-hot decoded output vector

31.2.3 Parameters

- **M** - Width of binary input (default: 4)
- **N** - Width of one-hot output (default: $2^M = 16$)

31.3 Implementation Details

31.3.1 Core Logic

The decoder uses a generate loop to create comparison logic for each output bit:

```
assign data = 0; // Initialize all outputs to 0

genvar i;
generate
    for (i = 0; i < N; i++) begin : gen_DECODER_LOOP
        assign data[i] = (encoded == i) ? 1'b1 : 1'b0;
    end
endgenerate
```

31.3.2 Operation Principle

- Each output bit corresponds to one possible input value
- Only one output bit is high at any time (one-hot encoding)
- Output bit *i* is high when encoded input equals *i*

31.4 Functional Behavior

31.4.1 Truth Table Example (M=2, N=4)

encoded[1:0]	data[3]	data[2]	data[1]	data[0]
00	0	0	0	1
01	0	0	1	0
10	0	1	0	0
11	1	0	0	0

31.4.2 Key Characteristics

- **Combinational logic:** No clock dependency, immediate response

- **One-hot output:** Exactly one bit high for valid inputs
- **Complete decoding:** All possible input combinations decoded
- **Default initialization:** All outputs start at 0

31.5 Design Features

31.5.1 Parameterization

- **Scalable width:** M parameter controls input width
- **Automatic sizing:** N automatically calculated as 2^M
- **Generate loops:** Synthesizes efficiently for any size

31.5.2 Synthesis Considerations

- **Resource usage:** Typically implements as LUT-based logic
- **Propagation delay:** Single LUT delay in most FPGA architectures
- **Fan-out:** Each input bit fans out to multiple output comparisons

31.6 Use Cases

- **Address decoding:** Memory or register select signals
- **State machine outputs:** Decode binary state to control signals
- **Multiplexer control:** Generate select signals for data routing
- **Interrupt controllers:** Decode interrupt vectors
- **Bus decoding:** Generate chip select signals

31.7 Common Applications

31.7.1 Memory Address Decoding

```
decoder #(.M(2), .N(4)) addr_decoder (
    .encoded(address[1:0]),
    .data(chip_select) // 4 chip selects
);
```

31.7.2 State Machine Output Decoding

```
decoder #(.M(3), .N(8)) state_decoder (  
    .encoded(current_state),  
    .data(state_outputs) // 8 control signals  
);
```

31.8 Timing Characteristics

- **Propagation delay:** Typically 1 LUT delay
- **Setup/hold:** None (purely combinational)
- **Output changes:** Immediately follow input changes

31.9 Related Modules

- **Encoder:** Performs inverse operation (one-hot to binary)
- **Priority Encoder:** Handles multiple simultaneous inputs
- **Multiplexer:** Often used together for data routing

31.10 Design Notes

- No error checking for invalid inputs (though all combinations are valid)
- Output initialization ensures clean power-up behavior
- Generate loop structure scales efficiently to any required size

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32 Encoder Module (encoder .sv)

32.1 Purpose

Converts a one-hot N-bit input to a binary encoded output. Performs the inverse operation of a decoder, encoding the position of the asserted bit into binary format.

32.2 Ports

32.2.1 Input Ports

- **decoded[N-1:0]** - One-hot input vector

32.2.2 Output Ports

- **data[\$clog2(N)-1:0]** - Binary encoded output representing the position of the set bit

32.2.3 Parameters

- **N** - Width of one-hot input vector (default: 8)

32.3 Implementation Details

32.3.1 Core Algorithm

The encoder uses a combinational always block with a priority search:

```
always_comb begin
    data = 0; // Default output when no bits set
    for (int i = 0; i < N; i++) begin
        if (decoded[i]) data = $clog2(N)'(i);
    end
end
```

32.3.2 Operation Principle

- Searches through input vector from LSB to MSB
- Outputs the binary representation of the highest indexed set bit
- Acts as a **priority encoder** when multiple bits are set (higher index wins)

32.4 Functional Behavior

32.4.1 Truth Table Example (N=8, output width=3)

decoded[7:0]	data[2:0]	Notes
00000001	000	Bit 0 set
00000010	001	Bit 1 set
00000100	010	Bit 2 set
00001000	011	Bit 3 set
00010000	100	Bit 4 set

decoded[7:0]	data[2:0]	Notes
00100000	101	Bit 5 set
01000000	110	Bit 6 set
10000000	111	Bit 7 set
00000000	000	No bits set
10000001	111	Multiple bits - highest wins

32.4.2 Key Characteristics

- **Priority encoding:** Higher-indexed bits take precedence
- **Zero default:** Outputs 0 when no input bits are set
- **Combinational:** Immediate response to input changes
- **Self-sizing:** Output width automatically calculated using $\lceil \log_2(N) \rceil$

32.5 Design Features

32.5.1 Automatic Width Calculation

- **Output width:** $\lceil \log_2(N) \rceil$ bits
- **Example:** $N=8 \rightarrow 3$ -bit output, $N=16 \rightarrow 4$ -bit output
- **Optimal sizing:** No wasted bits in output

32.5.2 Priority Behavior

The for-loop structure creates implicit priority: - Later iterations overwrite earlier results - Highest-indexed set bit determines final output - Behaves as priority encoder for invalid one-hot inputs

32.6 Use Cases

32.6.1 Valid One-Hot Inputs

- **Interrupt acknowledgment:** Convert interrupt vector to binary ID
- **Resource arbitration:** Encode granted request to requester ID
- **State machine encoding:** Convert one-hot state to binary

- **Position encoding:** Find position of active element

32.6.2 Priority Encoding Applications

- **Multiple interrupt handling:** Encode highest priority interrupt
- **Error reporting:** Report highest severity error condition
- **Resource allocation:** Grant to highest priority requester

32.7 Timing Characteristics

- **Propagation delay:** Depends on input width and synthesis
- **Critical path:** Through the for-loop comparisons
- **Setup/hold:** None (purely combinational)

32.8 Design Considerations

32.8.1 Input Validation

- **Assumes one-hot input** for standard encoder behavior
- **Handles multiple bits** gracefully (acts as priority encoder)
- **Zero input** produces zero output (may or may not be desired)

32.8.2 Synthesis Implications

- **Resource usage:** Typically synthesizes to multiplexer logic
- **Optimization:** Modern synthesizers optimize the for-loop efficiently
- **Scalability:** Performance degrades gradually with increasing N

32.9 Common Usage Patterns

32.9.1 Interrupt Controller

```
encoder #(.N(8)) int_encoder (  
    .decoded(interrupt_pending),  
    .data(interrupt_id)  
);
```

32.9.2 Arbiter Output Encoding

```
encoder #(.N(16)) grant_encoder (  
    .decoded(grant_vector),  
    .data(granted_id)  
);
```

32.10 Related Modules

- **Decoder:** Performs inverse operation (binary to one-hot)
- **Priority Encoder with Enable:** Enhanced version with enable control
- **Find First Set:** Similar functionality with different search order

32.11 Behavioral Notes

- **For-loop order:** Creates LSB-to-MSB priority (higher index wins)
- **Default assignment:** Ensures output has known value for all inputs
- **Type casting:** `$clog2(N) ' (i)` ensures proper bit width matching

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33 Priority Encoder with Enable (encoder_priority_enable.sv)

33.1 Purpose

Encodes the highest priority (highest index) asserted bit in the input vector, with an enable control signal. Unlike the basic encoder, this module explicitly searches from MSB to LSB for true priority encoding.

33.2 Ports

33.2.1 Input Ports

- **priority_in[WIDTH-1:0]** - Priority input vector (higher index = higher priority)

- **enable** - Enable signal (active high)

33.2.2 Output Ports

- **encode[\$clog2(WIDTH)-1:0]** - Binary encoded output of highest priority bit position

33.2.3 Parameters

- **WIDTH** - Input vector width (default: 8)

33.3 Implementation Details

33.3.1 Core Algorithm

logic found;

always_comb begin

// Default assignments to prevent latches

encode = '0;

found = 1'b0;

if (enable == 1'b1) **begin**

// Find the highest priority bit using found flag to prevent overwrites

for (int i = WIDTH-1; i >= 0; i--) **begin**

if (priority_in[i] == 1'b1 && !found) **begin**

encode = \$clog2(WIDTH)'(i);

found = 1'b1;

end

end

end

end

33.3.2 Operation Principle

- **True priority encoding:** Searches from MSB (highest priority) to LSB
- **Enable control:** Only operates when enable is asserted
- **Found flag:** Prevents multiple assignments, ensuring highest priority wins
- **Clean disable:** Outputs zero when disabled

33.4 Functional Behavior

33.4.1 Priority Resolution Example (WIDTH=8)

priority_in[7:0]	enable	encode[2:0]	Notes
00000001	1	000	Only bit 0 set
10000001	1	111	Bit 7 has highest priority
01010101	1	110	Bit 6 is highest set
11111111	1	111	Bit 7 wins among all
10000001	0	000	Disabled - output zero
00000000	1	000	No bits set

33.4.2 Key Differences from Basic Encoder

Feature	Basic Encoder	Priority Encoder with Enable
Search Direction	LSB to MSB	MSB to LSB
Priority	Higher index overwrites	True highest priority
Enable Control	None	Enable signal required
Found Flag	Implicit	Explicit prevention of overwrites

33.5 Design Features

33.5.1 True Priority Encoding

- **MSB-first search:** for (int i = WIDTH-1; i >= 0; i--)
- **Single assignment:** Found flag prevents multiple writes
- **Highest wins:** First match (highest index) determines output

33.5.2 Enable Control

- **Conditional operation:** Only active when enable = 1
- **Clean disable:** Output forced to zero when disabled
- **Power savings:** Can gate operation when not needed

33.5.3 Latch Prevention

// Default assignments to prevent latches

```
encode = '0;  
found = 1'b0;
```

- Ensures all signal paths have defined values
- Prevents synthesis warnings about incomplete assignments

33.6 Use Cases

33.6.1 Interrupt Controllers

```
encoder_priority_enable #(.WIDTH(8)) irq_encoder (  
    .priority_in(interrupt_requests),  
    .enable(interrupt_enable),  
    .encode(highest_priority_irq)  
);
```

33.6.2 Resource Arbitration

```
encoder_priority_enable #(.WIDTH(16)) arbiter (  
    .priority_in(request_vector),  
    .enable(arbitration_enable),  
    .encode(granted_requestor)  
);
```

33.6.3 Error Reporting

```
encoder_priority_enable #(.WIDTH(4)) error_encoder (  
    .priority_in({fatal_error, severe_error, warning, info}),  
    .enable(error_reporting_on),  
    .encode(error_level)  
);
```

33.7 Timing Characteristics

- **Propagation delay:** Depends on WIDTH and synthesis optimization
- **Critical path:** Through the priority resolution loop

- **Enable response:** Immediate when enable changes
- **Setup/hold:** None (purely combinational)

33.8 Design Considerations

33.8.1 Priority Assignment

- **Bit assignment:** Higher index = higher priority
- **Planning:** Assign critical functions to higher-indexed bits
- **Documentation:** Clear priority hierarchy essential

33.8.2 Performance Scaling

- **Linear complexity:** Search time increases with WIDTH
- **Synthesis optimization:** Modern tools optimize priority encoding well
- **Resource usage:** Typically maps to priority encoder primitives

33.8.3 Enable Usage Patterns

- **Power management:** Disable when priority encoding not needed
- **Conditional operation:** Enable based on system state
- **Test/debug:** Disable during certain test modes

33.9 Common Applications

33.9.1 Multi-Level Interrupt Controllers

- CPU interrupt prioritization
- Nested interrupt handling
- Real-time system scheduling

33.9.2 Bus Arbitration

- Multi-master bus systems
- DMA channel arbitration
- Memory access prioritization

33.9.3 Error and Exception Handling

- Exception priority resolution
- Multi-source error reporting
- Diagnostic priority encoding

33.10 Synthesis Considerations

- **Resource efficient:** Maps well to FPGA priority encoder resources
- **Timing closure:** Generally meets timing with proper constraints
- **Area scaling:** Grows approximately linearly with WIDTH

33.11 Related Modules

- **Basic Encoder:** Simpler version without enable or true priority
- **Find Last Set:** Similar MSB-first search functionality
- **Arbiter:** Often uses priority encoders for grant generation

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34 Asynchronous FIFO - Power of 2 (`fifo_async.sv`)

34.1 Purpose

Implements an asynchronous FIFO for safe clock domain crossing between different clock domains. **Restricted to power-of-2 depths only** due to Gray code pointer implementation.

34.2 Ports

34.2.1 Write Domain

- **wr_clk** - Write domain clock
- **wr_rst_n** - Write domain active-low reset
- **write** - Write enable signal

- **wr_data[DATA_WIDTH-1:0]** - Data to write
- **wr_full** - Write domain full flag
- **wr_almost_full** - Write domain almost full flag

34.2.2 Read Domain

- **rd_clk** - Read domain clock
- **rd_rst_n** - Read domain active-low reset
- **read** - Read enable signal
- **rd_data[DATA_WIDTH-1:0]** - Data read from FIFO
- **rd_empty** - Read domain empty flag
- **rd_almost_empty** - Read domain almost empty flag

34.2.3 Parameters

- **REGISTERED** - 0=mux mode, 1=flop mode for read output
- **DATA_WIDTH** - Width of data (default: 8)
- **DEPTH** - FIFO depth - **must be power of 2** (default: 16)
- **N_FLOP_CROSS** - Number of synchronizer stages (default: 2)
- **ALMOST_WR_MARGIN** - Almost full threshold (default: 1)
- **ALMOST_RD_MARGIN** - Almost empty threshold (default: 1)

34.3 Architecture Overview

34.3.1 Clock Domain Crossing Strategy

The FIFO uses **Gray code pointers** for safe clock domain crossing:

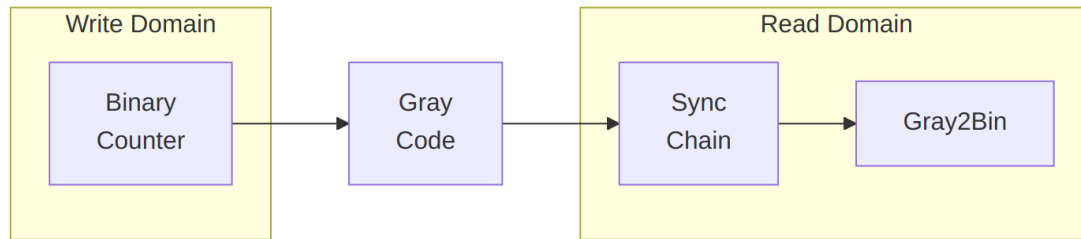


Diagram 2

34.3.2 Core Components

1. **Binary-Gray counters** (counter_bingray) for pointer generation
2. **Multi-stage synchronizers** (glitch_free_n_dff_arn) for CDC
3. **Gray-to-binary converters** (gray2bin) for pointer comparison
4. **Shared memory array** accessible from both domains
5. **FIFO control logic** for status flag generation

34.4 Implementation Deep Dive

34.4.1 Gray Code Pointer System

34.4.1.1 Why Gray Codes?

Gray codes ensure **only one bit changes** per increment: - **Binary**: 011 → 100 (3 bits change simultaneously) - **Gray**: 010 → 110 (only 1 bit changes) - **Benefit**: Eliminates metastability from multi-bit transitions

34.4.1.2 Pointer Architecture

```
// Write domain Gray counter
counter_bingray #(.WIDTH(AW + 1)) wr_ptr_counter_gray (
    .clk(wr_clk),
    .rst_n(wr_rst_n),
    .enable(write && !wr_full),
    .counter_bin(r_wr_ptr_bin),           // Binary for addressing
    .counter_bin_next(w_wr_ptr_bin_next), // Next binary value
    .counter_gray(r_wr_ptr_gray)         // Gray for CDC
);
```

34.4.1.3 Pointer Width Calculation

- **Address width**: $AW = \lceil \log_2(DEPTH) \rceil$
- **Pointer width**: $AW + 1$ (extra bit for wrap detection)

- **Example:** DEPTH=16 → AW=4, Pointer=5 bits

34.4.2 Clock Domain Crossing

34.4.2.1 Synchronizer Chains

```
// Cross read pointer to write domain
glitch_free_n_dff_arn #(
    .FLOP_COUNT(N_FLOP_CROSS),
    .WIDTH(AW + 1)
) rd_ptr_gray_cross_inst (
    .q(r_wdom_rd_ptr_gray),    // Synchronized output
    .d(r_rd_ptr_gray),        // Gray pointer input
    .clk(wr_clk),              // Destination clock
    .rst_n(wr_rst_n)           // Destination reset
);
```

34.4.2.2 Multi-Stage Synchronization

- **Default stages:** 2 flip-flops (N_FLOP_CROSS=2)
- **MTBF improvement:** Each stage reduces metastability probability
- **Latency trade-off:** More stages = better MTBF but higher latency

34.4.3 Memory Organization

34.4.3.1 Dual-Port Memory

```
logic [DW-1:0] r_mem[0:((1<<AW)-1)]; // Memory array

// Write port (write domain)
always_ff @(posedge wr_clk) begin
    if (write) begin
        r_mem[r_wr_addr] <= wr_data;
    end
end

// Read port (read domain)
assign w_rd_data = r_mem[r_rd_addr];
```

34.4.3.2 Address Generation

- **Write address:** r_wr_addr = r_wr_ptr_bin[AW-1:0]
- **Read address:** r_rd_addr = r_rd_ptr_bin[AW-1:0]
- **Truncation:** Uses only lower bits for memory indexing

34.4.4 Full/Empty Detection

34.4.4.1 Full Detection Logic

```
// In write domain
assign wr_full = (w_wdom_ptr_xor &&
                  (wr_ptr_bin[AW-1:0] == wdom_rd_ptr_bin[AW-1:0]));

// Where:
assign w_wdom_ptr_xor = wr_ptr_bin[AW] ^ wdom_rd_ptr_bin[AW];
```

34.4.4.2 Empty Detection Logic

```
// In read domain
assign rd_empty = (!w_rdom_ptr_xor_for_empty &&
                  (rd_ptr_bin[AW:0] == w_wr_ptr_for_empty[AW:0]));
```

34.4.4.3 Full/Empty Algorithm

- **Full condition:** MSBs differ AND LSBs equal (write caught up after wrap)
- **Empty condition:** All bits equal (pointers at same location)
- **MSB significance:** Indicates which pointer has wrapped around

34.5 Power-of-2 Requirement

34.5.1 Why Power-of-2 Only?

1. **Gray code properties:** Natural binary-Gray relationship
2. **Wraparound behavior:** Clean modulo- 2^n arithmetic
3. **Address truncation:** Simple bit slicing for memory addressing
4. **Pointer comparison:** Efficient full/empty detection

34.5.2 Limitation Impact

```
// Valid depths: 2, 4, 8, 16, 32, 64, 128, 256, ...
// Invalid depths: 3, 5, 6, 7, 9, 10, 12, 15, ...
```

34.6 Timing Considerations

34.6.1 Clock Domain Crossing Latency

- **Pointer propagation:** 2-3 clock cycles (depending on N_FLOP_CROSS)

- **Status flag delay:** Flags reflect state with synchronizer latency
- **Conservative design:** Prevents overflow/underflow despite delay

34.6.2 Metastability Protection

- **Gray code transitions:** Single bit changes only
- **Multi-stage sync:** Reduces MTBF exponentially
- **Setup/hold margins:** Proper timing constraints essential

34.7 Use Cases

34.7.1 Typical Applications

- **Video processing:** Different pixel and memory clock domains
- **Networking:** Packet buffers between different rate domains
- **Audio systems:** Sample rate conversion buffers
- **Microprocessor interfaces:** CPU and peripheral clock domains

34.7.2 When to Use vs. Alternatives

- **Use async FIFO when:** Clock domains are truly independent
- **Use sync FIFO when:** Single clock domain sufficient
- **Use async_div2 when:** Need non-power-of-2 depth

34.8 Design Guidelines

34.8.1 Depth Sizing

```
// Calculate minimum depth for burst handling
// DEPTH ≥ burst_size + synchronizer_latency + margin
parameter int MIN_DEPTH = 16; // Typical minimum for async FIFOs
```

34.8.2 Clock Relationship

- **Asynchronous clocks:** No phase relationship assumed
- **Clock gating:** Avoid gating clocks used by FIFO
- **Reset deassertion:** Ensure proper reset release sequencing

34.8.3 Almost Full/Empty Settings

- **Almost full:** DEPTH - ALMOST_WR_MARGIN
- **Almost empty:** ALMOST_RD_MARGIN
- **Guideline:** Set margins > synchronizer latency

34.9 Performance Characteristics

- **Throughput:** Up to 1 operation per clock per domain
- **Latency:** 0-1 cycles (depending on REGISTERED mode)
- **CDC latency:** 2-3 cycles for status propagation
- **Efficiency:** ~100% utilization possible

34.10 Error Detection Features

```
// Simulation-only error checking
always_ff @(posedge wr_clk) begin
    if (!wr_rst_n && (write && wr_full) == 1'b1) begin
        $display("Error: %s write while fifo full, %t", INSTANCE_NAME,
$time);
    end
end
```

34.11 WaveDrom Visualization

High-quality waveforms showcasing Gray code CDC mechanism are available!

Run the WaveDrom test to generate detailed timing diagrams:

```
# Generate Gray code waveforms (standard async FIFO implementation)
pytest val/common/test_fifo_async_wavedrom.py -v
```

Waveform Scenarios Generated:

1. **Write-Fill-Read-Empty Cycle**
 - Standard async FIFO operation
 - Gray code pointer progression
 - Power-of-2 depth utilization

2. Gray Code Synchronization

- Efficient Gray code CDC
- Logarithmic pointer width (vs. linear for Johnson)
- Cross-domain pointer transitions

3. Power-of-2 Depth Utilization

- Efficient addressing with power-of-2 depths
- Full depth wraparound behavior
- Resource-efficient pointer management

Key Differences from Johnson Counter Variant:

- **Pointer Width:** Logarithmic ($\lceil \log_2(\text{DEPTH}) \rceil + 1$) vs. linear (DEPTH) for Johnson
- **Depth Restriction:** Power-of-2 only vs. any even number for Johnson
- **Resource Efficiency:** Better for large depths (>32) vs. Johnson's flexibility for small depths

Comparison Tests:

- `test_fifo_sync_wavedrom.py` - Synchronous FIFO (single clock, no CDC)

34.12 Related Modules

- **USE_JOHNSON=1:** For non-power-of-2 depths, using Johnson counters (replaces the retired `fifo_async_div2`)
- **fifo_sync:** For single clock domain applications
- **counter_bingray:** Binary-Gray counter implementation
- **glitch_free_n_dff_arn:** Multi-stage synchronizer

34.13 Test and Verification

Comprehensive Test Suite: - `val/common/test_fifo_buffer_async.py` - Full functional verification - `val/common/test_fifo_async_wavedrom.py` - WaveDrom timing diagrams

Run Tests:

```
# Full functional test (basic/medium/full levels)
pytest val/common/test_fifo_buffer_async.py -v
```

```
# WaveDrom waveform generation
pytest val/common/test_fifo_async_wavedrom.py -v
```

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35 FIFO Control Logic (fifo_control.sv)

35.1 Purpose

Centralized control logic module that generates full/empty status flags for all FIFO variants (sync, async, async_div2). Handles the complex pointer arithmetic and mode-aware timing considerations.

35.2 Ports

35.2.1 Clock and Reset

- **wr_clk** - Write domain clock
- **wr_rst_n** - Write domain active-low reset
- **rd_clk** - Read domain clock
- **rd_rst_n** - Read domain active-low reset

35.2.2 Pointer Inputs

- **wr_ptr_bin[ADDR_WIDTH:0]** - Write pointer (binary, next value)
- **wdom_rd_ptr_bin[ADDR_WIDTH:0]** - Read pointer synchronized to write domain
- **rd_ptr_bin[ADDR_WIDTH:0]** - Read pointer (binary, next value)
- **rdom_wr_ptr_bin[ADDR_WIDTH:0]** - Write pointer synchronized to read domain

35.2.3 Status Outputs

- `count[ADDR_WIDTH:0]` - Current FIFO occupancy count
- `wr_full` - Write domain full flag
- `wr_almost_full` - Write domain almost full flag
- `rd_empty` - Read domain empty flag
- `rd_almost_empty` - Read domain almost empty flag

35.2.4 Parameters

- `ADDR_WIDTH` - Address width (default: 3)
- `DEPTH` - FIFO depth (default: 16)
- `ALMOST_WR_MARGIN` - Almost full threshold (default: 1)
- `ALMOST_RD_MARGIN` - Almost empty threshold (default: 1)
- `REGISTERED` - Output mode: 0=mux, 1=flop (default: 0)

35.3 Architecture Overview

35.3.1 Dual-Domain Design

The module operates across two clock domains: - **Write domain**: Generates full and almost_full flags - **Read domain**: Generates empty and almost_empty flags

35.3.2 Pointer Arithmetic Foundation

All status generation relies on **pointer comparison with wraparound detection**:

```
// XOR the MSBs to detect wraparound condition
assign w_wdom_ptr_xor = wr_ptr_bin[AW] ^ wdom_rd_ptr_bin[AW];
assign w_rdom_ptr_xor = rd_ptr_bin[AW] ^ rdom_wr_ptr_bin[AW];
```

35.4 Full Detection Logic

35.4.1 Basic Full Condition

```
assign w_wr_full_d = (w_wdom_ptr_xor &&
                     (wr_ptr_bin[AW-1:0] == wdom_rd_ptr_bin[AW-1:0]));
```

35.4.2 Full Detection Algorithm

- **Condition 1**: MSBs must differ (`w_wdom_ptr_xor = 1`)

- **Condition 2:** LSBs must be equal
- **Meaning:** Write pointer has “lapped” the read pointer

35.4.3 Visual Example (DEPTH=8, ADDR_WIDTH=3)

Pointers: [wrap_bit][address_bits]

Case 1: Not Full

wr_ptr = 0_101 (address 5, no wrap)

rd_ptr = 0_010 (address 2, no wrap)

→ MSBs same, not full

Case 2: Full

wr_ptr = 1_010 (address 2, wrapped once)

rd_ptr = 0_010 (address 2, not wrapped)

→ MSBs differ, addresses same = FULL

35.5 Almost Full Logic

35.5.1 Occupancy Calculation

```
assign w_almost_full_count = (w_wdom_ptr_xor) ?
    (AW'(D) - wdom_rd_ptr_bin[AW-1:0] + wr_ptr_bin[AW-1:0]) :
    (wr_ptr_bin[AW-1:0] - wdom_rd_ptr_bin[AW-1:0]);
```

35.5.2 Two Cases Handled

1. **No wraparound** ($w_wdom_ptr_xor = 0$):
 - Count = wr_ptr - rd_ptr
 - Simple subtraction
2. **Wraparound** ($w_wdom_ptr_xor = 1$):
 - Count = DEPTH - rd_ptr + wr_ptr
 - Accounts for circular buffer wraparound

35.5.3 Almost Full Threshold

```
assign w_wr_almost_full_d = w_almost_full_count >= AW'(AFT);
// Where AFT = DEPTH - ALMOST_WR_MARGIN
```

35.6 Empty Detection Logic - Mode Aware

35.6.1 Critical Innovation: Mode-Aware Write Pointer Selection

```
generate
  if (REGISTERED == 1) begin : gen_flop_mode
    // FLOP mode: Use previous cycle's write pointer
    logic [ADDR_WIDTH:0] r_rdom_wr_ptr_bin_delayed;

    always_ff @(posedge rd_clk or negedge rd_rst_n) begin
      if (!rd_rst_n) begin
        r_rdom_wr_ptr_bin_delayed <= '0;
      end else begin
        r_rdom_wr_ptr_bin_delayed <= rdom_wr_ptr_bin;
      end
    end

    assign w_wr_ptr_for_empty = r_rdom_wr_ptr_bin_delayed;
  end else begin : gen_mux_mode
    // MUX mode: Use current write pointer
    assign w_wr_ptr_for_empty = rdom_wr_ptr_bin;
  end
endgenerate
```

35.6.2 Why Mode-Aware Detection?

35.6.2.1 MUX Mode (REGISTERED = 0)

- **Data availability:** Immediate (combinational read)
- **Write pointer:** Use current value
- **Reasoning:** Data is immediately available when written

35.6.2.2 FLOP Mode (REGISTERED = 1)

- **Data availability:** Delayed by 1 cycle (registered read)
- **Write pointer:** Use delayed value
- **Reasoning:** Data isn't available until next clock cycle

35.6.3 Empty Detection Algorithm

```
assign w_rd_empty_d = (!w_rdom_ptr_xor_for_empty &&
  (rd_ptr_bin[AW:0] == w_wr_ptr_for_empty[AW:0]));
```

- **Condition 1:** MSBs must be same (no wrap difference)
- **Condition 2:** All bits must be equal

- **Meaning:** Read pointer has caught up to write pointer

35.7 Almost Empty Logic

35.7.1 Standard Timing (Mode-Independent)

Almost empty calculation uses standard timing regardless of FIFO mode:

```
assign w_almost_empty_count = (w_rdom_ptr_xor) ?
    (AW'(D) - rd_ptr_bin[AW-1:0] + rdom_wr_ptr_bin[AW-1:0]) :
    (rdom_wr_ptr_bin[AW-1:0] - rd_ptr_bin[AW-1:0]);
```

```
assign w_rd_almost_empty_d = w_almost_empty_count <= AW'(AET);
// Where AET = ALMOST_RD_MARGIN
```

35.8 Count Generation

35.8.1 Occupancy Count Logic

```
assign count = (w_rdom_ptr_xor) ?
    (rdom_wr_ptr_bin[AW-1:0] - rd_ptr_bin[AW-1:0] + AW'(D)) :
    (rdom_wr_ptr_bin[AW-1:0] - rd_ptr_bin[AW-1:0]);
```

35.8.2 Count Interpretation

- **Range:** 0 to DEPTH
- **Zero:** FIFO empty
- **DEPTH:** FIFO full
- **Uses:** Flow control, occupancy monitoring

35.9 Flag Registration

35.9.1 Write Domain Flags

```
always_ff @(posedge wr_clk, negedge wr_rst_n) begin
    if (!wr_rst_n) begin
        wr_full <= 'b0;
        wr_almost_full <= 'b0;
    end else begin
        wr_full <= w_wr_full_d;
        wr_almost_full <= w_wr_almost_full_d;
    end
end
```

35.9.2 Read Domain Flags

```
always_ff @(posedge rd_clk, negedge rd_rst_n) begin
    if (!rd_rst_n) begin
        rd_empty <= 'b1;           // Reset to empty
        rd_almost_empty <= 'b0;
    end else begin
        rd_empty <= w_rd_empty_d;
        rd_almost_empty <= w_rd_almost_empty_d;
    end
end
```

35.9.3 Reset Behavior

- **Full flags:** Reset to 0 (not full)
- **Empty flags:** Reset to 1 (empty)
- **Almost flags:** Reset to 0

35.10 Width Casting Fix

35.10.1 Type Width Matching

```
// Fixed: Cast D to AW-bit width to match other operands
assign w_almost_full_count = (w_wdom_ptr_xor) ?
    (AW'(D) - wdom_rd_ptr_bin[AW-1:0] + wr_ptr_bin[AW-1:0]) :
    (wr_ptr_bin[AW-1:0] - wdom_rd_ptr_bin[AW-1:0]);
```

The AW'(D) casting ensures all operands have matching bit widths, preventing synthesis warnings.

35.11 Key Design Insights

35.11.1 Wraparound Handling

- **MSB significance:** Extra bit detects wraparound events
- **Circular arithmetic:** Proper modulo DEPTH calculations
- **Comparison logic:** XOR-based wraparound detection

35.11.2 Timing Considerations

- **Synchronizer delay:** Accounted for in conservative design
- **Mode awareness:** Different timing for different output modes
- **Flag updates:** Registered for clean transitions

35.11.3 Conservative Design

- **Over-reporting:** Flags may assert slightly early
- **Safety margin:** Prevents overflow/underflow
- **Synchronizer latency:** Built into safety margins

35.12 Use Cases

- **All FIFO variants:** Shared by sync, async, and async_div2
- **Status monitoring:** Provides comprehensive FIFO state
- **Flow control:** Enables back-pressure and rate matching
- **Debug/verification:** Count output aids in debugging

35.13 Performance Impact

- **Minimal overhead:** Efficient pointer arithmetic
- **Low latency:** Single cycle flag updates
- **Resource efficient:** Shared across multiple FIFO types

35.14 Navigation

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36 Synchronous FIFO (fifo_sync.sv)

36.1 Purpose

Implements a synchronous First-In-First-Out buffer for single clock domain applications. Provides configurable depth, data width, and output modes with comprehensive status signaling.

36.2 Ports

36.2.1 Input Ports

- **clk** - System clock
- **rst_n** - Active-low reset

- **write** - Write enable signal
- **wr_data[DATA_WIDTH-1:0]** - Data to write into FIFO
- **read** - Read enable signal

36.2.2 Output Ports

- **wr_full** - Write domain full flag
- **wr_almost_full** - Write domain almost full flag
- **rd_data[DATA_WIDTH-1:0]** - Data read from FIFO
- **rd_empty** - Read domain empty flag
- **rd_almost_empty** - Read domain almost empty flag

36.2.3 Parameters

- **REGISTERED** - Output mode: 0=mux mode (combinational), 1=flop mode (registered)
- **DATA_WIDTH** - Width of data bus (default: 4)
- **DEPTH** - FIFO depth in words (default: 4)
- **ALMOST_WR_MARGIN** - Almost full threshold (default: 1)
- **ALMOST_RD_MARGIN** - Almost empty threshold (default: 1)
- **INSTANCE_NAME** - Debug identifier (default: "DEADF1F0")

36.3 Architecture Overview

36.3.1 Core Components

1. **Binary counters** for read/write pointers
2. **Memory array** for data storage
3. **FIFO control logic** for status generation
4. **Output multiplexing** based on REGISTERED parameter

36.3.2 Memory Organization

```
logic [DW-1:0] r_mem[0:((1<<AW)-1)]; // Memory array
assign r_wr_addr = r_wr_ptr_bin[AW-1:0]; // Write address
assign r_rd_addr = r_rd_ptr_bin[AW-1:0]; // Read address
```

36.4 Implementation Details

36.4.1 Pointer Management

Uses binary counters for both read and write pointers:

```
counter_bin #(.WIDTH(AW + 1), .MAX(D)) write_pointer_inst (
    .clk(clk),
    .rst_n(rst_n),
    .enable(write && !wr_full),
    .counter_bin_curr(r_wr_ptr_bin),
    .counter_bin_next(w_wr_ptr_bin_next)
);
```

36.4.1.1 Pointer Characteristics

- **Width:** $\$clog2(DEPTH) + 1$ bits (extra bit for wrap detection)
- **Increment:** Only when operation is valid (write && !full, read && !empty)
- **Wraparound:** Automatic modulo DEPTH counting

36.4.2 Memory Operations

36.4.2.1 Write Operation

```
always_ff @(posedge clk) begin
    if (write) begin
        r_mem[r_wr_addr] <= wr_data;
    end
end
```

36.4.2.2 Read Operation - Dual Mode

```
generate
    if (REGISTERED != 0) begin : gen_flop_mode
        // Flop mode - registered output
        always_ff @(posedge clk or negedge rst_n) begin
            if (!rst_n)
                rd_data <= 'b0;
            else
                rd_data <= w_rd_data;
        end
    end else begin : gen_mux_mode
        // Mux mode - non-registered output
```

```

        assign rd_data = w_rd_data;
    end
endgenerate

```

36.4.3 Status Flag Generation

Utilizes shared `fifo_control` module: - **Full detection**: Based on pointer comparison with wraparound handling - **Almost full**: When remaining space $\leq$ `ALMOST_WR_MARGIN` - **Empty detection**: When read pointer equals write pointer - **Almost empty**: When available data $\leq$ `ALMOST_RD_MARGIN`

36.5 Operating Modes

36.5.1 Mux Mode (`REGISTERED = 0`)

- **Read latency**: 0 cycles (combinational output)
- **Data availability**: Immediate after write
- **Use case**: Low-latency applications
- **Timing**: Read data changes combinatorially with address

36.5.2 Flop Mode (`REGISTERED = 1`)

- **Read latency**: 1 cycle (registered output)
- **Data availability**: 1 cycle after read enable
- **Use case**: High-speed designs, timing closure
- **Timing**: Read data stable for full clock cycle

36.6 Functional Behavior

36.6.1 Write Operations

- **Condition**: `write && !wr_full`
- **Action**: Store data at write pointer, increment pointer
- **Blocking**: Writes ignored when FIFO is full
- **Status**: Full flags update based on new occupancy

36.6.2 Read Operations

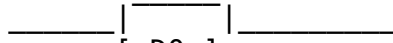
- **Condition:** read && !rd_empty
- **Action:** Advance read pointer (data handling depends on mode)
- **Blocking:** Reads ignored when FIFO is empty
- **Status:** Empty flags update based on new occupancy

36.6.3 Reset Behavior

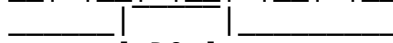
- **Pointers:** Reset to 0
- **Flags:** Full flags → 0, Empty flags → 1
- **Data:** Read data cleared in flop mode
- **Memory:** Contents undefined but not critical

36.7 Timing Diagrams

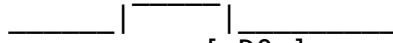
36.7.1 Write Sequence

clk 
write 
wr_data ==[D0]==
wr_full _____|⁻ (when FIFO becomes full)

36.7.2 Read Sequence - Mux Mode

clk 
read 
rd_data ==[D0]== (immediate)

36.7.3 Read Sequence - Flop Mode

clk 
read 
rd_data ==[D0]== (1 cycle delay)

36.8 Use Cases

36.8.1 Low Latency Applications (Mux Mode)

- Data streaming with minimal delay
- Real-time processing pipelines

- Clock domain buffers

36.8.2 High Speed Applications (Flop Mode)

- High-frequency designs
- Timing-critical paths
- Pipelined architectures

36.9 Design Considerations

36.9.1 Mode Selection Guidelines

- **Choose Mux Mode when:** Latency is critical, moderate clock speeds
- **Choose Flop Mode when:** High clock speeds, timing closure issues
- **Performance impact:** Flop mode adds 1 cycle latency but improves fmax

36.9.2 Sizing Considerations

- **Depth:** Must accommodate worst-case burst sizes
- **Almost flags:** Set margins based on producer/consumer response times
- **Data width:** Should match datapath requirements

36.9.3 Error Detection

Built-in simulation checks:

```
always_ff @(posedge clk) begin
    if ((write && wr_full) == 1'b1) begin
        $display("Error: %s write while fifo full, %t", INSTANCE_NAME,
$time);
    end
end
```

36.10 Performance Characteristics

- **Throughput:** 1 operation per clock cycle (when not full/empty)
- **Latency:** 0 cycles (mux) or 1 cycle (flop) for read
- **Occupancy:** 0 to DEPTH words
- **Efficiency:** 100% bandwidth utilization possible

36.11 WaveDrom Visualization

High-quality waveforms showcasing synchronous FIFO operation are available!

Run the WaveDrom test to generate detailed timing diagrams:

```
# Generate synchronous FIFO waveforms (single clock domain)
pytest val/common/test_fifo_sync_wavedrom.py -v
```

Waveform Scenarios Generated:

1. Write-Fill-Read-Empty Cycle

- Basic synchronous FIFO operation
- Simple binary pointer management
- No CDC complexity (single clock domain)

2. Back-to-Back Operations

- Maximum throughput demonstration
- Minimal inter-transaction delays
- Sequential write and read bursts

3. Simultaneous Write-Read (Ping-Pong)

- Read and write in same cycle
- Steady-state FIFO operation
- Demonstrating simultaneous access capability

4. Flag Transitions

- Full/empty flag behavior at boundaries
- Almost-full/almost-empty thresholds
- Flow control signaling

Key Characteristics vs. Async FIFOs:

- **Single Clock Domain:** No CDC complexity, simpler design
- **Binary Pointers:** Direct addressing, no Gray code conversion

- **Zero CDC Latency:** Flags update immediately (no synchronization delay)
- **Simultaneous Access:** Can read and write in same cycle

Comparison Tests:

- `test_fifo_async_wavedrom.py` - Gray code CDC (power-of-2 depths)

36.12 Related Modules

- **fifo_async:** For clock domain crossing applications
- **fifo_control:** Shared status flag generation logic
- **counter_bin:** Binary counter implementation

36.13 Test and Verification

Comprehensive Test Suite: - `val/common/test_fifo_buffer.py` - Full functional verification - `val/common/test_fifo_sync_wavedrom.py` - WaveDrom timing diagrams

Run Tests:

```
# Full functional test (basic/medium/full levels)
pytest val/common/test_fifo_buffer.py -v
```

```
# WaveDrom waveform generation
pytest val/common/test_fifo_sync_wavedrom.py -v
```

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37 Multi-Field Synchronous FIFO

Module: `fifo_sync_multi.sv` **Location:** `rtl/common/` **Status:** Production Ready

37.1 Overview

`fifo_sync_multi` is a thin convenience wrapper around `fifo_sync` that lets a caller push and pop several **named fields** through one synchronous FIFO without hand-packing them into a single wide bus. It concatenates an address field, a control field, and two data words into one `fifo_sync` payload on the write side, and splits the payload back out into the same named fields on the read side.

The storage, pointer management, and full/empty/almost flag generation are entirely provided by the underlying `fifo_sync` instance. This wrapper adds only the field-packing wiring.

37.1.1 Key Features

- **Field-oriented interface:** Separate `wr_addr` / `wr_ctrl` / `wr_data0` / `wr_data1` write ports and matching read ports
- **Single storage instance:** All fields ride one `fifo_sync` with a combined `AW + CW + DW + DW` payload
- **Inherited status flags:** `wr_full`, `wr_almost_full`, `rd_empty`, `rd_almost_empty` come straight from `fifo_sync`
- **Mux / flop output modes:** `REGISTERED` selects combinational (mux) or registered (flop) read output
- **Any depth:** Depth is passed through to `fifo_sync`

37.2 Module Purpose

Datapaths frequently move a small bundle of related fields together (for example an address plus a control tag plus a pair of data beats). Rather than making every caller concatenate and slice those fields by hand around a plain `fifo_sync`, this wrapper exposes the fields as individual ports and does the packing internally, keeping call sites readable and consistent.

Use Cases:

- Buffering an {address, control, data0, data1} bundle between two pipeline stages
- Queuing descriptor-like records where fields have distinct widths
- Any place a plain `fifo_sync` would be used but the payload is naturally several fields

Key Benefit: Callers work with named fields of distinct widths while still using a single, well-tested `fifo_sync` for all storage and flow control.

37.3 Parameters

Parameter	Type	Default	Description
REGISTERED	int	0	Read output mode: 0 = mux mode (combinational), 1 = flop

Parameter	Type	Default	Description
ADDR_WIDTH	int	4	mode (registered) Width of the address field (wr_addr / rd_addr)
CTRL_WIDTH	int	4	Width of the control field (wr_ctrl / rd_ctrl)
DATA_WIDTH	int	4	Width of each data word (wr_data0/1, rd_data0/1)
DEPTH	int	4	FIFO depth in entries
ALMOST_WR_MARGIN	int	1	Almost-full threshold, passed to fifo_sync
ALMOST_RD_MARGIN	int	1	Almost-empty threshold, passed to fifo_sync
AW	int	ADDR_WIDTH	Derived alias for ADDR_WIDTH (do not override)
CW	int	CTRL_WIDTH	Derived alias for CTRL_WIDTH (do not override)
DW	int	DATA_WIDTH	Derived alias for DATA_WIDTH (do not override)
D	int	DEPTH	Derived alias for DEPTH (do not override)

37.4 Port Groups

37.4.1 Clock and Reset

Port	Direction	Width	Description
clk	Input	1	System clock
rst_n	Input	1	Active-low reset

37.4.2 Write Interface

Port	Direction	Width	Description
write	Input	1	Write enable
wr_addr	Input	AW	Address field to enqueue
wr_ctrl	Input	CW	Control field to enqueue
wr_data0	Input	DW	Data word 0 to enqueue
wr_data1	Input	DW	Data word 1 to enqueue
wr_full	Output	1	Full flag
wr_almost_full	Output	1	Almost-full flag

37.4.3 Read Interface

Port	Direction	Width	Description
read	Input	1	Read enable
rd_addr	Output	AW	Dequeued address field
rd_ctrl	Output	CW	Dequeued control field
rd_data0	Output	DW	Dequeued data word 0
rd_data1	Output	DW	Dequeued data word 1
rd_empty	Output	1	Empty flag
rd_almost_empty	Output	1	Almost-empty flag

37.5 Functional Description

37.5.1 Field Packing

On the write side the four fields are concatenated into the single `fifo_sync` payload. The concatenation order places `wr_addr` in the most significant bits, then `wr_ctrl`, then `wr_data1`, then `wr_data0` in the least significant bits:

```
.wr_data ({wr_addr, wr_ctrl, wr_data1, wr_data0})
```

The `fifo_sync` payload width is therefore $AW + CW + DW + DW$.

37.5.2 Field Unpacking

On the read side the same concatenation order is used to split the payload back into named outputs, so each field re-emerges on its own port:

```
.rd_data ({rd_addr, rd_ctrl, rd_data1, rd_data0})
```

37.5.3 Storage and Flow Control

Everything else, binary read/write pointers, the memory array, and full/almost-full/empty/almost-empty flag generation, is delegated to the single `fifo_sync` instance. See `fifo_sync` for the pointer, memory, and flag details, including the mux-vs-flop read timing controlled by `REGISTERED`.

37.6 Usage Example

```
fifo_sync_multi #(
    .REGISTERED (0),      // 0 = mux (combinational read), 1 = flop
    (registered)
    .ADDR_WIDTH (8),
    .CTRL_WIDTH (4),
    .DATA_WIDTH (32),
    .DEPTH      (16)
) u_bundle_fifo (
    .clk          (clk),
    .rst_n        (rst_n),
    // write side
    .write         (wr_en),
    .wr_addr       (req_addr),
    .wr_ctrl       (req_tag),
    .wr_data0      (req_data_lo),
    .wr_data1      (req_data_hi),
    .wr_full       (fifo_full),
    .wr_almost_full (fifo_afull),
    // read side
    .read          (rd_en),
    .rd_addr       (out_addr),
    .rd_ctrl       (out_tag),
```

```

        .rd_data0      (out_data_lo),
        .rd_data1      (out_data_hi),
        .rd_empty       (fifo_empty),
        .rd_almost_empty (fifo_aempty)
    );

```

37.7 Design Notes

- **Pure wiring wrapper.** The module contains no storage or control logic of its own; it only concatenates and slices fields around one `fifo_sync`. All timing, flag, and depth behavior is that of `fifo_sync`.
- **Concatenation order matters.** Write and read use the same {`addr`, `ctrl`, `data1`, `data0`} order, so the fields map back correctly. Do not reorder one side without the other.
- **Two data words.** The wrapper is fixed at exactly two data words (`data0`, `data1`) plus an address and control field. For a different bundle shape, see the signal-map variant below or use `fifo_sync` directly.
- **REGISTERED read latency.** As with `fifo_sync`, mux mode gives 0-cycle combinational read and flop mode gives 1-cycle registered read.

37.8 Related Modules

37.8.1 Used By

- Pipeline stages that buffer an {address, control, `data0`, `data1`} bundle

37.8.2 Uses

- [fifo_sync](#) - The underlying single-payload synchronous FIFO providing storage and flow control

37.8.3 See Also

- [fifo_sync_multi_sigmap](#) - Same structure with a generic signal-map naming style (`sig_a..sig_h`)
- [fifo_control](#) - Shared full/empty flag generation used by `fifo_sync`
- [fifo_async](#) - Clock-domain-crossing FIFO variant

37.9 References

37.9.1 Source Code

- `rtl/common/testcode/fifo_sync_multi.sv`
- `rtl/common/fifo_sync.sv` (instantiated submodule)

37.9.2 Documentation

- `rtl/common/PRD.md`

Last Updated: 2026-07-15

37.10 Navigation

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38 Multi-Field Synchronous FIFO (Signal-Map Variant)

Module: `fifo_sync_multi_sigmap.sv` **Location:** `rtl/common/` **Status:** Production Ready

38.1 Overview

`fifo_sync_multi_sigmap` is a variant of `fifo_sync_multi` that carries the same four-field payload through one `fifo_sync` instance, but exposes the fields under **generic, position-oriented signal names** (`sig_a`, `sig_b`, `sig_c`, `sig_d` in, `sig_e`, `sig_f`, `sig_g`, `sig_h` out) instead of role-specific names like `addr` / `ctrl` / `data`. This makes it a convenient building block when the fields do not have fixed semantic roles, or when a generator maps an array of arbitrary signals onto FIFO slots by position.

As with the base wrapper, all storage, pointers, and status-flag generation come from the single underlying `fifo_sync`; this module only supplies the field-packing wiring.

38.1.1 Key Features

- **Generic field names:** Four write fields `wr_sig_a`..`wr_sig_d` and four read fields `rd_sig_e`..`rd_sig_h`
- **Single storage instance:** All fields packed into one `fifo_sync` payload of $AW + CW + DW + DW$ bits
- **Inherited status flags:** `wr_full`, `wr_almost_full`, `rd_empty`, `rd_almost_empty` from `fifo_sync`
- **Mux / flop output modes:** `REGISTERED` selects combinational (mux) or registered (flop) read output

- **Any depth:** Depth passed through to `fifo_sync`

38.2 Module Purpose

This module serves the same buffering role as `fifo_sync_multi`, but its port names are deliberately semantic-free so it can be dropped in wherever a bundle of same-shaped fields is mapped by position rather than by meaning. It is especially handy for generated or templated instantiations that assign an ordered list of signals to FIFO slots.

Use Cases:

- Buffering a bundle of positionally-mapped fields whose roles are assigned by the caller
- Generated / templated wiring that connects an ordered signal list to FIFO slots
- A neutral-named alternative to `fifo_sync_multi` when `addr` / `ctrl` / `data` labels do not fit

Key Benefit: Provides the same single-`fifo_sync` field bundling as `fifo_sync_multi` while keeping port names generic, so callers assign meaning by position.

38.3 Parameters

Parameter	Type	Default	Description
REGISTERED	int	0	Read output mode: 0 = mux mode (combinational), 1 = flop mode (registered)
ADDR_WIDTH	int	4	Width of the first field (<code>wr_siga</code> / <code>rd_sige</code>)
CTRL_WIDTH	int	4	Width of the second field (<code>wr_sigb</code> / <code>rd_sigf</code>)
DATA_WIDTH	int	4	Width of each of the two data-sized fields (<code>sigc</code> / <code>sigd</code> , <code>sigg</code> / <code>sigh</code>)
DEPTH	int	4	FIFO depth in entries
ALMOST_WR_MARGIN	int	1	Almost-full threshold, passed to <code>fifo_sync</code>
ALMOST_RD_MARGIN	int	1	Almost-empty threshold,

Parameter	Type	Default	Description
AW	int	ADDR_WIDTH	passed to fifo_sync Derived alias for ADDR_WIDTH (do not override)
CW	int	CTRL_WIDTH	Derived alias for CTRL_WIDTH (do not override)
DW	int	DATA_WIDTH	Derived alias for DATA_WIDTH (do not override)
D	int	DEPTH	Derived alias for DEPTH (do not override)

38.4 Port Groups

38.4.1 Clock and Reset

Port	Direction	Width	Description
clk	Input	1	System clock
rst_n	Input	1	Active-low reset

38.4.2 Write Interface

Port	Direction	Width	Description
write	Input	1	Write enable
wr_siga	Input	AW	First field to enqueue
wr_sigb	Input	CW	Second field to enqueue
wr_sigc	Input	DW	Third (data-sized) field to enqueue
wr_sigd	Input	DW	Fourth (data-sized) field to enqueue

Port	Direction	Width	Description
wr_full	Output	1	Full flag
wr_almost_full	Output	1	Almost-full flag

38.4.3 Read Interface

Port	Direction	Width	Description
read	Input	1	Read enable
rd_sige	Output	AW	Dequeued first field (maps from wr_siga)
rd_sigf	Output	CW	Dequeued second field (maps from wr_sigb)
rd_sigg	Output	DW	Dequeued third field (maps from wr_sigc)
rd_sigh	Output	DW	Dequeued fourth field (maps from wr_sigd)
rd_empty	Output	1	Empty flag
rd_almost_empty	Output	1	Almost-empty flag

38.5 Functional Description

38.5.1 Field Packing

The four write fields are concatenated into the single `fifo_sync` payload with `wr_siga` in the most significant bits, then `wr_sigb`, then `wr_sigd`, then `wr_sigc` in the least significant bits:

```
.wr_data ({wr_siga, wr_sigb, wr_sigd, wr_sigc})
```

The payload width is $AW + CW + DW + DW$.

38.5.2 Field Unpacking

The read side splits the payload back out using the matching order, so each field re-emerges on its own output. `sigc/sigd` map to `sigg/sigh` respectively:

```
.rd_data ({rd_sige, rd_sigf, rd_sigh, rd_sigg})
```

The packing order pairs the two data-sized fields identically on both sides (`sigd/sigc` in, `sigh/sigg` out), preserving the positional mapping through the FIFO.

38.5.3 Storage and Flow Control

As with `fifo_sync_multi`, all pointer management, memory, and full/almost-full/empty/almost-empty flag generation is provided by the single `fifo_sync` instance. See `fifo_sync` for the underlying storage and the mux-vs-flop read-timing behavior selected by `REGISTERED`.

38.6 Usage Example

```
fifo_sync_multi_sigmap #(
    .REGISTERED (1),           // 1 = registered (flop) read output
    .ADDR_WIDTH (10),
    .CTRL_WIDTH (2),
    .DATA_WIDTH (16),
    .DEPTH      (8)
) u_sigmap_fifo (
    .clk          (clk),
    .rst_n        (rst_n),
    // write side (positional fields)
    .write        (wr_en),
    .wr_siga      (field_a),
    .wr_sigb      (field_b),
    .wr_sigc      (field_c),
    .wr_sigd      (field_d),
    .wr_full      (fifo_full),
    .wr_almost_full (fifo_afull),
    // read side (positional fields)
    .read         (rd_en),
    .rd_sige      (out_a),
    .rd_sigf      (out_b),
    .rd_sigg      (out_c),
    .rd_sigh      (out_d),
    .rd_empty     (fifo_empty),
    .rd_almost_empty (fifo_aempty)
);
```

38.7 Design Notes

- **Naming is the only difference.** Structurally this is identical to `fifo_sync_multi`; only the port names change from role-based (`addr/ctrl/data`) to positional (`siga..sigh`).
- **Positional mapping.** `siga -> sige`, `sigb -> sigf`, `sigc -> sigg`, `sigd -> sigh`. The two data-sized fields keep their pairing because the write and read concatenations use the same order.
- **Pure wiring wrapper.** No storage or control logic lives here; all timing, flags, and depth behavior come from `fifo_sync`.

38.8 Related Modules

38.8.1 Used By

- Generated or templated instantiations that map an ordered signal list onto FIFO slots

38.8.2 Uses

- [fifo_sync](#) - The underlying single-payload synchronous FIFO providing storage and flow control

38.8.3 See Also

- [fifo_sync_multi](#) - The role-named (addr / ctrl / data0 / data1) equivalent
- [fifo_control](#) - Shared full/empty flag generation used by [fifo_sync](#)
- [fifo_async](#) - Clock-domain-crossing FIFO variant

38.9 References

38.9.1 Source Code

- rtl/common/testcode/fifo_sync_multi_sigmap.sv
- rtl/common/fifo_sync.sv (instantiated submodule)

38.9.2 Documentation

- rtl/common/PRD.md

Last Updated: 2026-07-15

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39 Find First Set (find_first_set.sv)

39.1 Purpose

Finds the index of the least significant bit (LSB) that is set to '1' in the input vector. Implements a priority encoder that favors lower-indexed bits.

39.2 Ports

39.2.1 Input Ports

- **data[WIDTH-1:0]** - Input data vector to search

39.2.2 Output Ports

- **index[\$clog2(WIDTH)-1:0]** - Index of the first (lowest) set bit

39.2.3 Parameters

- **WIDTH** - Width of input data vector (default: 32)
- **INSTANCE_NAME** - String identifier for debug (default: “FFS”)

39.3 Implementation Details

39.3.1 Core Algorithm

```
localparam int N = $clog2(WIDTH);
logic w_found;

always_comb begin
    index = {N{1'b0}}; // Default value if no bit is set
    w_found = 1'b0;

    for (int i = 0; i < WIDTH; i++) begin
        if (data[i] && !w_found) begin
            index = i[N-1:0]; // Ensure correct bit width
            w_found = 1'b1;
        end
    end
end
```

39.3.2 Search Strategy

- **Direction:** LSB to MSB (left to right, ascending index)
- **Priority:** Lower indices have higher priority
- **First match:** Stops at first ‘1’ bit encountered
- **Found flag:** Prevents multiple assignments

39.4 Functional Behavior

39.4.1 Truth Table Examples (WIDTH=8)

data[7:0]	index[2:0]	Description
00000001	000	Bit 0 is first set
00000010	001	Bit 1 is first set
00000100	010	Bit 2 is first set
00001000	011	Bit 3 is first set
00010000	100	Bit 4 is first set
00100000	101	Bit 5 is first set
01000000	110	Bit 6 is first set
10000000	111	Bit 7 is first set
00000000	000	No bits set - default
10000001	000	Multiple bits - LSB wins
11111111	000	All bits set - bit 0 wins

39.4.2 Key Characteristics

- **LSB priority:** Always returns lowest index when multiple bits set
- **Zero default:** Returns 0 when no bits are set
- **Deterministic:** Same input always produces same output
- **Combinational:** Immediate response to input changes

39.5 Design Features

39.5.1 Width Parameterization

- **Automatic sizing:** Output width calculated as $\lceil \log_2(\text{WIDTH}) \rceil$
- **Examples:**
 - WIDTH=8 → index[2:0] (3 bits)
 - WIDTH=16 → index[3:0] (4 bits)

- WIDTH=32 → index[4:0] (5 bits)

39.5.2 Found Flag Logic

```
if (data[i] && !w_found) begin
    index = i[N-1:0];
    w_found = 1'b1;
end
```

- **Purpose:** Prevents overwriting the first found result
- **Mechanism:** Once w_found is set, no further assignments occur
- **Result:** Only the lowest index is captured

39.5.3 Bit Width Handling

```
index = i[N-1:0]; // Ensure correct bit width
```

- **Type safety:** Explicit width matching prevents synthesis warnings
- **Truncation:** Safely handles loop variable width vs. output width

39.6 Use Cases

39.6.1 Interrupt Processing

```
find_first_set #(.WIDTH(8)) irq_ffs (
    .data(interrupt_pending),
    .index(lowest_priority_irq)
);
```

- Process lowest-numbered interrupt first
- Fair servicing of interrupt sources

39.6.2 Resource Allocation

```
find_first_set #(.WIDTH(16)) resource_ffs (
    .data(available_resources),
    .index(allocated_resource_id)
);
```

- Allocate lowest-numbered available resource
- Round-robin allocation when combined with masking

39.6.3 Error Detection

```
find_first_set #(.WIDTH(4)) error_ffs (  
    .data(error_status),  
    .index(first_error_type)  
);
```

- Report first error condition detected
- Prioritize error handling by position

39.6.4 Bit Manipulation

```
find_first_set #(.WIDTH(32)) bit_scan (  
    .data(search_pattern),  
    .index(first_set_position)  
);
```

- Bit scanning operations
- Pattern analysis and parsing

39.7 Timing Characteristics

39.7.1 Propagation Delay

- **Best case:** 1 LUT delay (bit 0 set)
- **Worst case:** Multiple LUT delays (higher bits set)
- **Average case:** Depends on typical input patterns

39.7.2 Critical Path

- **Path length:** Increases with WIDTH
- **Optimization:** Modern synthesizers create efficient priority encoders
- **Scaling:** Generally logarithmic complexity in hardware

39.8 Algorithm Comparison

39.8.1 vs. Find Last Set

Aspect	Find First Set	Find Last Set
Search direction	LSB → MSB	MSB → LSB
Priority	Lower index wins	Higher index wins

Aspect	Find First Set	Find Last Set
Use case	Fair allocation	Priority allocation

39.8.2 vs. Priority Encoder

Aspect	Find First Set	Priority Encoder
Priority direction	LSB has priority	MSB has priority
Application	Round-robin systems	Hierarchical systems

39.9 Design Considerations

39.9.1 Input Validation

- **All zeros:** Returns 0 (may need validation in application)
- **Valid range:** All input patterns handled gracefully
- **No error conditions:** Always produces a result

39.9.2 Performance Optimization

- **Synthesis:** Let tools optimize the priority encoding
- **Pipelining:** Consider registering for high-speed applications
- **Parallel:** Can process multiple vectors simultaneously

39.9.3 Width Selection

- **Power of 2:** Often most efficient for synthesis
- **Arbitrary width:** Supported but may have overhead
- **Large widths:** Consider hierarchical implementation

39.10 Common Usage Patterns

39.10.1 Round-Robin Arbitration

```
// Mask off already-serviced requests
assign masked_requests = pending_requests & ~service_mask;

find_first_set #(.WIDTH(N)) arbiter_ffs (
    .data(masked_requests),
    .index(next_grant)
);
```

```
// Update mask to exclude granted request  
assign next_mask = service_mask | (1 << next_grant);
```

39.10.2 Free List Management

```
find_first_set #(.WIDTH(P00L_SIZE)) free_finder (  
    .data(free_pool_bitmap),  
    .index(allocated_entry)  
);
```

```
// Mark entry as allocated  
assign updated_bitmap = free_pool_bitmap & ~(1 << allocated_entry);
```

39.10.3 Packet Classification

```
find_first_set #(.WIDTH(NUM_RULES)) rule_matcher (  
    .data(rule_match_vector),  
    .index(matched_rule_id)  
);
```

39.11 Synthesis Considerations

39.11.1 Resource Utilization

- **LUT usage:** Typically maps to priority encoder primitives
- **Efficiency:** Good utilization for most FPGA architectures
- **Scaling:** Linear increase in resources with WIDTH

39.11.2 Timing Optimization

- **Critical paths:** May need pipeline registers for large WIDTH
- **Clock frequency:** Generally achieves high fmax
- **Constraints:** Standard combinational timing constraints apply

39.12 Verification Considerations

39.12.1 Test Scenarios

- **Single bit:** Each bit position individually
- **Multiple bits:** Various combinations
- **All zeros:** Edge case handling

- **All ones:** Maximum case behavior

39.12.2 Coverage Metrics

- **Bit coverage:** Each input bit position
- **Pattern coverage:** Representative bit combinations
- **Boundary cases:** Min/max index values

39.13 Related Modules

- **find_last_set:** MSB-first search variant
- **encoder:** Basic priority encoder
- **encoder_priority_enable:** Enhanced priority encoder with enable

39.14 Navigation

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40 Find Last Set (find_last_set.sv)

40.1 Purpose

Finds the index of the most significant bit (MSB) that is set to ‘1’ in the input vector. Implements a priority encoder that favors higher-indexed bits.

40.2 Ports

40.2.1 Input Ports

- **data[WIDTH-1:0]** - Input data vector to search

40.2.2 Output Ports

- **index[\$clog2(WIDTH)-1:0]** - Index of the last (highest) set bit

40.2.3 Parameters

- **WIDTH** - Width of input data vector (default: 32)
- **INSTANCE_NAME** - String identifier for debug (default: “FLS”)

40.3 Implementation Details

40.3.1 Core Algorithm

```
localparam int N = $clog2(WIDTH);
logic w_found;

always_comb begin
    index = {N{1'b0}}; // Default value if no bit is set
    w_found = 1'b0;

    for (int i = WIDTH - 1; i >= 0; i--) begin
        if (data[i] && !w_found) begin
            index = i[N-1:0]; // Ensure correct bit width
            w_found = 1'b1;
        end
    end
end
```

40.3.2 Search Strategy

- **Direction:** MSB to LSB (right to left, descending index)
- **Priority:** Higher indices have higher priority
- **First match:** Stops at first '1' bit encountered from MSB
- **Found flag:** Prevents multiple assignments

40.4 Functional Behavior

40.4.1 Truth Table Examples (WIDTH=8)

data[7:0]	index[2:0]	Description
10000000	111	Bit 7 is last set
01000000	110	Bit 6 is last set
00100000	101	Bit 5 is last set
00010000	100	Bit 4 is last set
00001000	011	Bit 3 is last set
00000100	010	Bit 2 is last set
00000010	001	Bit 1 is last set
00000001	000	Bit 0 is last set
00000000	000	No bits set - default

data[7:0]	index[2:0]	Description
10000001	111	Multiple bits - MSB wins
11111111	111	All bits set - bit 7 wins

40.4.2 Key Characteristics

- **MSB priority:** Always returns highest index when multiple bits set
- **Zero default:** Returns 0 when no bits are set (same as find_first_set)
- **Deterministic:** Same input always produces same output
- **Combinational:** Immediate response to input changes

40.5 Design Features

40.5.1 Reverse Loop Implementation

```
for (int i = WIDTH - 1; i >= 0; i--) begin
```

- **MSB-first:** Starts from highest bit index
- **Decrementing:** Moves toward LSB
- **Early termination:** Stops at first match found

40.5.2 Width Parameterization

Same auto-sizing as find_first_set: - **Output width:** $\$clog2(WIDTH)$ bits - **Examples:** -
 WIDTH=8 → index[2:0] (3 bits) - WIDTH=16 → index[3:0] (4 bits)
 - WIDTH=32 → index[4:0] (5 bits)

40.5.3 Found Flag Logic

```
if (data[i] && !w_found) begin
    index = i[N-1:0];
    w_found = 1'b1;
end
```

- **Single assignment:** Prevents overwriting the highest found bit
- **Priority enforcement:** Only the first match (highest index) is captured

40.6 Use Cases

40.6.1 Interrupt Priority Handling

```
find_last_set #(.WIDTH(8)) irq_priority (
    .data(interrupt_pending),
    .index(highest_priority_irq)
);
```

- Service highest-priority interrupt first
- Preemptive interrupt handling

40.6.2 Leading Zero Count

```
// Find position of leading 1, then calculate leading zeros
find_last_set #(.WIDTH(32)) leading_one (
    .data(input_value),
    .index(msb_position)
);
```

```
assign leading_zero_count = (input_value == 0) ? 32 : (31 -
msb_position);
```

40.6.3 Bit Width Calculation

```
find_last_set #(.WIDTH(64)) width_calc (
    .data(value),
    .index(msb_pos)
);
```

```
assign required_bits = msb_pos + 1; // Minimum bits needed to
represent value
```

40.6.4 Cache Line Selection

```
find_last_set #(.WIDTH(16)) cache_priority (
    .data(cache_line_priorities),
    .index(selected_cache_line)
);
```

- Select highest priority cache line for replacement
- LRU policy implementation

40.7 Timing Characteristics

40.7.1 Propagation Delay

- **Best case:** 1 LUT delay (MSB set)
- **Worst case:** Multiple LUT delays (LSB only set)
- **Average case:** Depends on typical bit patterns

40.7.2 Critical Path Analysis

MSB check → MSB-1 check → ... → LSB check → Output

- **Path length:** Linear with WIDTH in worst case
- **Typical case:** Shorter path when higher bits are set
- **Optimization:** Synthesis tools create efficient implementations

40.8 Algorithm Comparison

40.8.1 Find Last Set vs. Find First Set

Aspect	Find Last Set	Find First Set
Search Direction	MSB → LSB	LSB → MSB
Priority	Higher index wins	Lower index wins
Loop Direction	i-- (decrementing)	i++ (incrementing)
Use Case	Priority systems	Fair allocation
Default Output	0 (same)	0 (same)

40.8.2 Typical Use Case Patterns

Application Type	Preferred Algorithm
Interrupt Controllers	Find Last Set (priority)
Round-Robin Arbiters	Find First Set (fairness)
Error Reporting	Find Last Set (severity)
Resource Allocation	Find First Set (efficiency)
Bit Manipulation	Either (depends on need)

40.9 Design Considerations

40.9.1 Zero Input Handling

Both modules return 0 for zero input:

```
// data = 8'b00000000
// Both find_first_set and find_last_set return 3'b000
```

Application consideration: May need explicit zero detection:

```
assign valid_result = |data; // OR-reduce to detect any bits set
assign result_index = valid_result ? index : INVALID_INDEX;
```

40.9.2 Performance Scaling

- **Small WIDTH (<16):** Excellent performance
- **Medium WIDTH (16-64):** Good performance, may need pipelining
- **Large WIDTH (>64):** Consider hierarchical implementation

40.9.3 Synthesis Optimization

Modern synthesis tools recognize these patterns: - **Priority encoder primitives:** Often map to dedicated hardware - **LUT optimization:** Efficient resource utilization - **Timing optimization:** Automatic critical path optimization

40.10 Advanced Usage Patterns

40.10.1 Hierarchical Priority Encoding

```
// For very wide inputs, use hierarchical approach
find_last_set #(.WIDTH(8)) level0_fls [7:0] (
    .data(data_chunks), // 8 chunks of 8 bits each
    .index(chunk_indices)
);

find_last_set #(.WIDTH(8)) level1_fls (
    .data(chunk_valid), // Which chunks have set bits
    .index(active_chunk)
);

// Combine results
assign final_index = {active_chunk, chunk_indices[active_chunk]};
```

40.10.2 Conditional Priority Selection

```
// Select between different priority schemes
assign priority_vector = use_high_priority ? high_pri_requests :
```

```

        use_fair_mode ? fair_requests :
        normal_requests;

find_last_set #(.WIDTH(N)) flexible_priority (
    .data(priority_vector),
    .index(selected_request)
);

40.10.3 Dynamic Width Processing
// Process variable-width data
logic [MAX_WIDTH-1:0] padded_data;
assign padded_data = {(MAX_WIDTH-actual_width){1'b0}}, input_data};

find_last_set #(.WIDTH(MAX_WIDTH)) variable_fl (
    .data(padded_data),
    .index(raw_index)
);

// Adjust result for actual width
assign adjusted_index = (raw_index >= actual_width) ? 0 : raw_index;

```

40.11 Verification Strategies

40.11.1 Comprehensive Test Coverage

```

// Test single bit at each position
for (int i = 0; i < WIDTH; i++) begin
    test_vector = 1 << i;
    expected_result = i;
    // Verify find_last_set returns i
end

// Test multiple bits with known MSB
test_vector = 8'b10101010;
expected_result = 7; // MSB position

// Test edge cases
test_vector = 8'b00000000; // All zeros
expected_result = 0; // Default

```

40.11.2 Random Testing

```

// Random bit patterns
repeat (1000) begin
    random_vector = $random();
    expected = calculate_msb_position(random_vector);

```

```
// Verify against reference function  
end
```

40.12 Related Modules

- **find_first_set**: LSB-priority search
- **encoder_priority_enable**: Enhanced priority encoder
- **leading_one_trailing_one**: Position detection for Johnson counters

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41 Glitch-Free N-DFF Synchronizer (glitch_free_n_dff_arn.sv)

41.1 Purpose

Provides a parameterized multi-stage synchronizer for safe clock domain crossing (CDC). Reduces metastability risk through configurable stages of flip-flops while maintaining data integrity across asynchronous clock domains.

41.2 Ports

41.2.1 Input Ports

- **clk** - Destination domain clock
- **rst_n** - Destination domain active-low reset
- **d[WIDTH-1:0]** - Input data from source clock domain

41.2.2 Output Ports

- **q[WIDTH-1:0]** - Synchronized output data in destination domain

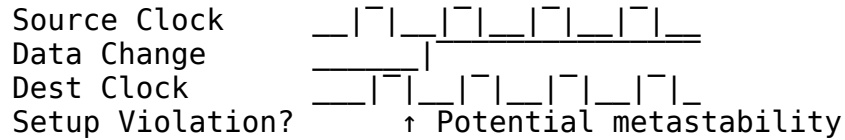
41.2.3 Parameters

- **FLOP_COUNT** - Number of synchronizer flip-flop stages (default: 3)
- **WIDTH** - Data width in bits (default: 4)

41.3 Clock Domain Crossing Theory

41.3.1 Metastability Problem

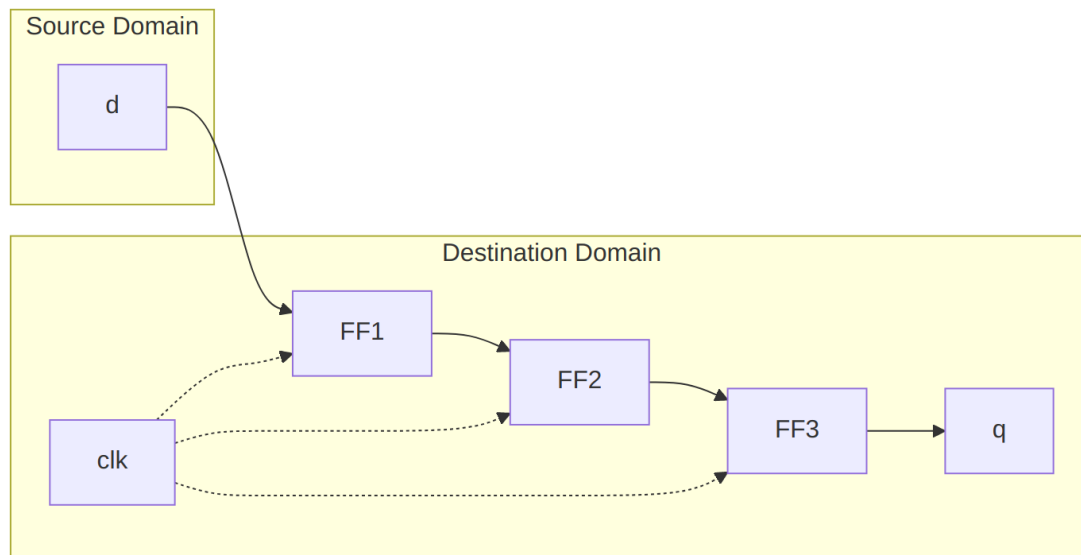
When signals cross between asynchronous clock domains:



41.3.2 Metastability Effects

- **Undefined output:** Neither 0 nor 1, but intermediate voltage
- **Extended resolution:** Takes longer than normal clock period
- **Propagation:** Can corrupt downstream logic
- **System failure:** Can cause entire design malfunction

41.4 Multi-Stage Synchronizer Solution



Architecture Overview

41.4.1 Stage-by-Stage Analysis

1. **Stage 1 (FF1):** Captures source data, may go metastable
2. **Stage 2 (FF2):** Allows metastability to resolve, low probability of propagation

3. **Stage 3+ (FFN):** Further reduces MTBF (Mean Time Between Failures)

41.5 Implementation Details

41.5.1 Packed Array Structure

logic [FC-1:0][WIDTH-1:0] r_q_array;

- **Efficient storage:** Single declaration for all stages
- **Clear indexing:** r_q_array[0] is first stage, r_q_array[FC-1] is last
- **Synthesis friendly:** Maps well to FPGA/ASIC resources

41.5.2 Shift Register Behavior

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        // Reset all flip-flops to zero
        r_q_array <= {FC{{DW{1'b0}}}};
    end else begin
        // Load new data into first stage
        r_q_array[0] <= d;
        // Shift existing data through remaining stages
        for (int i = 1; i < FC; i++) begin
            r_q_array[i] <= r_q_array[i-1];
        end
    end
end
```

41.5.3 Reset Behavior

- **Synchronous reset:** Uses destination domain reset
- **Complete reset:** All stages reset to zero
- **Nested replication:** {FC{{DW{1'b0}}}} creates FC copies of DW zeros

41.6 MTBF (Mean Time Between Failures) Analysis

41.6.1 MTBF Calculation

$$\text{MTBF} = (e^{(t_{\text{res}}/\tau)}) / (T_0 \times f_{\text{clk}} \times f_{\text{data}})$$

Where:

- t_{res} = Resolution time available (clock period - setup - routing)
- τ = Flip-flop metastability resolution time constant (~200ps typical)

- T_0 = Metastability aperture / window constant (device-specific, ~1e-12 s)
- f_{clk} = Destination clock frequency
- f_{data} = Data transition frequency

τ appears **ONLY in the exponent**. The denominator constant is T_0 , not τ – they are different device parameters and are not interchangeable. Both are process/library specific: take them from the vendor’s metastability characterization data, not from a datasheet timing table.

The stage-count table below is a **rule of thumb** for relative improvement, not a result derived from this formula. Use the formula only with real τ/T_0 values for your target device; otherwise rely on the table for sizing.

41.6.2 Stage Count Impact

Stages	Relative MTBF	Typical MTBF
1	1× (baseline)	Minutes
2	~1000×	Hours
3	~1,000,000×	Years
4	~1,000,000,000×	Millennia

41.6.3 Practical Guidelines

- **2 stages:** Minimum for most applications
- **3 stages:** Recommended for high-reliability systems
- **4+ stages:** Extreme reliability requirements (aerospace, medical)

41.7 Latency vs. Reliability Trade-off

41.7.1 Synchronizer Latency

- **Minimum latency:** 2 clock cycles (2-stage synchronizer)
- **General latency:** FLOP_COUNT clock cycles
- **Fixed delay:** Independent of data pattern

41.7.2 Performance Impact

```
// 2-stage: 2 cycle latency, good MTBF
glitch_free_n_dff_arn #(.FLOP_COUNT(2)) fast_sync (...);
```

```
// 3-stage: 3 cycle latency, excellent MTBF
```

```
glitch_free_n_dff_arn #(.FLOP_COUNT(3)) safe_sync (...);
```

```
// 4-stage: 4 cycle latency, extreme MTBF
```

```
glitch_free_n_dff_arn #(.FLOP_COUNT(4)) ultra_safe_sync (...);
```

41.8 Usage Guidelines

41.8.1 Appropriate Use Cases

- **Control signals:** Reset, enable, mode switches
- **Slow status signals:** Error flags, configuration bits
- **Gray code pointers:** FIFO read/write pointers
- **Single-bit signals:** Generally safe for multi-bit if Gray coded

41.8.2 Inappropriate Use Cases

- **High-speed data buses:** Use proper CDC techniques (handshaking, FIFOs)
- **Binary counters:** Multiple bits can transition simultaneously
- **Address buses:** Binary addresses are not CDC-safe
- **Clock signals:** Never synchronize clocks this way

41.8.3 Multi-Bit Considerations

```
// SAFE: Gray-coded FIFO pointers
```

```
glitch_free_n_dff_arn #(.WIDTH(5)) gray_sync (  
    .d(gray_pointer),      // Gray code - only 1 bit changes  
    .q(sync_gray_pointer)  
);
```

```
// UNSAFE: Binary counter synchronization
```

```
// Don't do this - multiple bits can change simultaneously
```

```
glitch_free_n_dff_arn #(.WIDTH(8)) bad_sync (  
    .d(binary_counter),    // Binary - multiple bit transitions  
    .q(sync_counter)      // May capture inconsistent values  
);
```

41.9 Advanced Applications

41.9.1 Reset Synchronizer Pattern

```
// Asynchronous assert, synchronous deassert
```

```
logic reset_sync;
```

```
glitch_free_n_dff_arn #(.FLOP_COUNT(2), .WIDTH(1)) reset_sync_inst (
    .clk(dest_clk),
    .rst_n(1'b1),           // No reset feedback
    .d(~async_reset),       // Invert for active-low
    .q(reset_sync)
);
```

```
assign sync_reset_n = reset_sync & ~async_reset;
```

41.9.2 Enable Signal Synchronization

```
glitch_free_n_dff_arn #(.FLOP_COUNT(3), .WIDTH(1)) enable_sync (
    .clk(fast_clk),
    .rst_n(rst_n),
    .d(slow_enable),
    .q(fast_enable_sync)
);
```

41.9.3 Status Flag Collection

```
// Synchronize multiple independent status signals
glitch_free_n_dff_arn #(.FLOP_COUNT(2), .WIDTH(4)) status_sync (
    .clk(dest_clk),
    .rst_n(rst_n),
    .d({error_flag, done_flag, busy_flag, ready_flag}),
    .q({sync_error, sync_done, sync_busy, sync_ready})
);
```

41.10 Synthesis Considerations

41.10.1 FPGA Implementation

- **Dedicated resources:** Maps to flip-flop primitives
- **Placement:** Tools may place stages in different locations
- **Timing constraints:** Require proper CDC constraints

41.10.2 Timing Constraints

```
# Constrain the synchronizer input as asynchronous
set_false_path -to [get_pins sync_inst/r_q_array[0]/D]

# Or set maximum delay if there's some relationship
set_max_delay -to [get_pins sync_inst/r_q_array[0]/D] 2.0
```


41.10.3 Optimization Prevention

```
// Synthesis attributes to prevent optimization
(* ASYNC_REG = "TRUE" *) logic [FC-1:0][WIDTH-1:0] r_q_array;
(* DONT_TOUCH = "TRUE" *) logic [FC-1:0][WIDTH-1:0] r_q_array;
```

41.11 Verification Strategies

41.11.1 Functional Verification

- **Data integrity:** Verify output eventually matches input
- **Reset behavior:** Check proper reset operation
- **Latency verification:** Confirm expected delay

41.11.2 Metastability Testing

```
// Stress test with rapid data changes
initial begin
    forever begin
        @(posedge source_clk);
        d <= $random();
        #($random() % 100); // Random timing
    end
end
```

41.11.3 Formal Verification

- **Eventually properties:** Data will eventually propagate
- **Reset properties:** Reset behavior verification
- **No X propagation:** Ensure no unknown values propagate

41.12 Debug Features

41.12.1 Waveform Analysis

```
// Flatten array for waveform viewing
wire [(DW*FC)-1:0] flat_r_q;
genvar i;
generate
    for (i = 0; i < FC; i++) begin : gen_flatten_memory
        assign flat_r_q[i*DW+:DW] = r_q_array[i];
    end
endgenerate
```

41.12.2 Simulation Visibility

- **All stages visible:** Can observe data propagation through stages
- **Metastability injection:** Can inject X values for testing
- **Timing analysis:** Observe setup/hold violations

41.13 Related Modules

- **FIFO CDC modules:** Use this for pointer synchronization
- **Reset synchronizers:** Specialized CDC for reset signals
- **Handshake synchronizers:** For complex data CDC
- **Pulse synchronizers:** For single-cycle pulse CDC

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42 Gray-to-Binary Converter (gray2bin.sv)

42.1 Purpose

Converts Gray code (reflected binary code) to standard binary representation using XOR reduction. Essential component for asynchronous FIFO pointer comparison and other CDC applications.

42.2 Ports

42.2.1 Input Ports

- **gray[WIDTH-1:0]** - Gray code input

42.2.2 Output Ports

- **binary[WIDTH-1:0]** - Binary code output

42.2.3 Parameters

- **WIDTH** - Data width in bits (default: 4)

42.3 Gray Code Theory

42.3.1 What is Gray Code?

Gray code (also called reflected binary code) is a binary numeral system where **only one bit changes** between consecutive values:

Binary	Gray	Transitions
000	→ 000	
001	→ 001	1 bit change
010	→ 011	1 bit change
011	→ 010	1 bit change
100	→ 110	1 bit change
101	→ 111	1 bit change
110	→ 101	1 bit change
111	→ 100	1 bit change

42.3.2 Why Gray Codes Matter for CDC

- **Metastability protection:** Only one bit transition eliminates multi-bit race conditions
- **Safe sampling:** Intermediate values during transition are still valid Gray codes
- **FIFO pointers:** Essential for async FIFO full/empty detection

42.4 Conversion Algorithm

42.4.1 Mathematical Foundation

For Gray-to-binary conversion, each binary bit is the XOR of all Gray bits from that position to the MSB:

$$\text{binary}[i] = \text{gray}[\text{MSB}] \oplus \text{gray}[\text{MSB}-1] \oplus \dots \oplus \text{gray}[i]$$

42.4.2 Implementation

```
genvar i;
generate
  for (i = 0; i < WIDTH; i++) begin : gen_gray_to_bin
    assign binary[i] = ^(gray >> i);
  end
endgenerate
```

42.4.3 Bit-by-Bit Breakdown

```
// For 4-bit example:
assign binary[3] = gray[3];                                // MSB unchanged
```

```

assign binary[2] = gray[3] ^ gray[2];           // XOR from MSB
down
assign binary[1] = gray[3] ^ gray[2] ^ gray[1];   // XOR from MSB
down
assign binary[0] = gray[3] ^ gray[2] ^ gray[1] ^ gray[0]; // XOR all
bits

```

42.5 Functional Examples

42.5.1 4-Bit Conversion Table

Gray[3:0]	Binary[3:0]	Calculation
0000	0000	$0 \oplus 0 \oplus 0 \oplus 0 = 0,$ $0 \oplus 0 \oplus 0 = 0, 0 \oplus 0 = 0,$ $0 = 0$
0001	0001	$0 \oplus 0 \oplus 0 \oplus 1 = 1,$ $0 \oplus 0 \oplus 0 = 0, 0 \oplus 0 = 0,$ $0 = 0$
0011	0010	$0 \oplus 0 \oplus 1 \oplus 1 = 0,$ $0 \oplus 0 \oplus 1 = 1, 0 \oplus 0 = 0,$ $0 = 0$
0010	0011	$0 \oplus 0 \oplus 1 \oplus 0 = 1,$ $0 \oplus 0 \oplus 1 = 1, 0 \oplus 0 = 0,$ $0 = 0$
0110	0100	$0 \oplus 1 \oplus 1 \oplus 0 = 0,$ $0 \oplus 1 \oplus 1 = 0, 0 \oplus 1 = 1,$ $0 = 0$
0111	0101	$0 \oplus 1 \oplus 1 \oplus 1 = 1,$ $0 \oplus 1 \oplus 1 = 0, 0 \oplus 1 = 1,$ $0 = 0$
0101	0110	$0 \oplus 1 \oplus 0 \oplus 1 = 0,$ $0 \oplus 1 \oplus 0 = 1, 0 \oplus 1 = 1,$ $0 = 0$
0100	0111	$0 \oplus 1 \oplus 0 \oplus 0 = 1,$ $0 \oplus 1 \oplus 0 = 1, 0 \oplus 1 = 1,$ $0 = 0$

42.5.2 Step-by-Step Example (Gray 0110 → Binary)

Gray input: 0110

```
binary[3] = gray[3] = 0
binary[2] = gray[3] ^ gray[2] = 0 ^ 1 = 1
binary[1] = gray[3] ^ gray[2] ^ gray[1] = 0 ^ 1 ^ 1 = 0
binary[0] = gray[3] ^ gray[2] ^ gray[1] ^ gray[0] = 0 ^ 1 ^ 1 ^ 0 = 0
```

Result: binary = 0100 (decimal 4)

42.6 Implementation Analysis

42.6.1 XOR Reduction Technique

```
assign binary[i] = ^(gray >> i);
```

How it works: - gray >> i: Right-shift gray by i positions - ^(...): XOR-reduce all bits in the shifted value - **Effect:** XORs all bits from position i to MSB

42.6.2 Generate Loop Benefits

- **Parameterizable:** Works for any WIDTH
- **Synthesis friendly:** Tools recognize and optimize pattern
- **Parallel computation:** All bits calculated simultaneously
- **Resource efficient:** Maps to XOR tree structures

42.7 Use Cases in Digital Design

42.7.1 Asynchronous FIFO Pointers

```
// Convert synchronized Gray pointers back to binary for comparison
```

```
gray2bin #(.WIDTH(5)) wr_ptr_conv (  
    .gray(sync_wr_ptr_gray),  
    .binary(sync_wr_ptr_bin)  
);
```

```
gray2bin #(.WIDTH(5)) rd_ptr_conv (  
    .gray(sync_rd_ptr_gray),  
    .binary(sync_rd_ptr_bin)  
);
```

```
// Now can perform binary arithmetic for occupancy calculation
```

```
assign occupancy = sync_wr_ptr_bin - sync_rd_ptr_bin;
```

42.7.2 Clock Domain Crossing Counters

```
// Convert Gray counter back to binary for address generation
gray2bin #(.WIDTH(AW)) addr_converter (
    .gray(gray_counter),
    .binary(memory_address)
);
```

42.7.3 Position Encoding

```
// Convert Gray-encoded position to binary for processing
gray2bin #(.WIDTH(8)) position_decoder (
    .gray(gray_position),
    .binary(bin_position)
);
```

42.8 Timing Characteristics

42.8.1 Propagation Delay

- **XOR tree depth:** $\log_2(\text{WIDTH})$ levels
- **Typical delay:** 1-2 LUT delays for most widths
- **Critical path:** From MSB input to LSB output

42.8.2 Delay Analysis by Width

WIDTH	XOR Levels	Typical Delay
4	2	1 LUT delay
8	3	2 LUT delays
16	4	2 LUT delays
32	5	3 LUT delays

42.8.3 Synthesis Optimization

Modern synthesis tools: - **Recognize pattern:** Optimize XOR reduction automatically - **Balance trees:** Create balanced XOR structures - **Resource sharing:** Reuse XOR gates where possible

42.9 Design Considerations

42.9.1 Width Scaling

- **Linear resource growth:** XOR gates increase with WIDTH
- **Logarithmic delay growth:** Delay scales with $\log(\text{WIDTH})$

- **Practical limits:** Works well up to 64+ bits

42.9.2 Input Validation

```
// Gray codes have no invalid states - all inputs are valid
// However, may want to validate source of Gray code
always_comb begin
    if (gray_code_valid) begin
        binary_out = converted_binary;
    end else begin
        binary_out = '0; // Default when invalid
    end
end
```

42.9.3 Pipeline Considerations

For very high-speed applications:

```
// Optional pipeline stage for timing closure
always_ff @(posedge clk) begin
    if (!rst_n) begin
        binary_reg <= '0;
    end else begin
        binary_reg <= binary_comb;
    end
end
```

42.10 Verification Strategies

42.10.1 Exhaustive Testing

```
// For reasonable widths, test all possible inputs
for (int i = 0; i < (1 << WIDTH); i++) begin
    gray_input = int_to_gray(i); // Convert integer to Gray
    expected_binary = i; // Expected result
    #1;
    assert(binary_output == expected_binary);
end
```

42.10.2 Property-Based Verification

```
// Verify round-trip conversion
property round_trip;
    @(posedge clk)
        binary_output == original_binary;
endproperty

// Apply after binary→Gray→binary conversion
assert property (round_trip);
```

42.10.3 Random Testing

```
repeat (10000) begin
    random_gray = $random();
    expected_result = reference_gray2bin(random_gray);
    #1;
    assert(binary_output == expected_result);
end
```

42.11 Common Mistakes and Pitfalls

42.11.1 Bit Order Confusion

```
// WRONG: Bit order matters in Gray codes
assign binary[i] = ^(gray << i); // Left shift instead of right
```

```
// CORRECT:
assign binary[i] = ^(gray >> i); // Right shift
```

42.11.2 Width Mismatches

```
// WRONG: Input/output width mismatch
gray2bin #(.WIDTH(4)) converter (
    .gray(gray_5bit), // 5-bit input
    .binary(binary_4bit) // 4-bit output
);
```

```
// CORRECT: Match widths
gray2bin #(.WIDTH(5)) converter (
    .gray(gray_5bit),
    .binary(binary_5bit)
);
```

42.11.3 Timing Assumptions

```
// WRONG: Assuming zero delay
gray_in <= new_value;
binary_out <= converted_value; // May not be updated yet
```

```
// CORRECT: Account for combinational delay
gray_in <= new_value;
#1; // Wait for conversion
binary_out <= converted_value;
```

42.12 Related Modules

- **bin2gray**: Performs inverse conversion (binary to Gray)
- **counter_bingray**: Combined binary/Gray counter

- **fifo_async**: Uses Gray codes for CDC
- **johnson2bin**: Johnson counter to binary converter (different algorithm)

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43 Hex to 7-Segment Display (hex_to_7seg.sv)

43.1 Purpose

Converts 4-bit hexadecimal values (0-F) to 7-segment display patterns for common anode displays. Provides human-readable display of digital values for debugging, user interfaces, and embedded systems.

43.2 Ports

43.2.1 Input Ports

- **hex[3:0]** - 4-bit hexadecimal input (0x0 to 0xF)

43.2.2 Output Ports

- **seg[6:0]** - 7-segment display pattern

43.2.3 Parameters

None - fixed functionality for standard 7-segment displays.

43.3 7-Segment Display Layout

43.3.1 Segment Naming Convention

```

aaa
f   b
f   b
ggg
e   c
e   c
ddd

```

43.3.2 Bit Assignment

- **seg[6:0]** = {g, f, e, d, c, b, a}

- **Common Anode:** 0 = segment ON, 1 = segment OFF
- **Active Low:** LED segments illuminate when driven low

43.4 Character Patterns

43.4.1 Complete Truth Table

Hex	Display	seg[6:0]	Segments ON	Pattern
0x0	0	1000000	abcdef	Classic 0
0x1	1	1111001	bc	Right side
0x2	2	0100100	abged	Snake pattern
0x3	3	0110000	abgcd	Right-heavy
0x4	4	0011001	fgbc	Left + right
0x5	5	0010010	afgcd	S-curve
0x6	6	0000010	afgdce	Almost all
0x7	7	1111000	abc	Top + right
0x8	8	0000000	all	All segments
0x9	9	0011000	abgcdf	Top + middle + right
0xA	A	0001000	abcefg	Letter A
0xB	b	0000011	fgedc	Lowercase b
0xC	C	1000110	afed	Letter C
0xD	d	0100001	bgedc	Lowercase d
0xE	E	0000110	afged	Letter E
0xF	F	0001110	afge	Letter F

43.4.2 Visual Representation

```

###  #  ###  ###  #  #  ###  ###  ###  ###  ###
#  #  #  #  #  #  #  #  #  #  #  #  #  #  #
#  #  #  ###  ###  ###  ###  ###  #  ###  ###
#  #  #  #  #  #  #  #  #  #  #  #  #

```

###	#	###	###	#	###	###	#	###	#
0	1	2	3	4	5	6	7	8	9

###	###	##	###	###	###
#	#	#	#	#	#
###	###	#	#	#	###
#	#	#	#	#	#
#	#	###	##	###	###
A	b	C	d	E	F

43.5 Implementation Details

43.5.1 Case Statement Structure

```

always_comb begin
  casez (hex)
    4'h0: seg = 7'b1000000; // 0: abcdef (not g)
    4'h1: seg = 7'b1111001; // 1: bc
    4'h2: seg = 7'b0100100; // 2: abged
    4'h3: seg = 7'b0110000; // 3: abgcd
    4'h4: seg = 7'b0011001; // 4: fgbc
    4'h5: seg = 7'b0010010; // 5: afgcd
    4'h6: seg = 7'b0000010; // 6: afgdce
    4'h7: seg = 7'b1111000; // 7: abc
    4'h8: seg = 7'b0000000; // 8: all segments
    4'h9: seg = 7'b0011000; // 9: abgcdf
    4'ha: seg = 7'b0001000; // A: abcefg (not d)
    4'hb: seg = 7'b0000011; // b: fgedc (not ab)
    4'hc: seg = 7'b1000110; // C: afed (not bcd)
    4'hd: seg = 7'b0100001; // d: bgedc (not af)
    4'he: seg = 7'b0000110; // E: afged (not bc)
    4'hf: seg = 7'b0001110; // F: afge (not bcd)
    default: seg = 7'bxxxxxxx; // Invalid - undefined output
  endcase
end

```

43.5.2 Design Choices

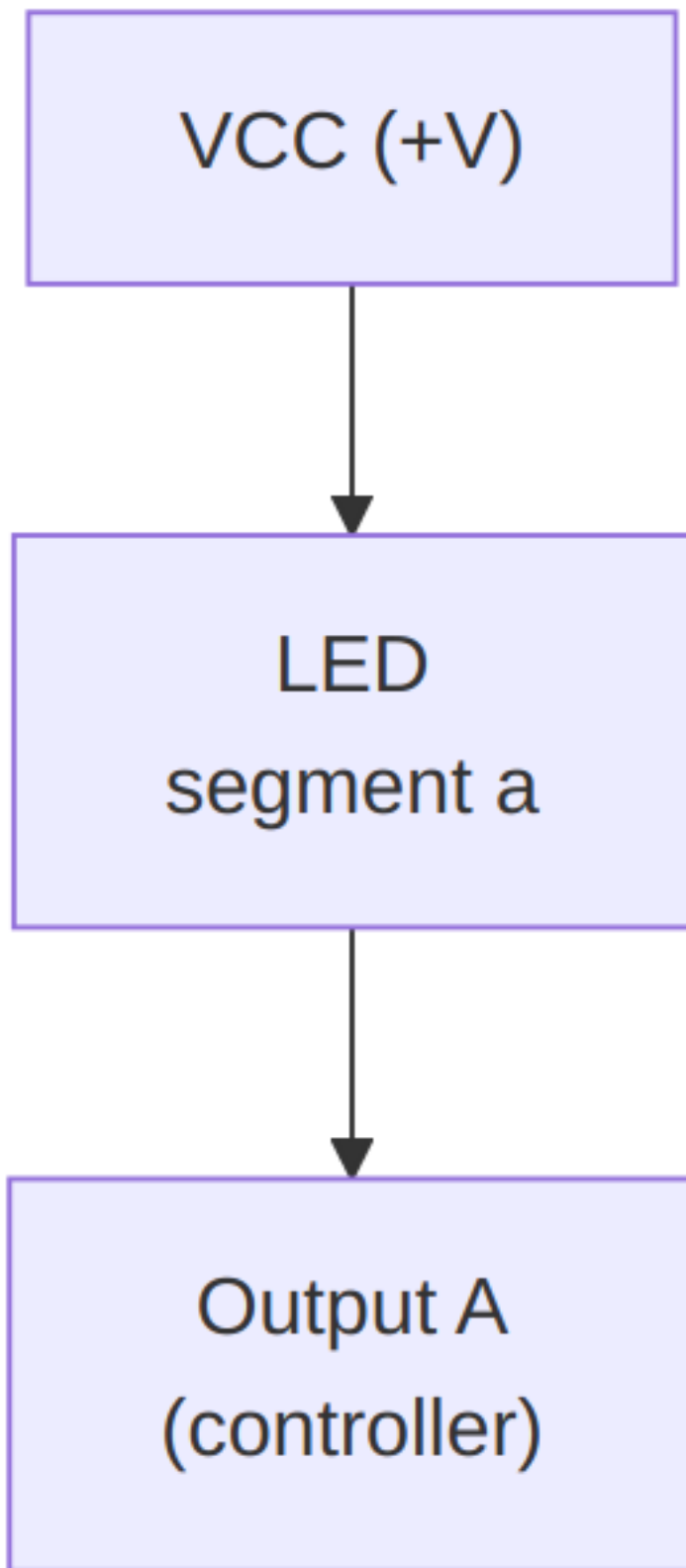
43.5.2.1 Lowercase vs. Uppercase

- **A**: Uppercase (more readable)
- **b, d**: Lowercase (distinguishable from 8, 0)
- **C**: Uppercase (standard representation)
- **E, F**: Uppercase (clear representation)

43.5.2.2 *Common Anode Configuration*

- **Active low outputs:** 0 = LED on, 1 = LED off
- **Current sourcing:** Display provides +V, controller sinks current
- **Logic inversion:** Many controllers expect active-high, requiring external inversion

43.6 Display Technology Integration



Common Anode Connection

43.6.1 Common Cathode Alternative

For common cathode displays, invert all outputs:

```
// For common cathode displays:  
assign seg_common_cathode = ~seg;
```

43.6.2 Current Limiting

Controller → [Resistor] → LED Segment → Common Anode (+V)

Typical resistor values: 220Ω - 1kΩ depending on LED specifications.

43.7 Use Cases and Applications

43.7.1 Debug Displays

```
// Display processor status  
hex_to_7seg status_display (  
    .hex(processor_status[3:0]),  
    .seg(status_segments)  
);
```

43.7.2 Multi-Digit Displays

```
// 4-digit hex display  
hex_to_7seg digit0 (.hex(value[3:0]), .seg(seg0));  
hex_to_7seg digit1 (.hex(value[7:4]), .seg(seg1));  
hex_to_7seg digit2 (.hex(value[11:8]), .seg(seg2));  
hex_to_7seg digit3 (.hex(value[15:12]), .seg(seg3));
```

43.7.3 Memory/Register Viewers

```
// Display memory contents  
hex_to_7seg mem_display (  
    .hex(memory_data[address]),  
    .seg(memory_segments)  
);
```

43.7.4 Error Code Display

```
// Show error codes  
hex_to_7seg error_display (  
    .hex(error_code),  
    .seg(error_segments)  
);
```

43.8 Timing Characteristics

43.8.1 Propagation Delay

- **Combinational logic:** Case statement maps to LUT
- **Typical delay:** 1 LUT delay (sub-nanosecond)
- **No clock dependency:** Immediate response to input changes

43.8.2 Display Update Rate

- **Electronic response:** Immediate (propagation delay)
- **Visual response:** Limited by LED rise/fall times (~microseconds)
- **Human perception:** >30Hz refresh rate for flicker-free display

43.9 Design Variations

43.9.1 Extended Character Set

Some implementations add additional characters:

```
// Extended characters (implementation dependent)
4'h10: seg = 7'b1111111; // Blank/space
4'h11: seg = 7'b0111111; // Dash/minus
4'h12: seg = 7'b1110111; // Underscore
```

43.9.2 Alternative Fonts

Different character representations for better readability:

```
// Alternative 'b' (uppercase-like)
4'hb: seg = 7'b0000111; // More symmetric b
```

43.9.3 Brightness Control

```
// PWM brightness control
logic [6:0] seg_raw;
hex_to_7seg decoder (.hex(hex_in), .seg(seg_raw));

// Apply PWM for brightness
assign seg_out = brightness_enable ? seg_raw : 7'b1111111;
```


43.10 Multiplexed Display Support

43.10.1 Time-Division Multiplexing

```
// 4-digit multiplexed display
logic [1:0] digit_select;
logic [3:0] current_hex;

// Select current digit data
always_comb begin
    case (digit_select)
        2'b00: current_hex = value[3:0];    // Digit 0
        2'b01: current_hex = value[7:4];    // Digit 1
        2'b10: current_hex = value[11:8];   // Digit 2
        2'b11: current_hex = value[15:12];  // Digit 3
    endcase
end

hex_to_7seg display_decoder (
    .hex(current_hex),
    .seg(segments)
);

// Digit enable signals (one active at a time)
assign digit_enable = ~(4'b0001 << digit_select);
```

43.10.2 Refresh Rate Calculation

```
// For 4-digit display at 1kHz total refresh:
// Each digit displayed for 250µs
// Switching frequency: 4kHz

localparam REFRESH_DIVIDER = CLOCK_FREQ / 4000;
```

43.11 Verification and Testing

43.11.1 Exhaustive Testing

```
// Test all 16 hex values
for (int i = 0; i < 16; i++) begin
    hex_input = i[3:0];
    #1;
    expected_segments = reference_table[i];
    assert(seg_output == expected_segments);
end
```

43.11.2 Visual Verification

- **Physical testing:** Connect to actual 7-segment display
- **Simulation:** Use testbench with visual output
- **Reference comparison:** Compare with known-good implementations

43.11.3 Pattern Validation

```
// Verify no segments are permanently on/off
logic [6:0] all_segments_used = '0;
for (int i = 0; i < 16; i++) begin
    all_segments_used |= ~reference_table[i]; // Accumulate active
segments
end
assert(all_segments_used == 7'b1111111); // All segments used
somewhere
```

43.12 Common Issues and Debugging

43.12.1 Segment Mapping Errors

- **Bit order:** Ensure {g,f,e,d,c,b,a} ordering matches hardware
- **Active level:** Verify common anode vs. common cathode
- **Pin assignment:** Check physical connections match bit assignments

43.12.2 Display Quality Issues

- **Ghost segments:** Check current limiting and multiplexing timing
- **Dim display:** Verify current levels and refresh rates
- **Flicker:** Ensure adequate refresh rate (>30Hz visible, >60Hz preferred)

43.12.3 Character Readability

- **Ambiguous characters:** 6 vs b, 8 vs B, 0 vs O
- **Context:** Use surrounding digits/letters for disambiguation
- **Alternative fonts:** Consider modified patterns for better distinction

43.13 Related Modules

- **BCD to 7-segment:** For decimal-only displays

- **ASCII to 7-segment:** Extended character set support
- **Display multiplexer:** Time-division multiplexing controller
- **PWM brightness controller:** Dynamic brightness adjustment

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44 icg

An Integrated Clock Gating (ICG) module that provides controlled clock gating for power optimization in digital designs.

44.1 Overview

The `icg` module implements a latch-based clock gating cell that allows selective disabling of clock signals to reduce dynamic power consumption. This is a fundamental power management technique used extensively in modern ASIC and FPGA designs.

44.2 Module Declaration

```
module icg(
    input  logic en,
    input  logic clk,
    output logic gclk
);
```

44.3 Ports

44.3.1 Inputs

Port	Width	Description
en	1	Enable signal - when high, allows clock to pass through
clk	1	Input clock signal to be gated

44.3.2 Outputs

Port	Width	Description
gclk	1	Gated clock output - clk when en=1, low when en=0

44.4 Functionality

44.4.1 Clock Gating Operation

The ICG module implements a safe clock gating scheme that:

1. **Latches Enable Signal:** Uses a negative-edge triggered latch to capture the enable signal
2. **Prevents Glitches:** Ensures the enable signal is stable during clock high periods
3. **Generates Gated Clock:** ANDs the latched enable with the input clock

44.4.2 Implementation Details

```
logic en_out;
```

```
// Latch-based approach - enable is latched on clock low
always_latch begin
    if (!clk) begin
        en_out = en;
    end
end
```

```
// Generate gated clock
assign gclk = en_out && clk;
```

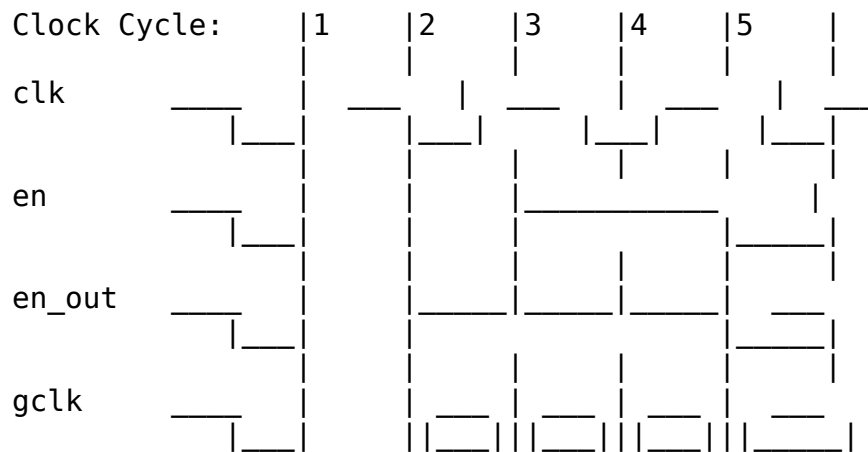
44.5 Timing Characteristics

44.5.1 Critical Timing Requirements

Characteristic	Description
Setup Time	en must be stable before clk rising edge
Hold Time	en must remain stable after clk rising edge

Characteristic	Description
Propagation Delay	Delay from clk to gclk through AND gate
Enable Latching	en is latched when clk is low

44.5.2 Timing Diagram



44.6 Power Optimization Benefits

44.6.1 Dynamic Power Reduction

The ICG module reduces dynamic power consumption by:

- **Clock Tree Power:** Eliminates switching activity in downstream clock networks
- **Register Power:** Prevents unnecessary flip-flop transitions when data is not changing
- **Combinational Power:** Reduces spurious switching in logic fed by gated registers

44.6.2 Power Savings Calculation

Typical power savings achievable: - **Clock Network:** 20-40% reduction in clock tree power - **Sequential Logic:** 30-60% reduction in flip-flop power - **Overall Dynamic Power:** 10-30% depending on gating efficiency

44.7 Design Considerations

44.7.1 Safety Requirements

- **Glitch-Free Operation:** Latch-based design prevents glitches on gated clock
- **Setup/Hold Timing:** Enable signal must meet timing requirements relative to clock
- **Clock Domain:** All logic using gclk must be in the same clock domain

44.7.2 Enable Signal Guidelines

```
// Good: Enable changes during clock low period
always_ff @(posedge clk) begin
    if (rst_n) begin
        enable_reg <= new_enable_value; // Changes on clock edge
    end
end

// Caution: Ensure enable is stable during clock high
assign icg_enable = enable_reg && additional_condition;
```

44.8 Usage Examples

44.8.1 Basic Register Bank Gating

```
// Clock gating for a register bank
logic bank_enable;
logic gated_clk;

icg u_bank_icg (
    .en    (bank_enable),
    .clk   (sys_clk),
    .gclk  (gated_clk)
);

// Register bank using gated clock
always_ff @(posedge gated_clk or negedge rst_n) begin
    if (!rst_n) begin
        register_bank <= '0;
    end else begin
        register_bank <= next_register_values;
    end
end
```

44.8.2 CPU Pipeline Stage Gating

```
// Pipeline stage clock gating
logic pipe_stage_valid;
logic stage_clk;

icg u_pipe_icg (
    .en    (pipe_stage_valid),
    .clk   (cpu_clk),
    .gclk  (stage_clk)
);

// Pipeline stage registers
always_ff @(posedge stage_clk or negedge rst_n) begin
    if (!rst_n) begin
        pipe_stage_regs <= '0;
    end else begin
        pipe_stage_regs <= pipe_stage_inputs;
    end
end
```

44.8.3 Memory Interface Gating

```
// Memory controller clock gating
logic mem_access_active;
logic mem_clk;

icg u_mem_icg (
    .en    (mem_access_active),
    .clk   (system_clk),
    .gclk  (mem_clk)
);

assign mem_access_active = mem_read_req || mem_write_req || mem_busy;
```

44.9 Implementation Alternatives

44.9.1 Standard Cell ICG

Many ASIC libraries provide optimized ICG cells:

```
// Using library ICG cell (example)
CLOCK_GATE_ICG u_lib_icg (
    .CLK    (clk),
    .EN     (en),
    .GCLK   (gclk)
);
```

44.9.2 FPGA Implementation

For FPGA targets, use dedicated clock control primitives:

```
// Xilinx BUFGCE example
BUFGCE u_bufgce (
    .I   (clk),
    .CE  (en),
    .O   (gclk)
);
```

44.10 Verification Considerations

44.10.1 Functional Tests

- **Enable/Disable:** Verify correct gating behavior for enable transitions
- **Clock Integrity:** Ensure gated clock maintains proper duty cycle when enabled
- **Reset Behavior:** Verify proper operation during reset sequences

44.10.2 Timing Tests

```
// Timing verification example
initial begin
    // Test enable setup/hold timing
    @(negedge clk);
    en = 1'b1; // Enable during clock low
    @(posedge clk);
    // Verify gclk follows clk

    @(negedge clk);
    en = 1'b0; // Disable during clock low
    @(posedge clk);
    // Verify gclk remains low
end
```

44.10.3 Power Analysis

- **Activity Monitoring:** Track switching activity reduction in gated domains
- **Power Estimation:** Measure power savings using EDA power analysis tools
- **Efficiency Metrics:** Calculate gating efficiency (% time clock is disabled)

44.11 Synthesis Guidelines

44.11.1 Technology Mapping

- **ASIC:** Use library-specific ICG cells for optimal area/power/timing
- **FPGA:** Map to dedicated clock control primitives (BUFGCE, etc.)
- **Timing Closure:** Account for additional setup/hold requirements

44.11.2 Clock Tree Synthesis

```
# Example synthesis constraints for ICG
set_case_analysis 0 [get_pins u_icg/en] ;# For power analysis
set_clock_gating_check -setup 0.1 [get_pins u_icg/gclk]
set_clock_gating_check -hold 0.1 [get_pins u_icg/gclk]
```

44.12 Design Integration

44.12.1 Clock Tree Integration

The ICG should be placed strategically in the clock tree:

Main Clock → ICG → Local Clock Tree → Registers
 ↑
 Enable Logic

44.12.2 Power Domain Considerations

- **Retention:** Ensure enable logic remains powered during clock gating
- **Isolation:** Properly isolate gated domains in multi-voltage designs
- **Power Sequencing:** Coordinate with power management units

44.13 Related Modules

- `clock_gate_ctrl`: Enhanced clock gating controller with additional features
- `clock_divider`: Clock frequency division functionality
- `reset_sync`: Reset synchronization for gated clock domains

The `icg` module provides a fundamental building block for power-efficient digital design, enabling significant power savings through controlled clock gating while maintaining design safety and functionality.

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45 Johnson-to-Binary Converter (johnson2bin.sv)

45.1 Purpose

Converts Johnson counter codes to binary representation for use in asynchronous FIFOs with non-power-of-2 depths. Unlike standard Gray-to-binary conversion, this module handles the unique properties of Johnson counter sequences.

45.2 Ports

45.2.1 Input Ports

- **clk** - Clock signal (required for leading/trailing one detection)
- **rst_n** - Active-low reset
- **gray[JCW-1:0]** - Johnson counter input (misleadingly named “gray”)

45.2.2 Output Ports

- **binary[WIDTH-1:0]** - Binary output representing position in sequence

45.2.3 Parameters

- **JCW** - Johnson Counter Width (equals FIFO DEPTH)
- **WIDTH** - Binary output width (typically $\lceil \log_2(\text{DEPTH}) \rceil + 1$)
- **INSTANCE_NAME** - String identifier for debug

45.3 Johnson Counter Sequence Review

45.3.1 Johnson Counter Characteristics

Johnson counters are **shift registers with inverted feedback**:

For JCW=6 (DEPTH=6):

State 0: 000000 ← Initial state
State 1: 100000 ← Shift left, insert 1
State 2: 110000
State 3: 111000

```

State 4: 111100
State 5: 111110
State 6: 111111 ← All 1s (special case)
State 7: 011111 ← Shift left, insert ~MSB (0)
State 8: 001111
State 9: 000111
State 10: 000011
State 11: 000001
State 0: 000000 ← Cycle complete (12 states total)

```

45.3.2 Key Properties

- **Single bit transitions:** Only one bit changes per state
- **Two phases:**
 - **First half** (0 to DEPTH-1): Filling with 1s from left
 - **Second half** (DEPTH to 2×DEPTH-1): Emptying 1s from left
- **Wrap indicator:** MSB indicates which half of cycle

45.4 Conversion Algorithm

45.4.1 Strategy Overview

The conversion uses **position detection** of the transition between 1s and 0s:

```

if (w_all_zeroes || w_all_ones) begin
    w_binary = {WIDTH{1'b0}};
end else if (gray[JCW-1]) begin
    // Second half: use leading one position directly
    w_binary = {{(WIDTH-N){1'b0}}, w_trailing_one};
end else begin
    // First half: use trailing one + 1
    w_binary = {{(WIDTH-N){1'b0}}, (w_leading_one + 1'b1)};
end

```

45.4.2 Position Detection Module

```

leading_one_trailing_one #(
    .WIDTH(JCW),
    .INSTANCE_NAME(INSTANCE_NAME)
) u_leading_one_trailing_one (
    .data(gray),
    .leadingone(w_leading_one),
    .trailingone(w_trailing_one),
    .all_zeroes(w_all_zeroes),
    .all_ones(w_all_ones),

```

```
        .valid(w_valid)
    );
```

45.5 Detailed Conversion Examples

45.5.1 First Half Conversion (MSB = 0)

Johnson: 001111 ($w_leading_one = 5$, $w_trailing_one = 2$)
 Logic: First half, so $position = w_leading_one + 1 = 5 + 1 = 6$
 Binary: 000110 (with MSB=0 indicating first half)

45.5.2 Second Half Conversion (MSB = 1)

Johnson: 111000 ($w_leading_one = 0$, $w_trailing_one = 2$)
 Logic: Second half, so $position = w_trailing_one = 2$
 Binary: 100010 (with MSB=1 indicating second half)

45.5.3 Special Cases

Johnson: 000000 → Binary: 000000 (all zeros case)
 Johnson: 111111 → Binary: 000000 (all ones case - same as zeros)

45.6 Implementation Deep Dive

45.6.1 Three-Part Binary Construction

```
assign binary[WIDTH-1] = gray[JCW-1];           // MSB = wrap
indicator
assign binary[WIDTH-2:0] = w_binary[WIDTH-2:0]; // Lower bits
= position
```

45.6.2 Width Calculations

```
localparam int N = $clog2(JCW);                 // Address
bits needed
localparam int PAD_WIDTH = (WIDTH > N+1) ? WIDTH-N-1 : 0; // Padding
if needed
```

45.6.3 Why This Algorithm Works

45.6.3.1 First Half Logic (MSB = 0)

- **Pattern:** 000...0111...1 (0s followed by 1s)
- **Leading one:** Position of leftmost 1
- **Conversion:** $Position = leading_one + 1$
- **Reasoning:** We've filled $(leading_one + 1)$ positions with 1s

45.6.3.2 Second Half Logic (MSB = 1)

- **Pattern:** 111...1000...0 (1s followed by 0s)
- **Trailing one:** Position of rightmost 1
- **Conversion:** Position = trailing_one
- **Reasoning:** We're emptying from the left, trailing_one shows how far

45.7 Use in Asynchronous FIFO

45.7.1 Context in FIFO Operation

```
// fifo_async.sv (USE_JOHNSON=1) usage:
johnson2bin #(
    .JCW(JCW),                // = DEPTH
    .WIDTH(AW + 1),           // Address width + wrap bit
    .INSTANCE_NAME("rd_ptr_gray2bin_inst")
) rd_ptr_gray2bin_inst(
    .binary(w_wdom_rd_ptr_bin), // Binary for arithmetic
    .gray(r_wdom_rd_ptr_gray),  // Johnson counter from CDC
    .clk(wr_clk),
    .rst_n(wr_rst_n)
);
```

45.7.2 Why Clock is Required

Unlike pure combinational Gray-to-binary conversion, Johnson-to-binary needs the leading_one_trailing_one helper, which may use sequential logic for complex position detection.

45.8 Comparison with Standard Gray2Bin

Aspect	Standard Gray2Bin	Johnson2Bin (johnson2bin)
Input type	Traditional Gray code	Johnson counter sequence
Algorithm	XOR reduction	Position detection
Complexity	Simple combinational	Complex position logic
Clock requirement	None	Required for helper modules
Width scaling	Logarithmic	Linear with JCW

Aspect	Standard Gray2Bin	Johnson2Bin (johnson2bin)
Use case	Power-of-2 sequences	Any even sequences

45.9 Performance Characteristics

45.9.1 Timing Analysis

- **Critical path:** Through position detection logic
- **Delay components:**
 - Leading/trailing one detection
 - Binary arithmetic (addition)
 - Output multiplexing
- **Synthesis complexity:** Higher than standard Gray conversion

45.9.2 Resource Utilization

Resources scale with JCW (Johnson Counter Width):

- Small FIFOs (JCW ≤ 16): Reasonable overhead
- Medium FIFOs (JCW = 32-64): Significant resources
- Large FIFOs (JCW > 64): May become limiting factor

45.10 Design Considerations

45.10.1 When to Use Johnson vs. Gray

// Use Johnson counter approach when:

```
parameter int DEPTH = 10; // Non-power-of-2 required
parameter int DEPTH = 6;  // Specific size needed
parameter int DEPTH = 14; // Standard Gray won't work
```

// Use standard Gray approach when:

```
parameter int DEPTH = 8;  // Power-of-2 is acceptable
parameter int DEPTH = 16; // Efficiency more important
parameter int DEPTH = 64; // Large depth, resource conscious
```

45.10.2 Resource Planning

// Resource overhead estimation:

// Standard Gray: $\sim \log_2(\text{DEPTH})$ XOR gates

```
// Johnson: ~DEPTH comparators + position logic
```

```
// Break-even point typically around DEPTH = 16-32
```

45.10.3 Verification Challenges

```
// More complex verification due to:  
// 1. Two-phase sequence behavior  
// 2. Position detection correctness  
// 3. Special case handling (all 0s, all 1s)  
// 4. Wraparound boundary conditions
```

45.11 Error Conditions and Debug

45.11.1 Invalid Johnson Sequences

The `leading_one_trailing_one` module provides validation:

```
.valid(w_valid) // Indicates valid Johnson pattern
```

Valid Johnson patterns have exactly one transition from 1s to 0s (or vice versa).

45.11.2 Debug Visibility

```
// Monitor internal state for debugging:  
// - w_leading_one: Position of leftmost 1  
// - w_trailing_one: Position of rightmost 1  
// - w_all_zeroes: Special case flag  
// - w_all_ones: Special case flag  
// - w_valid: Sequence validity
```

45.11.3 Common Issues

1. **Invalid sequences:** Non-Johnson patterns on input
2. **Width mismatches:** JCW vs. actual Johnson counter width
3. **Phase confusion:** Misunderstanding first vs. second half logic
4. **Boundary conditions:** All-zeros and all-ones handling

45.12 Related Modules

- **counter_johnson:** Generates Johnson counter sequences
- **leading_one_trailing_one:** Position detection helper
- **fifo_async / gaxi_fifo_async (USE_JOHNSON=1):** Primary users of this conversion

- **gray2bin**: Standard Gray-to-binary for comparison
- **fifo_control**: Uses converted binary values for status generation

45.13 Advanced Topics

45.13.1 Hierarchical Johnson Counters

For very large depths, consider hierarchical approach:

```
// Break large Johnson counter into smaller segments
// Use multiple johnson2bin instances with higher-level arbitration
```

45.13.2 Alternative Position Detection

Some implementations use different position detection algorithms: - **Priority encoders**: Find first/last set bit - **Thermometer decoders**: Convert 1-hot positions - **LUT-based**: For small, fixed widths

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46 Leading One Trailing One Module

46.1 Purpose

The `leading_one_trailing_one` module identifies the positions of the most significant bit (leading one) and least significant bit (trailing one) in an input data vector. It provides both index outputs and one-hot vector representations of these positions, along with status flags for edge cases.

46.2 Key Features

- Finds the highest set bit position (leading one)
- Finds the lowest set bit position (trailing one)
- Provides one-hot vector outputs for both positions
- Includes validity and edge case detection
- Parameterizable width with automatic index sizing

46.3 Port Description

46.3.1 Parameters

- **WIDTH:** Width of input data vector (default: 8)
- **INSTANCE_NAME:** String identifier for debugging (default: “”)

46.3.2 Inputs

Port	Width	Description
data	WIDTH	Input data vector to analyze

46.3.3 Outputs

Port	Width	Description
leadingone	$\$clog2(WIDTH)$	Index of the most significant set bit
leadingone_vector	WIDTH	One-hot vector with leading one position marked
trailingone	$\$clog2(WIDTH)$	Index of the least significant set bit
trailingone_vector	WIDTH	One-hot vector with trailing one position marked
all_zeroes	1	High when input data is all zeros
all_ones	1	High when input data is all ones
valid	1	High when input data contains at least one set bit

46.4 Implementation Details

46.4.1 Submodules

The module instantiates two specialized finder modules:

1. **find_last_set:** Locates the position of the most significant set bit
 - Connected to leadingone output
 - Implements priority encoding from MSB to LSB
2. **find_first_set:** Locates the position of the least significant set bit

- Connected to trailingone output
- Implements priority encoding from LSB to MSB

46.4.2 One-Hot Vector Generation

```
always_comb begin
    leadingone_vector = '0;
    trailingone_vector = '0;

    // Only set vector bits if there is valid data
    if (|data) begin
        if (int'(leadingone) < WIDTH) begin
            leadingone_vector[leadingone] = 1'b1;
        end

        if (int'(trailingone) < WIDTH) begin
            trailingone_vector[trailingone] = 1'b1;
        end
    end
end
```

46.4.3 Status Flag Logic

- **all_ones:** Uses reduction AND (&data) to detect when all bits are set
- **all_zeros:** Uses reduction OR negation (~(|data)) to detect when no bits are set
- **valid:** Uses reduction OR (|data) to indicate presence of at least one set bit

46.5 Special Implementation Notes

1. **Index Width Calculation:** Uses $\$clog2(WIDTH)$ to automatically size index outputs for any input width
2. **Bounds Checking:** Includes explicit bounds checking ($int'(leadingone) < WIDTH$) to prevent out-of-range indexing
3. **Zero Input Handling:** When input is all zeros:
 - One-hot vectors remain all zeros
 - Index values are undefined but bounded
 - valid flag indicates invalid state

4. **Type Casting:** Uses `int'()` casting for index comparisons to ensure proper integer arithmetic

46.6 Usage Examples

46.6.1 8-bit Example

```
Input:  data = 8'b00101000
Output: leadingone = 3'd5
        leadingone_vector = 8'b00100000
        trailingone = 3'd3
        trailingone_vector = 8'b00001000
        all_zeroes = 1'b0
        all_ones = 1'b0
        valid = 1'b1
```

46.6.2 Edge Cases

```
Input:  data = 8'b00000000
Output: leadingone = undefined
        leadingone_vector = 8'b00000000
        trailingone = undefined
        trailingone_vector = 8'b00000000
        all_zeroes = 1'b1
        all_ones = 1'b0
        valid = 1'b0
```

```
Input:  data = 8'b11111111
Output: leadingone = 3'd7
        leadingone_vector = 8'b10000000
        trailingone = 3'd0
        trailingone_vector = 8'b00000001
        all_zeroes = 1'b0
        all_ones = 1'b1
        valid = 1'b1
```

46.7 Applications

- Priority arbitration systems
- Bit manipulation operations
- Data compression algorithms
- Network packet processing
- Cache management systems

46.8 Navigation

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47 Modulo-3 Compressor

Module: `mod_3_compress.sv` **Location:** `rtl/common/` **Status:** Production Ready

47.1 Overview

`mod_3_compress` is a purely combinational block that computes $d_in \bmod 3$ for a 16-bit operand, returning the 2-bit remainder (0..2). It is built in the same carry-save-compressor style as `div_by_15_ceil_32compress.sv`, a 3:2 compressor tree feeding a final carry-propagate add and a small fold, but it emits only the remainder, which is exactly what the monbus burst-writer needs to round a beat count **down** to a whole number of 3-beat records: $rounded = X - (X \bmod 3)$.

It uses no `*` or `/` operator, so it infers no DSP block and no iterative divider, just a shallow tree of carry-save adders and a couple of small adds.

47.1.1 Key Features

- **Combinational, single-cycle:** No clock, no state; pure logic from `d_in` to `rem_out`
- **No multiply / divide:** Avoids inferred DSP and iterative dividers
- **Carry-save tree:** Eight base-4 digits reduced with 3:2 compressors instead of a ripple-add chain
- **Remainder only:** Returns just $d_in \bmod 3$ (0..2), the quantity the monbus record rounding needs
- **Shared style:** Same construction as `div_by_15_ceil_32compress.sv`, reusing `math_adder_carry_save_nbit`

47.2 Module Purpose

The monbus compressor packs monitor packets into fixed 3-beat records. To decide how many whole records a beat count covers, it needs $X - (X \bmod 3)$, which requires the remainder $X \bmod 3$. This module computes that remainder combinationaly without a divider, so the burst-writer can round the count down in a single cycle of logic.

Use Cases:

- Rounding a monbus beat count down to a whole number of 3-beat records ($X - (X \bmod 3)$)
- Any combinational $\bmod 3$ of a 16-bit value where a DSP-free, divider-free implementation is desired
- A worked reference for the base-4 digit-sum residue technique with carry-save reduction

Key Benefit: Produces $d_{in} \bmod 3$ in one combinational stage with a shallow carry-save tree, no multiplier, divider, or DSP block required.

47.3 Parameters

This module has no parameters. The operand width is fixed at 16 bits and the result at 2 bits.

(Internally, `localparam int BITS = 6` sizes the carry-save datapath so the weight-2 carry left-shifts have headroom.)

47.4 Port Groups

47.4.1 Data

Port	Direction	Width	Description
d_in	Input	16	Operand whose remainder $\bmod 3$ is computed
rem_out	Output	2	$d_{in} \bmod 3$, in the range 0..2

47.5 Functional Description

47.5.1 Base-4 Digit-Sum Method

The core identity is that $4^k \equiv 1 \pmod{3}$. If d_{in} is split into 2-bit groups, each group is a base-4 digit with weight 4^k , and since every weight is $1 \pmod{3}$, the **sum of the eight digits is congruent to $d_{in} \pmod{3}$** . The problem therefore reduces to summing eight small digits and taking that sum $\bmod 3$.

The eight 2-bit groups are zero-extended to the 6-bit carry-save width:

```

assign w_g0 = {{(BITS-2){1'b0}}, d_in[1:0]};
assign w_g1 = {{(BITS-2){1'b0}}, d_in[3:2]};
// ... through ...
assign w_g7 = {{(BITS-2){1'b0}}, d_in[15:14]};

```

47.5.23:2 Compressor Tree

Rather than a ripple-add chain, the eight digits are reduced with a tree of math_adder_carry_save_nbit 3:2 compressors. Each carry output has weight 2, so wherever a carry feeds the next level it is left-shifted by one ({carry[BITS-2:0], 1'b0}); the always-zero top carry bit is dropped, exactly as the div-by-15 reference does:

- **Level 1:** three compressors reduce the eight digits three at a time (the last triple padded with 0)
- **Level 2:** two compressors combine the level-1 sums and shifted carries
- **Level 3:** one compressor, holding w_carryE2 for the next level
- **Level 4:** one compressor folds in the held carry

A final carry-propagate add collapses the last sum/carry pair into the digit sum (0..24, still $\equiv d_in \pmod{3}$):

```

assign w_grp_sum = w_sumG4 + {w_carryG4[BITS-2:0], 1'b0};

```

47.5.3 Final Fold and Conditional Subtract

The digit sum w_grp_sum is at most 24, so it is folded once more onto its own base-4 digits (same residue mod 3), giving a value in 0..7, and then a conditional subtract brings it into the final 0..2 range:

```

assign w_fold = {2'b0, w_grp_sum[1:0]} + {2'b0, w_grp_sum[3:2]} +
{3'b0, w_grp_sum[4]};
assign rem_out = 2'((w_fold >= 4'd6) ? (w_fold - 4'd6)
                  : (w_fold >= 4'd3) ? (w_fold - 4'd3)
                  : w_fold);

```

The two-branch subtract handles w_fold values up to 7 (subtract 6, subtract 3, or pass through), yielding the exact remainder 0, 1, or 2.

47.6 Usage Example

```

// Round a beat count down to a whole number of 3-beat monbus records.
logic [15:0] beat_count;
logic [1:0]   beat_rem;
logic [15:0] rounded_count;

mod_3_compress u_mod3 (

```

```

        .d_in      (beat_count),
        .rem_out   (beat_rem)
    );

// rounded = X - (X mod 3)
assign rounded_count = beat_count - {14'b0, beat_rem};

```

47.7 Design Notes

- **Why remainder only?** The div-by-15 reference computes a quotient; here only the residue is needed, so the quotient machinery is dropped, keeping the logic shallow.
- **Datapath width headroom.** BITS = 6 (rather than the minimum 5 needed for a 0..24 sum) gives the weight-2 carry left-shifts room so the dropped top carry is always 0, matching the div-by-15 construction.
- **No inferred arithmetic primitives.** Because there is no * or /, synthesis produces adders and LUTs only, no DSP slices, no multi-cycle divider.
- **Purely combinational.** rem_out tracks d_in with no clock; register the result externally if a pipeline stage is desired.

47.8 Related Modules

47.8.1 Used By

- [monbus_compressor](#) - Rounds a beat count down to whole 3-beat records when packing monitor packets
- [monbus_group_core](#) - Shared filter/FIFO core of the monbus group wrappers

47.8.2 Uses

- [math_adder_carry_save](#) - The N-bit carry-save (3:2) compressor instantiated throughout the tree

47.8.3 See Also

- `div_by_15_ceil_32compress.sv` - The reference implementation this module's compressor style follows

47.9 References

47.9.1 Source Code

- `rtl/common/mod_3_compress.sv`
- `rtl/common/math_adder_carry_save.sv` (instantiated submodule)
- `rtl/common/div_by_15_ceil_32compress.sv` (style reference)

47.9.2 Documentation

- `rtl/common/PRD.md`

Last Updated: 2026-07-15

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48 PWM (Pulse Width Modulation) Module

48.1 Purpose

The PWM module generates multiple independent pulse width modulation signals with configurable duty cycles, periods, and repeat counts. Each channel operates as an independent state machine with precise timing control and completion detection.

48.2 Key Features

- Multiple independent PWM channels (parameterizable)
- Individual duty cycle, period, and repeat count control per channel
- Edge-triggered start control with automatic edge detection
- Completion status reporting per channel
- Support for continuous operation (infinite repeat)
- Precise timing with configurable counter width

48.3 Port Description

48.3.1 Parameters

- **WIDTH:** Bit width of counters and timing parameters (default: 8)

- **CHANNELS:** Number of independent PWM channels (default: 4)

48.3.2 Inputs

Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
start	CHANNELS	Start trigger for each channel (edge-triggered)
duty	CHANNELS*WIDTH	Concatenated duty cycle values for all channels
period	CHANNELS*WIDTH	Concatenated period values for all channels
repeat_count	CHANNELS*WIDTH	Concatenated repeat counts (0 = infinite)

48.3.3 Outputs

Port	Width	Description
done	CHANNELS	Completion status for each channel
pwm_out	CHANNELS	PWM output signals for each channels

48.4 State Machine

Each PWM channel implements a 3-state finite state machine:

48.4.1 States

```
typedef enum logic [1:0] {
    IDLE = 2'b00,      // Waiting for start trigger
    RUNNING = 2'b01,   // Actively generating PWM
    DONE = 2'b10       // Completed all repeats
} state_t;
```

48.4.2 State Transitions

48.4.2.1 IDLE → RUNNING

- **Condition:** w_start_edge && local_period > 0
- **Action:** Reset counters and begin PWM generation

48.4.2.2 *RUNNING → DONE*

- **Condition:** `w_period_complete && w_all_repeats_done`
- **Action:** Stop PWM generation and assert done flag

48.4.2.3 *DONE → RUNNING*

- **Condition:** `w_start_edge`
- **Action:** Restart PWM generation with reset counters

48.4.2.4 *Any State → IDLE*

- **Condition:** System reset
- **Action:** Initialize all registers

48.5 Implementation Details

48.5.1 Per-Channel Parameter Extraction

```
localparam int EndIdx = (i + 1) * WIDTH - 1;
assign local_duty = duty[EndIdx:WIDTH];
assign local_period = period[EndIdx:WIDTH];
assign local_repeat = repeat_count[EndIdx:WIDTH];
```

48.5.2 Edge Detection for Start Signal

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) r_start_prev <= 1'b0;
    else r_start_prev <= start[i];
end
assign w_start_edge = start[i] && !r_start_prev;
```

48.5.3 PWM Output Generation Logic

```
always_comb begin
    case (r_state)
        RUNNING: begin
            if (local_duty == 0) begin
                // 0% duty cycle
                pwm_out[i] = 1'b0;
            end else if (local_duty >= local_period) begin
                // 100% duty cycle (duty >= period)
                pwm_out[i] = 1'b1;
            end else begin
                // Normal case: PWM high for first 'duty' cycles
                pwm_out[i] = (r_count < local_duty);
            end
        end
    end
```

```

        default: begin
            pwm_out[i] = 1'b0;
        end
    endcase
end

```

48.5.4 Counter Management

48.5.4.1 Main Counter (Period Timing)

- Increments each clock cycle during RUNNING state
- Resets to 0 when reaching `local_period - 1`
- Controls PWM frequency

48.5.4.2 Repeat Counter

- Increments each time a period completes
- Compared against `repeat_count` to determine completion
- Special case: `repeat_count == 0` means infinite operation

48.5.5 Critical Control Signals

48.5.5.1 Period Completion Detection

```
assign w_period_complete = (r_count == local_period - 1) && (r_state == RUNNING);
```

48.5.5.2 All Repeats Done Detection

```
assign w_all_repeats_done = (local_repeat == 0) ? 1'b0 : // 0 means
infinite
(r_repeat_value >= local_repeat);
```

48.6 Special Implementation Notes

48.6.1 1. Edge-Triggered Start Control

- Prevents continuous restart while start signal is held high
- Enables precise control of PWM initiation
- Each channel has independent edge detection

48.6.2 2. Duty Cycle Edge Cases

- **0% Duty:** Output permanently low during RUNNING state

- **100% Duty:** Output permanently high when `duty >= period`
- **Normal Operation:** High for first duty clock cycles of each period

48.6.33. Infinite Repeat Mode

- `repeat_count = 0` enables continuous operation
- PWM runs indefinitely until new start trigger
- Useful for continuous motor control or LED dimming

48.6.44. Simultaneous Multi-Channel Operation

- All channels operate independently
- No inter-channel dependencies or conflicts
- Scalable to any number of channels via parameters

48.6.55. Reset Behavior

- Asynchronous reset immediately stops all PWM outputs
- All internal counters and state machines reset to IDLE
- Clean startup guaranteed after reset deassertion

48.7 Timing Examples

48.7.1 Single Period Example (`WIDTH=8, duty=3, period=8`)

```
Clock:  0 1 2 3 4 5 6 7 0 1 2 3 4 5 6 7 ...
Count:  0 1 2 3 4 5 6 7 0 1 2 3 4 5 6 7 ...
PWM:    1 1 1 0 0 0 0 0 1 1 1 0 0 0 0 0 ...
```

48.7.2 Multi-Repeat Example (`repeat_count=2`)

```
Period:  1      2      DONE
PWM:     [Solid][Hatched][Solid][Hatched]
```

48.8 Usage Scenarios

48.8.1 Motor Speed Control

```
// 50% duty cycle, 1kHz frequency (assuming 1MHz clock)
duty = 500;
period = 1000;
repeat_count = 0; // Continuous operation
```

48.8.2 LED Dimming Sequence

```
// Fade effect with 10 brightness levels  
duty = brightness_level * 10;  
period = 100;  
repeat_count = fade_duration;
```

48.8.3 Servo Motor Control

```
// 20ms period with 1-2ms pulse width  
duty = servo_position;           // 1000-2000 for 1-2ms  
period = 20000;                  // 20ms at 1MHz clock  
repeat_count = 1;                // Single pulse
```

48.9 Applications

- Motor speed control
- LED brightness control
- Servo motor positioning
- Audio signal generation
- Power supply switching
- Heating element control

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49 Reset Synchronizer Module

49.1 Purpose

The `reset_sync` module implements a parameterized reset synchronizer that provides asynchronous assertion and synchronous deassertion of reset signals. This is a critical circuit for proper reset distribution in synchronous digital systems, ensuring that reset release occurs synchronously with the clock to prevent metastability and timing violations.

49.2 Key Features

- Asynchronous reset assertion (immediate)

- Synchronous reset deassertion (clocked)
- Parameterizable synchronization depth
- Metastability resolution
- Standard reset synchronizer topology
- Clean reset release timing

49.3 Port Description

49.3.1 Parameters

- N: Number of synchronization stages (default: 3)

49.3.2 Inputs

Port	Width	Description
clk	1	System clock for synchronization
rst_n	1	Asynchronous input reset (active-low)

49.3.3 Outputs

Port	Width	Description
sync_rst_n	1	Synchronized reset output (active-low)

49.4 Reset Synchronizer Theory

49.4.1 The Reset Problem

In synchronous digital systems, reset signals must be carefully managed to avoid: - **Metastability**: When reset is released asynchronously relative to clock - **Timing Violations**: Setup/hold violations during reset release - **Race Conditions**: Different flip-flops releasing from reset at different times

49.4.2 Asynchronous Assert, Synchronous Deassert

The ideal reset behavior is: - **Assert**: Immediate (asynchronous) when reset input becomes active - **Deassert**: Synchronized to clock edge to ensure clean timing

Synchronized deassertion
(N clock cycles after rst_n release)

49.6.3 State Sequence for N=3

Clock	rst_n	r_sync_reg	sync_rst_n	Description
-	0	000	0	Reset active, output low
1	0	001	0	First 1 shifted in
2	0	011	0	Second 1 shifted in
3	0	111	1	Third 1 reaches output
4+	0	111	1	Reset deasserted, stable

49.7 Special Implementation Notes

49.7.1 1. Inverted Logic Explanation

The seemingly counterintuitive reset condition logic serves a specific purpose: - During reset: shift register accumulates 1's - After reset release: shift register is cleared, then accumulates 1's again - This ensures N clock cycles of delay before output goes high

49.7.22. Metastability Resolution

The N-stage design provides multiple clock cycles for metastability resolution: - Each flip-flop stage resolves potential metastability - Probability of metastability propagation: $(MTBF)^{(-N)}$ - N=2 typically sufficient, N=3 provides extra margin

49.7.33. Initialization

```
logic [N-1:0] r_sync_reg = {N{1'b0}};
```

Explicit initialization ensures predictable power-on behavior.

49.7.44. Parameterization Benefits

Configurable N allows optimization for different requirements: - N=2: Minimum recommended, fast reset release - N=3: Standard choice, good metastability margin - N≥4: Conservative, slower but very robust

49.8 Parameter Selection Guidelines

49.8.1 Choosing N Value

49.8.1.1 $N=2$ (Minimum)

```
reset_sync #(.N(2)) rst_sync_inst (  
    .clk(sys_clk),  
    .rst_n(external_rst_n),  
    .sync_rst_n(internal_rst_n)  
);
```

- **Use when:** Fast reset release required
- **Risk:** Minimal metastability margin
- **Applications:** High-speed designs with good clock quality

49.8.1.2 $N=3$ (Recommended)

```
reset_sync #(.N(3)) rst_sync_inst (  
    .clk(sys_clk),  
    .rst_n(external_rst_n),  
    .sync_rst_n(internal_rst_n)  
);
```

- **Use when:** Standard applications
- **Benefit:** Good balance of speed vs. robustness
- **Applications:** Most digital designs

49.8.1.3 $N \geq 4$ (Conservative)

```
reset_sync #(.N(4)) rst_sync_inst (  
    .clk(sys_clk),  
    .rst_n(external_rst_n),  
    .sync_rst_n(internal_rst_n)  
);
```

- **Use when:** Critical applications, poor clock quality
- **Benefit:** Maximum metastability protection
- **Trade-off:** Slower reset release

49.9 Applications

49.9.1 Global Reset Distribution

```
// Synchronize external reset for internal use
reset_sync #(.N(3)) global_rst_sync (
    .clk(system_clock),
    .rst_n(external_reset_n),
    .sync_rst_n(global_reset_n)
);

// Use synchronized reset throughout design
always_ff @(posedge system_clock or negedge global_reset_n) begin
    if (!global_reset_n) begin
        // Reset state
    end else begin
        // Normal operation
    end
end
```

49.9.2 Clock Domain Crossing

```
// Separate reset synchronizers for each clock domain
reset_sync #(.N(3)) clk_a_rst_sync (
    .clk(clk_a),
    .rst_n(system_rst_n),
    .sync_rst_n(clk_a_rst_n)
);

reset_sync #(.N(3)) clk_b_rst_sync (
    .clk(clk_b),
    .rst_n(system_rst_n),
    .sync_rst_n(clk_b_rst_n)
);
```

49.9.3 Power-On Reset

```
// Synchronize power-on reset
reset_sync #(.N(3)) por_sync (
    .clk(main_clk),
    .rst_n(power_on_rst_n),
    .sync_rst_n(system_rst_n)
);
```

49.9.4 Reset Controller Integration

```
module reset_controller (
    input logic sys_clk,
    input logic external_rst_n,
    input logic watchdog_rst_n,
```

```

    input logic sw_rst,
    output logic sync_rst_n
);

    logic combined_rst_n;

    // Combine all reset sources
    assign combined_rst_n = external_rst_n & watchdog_rst_n & ~sw_rst;

    // Synchronize the combined reset
    reset_sync #(.N(3)) rst_sync_inst (
        .clk(sys_clk),
        .rst_n(combined_rst_n),
        .sync_rst_n(sync_rst_n)
    );

endmodule

```

49.10 Design Considerations

49.10.1 Reset Tree Planning

```

// Plan reset distribution hierarchy
// Level 1: Global reset synchronizer
// Level 2: Domain-specific reset synchronizers
// Level 3: Module-level reset distribution

```

49.10.2 Clock Quality Requirements

- **Clean Clock:** Stable, low-jitter clock required
- **Clock Frequency:** Must be running during reset release
- **Clock Gating:** Avoid gating clock used for reset synchronization

49.10.3 Reset Source Management

```

// Multiple reset sources need careful combination
wire combined_reset = por_n & external_rst_n & watchdog_rst_n &
~soft_reset;

```

49.11 Common Design Mistakes

49.11.1 Incorrect Sensitivity List

```

// WRONG: Missing async reset in sensitivity list
always_ff @(posedge clk) begin
    if (!rst_n) r_sync_reg <= '0;
    else r_sync_reg <= {r_sync_reg[N-2:0], 1'b1};
end

```

end

```
// CORRECT: Include async reset  
always_ff @(posedge clk or negedge rst_n) begin  
    // ... correct implementation  
end
```

49.11.2 Wrong Reset Polarity

```
// Ensure consistent reset polarity throughout design  
// This module uses active-low reset (rst_n)
```

49.11.3 Clock Domain Issues

```
// Each clock domain needs its own reset synchronizer  
// Don't share synchronized reset across clock domains
```

49.12 Verification Considerations

49.12.1 Test Scenarios

- Reset assertion during various clock phases
- Reset deassertion timing verification
- Multiple reset assertion/deassertion cycles
- Clock jitter during reset release
- Power-on behavior

49.12.2 Assertions

```
// Verify synchronous deassertion  
property sync_deassert;  
    @(posedge clk)  
    $rose(rst_n) |-> ##N $rose(sync_rst_n);  
endproperty
```

```
// Verify asynchronous assertion  
property async_assert;  
    $fell(rst_n) |-> $fell(sync_rst_n);  
endproperty
```

49.12.3 Coverage Points

- All reset assertion scenarios
- Reset release with different clock phases

- Multiple consecutive reset cycles
- Parameter variation coverage (different N values)

49.13 Test Verification

49.13.1 Test Coverage

The reset_sync module has comprehensive test coverage via: - **Testbench Class:** bin/TBClasses/reset_sync_tb.py - **Test Runner:** val/common/test_reset_sync.py

49.13.2 Test Scenarios

✓ **Basic Reset Synchronization** - Verifies N-cycle synchronization delay ✓ **Immediate Reset Assertion** - Verifies asynchronous assertion behavior ✓ **Multiple Reset Cycles** - Tests repeated reset/release sequences ✓ **Reset Glitch Filtering** - Validates recovery from short reset pulses

49.13.3 Running Tests

```
# Run all reset_sync tests (4 configurations: N=2,3,4,5)
pytest val/common/test_reset_sync.py -v
```

```
# Run specific configuration
pytest val/common/test_reset_sync.py::test_reset_sync[2-min] -v
```

```
# Run with waveform generation
pytest val/common/test_reset_sync.py -v -s
```

49.13.4 Test Results

All 4 parameter configurations passing (100% success rate): - N=2 (min) - ✓ PASSED - N=3 (typical) - ✓ PASSED - N=4 (safe) - ✓ PASSED - N=5 (max) - ✓ PASSED

Bug Discovery: The comprehensive test suite discovered a critical RTL bug during initial test run (inverted reset polarity), demonstrating the value of thorough verification.

49.14 Related Modules

49.14.1 Reset Synchronizer Variants

- **Positive-edge reset:** For active-high reset systems
- **Multi-domain reset:** For complex clock domain systems
- **Reset pulse generator:** For generating reset pulses
- **Reset debouncer:** For mechanical switch inputs

49.14.2 Integration with Other Modules

- Clock domain crossing modules
- PLL/clock management units
- Power management controllers
- System controllers and reset managers

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50 Reverse Vector Module

50.1 Purpose

The `reverse_vector` module performs bit-order reversal on an input vector, swapping the most significant bit with the least significant bit, second MSB with second LSB, and so on. This is a purely combinational operation that reverses the bit ordering of the input data.

50.2 Key Features

- Combinational bit reversal operation
- Parameterizable vector width
- Zero propagation delay (combinational logic only)
- Simple and efficient implementation
- No clock or reset required

50.3 Port Description

50.3.1 Parameters

- **WIDTH:** Width of the input/output vectors (default: 32)

50.3.2 Inputs

Port	Width	Description
<code>vector_in</code>	WIDTH	Input vector to be

Port	Width	Description
		reversed

50.3.3 Outputs

Port	Width	Description
vector_rev	WIDTH	Bit-reversed output vector

50.4 Implementation Details

50.4.1 Bit Reversal Logic

```

always_comb begin
    for (integer i = 0; i < WIDTH; i++) begin
        vector_rev[(WIDTH-1)-i] = vector_in[i];
    end
end

```

The implementation uses a simple for-loop to map each bit from the input to its corresponding reversed position in the output: - Input bit 0 → Output bit (WIDTH-1) - Input bit 1 → Output bit (WIDTH-2) - Input bit i → Output bit (WIDTH-1-i) - Input bit (WIDTH-1) → Output bit 0

50.5 Operation Examples

50.5.1 8-bit Vector Reversal

Input: vector_in = 8'b10110001
Output: vector_rev = 8'b10001101

Bit mapping:

vector_in[7]	= 1	→	vector_rev[0]	= 1
vector_in[6]	= 0	→	vector_rev[1]	= 0
vector_in[5]	= 1	→	vector_rev[2]	= 1
vector_in[4]	= 1	→	vector_rev[3]	= 1
vector_in[3]	= 0	→	vector_rev[4]	= 0
vector_in[2]	= 0	→	vector_rev[5]	= 0
vector_in[1]	= 0	→	vector_rev[6]	= 0
vector_in[0]	= 1	→	vector_rev[7]	= 1

50.5.2 4-bit Vector Reversal

Input: vector_in = 4'b1011
Output: vector_rev = 4'b1101

Bit mapping:

vector_in[3]	= 1	→	vector_rev[0]	= 1
--------------	-----	---	---------------	-----

```
vector_in[2] = 0 → vector_rev[1] = 0
vector_in[1] = 1 → vector_rev[2] = 1
vector_in[0] = 1 → vector_rev[3] = 1
```

50.6 Special Implementation Notes

50.6.1 1. Combinational Operation

- No clock signal required
- No sequential logic elements
- Output changes immediately when input changes
- Zero propagation delay (limited only by gate delays)

50.6.2 2. Parameterizable Width

The module works with any vector width specified by the WIDTH parameter: - Automatically scales to any desired width - Synthesis tools optimize for the specific width - No hardcoded bit indices

50.6.3 3. For-Loop Synthesis

The for-loop is completely unrolled during synthesis, creating individual wire assignments for each bit position. This results in: - Parallel bit assignments (no sequential operation) - Optimal timing characteristics - Efficient hardware implementation

50.6.4 4. Bidirectional Property

Applying the reverse operation twice returns the original vector:

```
original_vector == reverse(reverse(original_vector))
```

50.7 Synthesis Considerations

50.7.1 Hardware Implementation

The synthesized hardware consists of: - Direct wire connections from input bits to output bits - No logic gates required (just routing) - Minimal area overhead - Excellent timing characteristics

50.7.2 Optimization

```
// Synthesis tools will optimize this to simple wire assignments:
assign vector_rev[WIDTH-1] = vector_in[0];
assign vector_rev[WIDTH-2] = vector_in[1];
// ...
assign vector_rev[1] = vector_in[WIDTH-2];
assign vector_rev[0] = vector_in[WIDTH-1];
```


50.8 Applications

50.8.1 Endianness Conversion

```
// Convert between big-endian and little-endian bit ordering  
little_endian_data = reverse_vector(big_endian_data);
```

50.8.2 Serial Communication

```
// Reverse bit order for serial transmission protocols  
// that send LSB first instead of MSB first  
serial_data = reverse_vector(parallel_data);
```

50.8.3 Cryptographic Operations

```
// Bit permutation operations in encryption algorithms  
permuted_key = reverse_vector(encryption_key);
```

50.8.4 Signal Processing

```
// Bit-reversed addressing for FFT algorithms  
reversed_address = reverse_vector(sequential_address);
```

50.8.5 Data Format Conversion

```
// Convert between different data representation formats  
converted_data = reverse_vector(input_format);
```

50.8.6 Mirror Pattern Generation

```
// Create mirror patterns for display or test applications  
mirror_pattern = reverse_vector(original_pattern);
```

50.9 Usage Examples

50.9.1 Instantiation with Different Widths

```
// 16-bit reversal  
reverse_vector #(.WIDTH(16)) rev16 (  
    .vector_in(data_16bit),  
    .vector_rev(reversed_16bit)  
);
```

```
// 64-bit reversal  
reverse_vector #(.WIDTH(64)) rev64 (  
    .vector_in(data_64bit),  
    .vector_rev(reversed_64bit)  
);
```

50.9.2 Pipeline Integration

```
// Use in data processing pipeline  
always_ff @(posedge clk) begin  
    stage1_data <= input_data;  
    stage2_data <= reverse_vector_inst.vector_rev;  
    stage3_data <= process(stage2_data);  
end
```

50.9.3 Conditional Reversal

```
// Conditional bit reversal based on control signal  
assign output_data = reverse_enable ?  
    reverse_vector_inst.vector_rev :  
    vector_in;
```

50.10 Design Considerations

50.10.1 Timing Analysis

- **Propagation Delay:** Minimal (wire delay only)
- **Critical Path:** Direct wire connections
- **Setup/Hold:** Not applicable (combinational)
- **Clock Frequency:** Unlimited (no sequential logic)

50.10.2 Power Consumption

- **Static Power:** Minimal (no storage elements)
- **Dynamic Power:** Only during input transitions
- **Switching Activity:** Proportional to input bit toggles

50.10.3 Area Utilization

- **Logic Elements:** Zero (pure routing)
- **Routing Resources:** WIDTH wire connections
- **Memory:** None required
- **Scalability:** Linear with WIDTH

50.11 Verification Considerations

50.11.1 Test Cases

```
// Test all zeros
vector_in = '0;
assert(vector_rev == '0);

// Test all ones
vector_in = '1;
assert(vector_rev == '1);

// Test alternating pattern
vector_in = 8'b10101010;
assert(vector_rev == 8'b01010101);

// Test double reversal
temp = reverse_vector_inst1.vector_rev;
assert(reverse_vector_inst2.vector_rev == vector_in);
```

50.11.2 Coverage Points

- All bit positions exercised
- Boundary conditions (all 0s, all 1s)
- Random patterns
- Double reversal verification

50.12 Related Modules and Alternatives

50.12.1 Byte Reversal

For byte-level endianness conversion, a separate module might be more appropriate:

```
// Reverse byte order instead of bit order
assign byte_reversed = {data[7:0], data[15:8], data[23:16],
data[31:24]};
```

50.12.2 Configurable Reversal

An enhanced version could include enable control:

```
assign vector_out = enable ? vector_rev : vector_in;
```

50.12.3 Multi-Width Reversal

Support multiple data widths in a single module:

```

generate
  case (WIDTH)
    8: assign vector_rev = {vector_in[0], vector_in[1], ...};
    16: assign vector_rev = {vector_in[0], vector_in[1], ...};
    32: assign vector_rev = {vector_in[0], vector_in[1], ...};
  endcase
endgenerate

```

50.13 Performance Metrics

50.13.1 Speed

- **Combinational Delay:** < 1ns (typical)
- **Maximum Frequency:** Limited by surrounding logic
- **Throughput:** One operation per clock cycle (if clocked)

50.13.2 Area

- **Gate Count:** 0 (wire-only implementation)
- **Equivalent Gates:** WIDTH (for routing complexity)
- **Memory Bits:** 0

50.13.3 Power

- **Leakage:** Negligible
- **Dynamic:** Proportional to toggle rate
- **Peak Power:** During simultaneous bit transitions

50.14 Common Pitfalls and Solutions

50.14.1 Simulation Warnings

```

// Use 'integer' instead of 'int' for compatibility
for (integer i = 0; i < WIDTH; i++) begin

```

50.14.2 Synthesis Issues

```

// Ensure WIDTH is properly parameterized
localparam int W = WIDTH; // Local parameter for clarity

```

50.14.3 Timing Closure

```

// Add pipeline stage if needed for timing
always_ff @(posedge clk) begin

```

```
        vector_rev_reg <= vector_rev;
end
```

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51 Barrel Shifter Module

51.1 Purpose

The `shifter_barrel` module implements a combinational barrel shifter that can perform multiple types of shift operations in a single clock cycle. It supports logical left/right shifts with and without wrap-around, arithmetic right shifts, and no-shift operations, all controlled by a 3-bit control signal.

51.2 Key Features

- Single-cycle shift operations for any shift amount
- Multiple shift modes: logical, arithmetic, and rotational
- Configurable shift amount up to data width
- Wrap-around (rotation) capabilities
- Purely combinational logic for maximum speed

51.3 Port Description

51.3.1 Parameters

- **WIDTH:** Width of the data bus (default: 8)

51.3.2 Inputs

Port	Width	Description
data	WIDTH	Input data to be shifted
ctrl	3	Control signal for shift operation mode
shift_amount	$\$clog2(WIDTH)+1$	Number of positions

Port	Width	Description
		to shift

51.3.3 Outputs

Port	Width	Description
data_out	WIDTH	Shifted output data

51.4 Control Signal Encoding

The 3-bit ctrl signal determines the shift operation:

ctrl	Operation	Description
3'b000	No shift	Output equals input
3'b001	Logical Right Shift (no wrap)	Fill with zeros from left
3'b010	Arithmetic Right Shift	Preserve sign bit (MSB)
3'b011	Logical Right Shift (wrap)	Rotate right
3'b100	Logical Left Shift (no wrap)	Fill with zeros from right
3'b110	Logical Left Shift (wrap)	Rotate left
Others	No shift	Default case

51.5 Implementation Details

51.5.1 Shift Amount Modulation

```
logic [$clog2(WIDTH)-1:0] shift_amount_mod;
assign shift_amount_mod = shift_amount[$clog2(WIDTH)-1:0];
```

The shift amount is automatically reduced modulo WIDTH to prevent out-of-bounds operations.

51.5.2 Wrap-Around Implementation

```
logic [WIDTH*2-1:0] w_data_double;
assign w_data_double = {data, data};
```

```
// Generate lookup values for rotating shifts
genvar i;
```

```

generate
  for (i = 0; i < WIDTH; i++) begin : gen_unrolled_shifts
    assign w_array_rs[i] = w_data_double[WIDTH-1+i:i];      //
    Right shift
    assign w_array_ls[i] = w_data_double[WIDTH*2-1-i:WIDTH-i]; //
    Left shift
  end
endgenerate

```

51.5.3 Pre-computed Rotation Arrays

The module uses two lookup arrays for efficient rotation: - **w_array_rs**[WIDTH](#): Pre-computed right rotation results for all possible shift amounts - **w_array_ls**[WIDTH](#): Pre-computed left rotation results for all possible shift amounts

This approach eliminates the need for complex barrel shifting logic and provides constant-time access to any rotation amount.

51.5.4 Main Shift Logic

```

always_comb begin
  case (ctrl)
    3'b000: // No shift
      data_out = data;

    3'b001: // Logical Right Shift (no wrap)
      data_out = (shift_amount_mod == 0) ? data : data >>
shift_amount;

    3'b010: // Arithmetic Right Shift (preserve sign)
      data_out = $signed(data) >>> shift_amount_mod;

    3'b011: // Logical Right Shift with wrap
      data_out = w_array_rs[shift_amount_mod];

    3'b100: // Logical Left Shift (no wrap)
      data_out = (shift_amount_mod == 0) ? data : data <<
shift_amount;

    3'b110: // Logical Left Shift with wrap
      data_out = w_array_ls[shift_amount_mod];

    default:
      data_out = data;
  endcase
end

```

51.6 Special Implementation Notes

51.6.11. Combinational Design

- All operations complete in zero clock cycles
- No state machines or sequential logic
- Suitable for high-frequency operations

51.6.22. Arithmetic Right Shift

Uses SystemVerilog's `>>>` operator with `$signed()` casting to properly handle sign extension for two's complement numbers.

51.6.33. Zero Shift Optimization

Explicit checks for `shift_amount_mod == 0` optimize the common case of no shifting, avoiding unnecessary computation.

51.6.44. Double-Width Concatenation

The `{data, data}` concatenation creates a seamless circular buffer for rotation operations, enabling simple slice extraction for any rotation amount.

51.6.55. Generate Block Optimization

Pre-computing all possible rotation results at compile time eliminates runtime computation, trading minimal area for maximum speed.

51.7 Timing Examples

51.7.1 8-bit Examples (WIDTH=8)

51.7.1.1 *Logical Right Shift (no wrap)*

Input: `data = 8'b11010110`, `shift_amount = 3`, `ctrl = 3'b001`
Output: `8'b00011010`

51.7.1.2 *Arithmetic Right Shift*

Input: `data = 8'b11010110`, `shift_amount = 3`, `ctrl = 3'b010`
Output: `8'b11111010` (sign extended)

51.7.1.3 *Right Rotation*

Input: `data = 8'b11010110`, `shift_amount = 3`, `ctrl = 3'b011`
Output: `8'b11011010` (bits wrap around)

51.7.1.4 *Logical Left Shift (no wrap)*

Input: data = 8'b11010110, shift_amount = 3, ctrl = 3'b100
Output: 8'b10110000

51.7.1.5 *Left Rotation*

Input: data = 8'b11010110, shift_amount = 3, ctrl = 3'b110
Output: 8'b10110110 (bits wrap around)

51.8 Resource Utilization

51.8.1 Area Considerations

- **Lookup Arrays:** Require $\text{WIDTH} \times \text{WIDTH}$ bits of combinational logic
- **Multiplexers:** Large mux structures for final output selection
- **Total:** Approximately $O(\text{WIDTH}^2)$ area complexity

51.8.2 Speed Characteristics

- **Propagation Delay:** Constant regardless of shift amount
- **Critical Path:** Through lookup array and final output mux
- **Frequency:** Limited by combinational delay, not shift complexity

51.9 Applications

- ALU implementations
- Cryptographic operations
- Signal processing
- Bit manipulation engines
- Network packet processing
- Data alignment circuits

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52 Beat-Packing Shifter

Module: shifter_beat_pack.sv **Location:** rtl/common/ **Status:** Production Ready

52.1 Overview

shifter_beat_pack is a bit-granular packing/aligning shifter. It accepts fixed-width `CHUNK_BITS` entries on a push handshake, accumulates them in a multi-chunk staging register, and drains **configurable-width “beats”** out the low end on a pop handshake. The beat width is a runtime input (bytes minus one), so the same instance can emit different beat sizes across bursts.

All of the shift/load/mux logic lives inside this one module so that callers, such as an aligner that needs to repack DFI cycles into DRAM beats, do not have to invent their own shift-and-compensate scheme. Push and pop can occur in the same cycle; the next-state logic applies the pop first and the push second, and a single non-blocking assignment commits the result so no internal last-assignment race is possible.

52.1.1 Key Features

- **Bit-granular staging:** A `DEPTH_CHUNKS × CHUNK_BITS` register packs chunks and drains beats at bit resolution
- **Runtime beat width:** `cfg_beat_bytes_m1` selects the beat size (bytes – 1) per burst without re-elaboration
- **Independent ingress / egress sizing:** Chunk width and beat cap are separate parameters
- **Same-cycle push + pop:** Pop-then-push ordering with one NBA eliminates internal races
- **Elaboration-time contract checks:** `$error` guards enforce `DEPTH_CHUNKS >= 2` and `MAX_BEAT_BITS < STORAGE_BITS`
- **Status outputs:** `empty` and `count_bits_o` expose occupancy to the wrapper

52.2 Module Purpose

The module solves the recurring problem of repacking a stream of one fixed width into beats of another (often runtime-selected) width. Data arrives one whole chunk at a time and leaves as beats whose width is chosen at run time; the module holds the partial-beat residue between cycles and shifts it down as beats drain, so the caller sees clean valid/ready handshakes on both sides.

Use Cases:

- Packing DFI cycles into DRAM beats in a memory-controller aligner

- Width adaptation where ingress chunk width and egress beat width differ and the beat width is runtime-configurable
- Any packer that must hold partial-beat residue while new fixed-width data lands

Key Benefit: Centralizes the shift, load, and residue-compensation logic behind two simple handshakes, so callers get runtime-variable beat repacking without hand-rolling their own barrel shifter.

52.3 Parameters

Parameter	Type	Default	Description
CHUNK_BITS	int	128	Width of one push entry (ingress granularity)
MAX_BEAT_BYTES	int	16	Maximum beat width in bytes; caps what reaches pop_data
CFG_BITS	int	8	Width of the runtime <code>cfg_beat_bytes_m1</code> field. 8 bits gives <code>beat_bytes</code> $\in$ 1..256; 4 bits gives 1..16, etc.
DEPTH_CHUNKS	int	2	Staging-register depth in chunks. Default 2 = “current + prefetched-next”. Must be ≥ 2 . Larger depths absorb more before draining but widen the pop-side barrel shifter (timing cost).
MAX_BEAT_BITS	int	$\text{MAX_BEAT_BYTES} * 8$	Derived (do not override): maximum beat width in bits
STORAGE_BITS	int	$\text{DEPTH_CHUNKS} * \text{CHUNK_BITS}$	Derived (do not override): total staging-register width

Parameter	Type	Default	Description
COUNT_BITS	int	$\$clog2(STORAGE_BITS + 1)$	Derived (do not override): width of the bit-occupancy counter
IDX_BITS	int	$\$clog2(STORAGE_BITS)$	Derived (do not override): index width for r_data part-selects

52.4 Port Groups

52.4.1 Clock and Reset

Port	Direction	Width	Description
clk	Input	1	Clock
rst_n	Input	1	Active-low reset (via reset_defs.svh macros)

52.4.2 Configuration

Port	Direction	Width	Description
cfg_beat_bytes_m1	Input	CFG_BITS	Beat width in bytes – 1 (0 → 1 byte). Held stable per burst.

52.4.3 Push Interface (ingress)

Port	Direction	Width	Description
push_valid	Input	1	New chunk valid
push_ready	Output	1	Room for another whole chunk (post-push count ≤ STORAGE_BITS)
push_data	Input	CHUNK_BITS	The chunk to pack

52.4.4 Pop Interface (egress)

Port	Direction	Width	Description
pop_valid	Output	1	At least one whole beat is

Port	Direction	Width	Description
pop_ready	Input	1	available
pop_data	Output	MAX_BEAT_BITS	Consumer accepts a beat Low MAX_BEAT_BITS of the staging register; consumer reads only the low (cfg_beat_bytes_m1+1)*8 bits

52.4.5 Status

Port	Direction	Width	Description
empty	Output	1	Staging register holds no bits (r_count == 0)
count_bits_0	Output	COUNT_BITS	Current occupancy in bits

52.5 Functional Description

52.5.1 Runtime Beat Width

The runtime beat width in bits is derived from the byte-minus-one config field, widened by 4 bits so the counter math can represent the largest beat any CFG_BITS-wide value can encode:

```
localparam int BEAT_BITS_W = CFG_BITS + 4;
assign w_beat_bits = ({..., cfg_beat_bytes_m1} + 1) << 3; //
(bytes_m1 + 1) * 8
```

MAX_BEAT_BYTES caps what actually appears on pop_data; w_beat_bits is used only for the occupancy accounting.

52.5.2 Handshake and Status

The combinational handshakes are driven directly from the bit-occupancy counter r_count:

- empty = (r_count == 0)
- push_ready = there is room for another whole chunk, i.e. r_count <= STORAGE_BITS - CHUNK_BITS
- pop_valid = the register holds at least one whole beat, i.e. r_count >= w_beat_bits and non-zero

- `pop_data` = the low `MAX_BEAT_BITS` of `r_data` (a plain slice, well-defined because a beat is always strictly narrower than the staging window)

When the configured beat is narrower than `MAX_BEAT_BITS`, the upper bits of `pop_data` are stale bytes belonging to the next beat; the consumer already knows the width from `cfg` and reads only the low valid bits.

52.5.3 Next-State: Pop First, Push Second

Each cycle the combinational block computes next-state `v_data` / `v_count` in two ordered steps:

1. **Pop** (when `pop_valid` && `pop_ready`): shift the register down by `w_beat_bits` and subtract that many bits from the count, draining the low beat.
2. **Push** (when `push_valid` && `push_ready`): land the new chunk at [`v_count` +: `CHUNK_BITS`] using the *post-pop* occupancy, so the chunk sits right above the still-valid bits whether or not a pop fired, then add `CHUNK_BITS` to the count.

```

if (pop_valid && pop_ready) begin
    v_data = v_data >> w_beat_bits;
    v_count = v_count - COUNT_BITS'(w_beat_bits);
end
if (push_valid && push_ready) begin
    v_data[v_count[IDX_BITS-1:0] +: CHUNK_BITS] = push_data;
    v_count = v_count + COUNT_BITS'(CHUNK_BITS);
end

```

A single non-blocking assignment then writes `r_data` / `r_count` from `v_data` / `v_count`, so same-cycle push + pop has no internal last-assignment race.

52.5.4 Elaboration-Time Contract

Two \$error guards enforce the sizing rules that keep forward progress guaranteed:

- `DEPTH_CHUNKS` $\geq$ 2: a single-chunk register cannot hold partial-beat residue while a new chunk lands.
- `MAX_BEAT_BITS` < `STORAGE_BITS`: a beat must be strictly narrower than the staging window.

52.5.5 Sizing Guidance

Pick `CHUNK_BITS` and `MAX_BEAT_BYTES` so any runtime beat width fits in the 2N (or deeper) storage, i.e. `cfg_beat_bytes` $\times$ 8 $\leq$ `DEPTH_CHUNKS` $\times$ `CHUNK_BITS`. Ingress chunk width and egress beat cap are independent, so callers can right-size both.

52.6 Usage Example

```
// Repack 128-bit DFI cycles into runtime-sized DRAM beats (up to 16
bytes).
shifter_beat_pack #(
    .CHUNK_BITS      (128),
    .MAX_BEAT_BYTES  (16),
    .CFG_BITS        (8),
    .DEPTH_CHUNKS    (2)
) u_beat_pack (
    .clk              (clk),
    .rst_n            (rst_n),
    .cfg_beat_bytes_m1 (beat_bytes - 8'd1),    // e.g. 8 bytes -> 7

    // ingress: one 128-bit chunk per handshake
    .push_valid       (dfi_valid),
    .push_ready       (dfi_ready),
    .push_data        (dfi_cycle),

    // egress: one beat per handshake
    .pop_valid        (beat_valid),
    .pop_ready        (beat_ready),
    .pop_data         (beat_data),              // read low
    (beat_bytes*8) bits

    // status
    .empty            (pack_empty),
    .count_bits_o     (pack_bits)
);
```

52.7 Design Notes

- **Bit occupancy, not entry count.** `r_count` tracks bits, not entries, because chunks land whole but beats drain at a runtime-variable bit width.
- **Pop-then-push ordering is deliberate.** Draining first frees the low bits so the incoming chunk lands at the correct post-pop position, which is what makes same-cycle push + pop safe.
- **pop_data upper bits may be stale.** Only the low $(\text{cfg_beat_bytes_m1}+1)*8$ bits are meaningful; the consumer must mask to the configured width.

- **Depth vs. timing.** Increasing DEPTH_CHUNKS lets the packer absorb more data before draining but widens the pop-side barrel shifter, degrading timing.
- **Reset macros.** The module uses the project `reset_defs.svh` ALWAYS_FF_RST / RST_ASSERTED macros for the registered state.

52.8 Related Modules

52.8.1 Used By

- Memory-controller aligners that repack DFI cycles into DRAM beats

52.8.2 Uses

- None (self-contained; relies only on the `reset_defs.svh` reset macros)

52.8.3 See Also

- [fifo_sync](#) - Whole-entry synchronous FIFO (contrast: this module repacks at bit granularity)

52.9 References

52.9.1 Source Code

- `rtl/common/shifter_beat_pack.sv`

52.9.2 Documentation

- `rtl/common/PRD.md`

Last Updated: 2026-07-15

52.10 Navigation

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53 Fibonacci LFSR Module

53.1 Purpose

The `shifter_lfsr_fibonacci` module implements a Fibonacci Linear Feedback Shift Register, which is a specific type of LFSR where the feedback is applied only to the most significant bit position. This design is complementary to Galois LFSRs and offers different implementation trade-offs.

53.2 Key Features

- Fibonacci (external XOR) LFSR architecture
- Configurable tap positions and polynomial
- Right-shift operation with feedback to MSB
- Seed loading and cycle detection
- Non-zero state enforcement
- Parameterizable width and tap configuration

53.3 Port Description

53.3.1 Parameters

- **WIDTH:** Width of the LFSR register (default: 8)
- **TAP_INDEX_WIDTH:** Width of each tap index (default: 12)
- **TAP_COUNT:** Number of feedback taps (default: 4)
- **TIW:** Shorthand for TAP_INDEX_WIDTH

53.3.2 Inputs

Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
enable	1	Enable LFSR operation
seed_load	1	Load seed value into LFSR
seed_data	WIDTH	Seed value for LFSR initialization
taps	TAP_COUNT*TIW	Concatenated tap positions

53.3.3 Outputs

Port	Width	Description
lfsr_out	WIDTH	Current LFSR value
lfsr_done	1	High when LFSR returns to seed value

53.4 Fibonacci vs Galois LFSR Architecture

53.4.1 Fibonacci LFSR Characteristics

- **Feedback Location:** Single XOR gate feeding MSB
- **Data Path:** Simple right-shift with external feedback
- **XOR Gate Count:** One multi-input XOR gate
- **Critical Path:** Through feedback XOR to MSB



Implementation Comparison

53.5 Implementation Details

53.5.1 Tap Processing (Same as Standard LFSR)

```
// Split concatenated tap positions
always_comb begin
    for (int i = 0; i < TAP_COUNT; i++)
        w_tap_positions[i] = taps[i*TIW+:TIW];
end

// Convert to bit mask
always_comb begin
    w_taps = 'b0;
    for (int i = 0; i < TAP_COUNT; i++)
        if (w_tap_positions[i] > 0)
            w_taps[w_tap_positions[i]-1'b1] = 1'b1;
end
```

53.5.2 Fibonacci Feedback Calculation

```
assign w_feedback = ^(r_lfsr & w_taps);
```

Key Difference: Uses XOR (^) instead of XNOR (~^) compared to the standard LFSR module.

53.5.3 Right-Shift with MSB Feedback

```
always_ff @(posedge clk or negedge rst_n) begin
    if (~rst_n) begin
        r_lfsr <= 'b0; // initialization to all 0's
    end else begin
        if (enable) begin
            if (seed_load) begin
                r_lfsr <= seed_data; // Load seed
            end else if (!r_lfsr) begin // Only shift if we have non-
zero value
                // Fibonacci LFSR: Shift right, feedback to MSB
                r_lfsr <= {w_feedback, r_lfsr[WIDTH-1:1]};
            end
        end
    end
end
```

53.6 Special Implementation Notes

53.6.1 1. XOR vs XNOR Feedback

- **Fibonacci LFSR:** Uses XOR (^) for feedback calculation
- **Standard LFSR:** Uses XNOR (~^) for feedback calculation

- Both produce maximal-length sequences with appropriate polynomials

53.6.22. Non-Zero State Protection

```
end else if (!r_lfsr) begin // Only shift if we have non-zero value
```

Prevents shifting when LFSR contains all zeros, which would lock the sequence in the zero state permanently.

53.6.33. Right-Shift Architecture

```
r_lfsr <= {w_feedback, r_lfsr[WIDTH-1:1]};
```

- Feedback enters at MSB position
- Data shifts right (towards LSB)
- LSB data is discarded each cycle

53.6.44. Polynomial Compatibility

The tap positions should correspond to primitive polynomials appropriate for Fibonacci LFSR implementation. The polynomial form is:

$$P(x) = x^n + x^{(\text{tap1})} + x^{(\text{tap2})} + \dots + x^{(\text{tapk})} + 1$$

53.6.55. Reset to All Zeros

Unlike some LFSR implementations that reset to all ones, this module resets to all zeros and relies on seed loading for proper initialization.

53.7 Timing Example (4-bit Fibonacci LFSR)

53.7.1 Configuration

- WIDTH = 4
- Polynomial: $x^4 + x^3 + 1$ (taps at positions 4,3)
- Seed: 4'b0001

53.7.2 Sequence Generation

Cycle	LFSR	Tap Bits	Feedback	Next LFSR
0	0001	00	0	0000 (shift right)
1	0000	00	0	0000 (locked - need seed)

After seed reload with 4'b1001:

0	1001	10	$1 \oplus 0 = 1$	1100
---	------	----	------------------	------

1		1100		11		$1^1=0$		0110
2		0110		01		$0^1=1$		1011
3		1011		10		$1^0=1$		1101
4		1101		11		$1^1=0$		0110
...								

53.8 Comparison with Galois LFSR

53.8.1 Advantages of Fibonacci LFSR

- **Simpler Data Path:** Only one shift register with external feedback
- **Easy to Understand:** Classical LFSR textbook implementation
- **Single XOR Gate:** All feedback logic concentrated in one location

53.8.2 Disadvantages of Fibonacci LFSR

- **Fan-out:** Feedback XOR gate may have high fan-out for many taps
- **Critical Path:** All tapped bits must route to single XOR gate
- **Timing:** May be slower than distributed Galois implementation

53.8.3 When to Use Fibonacci LFSR

- **Educational Purposes:** Easier to understand and debug
- **Few Taps:** When polynomial has only 2-3 taps
- **Area Constraints:** When minimizing XOR gate count is critical
- **Legacy Compatibility:** When interfacing with existing Fibonacci LFSR systems

53.9 Applications

- Pseudo-random sequence generation (identical to Galois LFSR output)
- CRC calculation (with appropriate polynomial)
- Data scrambling and encryption
- Test pattern generation
- Spread spectrum communications

- Error detection and correction

53.10 Polynomial Examples for Fibonacci LFSR

53.10.1 Common Primitive Polynomials

Width	Polynomial	Tap Positions	Period
3	x^3+x^2+1	[3,2]	7
4	x^4+x^3+1	[4,3]	15
5	x^5+x^3+1	[5,3]	31
8	$x^8+x^6+x^5+x^4+1$	[8,6,5,4]	255
16	$x^{16}+x^{15}+x^{13}+x^4+1$	[16,15,13,4]	65535

53.10.2 Usage Note

The same polynomial can be used for both Fibonacci and Galois LFSRs, but they will produce different (but equivalent) pseudo-random sequences.

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54 Galois LFSR Module

54.1 Purpose

The `shifter_lfsr_galois` module implements a Galois Linear Feedback Shift Register, which features distributed XOR gates at tap positions rather than a single feedback point. This architecture offers better timing characteristics and higher operating frequencies compared to Fibonacci LFSRs.

54.2 Key Features

- Galois (internal XOR) LFSR architecture
- Distributed feedback XOR gates at tap positions
- Right-shift operation with LSB feedback
- Superior timing characteristics for high-speed operation

- Configurable tap positions and polynomial
- Seed loading and cycle detection

54.3 Port Description

54.3.1 Parameters

- **WIDTH:** Width of the LFSR register (default: 8)
- **TAP_INDEX_WIDTH:** Width of each tap index (default: 12)
- **TAP_COUNT:** Number of feedback taps (default: 4)
- **TIW:** Shorthand for TAP_INDEX_WIDTH

54.3.2 Inputs

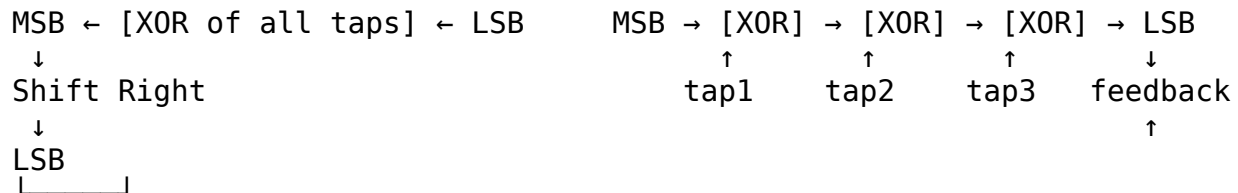
Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
enable	1	Enable LFSR operation
seed_load	1	Load seed value into LFSR
seed_data	WIDTH	Seed value for LFSR initialization
taps	TAP_COUNT*TIW	Concatenated tap positions

54.3.3 Outputs

Port	Width	Description
lfsr_out	WIDTH	Current LFSR value
lfsr_done	1	High when LFSR returns to seed value

54.4.1 Key Architectural Differences

Galois LFSR:



- **Distributed Logic:** XOR gates spread throughout the shift register
- **Better Timing:** Shorter critical paths between flip-flops
- **Higher Frequency:** Can operate at higher clock speeds
- **Localized Routing:** Reduced interconnect complexity

54.5.1 Tap Position Processing

```
// Split concatenated tap positions
always_comb begin
    for (int i = 0; i < TAP_COUNT; i++) begin
        w_tap_positions[i] = taps[i*TIW+:TIW];
    end
end
```

```
assign w_feedback = r_lfsr[0];
```

The feedback signal comes directly from the LSB (bit 0) of the current LFSR state.

```
always_comb begin
    // Start with right shift (include 0 in MSB)
    next_lfsr = {1'b0, r_lfsr[WIDTH-1:1]};

    // Apply Galois feedback taps if LSB is 1
    if (w_feedback) begin
        for (int j = 0; j < TAP_COUNT; j++) begin
            if (w_tap_positions[j] > 0 && w_tap_positions[j] <= WIDTH)

```



```

begin
    // Apply XOR to the tap positions
    next_lfsr[w_tap_positions[j]-1] =
next_lfsr[w_tap_positions[j]-1] ^ 1'b1;
    end
end
end
end

```

54.5.4 Sequential Update

```

always_ff @(posedge clk or negedge rst_n) begin
    if (~rst_n) begin
        r_lfsr <= {WIDTH{1'b1}}; // initialization to all 1's
    end else if (enable) begin
        if (seed_load) begin
            r_lfsr <= seed_data;
        end else begin
            r_lfsr <= next_lfsr;
        end
    end
end
end

```

54.6 Special Implementation Notes

54.6.1 1. Reset to All Ones

```
r_lfsr <= {WIDTH{1'b1}}; // initialization to all 1's
```

Unlike the Fibonacci version, this module resets to all ones, which is a valid non-zero state for immediate operation.

54.6.22. Combinational Next State Logic

The next LFSR state is calculated combinatorially in the `always_comb` block, then registered in the `always_ff` block. This separates timing concerns from logic complexity.

54.6.33. Conditional XOR Application

```

if (w_feedback) begin
    // Only apply tap XORs when LSB is 1

```

XOR operations at tap positions only occur when the LSB feedback bit is 1, implementing the proper Galois LFSR behavior.

54.6.44. Bounds Checking

```
if (w_tap_positions[j] > 0 && w_tap_positions[j] <= WIDTH) begin
```

Explicit bounds checking prevents out-of-range array access and handles unused tap positions (value 0).

54.6.55. Right-Shift with Distributed Feedback

The basic operation is a right shift, but with XOR gates inserted at tap positions that conditionally invert bits based on the LSB feedback.

54.7 Timing Example (4-bit Galois LFSR)

54.7.1 Configuration

- WIDTH = 4
- Polynomial: $x^4 + x^3 + 1$ (taps at positions 4,3)
- Seed: 4'b1001

54.7.2 Sequence Generation

Cycle	LFSR	LSB	Shift	XOR@pos4	XOR@pos3	Next LFSR
0	1001	1	0100	$0100 \wedge 1$	$0000 \wedge 1$	1110
1	1110	0	0111	no XOR	no XOR	0111
2	0111	1	0011	$0011 \wedge 1$	$0001 \wedge 1$	1010
3	1010	0	0101	no XOR	no XOR	0101
4	0101	1	0010	$0010 \wedge 1$	$0000 \wedge 1$	1011
...						

54.8 Performance Characteristics

54.8.1 Critical Path Analysis

Fibonacci LFSR Critical Path:

All tapped bits → Multi-input XOR → MSB flip-flop

Galois LFSR Critical Path:

Previous bit → XOR gate → Current bit flip-flop

54.8.2 Timing Advantages

- **Shorter Logic Depth:** Each bit only sees local XOR gate
- **Parallel Processing:** All XOR operations occur simultaneously
- **Reduced Fan-out:** No single XOR gate with multiple inputs
- **Better Scaling:** Critical path doesn't grow with tap count

54.9 Polynomial Considerations

54.9.1 Tap Position Mapping

For a polynomial $P(x) = x^n + x^a + x^b + \dots + 1$: - Galois LFSR taps: [n, a, b, ...] - Same polynomial works for both Fibonacci and Galois - Different sequences produced but same mathematical properties

54.9.2 Example Polynomials

Width	Polynomial	Galois Taps	Max Period
4	$x^4 + x^3 + 1$	[4,3]	15
8	$x^8 + x^6 + x^5 + x^4 + 1$	[8,6,5,4]	255
16	$x^{16} + x^{15} + x^{13} + x^4 + 1$	[16,15,13,4]	65535
32	$x^{32} + x^{22} + x^2 + x + 1$	[32,22,2,1]	$2^{32}-1$

54.10 Design Trade-offs

54.10.1 Galois LFSR Advantages

- **Speed:** Higher maximum clock frequency
- **Timing:** Better setup/hold characteristics
- **Scalability:** Performance doesn't degrade with tap count
- **Power:** Potentially lower dynamic power

54.10.2 Galois LFSR Disadvantages

- **Complexity:** More complex combinational logic
- **Area:** May require more logic gates for many taps
- **Debug:** Harder to trace intermediate states
- **Understanding:** Less intuitive than classical Fibonacci

54.11 Applications

54.11.1 High-Speed Systems

// Use Galois LFSR for high-frequency applications
// Clock rates > 500MHz where timing is critical

54.11.2 Communications

// Spread spectrum systems requiring fast sequence generation
// Real-time scrambling/descrambling applications

54.11.3 Cryptography

// Stream cipher implementations
// Key sequence generation

54.11.4 Testing

// High-speed BIST pattern generation
// Fast pseudo-random test vectors

54.12 When to Choose Galois over Fibonacci

54.12.1 Use Galois LFSR When:

- High clock frequency is required (>100MHz)
- Timing closure is challenging
- Multiple taps are needed (>3 taps)
- Power consumption is critical
- Synthesis tools favor distributed logic

54.12.2 Use Fibonacci LFSR When:

- Educational/demonstration purposes
- Legacy system compatibility required
- Very few taps (2 taps or less)
- Area is more critical than speed
- Debugging capabilities are important

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55 LFSR (Linear Feedback Shift Register) Module

55.1 Purpose

The `shifter_lfsr` module implements a generic Linear Feedback Shift Register (LFSR) with configurable tap positions for pseudo-random number generation. It supports various LFSR polynomials and provides cycle completion detection for deterministic sequence generation.

55.2 Key Features

- Configurable tap positions for different polynomial implementations
- Seed loading capability for sequence initialization
- Cycle completion detection (sequence wrap-around)
- Enable control for conditional operation
- Support for various LFSR widths and tap configurations

55.3 Port Description

55.3.1 Parameters

- **WIDTH:** Width of the LFSR register (default: 8)
- **TAP_INDEX_WIDTH:** Width of each tap index (default: 12)
- **TAP_COUNT:** Number of feedback taps (default: 4)
- **TIW:** Shorthand for TAP_INDEX_WIDTH

55.3.2 Inputs

Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
enable	1	Enable LFSR operation
seed_load	1	Load seed value into LFSR
seed_data	WIDTH	Seed value for LFSR

Port	Width	Description
taps	TAP_COUNT*TIW	initialization Concatenated tap positions

55.3.3 Outputs

Port	Width	Description
lfsr_out	WIDTH	Current LFSR value
lfsr_done	1	High when LFSR returns to seed value

55.4 LFSR Theory and Implementation

55.4.1 Feedback Polynomial

The LFSR implements a linear feedback shift register based on a primitive polynomial. The feedback is calculated by XORing bits at specified tap positions.

55.4.2 Tap Position Processing

```
// Split concatenated tap positions into separate groups
always_comb begin
    for (int i = 0; i < TAP_COUNT; i++)
        w_tap_positions[i] = taps[i*TIW+:TIW];
end

// Convert tap positions to bit mask
always_comb begin
    w_taps = 'b0;
    for (int i = 0; i < TAP_COUNT; i++)
        if (w_tap_positions[i] > 0)
            w_taps[w_tap_positions[i]-1'b1] = 1'b1;
end
```

55.4.3 Feedback Calculation

```
assign w_feedback = ~(r_lfsr[WIDTH-1:0] & w_taps[WIDTH-1:0]);
```

The feedback uses XNOR reduction (inverted XOR) of tapped bits, which is characteristic of maximal-length LFSR sequences.

55.4.4 Shift Register Operation

```
always_ff @(posedge clk or negedge rst_n) begin
    if (~rst_n) begin
        r_lfsr <= 'b0;
```

```

    end else begin
        if (enable) begin
            if (seed_load)
                r_lfsr <= seed_data;
            else
                r_lfsr <= {r_lfsr[WIDTH-2:0], w_feedback};
            end
        end
    end
end

```

55.5 LFSR Polynomial Reference

The module supports various standard LFSR polynomials. Here are some common configurations:

55.5.1 Popular LFSR Polynomials (XNOR taps)

Width	Tap Positions	Max Period
3	[3,2]	7
4	[4,3]	15
5	[5,3]	31
8	[8,6,5,4]	255
16	[16,15,13,4]	65535
32	[32,22,2,1]	$2^{32}-1$

55.5.2 Example Tap Configuration for 8-bit LFSR

```

// For WIDTH=8, polynomial  $x^8 + x^6 + x^5 + x^4 + 1$ 
// Taps at positions 8,6,5,4 (1-indexed)
parameter TAP_COUNT = 4;
parameter TIW = 4; // 4 bits enough for tap positions up to 8

// Concatenated tap positions: {8,6,5,4}
wire [TAP_COUNT*TIW-1:0] taps = {4'd8, 4'd6, 4'd5, 4'd4};

```

55.6 Special Implementation Notes

55.6.1 1. Tap Position Encoding

- Tap positions are 1-indexed (position 1 = bit 0)
- Zero tap position is ignored (allows unused taps)
- Multiple tap positions concatenated into single bus

55.6.22. XNOR Feedback

- Uses XNOR ($\sim^{\wedge}$) instead of XOR for feedback
- Produces maximal-length sequences for primitive polynomials
- Different from some LFSR implementations that use XOR

55.6.33. Cycle Detection

```
assign lfsr_done = (lfsr_out == seed_data) ? 1'b1 : 1'b0;
```

Detects when LFSR returns to its initial seed value, indicating completion of full pseudo-random sequence.

55.6.44. Left-Shift Architecture

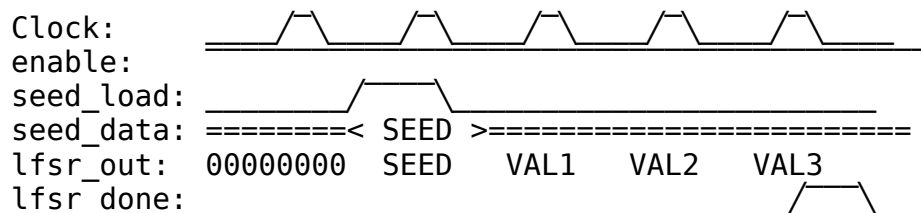
The implementation uses left-shift with feedback to LSB, which is equivalent to traditional right-shift LFSRs but with different bit ordering.

55.6.55. Zero State Handling

- Reset initializes LFSR to all zeros
- Seed loading required for proper operation
- All-zero state is typically avoided in LFSR operation

55.7 Timing Diagrams

55.7.1 Seed Loading and Operation



55.7.2 Sequence Example (3-bit LFSR, taps=[3,2])

Step	LFSR	Feedback	Next
0	001	$\sim(1^{\wedge}0)=1$	011
1	011	$\sim(1^{\wedge}1)=0$	110
2	110	$\sim(0^{\wedge}1)=1$	101
3	101	$\sim(1^{\wedge}0)=1$	011
4	011	$\sim(1^{\wedge}1)=0$	110
...	...	...	...

55.8 Applications

55.8.1 Pseudo-Random Number Generation

```
// Initialize with non-zero seed
seed_data = 8'hA5;
seed_load = 1'b1; // Load seed
// Then enable normal operation
enable = 1'b1;
seed_load = 1'b0;
// lfsr_out provides pseudo-random sequence
```

55.8.2 Built-In Self-Test (BIST)

```
// Use lfsr_done to detect test completion
if (lfsr_done) begin
    // Full test sequence completed
    test_complete = 1'b1;
end
```

55.8.3 Scrambling/Descrambling

```
// XOR data with LFSR output for scrambling
scrambled_data = input_data ^ lfsr_out;
```

55.9 Common Use Cases

- Pseudo-random number generation
- Data scrambling and encryption
- Built-in self-test pattern generation
- CRC calculation
- Spread spectrum communications
- Test vector generation
- Noise generation for audio applications

55.10 Navigation

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56 Universal Shifter Module

56.1 Purpose

The `shifter_universal` module implements a versatile shift register that can perform multiple operations including left shift, right shift, parallel load, and hold operations. It supports both serial and parallel data input/output, making it suitable for a wide range of digital system applications.

56.2 Key Features

- Four distinct operations: hold, right shift, left shift, parallel load
- Bidirectional serial data input/output
- Parallel data load capability
- Serial data capture during shift operations
- Synchronous operation with asynchronous reset
- Configurable width

56.3 Port Description

56.3.1 Parameters

- **WIDTH:** Width of the shift register (default: 4)

56.3.2 Inputs

Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
select	2	Operation select control
i_pdata	WIDTH	Parallel data input
i_sdata_lt	1	Serial data input (left side)
i_sdata_rt	1	Serial data input (right side)

56.3.3 Outputs

Port	Width	Description
o_pdata	WIDTH	Parallel data output
o_sdata_lt	1	Serial data output (left side)
o_sdata_rt	1	Serial data output (right side)

56.4 Operation Control

The 2-bit select signal controls the shifter operation:

select	Operation	Description
2'b00	Hold	Maintain current state
2'b01	Right Shift	Shift data right, input from left
2'b10	Left Shift	Shift data left, input from right
2'b11	Parallel Load	Load parallel input data

56.5 Implementation Details

56.5.1 Combinational Logic for Next State

```
always_comb begin
    // Default values - hold current state
    w_pdata = o_pdata;
    w_sdata_lt = 1'b0;
    w_sdata_rt = 1'b0;

    casez (select)
        2'b00: begin // Hold
            w_pdata = o_pdata;
            w_sdata_lt = 1'b0;
            w_sdata_rt = 1'b0;
        end

        2'b01: begin // Right Shift
            w_pdata = {i_sdata_rt, o_pdata[WIDTH-1:1]};
```

```

        w_sdata_lt = 1'b0;
        w_sdata_rt = o_pdata[0];
    end

    2'b10: begin // Left Shift
        w_pdata = {o_pdata[WIDTH-2:0], i_sdata_lt};
        w_sdata_lt = o_pdata[WIDTH-1];
        w_sdata_rt = 1'b0;
    end

    2'b11: begin // Parallel Load
        w_pdata = i_pdata;
        w_sdata_lt = 1'b0;
        w_sdata_rt = 1'b0;
    end
endcase
end

```

56.5.2 Sequential Logic for State Update

```

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        o_pdata    <= '0;
        o_sdata_lt <= 1'b0;
        o_sdata_rt <= 1'b0;
    end else begin
        o_pdata    <= w_pdata;
        o_sdata_lt <= w_sdata_lt;
        o_sdata_rt <= w_sdata_rt;
    end
end

```

56.6 Operation Details

56.6.1 Hold Operation (select = 2'b00)

- **Function:** Maintains current register state
- **Data Flow:** No data movement
- **Serial Outputs:** Both remain low (no shift occurring)
- **Use Case:** Pause operation while maintaining data integrity

56.6.2 Right Shift Operation (select = 2'b01)

- **Function:** Shifts data toward LSB
- **Data Flow:** i_sdata_rt → MSB, LSB → o_sdata_rt

- **Serial Input:** New data enters from `i_sdata_rt`
- **Serial Output:** Shifted-out data exits via `o_sdata_rt`
- **Use Case:** Serial-to-parallel conversion, delay lines

56.6.3 Left Shift Operation (`select = 2'b10`)

- **Function:** Shifts data toward MSB
- **Data Flow:** `i_sdata_lt` → LSB, MSB → `o_sdata_lt`
- **Serial Input:** New data enters from `i_sdata_lt`
- **Serial Output:** Shifted-out data exits via `o_sdata_lt`
- **Use Case:** Multiplication by 2, bit-serial processing

56.6.4 Parallel Load Operation (`select = 2'b11`)

- **Function:** Loads entire parallel word
- **Data Flow:** `i_pdata` → `o_pdata`
- **Serial Outputs:** Both remain low (no shift occurring)
- **Use Case:** Initialization, data injection

56.7 Special Implementation Notes

56.7.1 1. Bidirectional Serial Interface

The module provides separate serial inputs and outputs for both directions: - **Left side:** `i_sdata_lt` (input), `o_sdata_lt` (output) - **Right side:** `i_sdata_rt` (input), `o_sdata_rt` (output)

This allows chaining multiple shifters or connecting to bidirectional serial systems.

56.7.2.2. Clean Serial Output Management

Serial outputs are explicitly set to zero when not actively shifting in that direction, preventing glitches and providing well-defined states.

56.7.3.3. Asynchronous Reset

Reset immediately clears all outputs to zero, providing deterministic startup conditions.

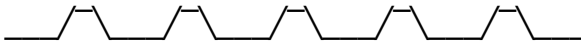
56.7.4.4. Combinational/Sequential Separation

The design cleanly separates: - **Combinational logic**: Calculates next state - **Sequential logic**: Registers the state on clock edge

This approach improves timing predictability and synthesis results.

56.8 Timing Diagrams

56.8.1 Right Shift Operation Example (WIDTH=4)

Clock: 
select: 01 01 01 01 01
i_sdata_rt: 1 0 1 1 0
o_pdata: 0000 1000 0100 1010 1101
o_sdata_rt: 0 0 0 0 1

56.8.2 Parallel Load followed by Left Shift

Clock: 
select: 11 10 10 10
i_pdata: 1010 ---- ---- ----
i_sdata_lt: ---- 1 0 1
o_pdata: 1010 0101 1010 0101
o_sdata_lt: ---- 1 0 1

56.9 Applications

56.9.1 Serial Communication Interface

```
// Receive shift register for UART
// Right shift to collect incoming serial data
select = 2'b01;
i_sdata_rt = uart_rx;
// When 8 bits received, parallel data available
received_byte = o_pdata;
```

56.9.2 Multiplication/Division by Powers of 2

```
// Multiply by 2: left shift
select = 2'b10;
i_sdata_lt = 1'b0; // Fill with zeros

// Divide by 2: right shift
select = 2'b01;
i_sdata_rt = 1'b0; // Fill with zeros (unsigned)
// or i_sdata_rt = o_pdata[WIDTH-1] for arithmetic shift
```

56.9.3 Data Delay Line

```
// Create N-cycle delay using N shifters
// Right shift with data input
select = 2'b01;
i_sdata_rt = data_input;
delayed_data = o_sdata_rt; // N cycles later
```

56.9.4 Pattern Generator

```
// Load pattern then circulate
// 1. Load initial pattern
select = 2'b11;
i_pdata = initial_pattern;

// 2. Circulate pattern
select = 2'b01; // Right shift
i_sdata_rt = o_sdata_rt; // Feedback for circulation
```

56.9.5 Scan Chain for Testing

```
// JTAG-style scan chain
// Load test data
select = 2'b11;
i_pdata = test_vector;

// Shift out for observation
select = 2'b01;
i_sdata_rt = next_scan_in;
scan_out = o_sdata_rt;
```

56.10 Design Considerations

56.10.1 Chaining Multiple Shifters

```
// Connect shifters in series
shifter1.i_sdata_rt = serial_input;
shifter2.i_sdata_rt = shifter1.o_sdata_rt;
shifter3.i_sdata_rt = shifter2.o_sdata_rt;
```

56.10.2 Bidirectional Operation

```
// Support both shift directions
if (shift_direction == RIGHT) begin
    select = 2'b01;
    i_sdata_rt = serial_data;
end else begin
    select = 2'b10;
    i_sdata_lt = serial_data;
end
```

56.10.3 Control Logic Integration

```
// State machine control
case (state)
  LOAD: select = 2'b11;
  SHIFT_RIGHT: select = 2'b01;
  SHIFT_LEFT: select = 2'b10;
  HOLD: select = 2'b00;
endcase
```

56.11 Common Use Cases

- Serial-to-parallel conversion
- Parallel-to-serial conversion
- Data delay and timing adjustment
- Arithmetic operations ($\times 2$, $\div 2$)
- Pattern generation and circulation
- Scan chain testing
- FIFO buffer implementation
- Digital signal processing pipelines

56.12 Navigation

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57 Sort Module RTL Documentation

57.1 Purpose

The Sort module is a pipelined hardware sorting engine that implements the odd-even sort algorithm to sort arrays of configurable size in descending order. The module is designed for high-throughput applications where new data can be accepted every clock cycle, with predictable latency equal to the number of elements being sorted.

57.2 Module Overview

```
module sort #(
    parameter int NUM_VALS = 5,    // Number of values to sort
    parameter int SIZE     = 16    // Size of each value in bits
) (
    input logic clk,              // Clock signal
    input logic rst_n,            // Active-low reset
    input logic [NUM_VALS*SIZE-1:0] data, // Packed input data
    input logic valid_in,         // Start sorting when
    asserted
    output logic [NUM_VALS*SIZE-1:0] sorted, // Packed sorted
    output
    output logic done             // Asserted when
    sorting complete
);
```

57.3 Port Descriptions

57.3.1 Input Ports

Port	Width	Type	Description
clk	1	Clock	System clock signal. All operations are synchronous to the rising edge.
rst_n	1	Reset	Active-low asynchronous reset. Clears all pipeline stages when deasserted.
data	NUM_VALS*SIZE	Data	Packed input array. Elements are packed as data[i*SIZE +: SIZE] for element i.
valid_in	1	Control	When asserted (high) for one cycle, starts the sorting process for the current data input.

57.3.2 Output Ports

Port	Width	Type	Description
sorted	NUM_VALS*SIZE	Data	Packed output array sorted in descending order. Same packing format as input.
done	1	Status	Asserted (high) when the

Port	Width	Type	Description
			sorting operation is complete and sorted output is valid.

57.3.3 Parameters

Parameter	Default	Range	Description
NUM_VALS	5	2-32	Number of values in the array to be sorted. Determines pipeline depth.
SIZE	16	1-64	Bit width of each individual value. All values must be unsigned integers.

57.4 Data Format

Input/Output Packing: Arrays are packed into a single wide bus where: - Element 0 occupies bits [SIZE-1:0] - Element 1 occupies bits [2*SIZE-1:SIZE] - Element i occupies bits [(i+1)*SIZE-1:i*SIZE]

Sort Order: Output is sorted in **descending order** (largest to smallest).

57.5 Algorithm: Odd-Even Sort

57.5.1 Overview

The module implements the **Odd-Even Sort** (also known as **Brick Sort**) algorithm, which is particularly well-suited for pipelined hardware implementation.

57.5.2 Algorithm Description

Odd-Even Sort works by alternating between two types of comparison passes:

1. **Odd Passes:** Compare and swap adjacent pairs at positions (0,1), (2,3), (4,5), etc.
2. **Even Passes:** Compare and swap adjacent pairs at positions (1,2), (3,4), (5,6), etc.

The algorithm requires exactly NUM_VALS passes to guarantee complete sorting, regardless of input data.

57.5.3 Example Execution

For input array [5, 3, 8, 1, 9] (5 elements):

Initial: [5, 3, 8, 1, 9]

Pass 1 (Odd): Compare (0,1), (2,3), (4,X)
[5, 3] → [5, 3] (no swap)
[8, 1] → [8, 1] (no swap)
[9, X] → [9, X] (no comparison)
Result: [5, 3, 8, 1, 9]

Pass 2 (Even): Compare (1,2), (3,4)
[3, 8] → [8, 3] (swap)
[1, 9] → [9, 1] (swap)
Result: [5, 8, 3, 9, 1]

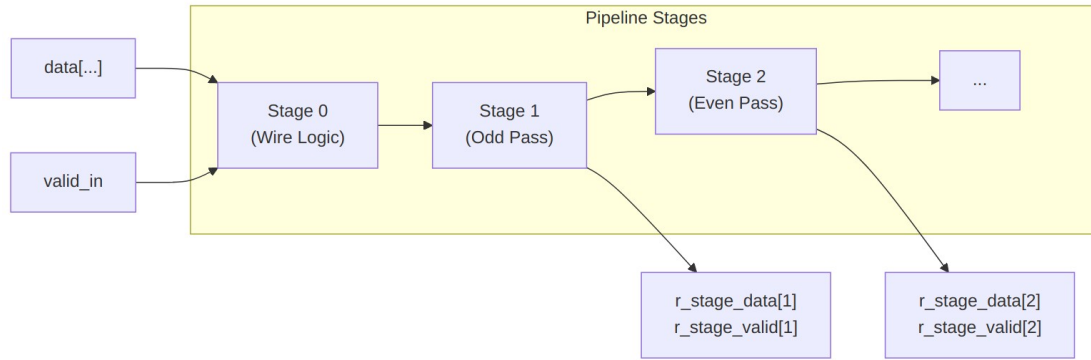
Pass 3 (Odd): Compare (0,1), (2,3), (4,X)
[5, 8] → [8, 5] (swap)
[3, 9] → [9, 3] (swap)
[1, X] → [1, X] (no comparison)
Result: [8, 5, 9, 3, 1]

Pass 4 (Even): Compare (1,2), (3,4)
[5, 9] → [9, 5] (swap)
[3, 1] → [3, 1] (no swap)
Result: [8, 9, 5, 3, 1]

Pass 5 (Odd): Compare (0,1), (2,3), (4,X)
[8, 9] → [9, 8] (swap)
[5, 3] → [5, 3] (no swap)
[1, X] → [1, X] (no comparison)
Final: [9, 8, 5, 3, 1] ✓ Sorted!

57.5.4 Algorithm Comparison

Algorithm	Time Complexity	Space Complexity	Hardware Efficiency	Pipeline Stages	Best Use Case
Odd-Even Sort	$O(n^2)$	$O(1)$	★★★★★ Excellent	n	Small arrays, predictable timing
Bubble Sort	$O(n^2)$	$O(1)$	★★★ Poor	Variable	Sequential processors only



Signal Flow

57.6.2 Naming Conventions

The module follows strict naming conventions for signal clarity:

- **Wire signals** (combinational): Prefixed with `w_`
 - `w_values[stage][element]`: Unpacked arrays for easy manipulation
 - `w_next_values[stage][element]`: Computed next values
 - `w_stage_data_0`: Input stage data wire
 - `w_stage_valid_0`: Input stage valid wire
- **Flopped signals** (sequential): Prefixed with `r_`
 - `r_stage_data[stage]`: Pipeline data registers
 - `r_stage_valid[stage]`: Pipeline valid registers

57.7 Implementation Details

57.7.1 Compare-Swap Logic

Each pipeline stage implements compare-swap operations based on the stage number:

```

// Determine if this is an odd or even pass
localparam bit IS_ODD_PASS = ((stage-1) % 2) == 0;

always_comb begin
    // Default: pass through unchanged
    for (int i = 0; i < NUM_VALS; i++) begin
        w_next_values[stage][i] = w_values[stage-1][i];
    end
end

```

```

if (IS_ODD_PASS) begin
    // Odd pass: compare (0,1), (2,3), (4,5), etc.
    for (int i = 0; i < NUM_VALS-1; i += 2) begin
        if (w_values[stage-1][i] < w_values[stage-1][i+1]) begin
            w_next_values[stage][i] = w_values[stage-1][i+1];
            w_next_values[stage][i+1] = w_values[stage-1][i];
        end
    end
end else begin
    // Even pass: compare (1,2), (3,4), (5,6), etc.
    for (int i = 1; i < NUM_VALS-1; i += 2) begin
        if (w_values[stage-1][i] < w_values[stage-1][i+1]) begin
            w_next_values[stage][i] = w_values[stage-1][i+1];
            w_next_values[stage][i+1] = w_values[stage-1][i];
        end
    end
end
end
end

```

57.7.2 Reset Behavior

- **Asynchronous Reset:** All pipeline stages are cleared when `rst_n` is deasserted
- **Data Reset:** All `r_stage_data` registers are cleared to zero
- **Valid Reset:** All `r_stage_valid` registers are cleared to zero
- **Output:** Both sorted and done outputs go to zero during reset

57.7.3 Timing Characteristics

- **Latency:** `NUM_VALS` clock cycles from `valid_in` assertion to done assertion
- **Throughput:** 1 array per clock cycle (once pipeline is full)
- **Critical Path:** One compare-swap operation (typically 1-2 gate delays)
- **Pipeline Depth:** `NUM_VALS` stages

57.8 State Machines

The Sort module does **not** use explicit Finite State Machines (FSMs). Instead, it uses a **pipeline-based control scheme**:

57.8.1 Control Flow

1. **Input Stage (Combinational):**

- Accepts new data when `valid_in` is asserted
- No state storage at input stage

2. Pipeline Progression (Sequential):

- Valid signal propagates through pipeline automatically
- Each stage processes data independently
- No centralized control or state machine needed

3. Output Indication:

- done signal is simply the valid signal from the final pipeline stage
- No complex output state management required

57.8.2 Valid Signal Flow

Cycle:	1	2	3	4	5	6	
<code>valid_in</code> :	1	0	0	0	0	0	
<code>r_stage_valid[1]</code> :	-		1	0	0	0	0
<code>r_stage_valid[2]</code> :	-		-	1	0	0	0
<code>r_stage_valid[3]</code> :	-		-	-	1	0	0
<code>r_stage_valid[4]</code> :	-		-	-	-	1	0
<code>r_stage_valid[5]</code> :	-		-	-	-	-	1
<code>done</code> :	0	0	0	0	0	1	

57.9 Special Features

57.9.1 1. Parameterizable Design

- **Flexible Array Size:** `NUM_VALS` can be any reasonable value (tested 3-32)
- **Configurable Data Width:** `SIZE` supports 1-64 bits per element
- **Automatic Pipeline Scaling:** Pipeline depth automatically adjusts to `NUM_VALS`

57.9.22. High Throughput Pipeline

- **Back-to-Back Operations:** New data can be accepted every cycle
- **Overlapped Processing:** Multiple arrays can be in the pipeline simultaneously
- **Predictable Performance:** Fixed latency regardless of data patterns

57.9.33. Synthesis-Friendly Design

- **Regular Structure:** All pipeline stages are nearly identical
- **Local Connectivity:** No long combinational paths across stages
- **Inference-Friendly:** Uses standard SystemVerilog constructs for reliable synthesis

57.9.44. Robust Data Handling

- **Packed Buses:** Efficient use of bus resources
- **Unpacked Internal Arrays:** Easy element-wise manipulation
- **Automatic Packing/Unpacking:** Transparent data format conversion

57.10 Verification Strategy

The module comes with a comprehensive CocoTB testbench that includes:

57.10.1 Test Levels

1. **Basic Tests:** Core functionality verification
2. **Random Tests:** Extensive random data patterns
3. **Boundary Tests:** Edge cases and limit conditions
4. **Pipeline Tests:** Throughput and overlapped operation verification
5. **Reset Tests:** Proper reset behavior validation

57.10.2 Test Coverage

- **Functional:** All possible input combinations for small arrays
- **Parametric:** Multiple NUM_VALS and SIZE combinations
- **Temporal:** Pipeline timing and valid signal propagation
- **Stress:** Back-to-back operations and maximum throughput

57.11 Usage Guidelines

57.11.1 Typical Use Cases

1. **Packet Processing:** Sorting priority fields or timestamps

2. **Signal Processing:** Median filtering, statistical operations
3. **Graphics:** Z-buffer sorting, polygon vertex ordering
4. **Control Systems:** Sorting sensor readings or control parameters

57.11.2 Integration Considerations

1. **Clock Domain:** Ensure clk is stable and meets timing requirements
2. **Reset Strategy:** Include proper reset sequencing in system design
3. **Data Formatting:** Pack input arrays correctly according to element ordering
4. **Flow Control:** Monitor done signal for output validity
5. **Resource Planning:** Consider pipeline depth impact on latency-sensitive applications

57.11.3 Performance Optimization

1. **Pipeline Depth:** Use smallest practical NUM_VALS for lowest latency
2. **Data Width:** Use smallest practical SIZE for best area efficiency
3. **Clock Frequency:** Critical path is typically one comparison operation
4. **Resource Sharing:** Multiple instances can share input/output buses if properly arbitrated

57.12 Synthesis Results

Typical synthesis results for common configurations:

Configuration	Pipeline Stages	Critical Path	Area (LUTs)	Frequency
3×8-bit	3	~1.2ns	~150	>800 MHz
5×16-bit	5	~1.5ns	~400	>650 MHz
8×32-bit	8	~2.0ns	~1200	>500 MHz

Results vary by target FPGA family and synthesis tool settings

57.13 Future Enhancements

Potential improvements for future versions:

1. **Bidirectional Sorting:** Parameter to select ascending vs descending order
 2. **Early Termination:** Detect when array is already sorted and terminate early
 3. **Variable Width Elements:** Support for mixed-width data elements
 4. **Priority Encoding:** Include secondary sort keys for equal elements
 5. **Power Optimization:** Clock gating for unused pipeline stages
-

This documentation describes the Sort module implementation. For the latest updates and additional examples, refer to the source code and testbench files.

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58 Pulse Synchronizer for Clock Domain Crossing

A safe pulse synchronizer that transfers single-cycle pulses between asynchronous clock domains using a toggle-based handshake with metastability filtering.

58.1 Overview

The `sync_pulse` module provides metastability-safe pulse synchronization across clock domains. It converts a single-cycle pulse in the source clock domain into a single-cycle pulse in the destination clock domain, handling arbitrary clock frequency ratios. The module uses a toggle-register approach with multi-stage synchronization to guarantee no pulse loss when minimum spacing requirements are met.

58.2 Module Declaration

```
module sync_pulse #(
    parameter int SYNC_STAGES = 3    // Synchronizer depth (range: 2-
4)
) (
    // Source clock domain
    input  logic i_src_clk,           // Source clock
    input  logic i_src_rst_n,        // Source reset (active-low)
    input  logic i_pulse,             // Input pulse (single-cycle high)
```

```

    // Destination clock domain
    input logic i_dst_clk,           // Destination clock
    input logic i_dst_rst_n,        // Destination reset (active-low)
    output logic o_pulse             // Output pulse (single-cycle high)
);

```

58.3 Parameters

58.3.1 User-Settable Parameters

Parameter	Type	Default	Description
SYNC_STAGES	int	3	Number of synchronizer stages for metastability filtering (range: 2-4)

SYNC_STAGES Guidelines: - **2 stages:** Minimum CDC-safe, basic reliability - **3 stages:** Recommended for most applications (>10¹² hours MTBF) - **4 stages:** Ultra-high reliability (aerospace, medical)

58.4 Ports

58.4.1 Source Clock Domain Ports

Port	Direction	Width	Description
i_src_clk	Input	1	Source clock (where input pulse originates)
i_src_rst_n	Input	1	Source domain active-low asynchronous reset
i_pulse	Input	1	Input pulse (must be single-cycle high)

58.4.2 Destination Clock Domain Ports

Port	Direction	Width	Description
i_dst_clk	Input	1	Destination clock (where output pulse appears)
i_dst_rst_n	Input	1	Destination domain active-low asynchronous reset
o_pulse	Output	1	Synchronized output pulse (single-cycle high)

58.5 Functionality

58.5.1 Toggle-Based Pulse Synchronization

The module uses a four-stage process to safely transfer pulses:

1. **Source Toggle:** Input pulse toggles r_src_toggle register
2. **Synchronization:** Toggle synchronized across clock domains via multi-stage synchronizer
3. **Edge Detection:** XOR detects toggle transitions to generate destination pulse
4. **Single-Cycle Output:** Output pulse is exactly one destination clock cycle wide

58.5.2 Timing Characteristics

- **Latency:** (SYNC_STAGES + 2) destination clock cycles
- **Minimum Pulse Spacing:** $3 \times T_{dst} + 2 \times T_{src}$ clock periods
- **MTBF:** $>10^{12}$ hours @ SYNC_STAGES=3, 100MHz

Where: - T_{dst} = Destination clock period - T_{src} = Source clock period

58.6 Implementation Details

58.6.1 Source Domain: Toggle Register

```
(* ASYNC_REG = "TRUE" *) logic r_src_toggle;

always_ff @(posedge i_src_clk or negedge i_src_rst_n) begin
    if (!i_src_rst_n) begin
        r_src_toggle <= 1'b0;
    end else if (i_pulse) begin
        r_src_toggle <= ~r_src_toggle; // Toggle on each pulse
    end
end
```

Key Points: - Toggle persists across clock domains (quasi-static) - Each pulse inverts toggle state (0→1 or 1→0) - Toggle visible to destination synchronizer for multiple cycles

58.6.2 Destination Domain: Multi-Stage Synchronizer

```
(* ASYNC_REG = "TRUE" *) logic [SYNC_STAGES-1:0] r_sync;
```

```

always_ff @(posedge i_dst_clk or negedge i_dst_rst_n) begin
    if (!i_dst_rst_n) begin
        r_sync <= '0;
    end else begin
        r_sync <= {r_sync[SYNC_STAGES-2:0], r_src_toggle}; // Shift
register
    end
end

```

Metastability Filtering: - Stage 0: Captures source toggle (may go metastable) - Stages 1 to N-1: Filter metastability exponentially - Stage N-1: Clean synchronized toggle output

58.6.3 Destination Domain: Edge Detection

```

logic r_sync_prev;

// Delay synchronized toggle by one cycle
always_ff @(posedge i_dst_clk or negedge i_dst_rst_n) begin
    if (!i_dst_rst_n) begin
        r_sync_prev <= 1'b0;
    end else begin
        r_sync_prev <= r_sync[SYNC_STAGES-1];
    end
end

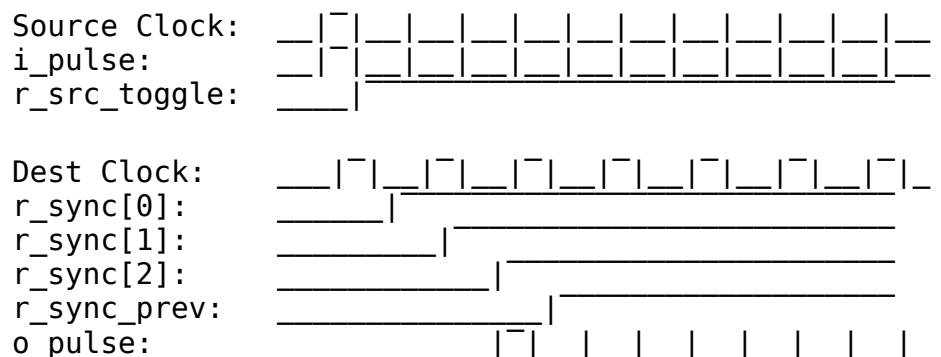
// XOR detects toggle edge (both rising and falling)
assign o_pulse = r_sync[SYNC_STAGES-1] ^ r_sync_prev;

```

Edge Detection Logic: - Compares current toggle with previous toggle - XOR: 0→1 or 1→0 transition generates pulse - Output: Single-cycle high pulse in destination domain

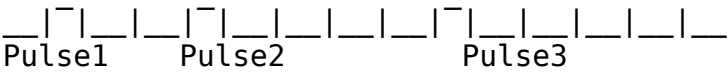
58.7 Protocol Timing

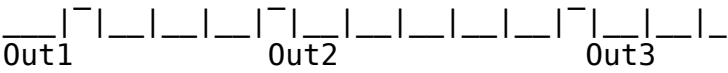
58.7.1 Single Pulse Transfer



Latency: 5 destination clocks (SYNC_STAGES=3 + 2)

58.7.2 Multiple Pulse Transfer

Source: 

Dest: 

Note: Minimum 3 destination clock spacing maintained

58.8 Usage Examples

58.8.1 Basic Pulse Synchronization

// Synchronize button press from slow to fast clock

```
sync_pulse #(
    .SYNC_STAGES(3)
) u_button_sync (
    .i_src_clk    (button_clk),           // Slow clock (debounced button)
    .i_src_rst_n (sys_rst_n),
    .i_pulse      (button_press),         // Single-cycle pulse
    .i_dst_clk    (core_clk),             // Fast core clock
    .i_dst_rst_n  (sys_rst_n),
    .o_pulse      (button_press_sync)
);
```

58.8.2 Interrupt Synchronization

// Synchronize peripheral interrupt to CPU clock

```
sync_pulse #(
    .SYNC_STAGES(3)
) u_irq_sync (
    .i_src_clk    (periph_clk),
    .i_src_rst_n  (periph_rst_n),
    .i_pulse      (periph_irq_pulse),    // Interrupt pulse
    .i_dst_clk    (cpu_clk),
    .i_dst_rst_n  (cpu_rst_n),
    .o_pulse      (cpu_irq)
);
```

// Use synchronized interrupt in CPU

```
always_ff @(posedge cpu_clk) begin
    if (cpu_irq) begin
        // Handle interrupt
        irq_pending <= 1'b1;
    end
end
```

58.8.3 Event Counter Synchronization

```
// Count events across clock domains
logic [15:0] event_count;

sync_pulse #(
    .SYNC_STAGES(3)
) u_event_sync (
    .i_src_clk    (sensor_clk),
    .i_src_rst_n (sys_rst_n),
    .i_pulse     (sensor_event),
    .i_dst_clk   (sys_clk),
    .i_dst_rst_n (sys_rst_n),
    .o_pulse     (event_sync)
);

// Count synchronized events
always_ff @(posedge sys_clk or negedge sys_rst_n) begin
    if (!sys_rst_n) begin
        event_count <= 16'h0;
    end else if (event_sync) begin
        event_count <= event_count + 16'h1;
    end
end
```

58.8.4 With Edge Detector for Multi-Cycle Signals

```
// Convert multi-cycle signal to single-cycle pulse first
logic multi_cycle_sig, single_cycle_pulse;

edge_detect u_edge_det (
    .clk      (src_clk),
    .rst_n    (src_rst_n),
    .sig_in   (multi_cycle_sig),
    .pulse_out(single_cycle_pulse) // Rising edge → pulse
);

// Then synchronize the pulse
sync_pulse #(
    .SYNC_STAGES(3)
) u_sync (
    .i_src_clk    (src_clk),
    .i_src_rst_n  (src_rst_n),
    .i_pulse     (single_cycle_pulse),
    .i_dst_clk   (dst_clk),
    .i_dst_rst_n  (dst_rst_n),
    .o_pulse     (pulse_sync)
);
```

58.9 Design Considerations

58.9.1 When to Use sync_pulse

✓ **Appropriate Use Cases:** - Single-cycle pulse events (interrupts, triggers, strobes) - Event counters across clock domains - Handshake acknowledgments - Sporadic control signals (button presses, sensor events) - Timeout/watchdog pulses

✗ **Inappropriate Use Cases:** - High-frequency continuous pulses (use async FIFO) - Data buses (use handshake or FIFO) - Multi-bit values (use glitch_free_n_dff_arn with Gray code) - Clocks themselves (use dedicated clock crossing primitives)

58.9.2 Minimum Pulse Spacing

To avoid pulse loss, ensure minimum spacing between source pulses:

Fast Source → Slow Destination:

$$\text{Min spacing} = 3 \times T_{\text{dst}} + 2 \times T_{\text{src}}$$

Example: 100MHz src → 25MHz dst

$T_{\text{src}} = 10\text{ns}$, $T_{\text{dst}} = 40\text{ns}$

$$\text{Min spacing} = 3 \times 40\text{ns} + 2 \times 10\text{ns} = 140\text{ns} = 14 \text{ source clocks}$$

Slow Source → Fast Destination:

$$\text{Min spacing} = 3 \times T_{\text{dst}} + 2 \times T_{\text{src}}$$

Example: 10MHz src → 100MHz dst

$T_{\text{src}} = 100\text{ns}$, $T_{\text{dst}} = 10\text{ns}$

$$\text{Min spacing} = 3 \times 10\text{ns} + 2 \times 100\text{ns} = 230\text{ns} = 2.3 \text{ source clocks}$$

(Round up: 3 source clocks minimum)

58.9.3 Reset Synchronization

Both resets should be properly synchronized to their respective clock domains:

// Proper reset synchronization

```
reset_sync u_src_rst_sync (  
    .clk          (i_src_clk),  
    .async_rst_n(global_rst_n),  
    .sync_rst_n  (i_src_rst_n)  
);
```

```
reset_sync u_dst_rst_sync (  
    .clk          (i_dst_clk),  
    .async_rst_n(global_rst_n),  
    .sync_rst_n  (i_dst_rst_n)  
);
```


58.10 Synthesis and Constraints

58.10.1 Synthesis Attributes

The module includes ASYNC_REG attributes to prevent optimization:

```
(* ASYNC_REG = "TRUE" *) logic r_src_toggle;  
(* ASYNC_REG = "TRUE" *) logic [SYNC_STAGES-1:0] r_sync;
```

These attributes: - Prevent register duplication - Ensure synchronizer stages stay together - Help tools recognize CDC paths

58.10.2 Timing Constraints (Xilinx XDC)

```
# Mark clock domain crossing paths as false paths  
set_false_path -from [get_clocks i_src_clk] -to [get_pins -  
hierarchical *r_sync_reg[0]/D]  
  
# Or set maximum delay for CDC analysis  
set_max_delay -from [get_clocks i_src_clk] -to [get_pins -hierarchical  
*r_sync_reg[0]/D] -datapath_only [get_property PERIOD [get_clocks  
i_dst_clk]]  
  
# Set ASYNC_REG property (if not in RTL)  
set_property ASYNC_REG TRUE [get_cells -hierarchical *r_sync_reg*]
```

58.10.3 Timing Constraints (Intel SDC)

```
# Set false path for CDC  
set_false_path -from [get_clocks i_src_clk] -to [get_registers  
*r_sync[0]]  
  
# Mark synchronizer chain  
set_instance_assignment -name SYNCHRONIZER_IDENTIFICATION "FORCED IF  
ASYNCHRONOUS" -to *r_sync*
```

58.11 Verification and Assertions

58.11.1 Formal Assertions (Built-in)

The module includes formal assertions for verification:

```
// Assert: Input pulse is single-cycle  
property p_single_cycle_pulse;  
    @(posedge i_src_clk) disable iff (!i_src_rst_n)  
    i_pulse |=> !i_pulse;  
endproperty  
  
// Assert: Output pulse is single-cycle
```

```

property p_output_single_cycle;
    @(posedge i_dst_clk) disable iff (!i_dst_rst_n)
    o_pulse |=> !o_pulse;
endproperty

```

58.11.2 Simulation Checks

```

// Check input pulse width
always @(posedge i_src_clk) begin
    if (i_pulse) begin
        @(posedge i_src_clk);
        assert (!i_pulse) else $error("Input pulse must be single-
cycle");
    end
end

// Check minimum pulse spacing
int last_pulse_time = 0;
always @(posedge i_dst_clk) begin
    if (o_pulse) begin
        int time_diff = $time - last_pulse_time;
        assert (time_diff >= 3*T_dst)
        else $warning("Pulse spacing too close: %0d ns",
time_diff);
        last_pulse_time = $time;
    end
end

```

58.12 Performance Analysis

58.12.1 Latency Breakdown

For SYNC_STAGES=3: 1. Source toggle register: 1 source clock 2. Synchronizer stages: 3 destination clocks 3. Edge detection: 1 destination clock 4. Output pulse generation: 1 destination clock

Total: 5 destination clocks + 1 source clock

58.12.2 Resource Utilization

Component	FFs	LUTs	Description
r_src_toggle	1	1	Source toggle register
r_sync	SYNC_STAGES	0	Synchronizer chain
r_sync_prev	1	0	Previous toggle storage

Component	FFs	LUTs	Description
o_pulse	0	1	XOR edge detector
Total	SYNC_STAGES +2	2	Minimal footprint

58.13 Common Pitfalls

✗ **Anti-Pattern 1:** Multi-cycle input pulses

```
// WRONG: Input pulse held for multiple cycles
i_pulse <= 1'b1;
#(2*CLK_PERIOD);
i_pulse <= 1'b0;
```

// Result: May generate multiple output pulses

```
// RIGHT: Single-cycle pulse
i_pulse <= 1'b1;
#CLK_PERIOD;
i_pulse <= 1'b0;
```

✗ **Anti-Pattern 2:** Pulses too close together

```
// WRONG: Back-to-back pulses in source domain
forever begin
    i_pulse <= 1'b1;
    #CLK_PERIOD;
    i_pulse <= 1'b0;
    #CLK_PERIOD; // Only 1 clock gap!
end
```

// Result: Pulse loss

```
// RIGHT: Adequate spacing
forever begin
    i_pulse <= 1'b1;
    #CLK_PERIOD;
    i_pulse <= 1'b0;
    #(MIN_SPACING*CLK_PERIOD); // Meet minimum spacing
end
```

✗ **Anti-Pattern 3:** Using for data transfer

```
// WRONG: Trying to transfer data with pulse
logic [7:0] data_to_sync;
```

```

always @(posedge src_clk) begin
    if (send_data) begin
        data_to_sync <= new_data; // Will be corrupted!
        i_pulse <= 1'b1;
    end
end
end

```

// RIGHT: Use handshake or async FIFO for data

58.14 Related Modules

- **glitch_free_n_dff_arn.sv** - Multi-bit signal synchronization (quasi-static)
- **reset_sync.sv** - Specialized reset synchronization
- **edge_detect.sv** - Edge-to-pulse conversion (single clock domain)
- **async_fifo.sv** - For high-throughput data CDC
- **handshake_sync.sv** - Multi-bit data CDC with handshake

58.15 References

- Cummings, Clifford E. “Clock Domain Crossing (CDC) Design & Verification Techniques Using SystemVerilog.” SNUG 2008.
- “Synthesis and Scripting Techniques for Designing Multi-Asynchronous Clock Designs.” Xilinx WP275.
- IEEE 1800-2017 SystemVerilog LRM - Clock Domain Crossing

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